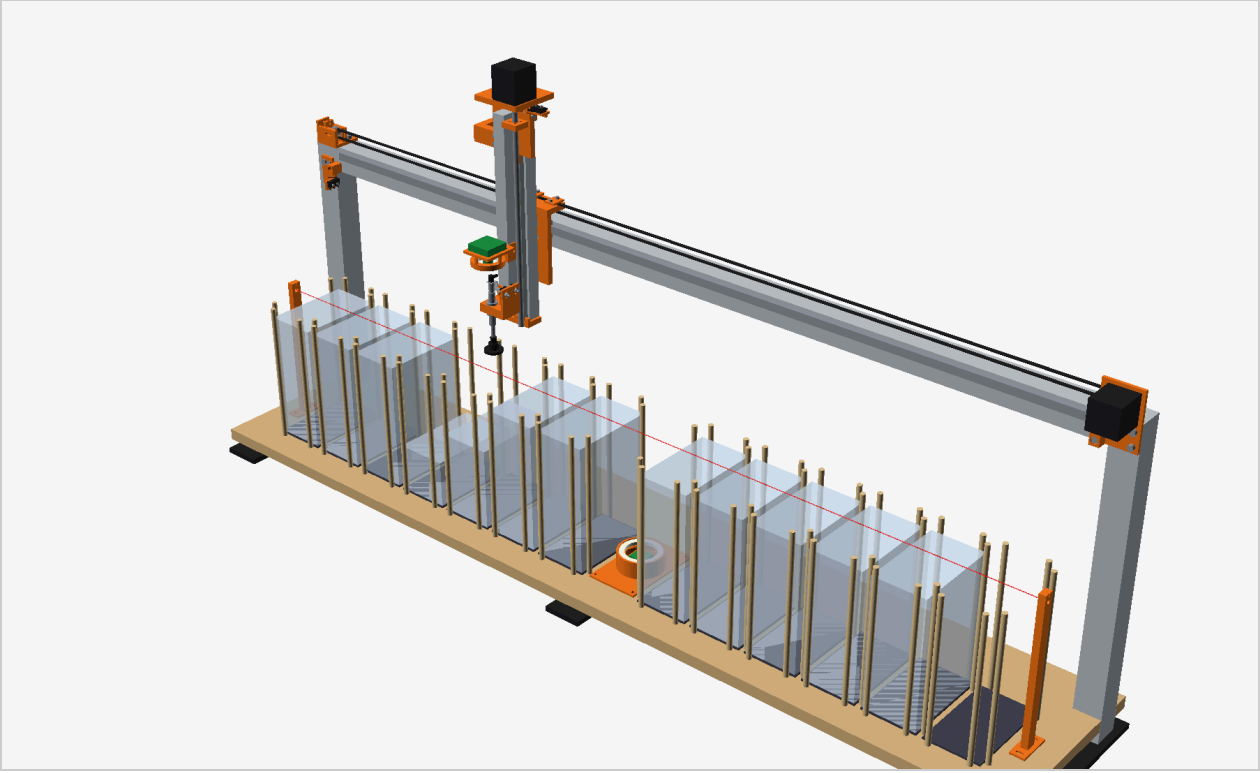
COMPLETE DOCUMENTATION

SlabRail: graded-slab batch scanner

Engineering design package – Rev A, draft for design review



Designed, CAD-checked and simulated. Not yet built.

Every document in the repo, in reading order, plus the drawing set.

2026-10-27 · built from git commit `47e3b4b + uncommitted changes`

https://github.com/Icevyy14/codex/tree/claude/psa-slab-batch-scanner-w2e2m9/psa-slab-scanner

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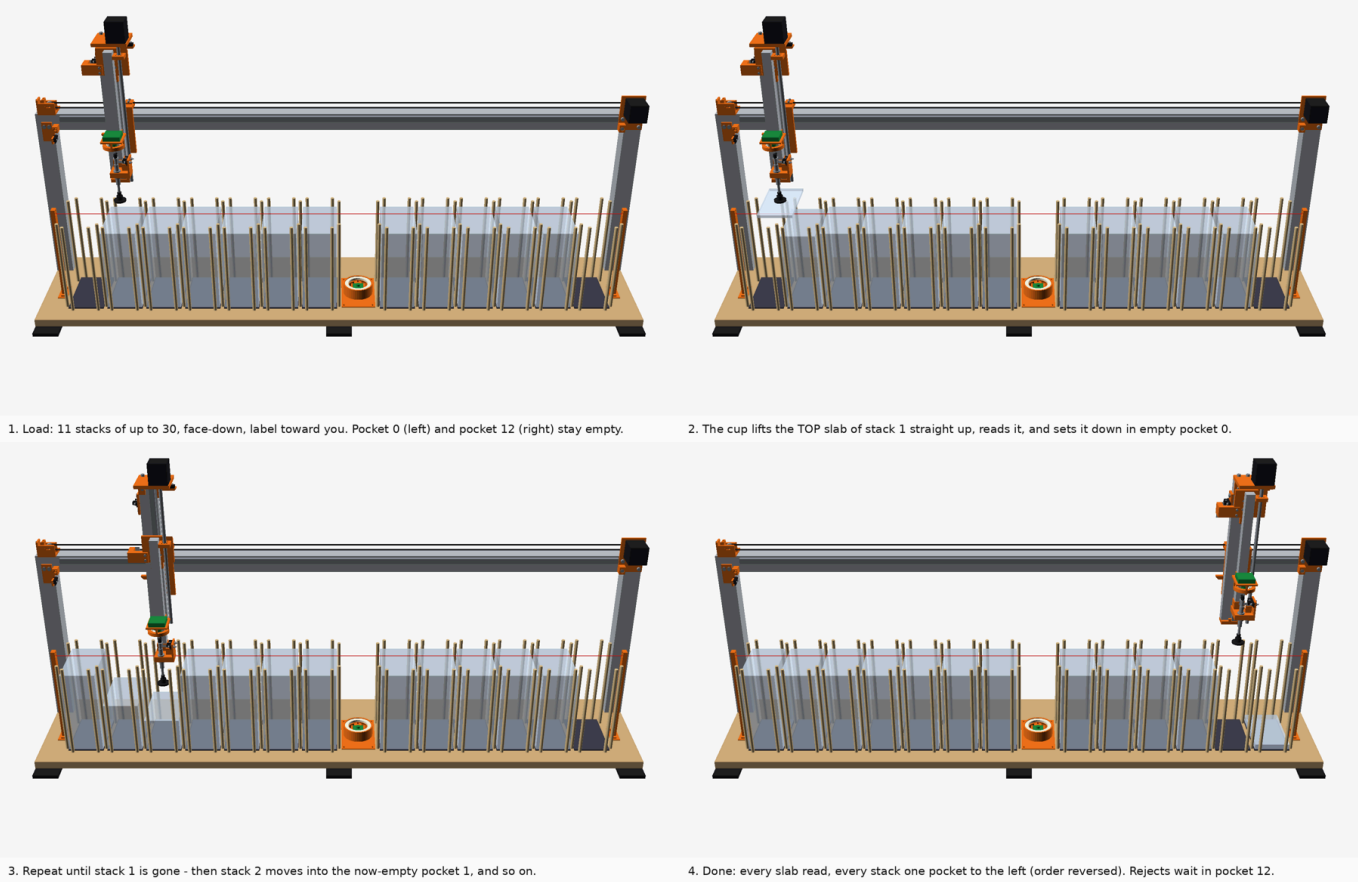
Page numbers are those of this PDF. The drawing sheets are ANSI B landscape and carry their own sheet numbers. Links to files that are not in this PDF open the repository.

SlabRail: unattended PSA / CGC slab cert scanner

Load up to **330 graded slabs** (11 stacks of 30) and walk away. About 80 minutes later (in the simulator: nothing has been built yet) you have `certs.csv` with every cert number, where each slab ended up, and a photo of each label. The machine has **two motors**: one moves the head left and right along a single beam, the other moves it up and down.

For engineers reviewing the design:

- Start with docs/REVIEWER_BRIEF.md: what is checked and what isn't, what is missing, and the review questions.
- `package/`: the review PDF, all the docs in one PDF, and the CAD zip (STEP, GLB, STL, drawing set).
- `web/`: an overview page, an interactive 3D model and a replay of a simulated run. GitHub shows them as source: download a file and open it in a browser.
- `docs/history/`: the design-round reports and the real-label evaluation. They are historical, so their numbers may be out of date.



1. Load: 11 stacks of up to 30, face-down, label toward you. Pocket 0 (left) and pocket 12 (right) stay empty.

2. The cup lifts the TOP slab of stack 1 straight up, reads it, and sets it down in empty pocket 0.

3. Repeat until stack 1 is gone - then stack 2 moves into the now-empty pocket 1, and so on.

4. Done: every slab read, every stack one pocket to the left (order reversed). Rejects wait in pocket 12.

Read this first

- **[Likely] Nothing like this is for sale.** No product at any price takes 200–300 slabs and returns cert numbers unattended. Raw-card sorters exclude slabs, and sheet-fed scanners can't take them. Details are in docs/RESEARCH.md.
- **[Likely] A $350–650 hands-free 2D scanner gets you 80% of the way with no build.** It reads 20–30 slabs per minute by hand, about 10 minutes per 250. This machine buys back those 10–15 minutes per batch and

adds a photo of every slab, which helps catch fake labels and swapped cards. Loading and unloading still take you about 10 minutes either way.

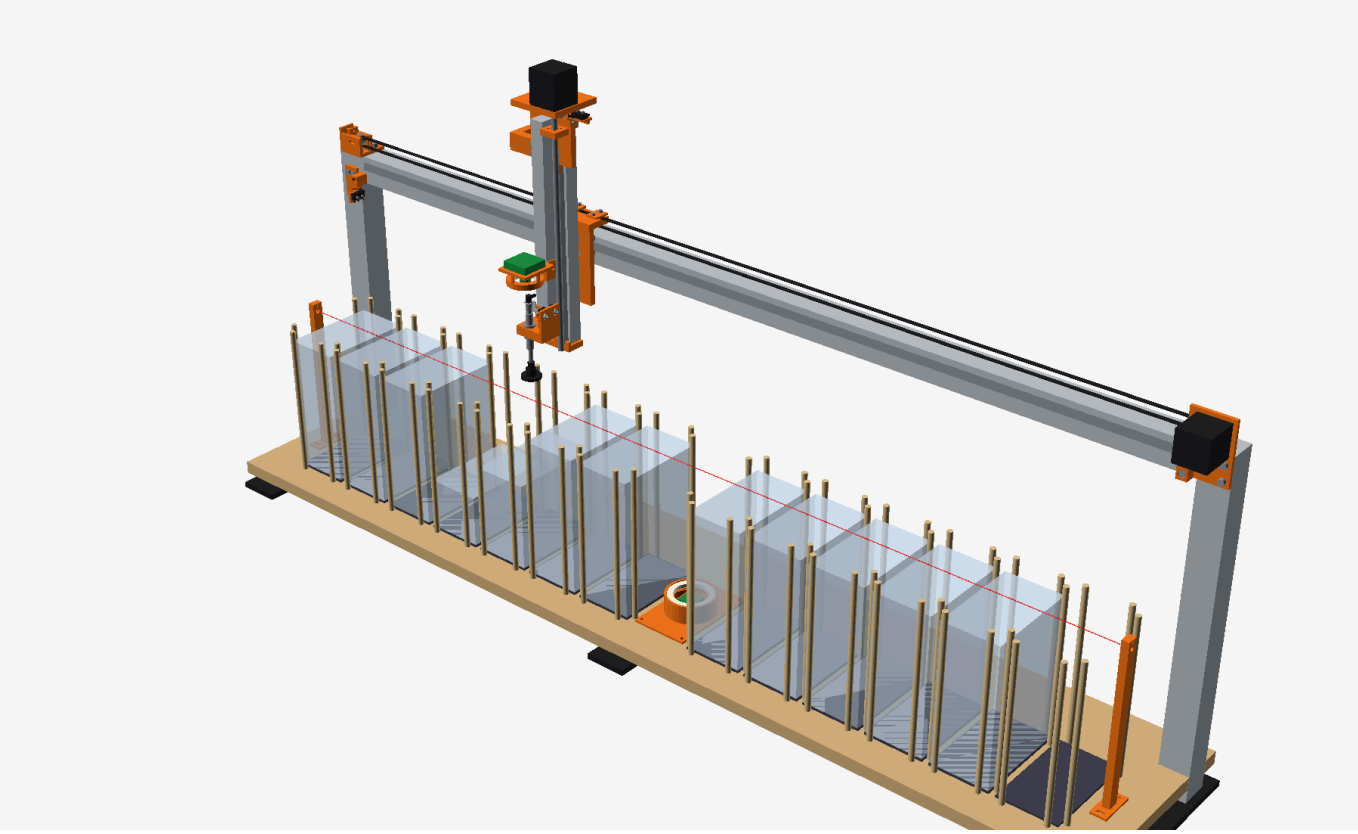
- **[Certain] It has not touched a real slab yet.** The geometry, the motion plan and the decoding were verified in CAD and in a simulator, and the label rules on public scans of 64 real slabs (below). Real cameras through the slab plastic and real cup behaviour still need to be tested. That is why the build starts with **two Phase 0 gates** before you buy any motion parts:
  - **0A, camera test:** can the cameras read *your* labels? About $281 with a borrowed school Raspberry Pi, about $470 if you buy one. About 6–7 h of hands-on work plus about 17 h of printing (BUILD_GUIDE §0.1).
  - **0B, cup test:** does the cup seal on your holders, and does it leave a mark after 24 h? This is the only part that touches your inventory. About $155 with a spare SMC cup (the one shop that lists the exact code shows it back-ordered), a hand vacuum pump on the bench, no mains wiring.
- **[Likely] The budget fits, with headroom.** The full list is about $2,182, and a realistic worst case is about $2,380, inside the $2,500–3,000 you said is fine. With the school's shop stock it drops to about $1,862 (worst case about $2,050). Upgrades (a genuine HIWIN rail, a stronger Z motor, a high-flex USB cable and others) are listed at qty 0: buy one only when the test it names fails. Details in docs/BOM.md; every order with a link per line in docs/SHOPPING_LIST.md.
- **[Certain, code and CAD] Slabs without a back-label code stop the run until the learned check takes over.** Before the cup touches a slab, the head camera must prove it is face-down: a code printed only on the back label (a PSA back QR, a CGC back QR or the CGC 2020–23 back barcode), or a card window that matches two card backs of slabs proven earlier. **The head camera now sees that window** (its lens sits 5 mm toward the label) [Certain, CAD frustum check], so after a Phase 0 GO and the first 5 code-proven slabs the learned check works: in the simulator's default mix 2 of 330 slabs stopped, against 43 with back-label codes only [Certain, sim; Guessing for real card backs through the plastic, which the machine's camera has not seen yet]. Until then PSA 2016 and older labels, CGC Authentic, BGS and SGC stop until you confirm each one, unless you start the run with `--all-face-down` (then a slab loaded face-up gets the cup on its front). CGC 2023+ and 2020–23 slabs no longer stop: their back QR proves them. A CGC slab loaded face-up stops with a "FACE-UP" message.
- **[Certain, CAD and simulator] A Z stall stops the run before the head moves sideways.** A second optical sensor on the Z spine checks that the head really is at carry height before every sideways move. It catches a lift that ended more than about 2.5 mm low; it does not make a weak Z motor strong, so the 100-lift test still has to pass.
- **[Certain, code] A pre-2016 PSA slab is `ok` only when OCR confirms its barcode.** Its only code is one front ITF. An 8-digit ITF whose printed number OCR reads the same names PSA; when OCR reads nothing or other digits, the slab is `needs_review` and stays in the output stack for you to check. Real set: 5 of 7 `ok` on the images as they are, 7 of 7 at machine scale; 20 of 21 in the simulator's default mix. **[Guessing] The rule assumes only PSA prints an 8-digit ITF.** If another grader does, its slab would come out `ok PSA` with the right digits and the wrong grader. Put any BGS or SGC slabs you own through Phase 0 first (docs/OPERATING.md §5.2).
- **Build effort:** a real mechatronics project, roughly 65–90 hours of hands-on work plus printing. Plan to use your school's instructors for the 120 V wiring, crimping and calibration.

How it works

1. You stack slabs **face-down, label toward you**, in pockets 1–11. Pocket 0 (far left) and pocket 12 (far right) stay empty.
2. The suction cup lifts the **top** slab of stack 1 straight up. No slab ever slides on another.
3. A camera on the head reads the back of the label: the QR code on PSA labels from 2017 on and on CGC labels, and the barcode on PSA labels from 2016 on.

4. The cup sets the slab down in pocket 0, and the next slab is now on top. When stack 1 is empty, stack 2 moves into pocket 1, and so on down the row.
5. Slabs whose back can't be read, or that carry only barcodes (no QR), are held over an **upward camera** in the middle of the row, which reads the front label and checks the printed digits with OCR. Anything still unreadable, read only by OCR, or from a grader other than PSA or CGC, goes to **pocket 12** for you to check by hand.

The full explanation, with the numbers, is in docs/DESIGN.md.



Capacity	330 per load (11 × 30 PSA). CGC reads too. BGS (its label has no code) and TAG go to pocket 12, but only load them if they measure ≤ 82.6 mm wide. SGC (≈ 86 mm) does not fit.
Footprint	about 1.44 m × 0.29 m × 0.72 m tall (1436 × 289.5 × 720 mm, the CAD assembly's bounding box over the feet, bed and Z motor; cables and drag chain not modelled). Fits a 6 ft bench.
Moving parts	X belt axis, Z lead-screw axis, a spring-loaded cup, 2 valves, 1 pump
Cycle	about 14 s per slab in the timing simulation: 79 min for a 330-slab load with the default label mix, including homing and the stack survey. Older front-only labels can take up to 30 s more each (OCR on the Pi), and each slab without a back-label code stops the run until the learned check is active, unless you use `--all-face-down`. Real speed will be measured in the acceptance tests.
Touches the slab	cup on the **back** only; edges against wooden dowels; slabs rest on each other as in any stack
Output	`certs.csv` (cert, grader, status, pocket and position), `certs.txt`, `report.html` with label photos
Budget	about **$2,182** expected for everything including a 15% allowance; about **$2,380** worst case (tools, the conditional HT1 cups and light roof, [Guessing] prices 20% high), inside the $2,500–3,000 ceiling. About **$1,862** (worst case about $2,050) if the school supplies filament, wire and common screws. Junk test slabs and the test-triggered upgrades are extra. Details in docs/BOM.md and docs/SHOPPING_LIST.md.

What's in this folder

Path	What it is
`docs/BUILD_GUIDE.md`	Step-by-step build, from Phase 0 to the first supervised batch
`docs/OPERATING.md`	Weekly use: loading rules, running, reading results, maintenance
`docs/BOM.md`, `docs/BOM.csv`	Every part with price and link
`docs/SHOPPING_LIST.md`	The same parts as the three orders you place (Phase 0A, Phase 0B, machine), with links
`docs/WIRING.md`, `docs/img/wiring.svg`	Wiring tables and diagram
`docs/FIRMWARE.md`, `firmware/fluidnc/config.yaml`	Motion controller (FluidNC) setup
`docs/REVIEWER_BRIEF.md`	For engineers: the review brief and questions
`docs/DESIGN.md`	Why this design, the damage analysis and the calculations
`docs/CONTRACT.md`	The interface contract: layout, axes, heights, pins
`docs/RESEARCH.md`	Existing products, PSA slab facts, the real PSA and CGC label formats, PSA API limits
`cad/`	Parametric CadQuery model. `layout.py` is the master dimensions file.
`cad/out/printed/`	**STL + STEP for every 3D-printed part**, plus `print_list.json`
`cad/out/assembly/slabrail_<pose>.step`, `.glb`	Full assembly in four poses (STEP opens in Fusion 360, Onshape or FreeCAD); `home` and `carry` also as GLB meshes for viewing. See `cad/README.md`.
`cad/out/drawings/`	The dimensioned drawing set (18 sheets, draft for design review)
`cad/out/cut_list.csv`, `cad/out/probe_flag_blank.svg`	Extrusion, dowel and flag-blank cut lengths; the 1:1 drawing of the probe flag
`software/`	Raspberry Pi software: run loop, decoding, simulator, CLI, web page, and `tools/real_eval.py` (the real-label check)
`package/`, `web/`, `docs/history/`	The engineering package; the overview, 3D and replay pages; the design-round history

Verification done so far (no hardware)

- **[Certain] All 177 CAD checks pass** (`cad/checks.py`, re-run on 23 Oct 2026, about 12–15 min; every number is in `cad/out/checks.txt`, so trust that file over this list).
  - Picking the last slab of every pocket with full neighbours clears them by ≥ 14.5 mm.
  - Carried slabs clear the dowels by ≥ 21 mm; a double pick of two thick holders by ≥ 5 mm.
  - A held slab clears every part fixed to the X carriage by ≥ 5 mm, even at the Z top hard stop.
  - The cameras have an unobstructed view of the label, and the head camera of the face-down card window: no part in view at hover over stacks of 1, 15 and 30, or with a slab held.
  - Both LED rings are modelled and seated; the head seat takes rings of 39.5–42.5 mm (every head ring check runs at both ends), and the barb elbow clears the head ring by 3.75 mm at the full stroke.
  - The X home and Z top switches sit on printed mounts: each clicks where the firmware expects (cup X 19, c = 258), and its striker holds the lever 0.6 mm past the click to the hard stop, inside the switch's 1.2 mm overtravel. A lever up to 23 mm from the hinge end fits.
  - The card window is checked on the card plane for PSA and CGC slabs pushed into a pocket corner: PSA keeps it all on the card, CGC has about 1.5% of it on the holder (limit 2%) [Likely].
  - The probe flag trips at 2.0 mm of buffer travel and clears its sensor and guide over the full 20 mm stroke; the buffer is modelled from SMC's catalog.
  - The carry-height sensor trips at c = 246.5 mm: clear at every height up to 242, blocked from 248 to the top stop, its flag 1.05 mm from the sensor.
  - The Phase 0 jig holds the head's foot, buffer and flag exactly where the machine does, relative to the lens.
  - Both carriages stay on their rails at every printed hard stop, and no two parts overlap at any checked pose.

    - **Not checked:** belt slack, cables and the vacuum tube, the real shapes of bought parts (which lever the BOM's switches have, among them), loads or deflection, the sideways push the switch strikers put on the lever, and the frame's corner brackets and plates. A camera tilt over about 1.3° toward the cup puts the cup pad in the card window (1.34° in the CAD check). The software checks a tilt toward the cup on every learned run; it does not see a tilt away from the cup, a roll along the row, or a head part as dark as the pad.
- **[Certain] Software tests:** `299 passed, 1 skipped in 1135.84s (0:18:55)` (full suite, 23 Oct 2026; the skipped test runs the rules on the real image set when it is pointed at it: on its own it gave `1 passed, 61 deselected in 137.29s (0:02:17)`). The simulator **raises a crash on any unsafe move**. Every slab is moved exactly once and read. The tests also cover:
    - the real payload strings of PSA and CGC labels (QR forms, ITF, Code128, the 13-digit CGC cert, the phantom read, which can no longer confirm itself or drop the real number),
    - the grader rule (a one-barcode slab is `ok` only when it is an 8-digit ITF that OCR confirms; a code in no grader's format never confirms one) and the OCR check of two-barcode reads,
    - the camera-aim check over an empty pocket (a tilt toward the cup turns the learned check off), also on a resume, with and without an empty pocket,
    - pre-2016 labels routed to the front camera,
    - double-pick detection, thick holders, overfill and seal failures stopping safely,
    - CGC slabs read, BGS to the exception pocket,
    - unexpected errors with a slab in the cup (the slab is put back or held, never dropped),
    - a Z stall caught before the sideways move, and a stuck or cut carry sensor,
    - the face-down check: a FACE-UP stop on a CGC front, a neighbour's label cropped out, a face-up slab still stops in learned mode, one wrong confirmation or one false code proof does not teach it a card front.
- **[Certain for these images] Real-label check** (`python -m tools.real_eval`, public scans of 64 real slabs, RESEARCH.md §4): **0 wrong certs in an `ok` verdict and 0 false face-down proofs**, both on the images as they are (60 of 64 `ok`; 2 pre-2016 PSA `needs_review` because OCR read other digits on one and nothing on the other, the BGS and SGC unread: their labels have no code) and rescaled and degraded to what the machine's cameras would see (60 of 61 `ok`, the BGS unread). The old rules gave one silent wrong cert on the same images. **Limits:** the rules were written from these same 64 slabs, and flat studio scans have no slab-plastic reflections; Phase 0 on your own slabs is the real test.
- **[Certain, synthetic labels in the real formats] Decode sweep.**
    - The back QR reads through blur and strong glare.
    - The front barcode needs 0.05–0.08 mm per pixel or finer; the station camera gives 0.039.
    - Not a single wrong cert was produced in any test.

Build order

1. **Phase 0, two gates. Order the machine only if both pass.** (BOM order 1 ≈ $626 in all.)
    - **1a. Camera test, ≈ $281 with a borrowed Pi (≈ $470 buying one), 6–7 h of hands-on work plus about 17 h of printing.** Build the camera test jig (`docs/img/phase0_jig.png`) and run `python -m tools.phase0` on 30–50 of your own slabs. Go on only if at least 95% read, 0 come out wrong, and `python -m tools.real_eval --phase0 phase0_results` shows 0 wrong certs and 0 false face-down proofs.
    - **1b. Cup test, ≈ $155 with the spare cup, 3–4 h of hands-on work plus the 24 h test.** Seal, leak and a 24 h marking test with a hand vacuum pump on the bench (BUILD_GUIDE §1.10). Go on only if the slab shows no mark after a dry wipe. If it does, stop and ask: the HT1 fallback cup needs an adapter nobody has sourced yet.
2. Order the rest of the BOM (order 2, ≈ $1,556; docs/SHOPPING_LIST.md). Print the parts in `cad/out/printed/`.

3. Follow `docs/BUILD_GUIDE.md`.
4. Calibrate. Run the acceptance tests on junk slabs, then do a supervised first batch.

SlabRail reviewer brief

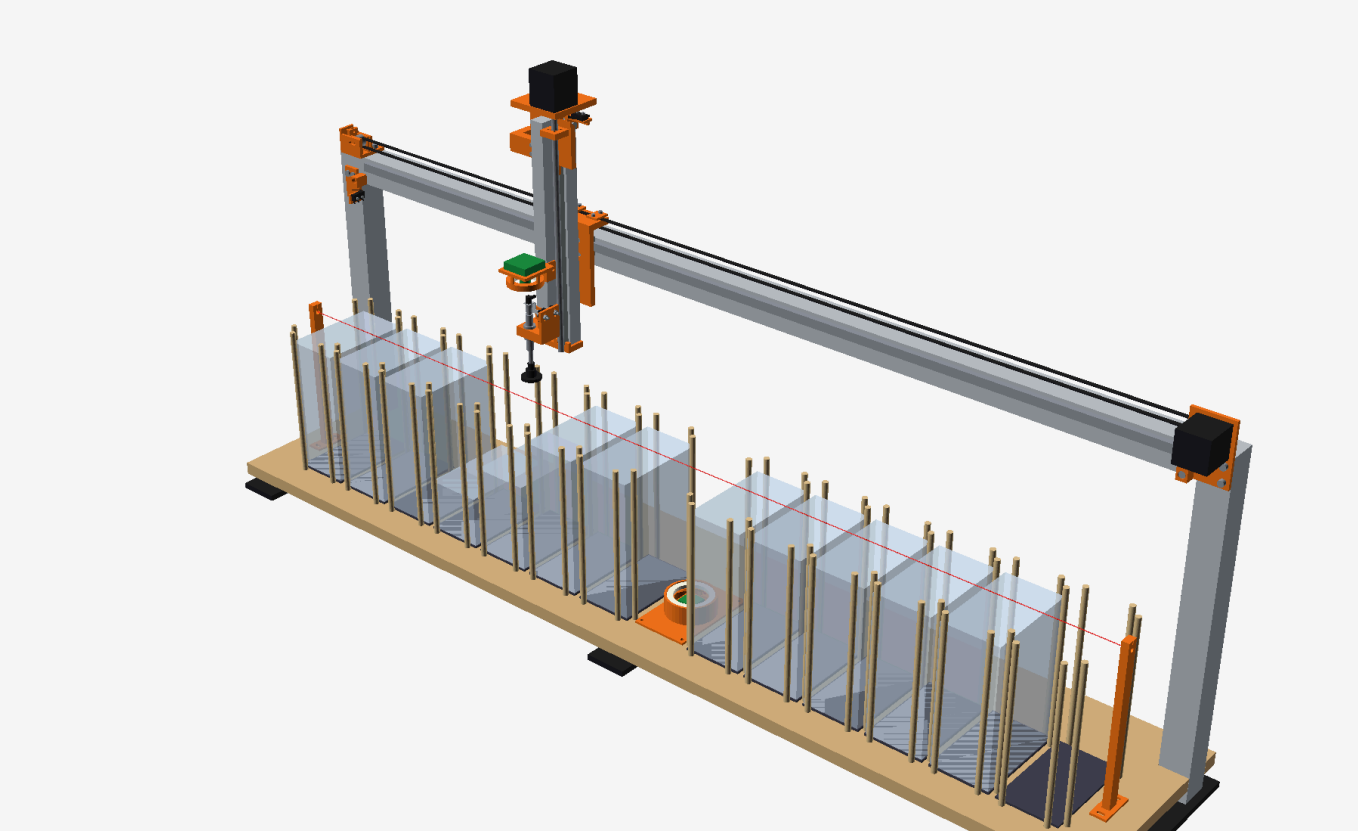
Read this first. It says what SlabRail is, what has been checked and how, where each kind of information lives, what is missing, and what we would like you to review (§9). The files are listed in §10.

Nothing has been built. Every result below comes from CAD checks, software tests, a simulator or public label scans. No real slab, camera, cup, motor or wire has been tested yet.

Confidence tags are copied from the doc each number comes from: **[Certain]** = measured, computed, or checked in the CAD or simulator; **[Likely]** = strong inference; **[Guessing]** = needs a real test. **[Certain] on a calculation or a CAD check means the arithmetic or the geometry is exact for the model and its inputs; it is not a test of the real part**, and many inputs are catalog values or estimates (§8). Units are mm unless stated. "DESIGN §4" is section 4 of `docs/DESIGN.md`; "D-05" is a drawing sheet (§10.2).

1. What it is

SlabRail is designed to read the cert numbers of graded trading-card slabs, PSA and CGC, unattended. One straight row of 14 cells sits under one 4040 gantry beam: 13 pockets formed by wooden dowels, and an upward read camera in cell 7. The owner loads up to 330 slabs face-down in pockets 1–11 (11 stacks of 30). A suction cup lifts the top slab of a stack straight up, a head camera reads the back label, and the cup sets the slab down in the empty pocket to its left, so every stack ends up one pocket left. A slab whose back can't be read is held over the upward camera to read its front label. Anything still unread or doubtful goes to pocket 12, the exception pocket. The output is `certs.csv` (cert, grader, status, pocket and position) and a photo of every label.



The owner is a student building it in a school shop. **[Likely] A $350–650 hands-free 2D scanner gets 80% of the way with no build** (README); the machine buys back about 10–15 minutes per batch and adds a photo of every slab.

2. Requirements as the owner gave them

Requirement	Where written	How the design answers it
Load 200–300+ slabs, walk away, get every cert number back	README; DESIGN §1–2	330 per load, one CSV row per slab, a count check; 79.4 min per load in the simulator. **But** until the learned face-down check is active, every slab without a back-label code stops the run (§5 step 3), and the build guide says not to leave it alone before three clean full runs [Guessing threshold].
PSA first, CGC too	README; CONTRACT	The rules read and stack both. BGS and TAG go to pocket 12 (only if ≤ 82.6 mm wide). SGC (about 86 mm) does not fit the 84 mm pocket.
Fewest moving parts	DESIGN §2	One row, no Y axis: 2 motors, a spring-loaded cup, 2 valves, 1 pump.
Budget about $2,000; $2,500–3,000 acceptable	CONTRACT	$2,181.94 with a 15% allowance [Likely ±10%]; worst case about $2,380 [Guessing] (BOM.md).
Built with school-shop tools	CONTRACT; BUILD_GUIDE	Extrusions bought cut (Misumi, ±0.5 mm); printed parts; rod holes drilled through a printed template that the machine positions (BUILD_GUIDE §5B). 65–90 h of hands-on work [Guessing].

Design rules: the cup touches only a slab's back and no slab slides on another (DESIGN §2–3); a wrong cert is worse than none, so the software never guesses (OPERATING).

3. Status and evidence

The next step is two bench gates that decide whether it gets built:

- **Phase 0A, camera test** (≈ $470, ≈ $281 with a borrowed Pi). Both cameras on a jig that holds the machine's head parts where the machine does, on 30–50 of the owner's slabs. GO: ≥ 95% read, 0 wrong certs, 0 false face-down proofs (BUILD_GUIDE §1.9).
- **Phase 0B, cup test** (≈ $155). The real SMC cup on a hand vacuum pump: seal, leak, and a 24 h marking test. GO: no mark that survives a dry microfiber wipe (BUILD_GUIDE §1.10).

Evidence	What it shows	What it does not show
CAD checks `cad/checks.py`: **177/177 pass** [Certain], 23 Oct 2026; numbers in `cad/out/checks.txt`	Clearance and interference of the model at every pick, place, travel, station, hard-stop and Phase 0 pose (last-slab pick 14.55 from full neighbours, loaded travel 21.4 over the rods, a double pick of two 11.5 holders 5.33); sight lines; switch and sensor trips.	Loads, deflection, belt slack, cables and tube, real shapes of bought parts, switch-lever side load, frame brackets (not in the CAD), print accuracy.
Software tests: **299 passed, 1 skipped** (23 Oct 2026; the skip is the real-image test, which passes alone) [Certain]	Parsing of real PSA and CGC payload strings, grader and face-down rules, carry-sensor logic, the stop paths with a slab in the cup.	Real cameras, GPIO electrics, serial timing. The bench-test tool has run only on the simulator.
Simulator: 330 slabs, seed 7, default mix [Certain, sim]; raises `Crash` on any unsafe move	`ok` 324, `needs_review` 6, no wrong cert or grader in an `ok`. Face-down: 287 by back QR, 41 learned, 2 confirmed. Double pick, seal failure, overfill, Z stall and sensor faults end without a simulated crash.	Real physics (air films, slip, leaks, stalls). The mix is [Guessing]. Every simulated slab is face-down.
Real labels: `tools/real_eval.py` on public scans of 64 real slabs [Certain for these images]; a decode sweep on synthetic labels	**0 wrong certs in an `ok`, 0 false face-down proofs.** Native: 60 `ok`, 2 `needs_review`, 2 `no_read` (BGS and SGC carry no code); at machine-camera scale 60 of 61 `ok`. Back QR reads at ≤ 0.14 mm/px through blur and glare.	The rules were written from these 64 slabs. Flat scans have no slab-plastic reflections.
Hardware	Nothing.	Everything physical.

4. The machine at a glance

Parameter	Value	Source
Overall size	**1436 × 289.5 × 720** (X × Y × Z): X −78 … 1358 over the feet, Y −117.5 … 172 (the bed), Z −49 (feet underside) … 671 (Z motor top); the same in every pose [Certain, CAD bounding box of the assembly STEP]	`cad/out/assembly/*.step`; D-01
Frame	Bed 1416 × 289.5 (18 mm plywood) on two 2040 × 1416 base rails; one portal: two 4040 × 401 uprights on the back base rail and a 4040 × 1416 X beam, centre 400 above the pad; 2040 × 340 Z spine; 2020 × 285 drop arm; joints not in the CAD	`cut_list.csv`, `layout.py`, BUILD_GUIDE §4
Pockets, slab	92 pitch, centres X = 42 + 92·cell; inside 84 × 139: ±1.85 / ±1.75 play on an 80.3 × 135.5 × 6.6 slab (design max 82.6 × 136.5 × 7.0, ≤ 65 g; thick holders 11.5); 82 dowels (5/16 in).	`layout.py`, `dims.py`
Heights	MAX fill line 210, overfill laser 214, rod tops 218; travel c = 248 (empty and loaded); station read c ≈ 171.6	`layout.py`, `machine.toml`
X axis	NEMA17, GT2 20T belt, 80 steps/mm; one MGN12H block on a 1300 rail; soft limits 22… 1253, printed hard stops at cup X 15 and 1262; 15000 mm/min, 2000 mm/s²; 1.0 A	CONTRACT, FIRMWARE
Z axis	NEMA17 (1.5–2.0 A, ≤ 4 mH), integrated T8×8 screw cut to 340; one MGN12H block on a 330 rail; c = −12…250 used, hard stops c = −14 and 262; 6000 mm/min, 1200 mm/s²; run 0.8 A, hold 0.5 A	CONTRACT, FIRMWARE
Drivers	TMC2160, 1/16 microstepping, spreadCycle, no StallGuard, never switched off; `must_home: true`. `r_sense_ohms` ships at 0.022 until the board's resistors are read: the wrong value gives 3.4 × or 0.29 × the set current.	FIRMWARE "Read this first", "What config.yaml sets"
Cup, vacuum	SMC `ZPT25UUK20-B5-A10`: Ø25 urethane cup, M10 spring buffer, 20 mm stroke; probe trips at 2.0 ± 0.5 of travel (about 1.2 N [Likely]). 19.1 N at −50 kPa, SF 30 on a 0.64 N slab; 13.4 N, SF 21 at the −35 kPa seal level (reached within 0.4 s) [Certain, arithmetic; the 78% effective area is an estimate]. Pump, reservoir, valves: BUILD_GUIDE §8.1.	DESIGN §4, `config.py`, BUILD_GUIDE §8.1
Sensors	Fail-safe optical contact probe (Rodent input); carry-height sensor (Pi GPIO 6, blocked from c = 246.5 up); vacuum sensor; overfill laser gate (GPIO 5); X home and Z top NC microswitches; optional lid switch (GPIO 25)	CONTRACT, WIRING §5–8
Cameras	Head: ELP 8 MP USB, 3.6 mm lens, 140 above the held slab, 5 toward the label from the cup axis, 0.055 mm/px. Station: Pi Camera Module 3, 140 below, 0.039 mm/px. The LED rings' mirror image misses the label band, so no polarisers [Certain, mirror geometry; real glare untested until Phase 0A].	CONTRACT, DESIGN §5
Control, power	BTT Rodent (FluidNC v4.1.1+) over USB to a Raspberry Pi 5 4 GB, which switches the pump, valves and rings. LRS-150-24 at 24.5 V behind a fused IEC inlet; E-stop in the +24 V line; fuses 5 / 3 / 1 A; Pi on its own supply.	CONTRACT, WIRING §1, `img/wiring.svg`
Cycle, cost, prints	14.2 s per slab, 79.4 min for 330 [Certain, sim; Guessing until measured]. Parts $1,897.34, **$2,181.94 with 15%** [Likely ±10%]. 25 printed parts (27 prints, 801 cm³, PETG, ≈ 44.5 h).	DESIGN §6, BOM.md, `print_list.json`

5. One slab's trip

Once per run: home (Z up, then X left), refuse to start if the overfill beam is blocked, survey every pocket with the probe (repeated and compared on resume), and in learned mode take one head frame over an empty pocket to check the camera's aim. Then for every slab (`runner.transfer`):

#	Step	Guarded by
1	Move over the source pocket at c = 248.	**Carry-height sensor** must read blocked before any sideways move.
2	Drop to just above the predicted stack top (`G38.3`).	**Contact probe** stops any downward move on contact.
3	One head frame: face-down check, first read of the back.	**Head camera.** Not proven face-down: stop untouched. Front code seen: FACE-UP stop.
4	Probe to the real top, at most 12 past the prediction.	Fail-safe probe (a cut wire reads as touching). > 4.5 taller than predicted stops the run.
5	Grip: seal at −35 kPa within 0.4 s, one retry.	**Carry-height sensor** must read clear; **vacuum sensor**.

#	Step	Guarded by
6	Peel at 2 mm/s through release, 0.5 s low over the stack; lift to c = 248; two head frames.	**Probe hard limit**; a Z stall over about 2.5 stops before any sideways move; vacuum alarm above −25 kPa.
7	If needed, front frames over the station (c ≈ 171.6); decode and OCR over the output stack.	**Station camera**; rules in `slabscan/certs.py`.
8	Probed set-down (1200 mm/min through the rod tops) on the output pocket or pocket 12.	Step ≥ 11.6: DOUBLE PICK; far too big: obstruction; under half a slab: cup empty.
9	Vent 0.25 s; back within 0.5 kPa of atmosphere within 1.5 s; write CSV row, photos, journal.	Failed vent: nothing moves. Count check at the end.

6. Safety and failure handling

No safety standard, safety category or performance level is claimed in any doc, and no risk assessment exists. What the docs have:

Hazard	What the design does	Gap	Source
Mains (120 V)	Fused IEC inlet (T3.15A [Likely]) and rocker in a closed ABS box; 115/230 selector set first; PE to the frame; pre-power checks (PE < 0.5 Ω); instructor check.	Fuse not checked against the LRS inrush. Anodised joints may not carry the earth [Likely].	WIRING §1.1, §12, §14
E-stop	Two latching NC blocks in series in the +24 V line: cuts the Rodent, pump, valves and LED buck, not the Pi, sensors or laser. Motors go limp; Z holds on detent [Likely].	Blocks rated DC-13 1.5 A at 24 V [Certain, BOM.csv]; WIRING §1.3 asks ≥ 5 A at 24 V DC; peak load ≈ 3.4 A [Guessing], capacitive. A weld shows only in the monthly check. **On release the pump and vacuum valve restart by themselves** (the Pi still drives them); motion needs `$H` first.	WIRING §1.3; OPERATING §8.1; BUILD_GUIDE §13.4
Pinch points	Downward moves are probing moves: contact that lifts the cup should stop the head (about 1.2 N at the trip, 1.6 N after a full-speed trip [Likely; never tested against a hand]). A "hands clear?" prompt on resume.	**No guard, enclosure or pinch analysis.** Not sensed: other head parts, the X axis, the open belt and pulleys at the beam ends, the X carriage at its bumpers, the Z carriage at its stops, the drag chain.	BUILD_GUIDE §14; DESIGN §3
Remote start	Start, Stop and Vent on the web page; Ctrl-C twice halts the motors but leaves them powered.	**The web page has no login and can start a run from a phone** [Certain, `web.py`]. The lid switch is a software check between slabs, not a power interlock [Certain, `runner._check_pause`].	OPERATING §3
Laser	≤ 1 mW Class 2 650 nm, 4 above the MAX line; the KY-008 (usually 5 mW, Class 3R [Likely]) excluded by name; labels.	On whenever the Pi is on, E-stop included, at about face height for someone bent over the stacks.	WIRING §8
Held slab, power cut	Both valves normally closed: the cup line stays sealed; the reservoir is on the pump side and does not help.	Hold time is the line's leak; ≥ 60 s is a [Guessing] target.	DESIGN §3; BUILD_GUIDE §13.4
Heat	Motors kept under about 50 °C.	No PETG temperature limit or check.	FIRMWARE step 6

Failure	Detected by	What happens	Designed / tested (bare § = BUILD_GUIDE)
Vacuum loss	Vacuum sensor	Carries the slab back slowly and probes it down. A slab is in the air about 2 s per cycle; a fall from carry height: ≤ about 240 mm, ≈ 0.15 J [Certain, arithmetic]; a corner on a dowel could chip [Guessing].	DESIGN §3 / simulator; BUILD_GUIDE §13.3, §13.4
Double pick	Placement step ≥ 11.6	`needs_review` "DOUBLE PICK" plus an "UNREAD" row; two 11.5 holders clear the rods by 5.33 [Certain, CAD].	DESIGN §2a / simulator; §13.5
Z stall	Carry sensor clear at travel height	Stops before any sideways move; a held slab is lowered onto its own pocket, or held. Does not prevent a stall.	WIRING §6a / simulator; §7.8

Failure	Detected by	What happens	Designed / tested (bare § = BUILD_GUIDE)
Misread, face-up	Consensus rules; hover frame	Doubtful reads go to pocket 12; a slab not proven face-down is not touched.	DESIGN §5 / unit tests, real labels; §13.1, §13.8
Sensor fault, error	Fail-safe probe wiring; the run's guard	FluidNC refuses to probe (`ALARM:4`); the slab is put back, or the pump holds it.	WIRING §6 / simulator; §13.6

7. What a reviewer will look for, and where it is

Item	Where	State
System block diagram	Electrical and control: `docs/img/wiring.svg`, WIRING §1. Vacuum: BUILD_GUIDE §8.1. Software: CONTRACT. Sequence: §5.	No single system diagram.
Moving-mass breakdown	DESIGN §4: CAD roll-up, 0.40 kg head and 1.62 kg X carriage, centre of mass 77 in front of the beam centre; the calculations use 0.75 / 1.9 kg	Heaviest items only, several masses [Guessing]; no full per-part list.
Motor and driver settings	FIRMWARE "What config.yaml sets"; `config.yaml`	Complete on paper; no torque curve for the BOM Z motor.
Power budget	WIRING §1.2	[Guessing] until parts are measured.
Failure modes	§6; DESIGN §3, §7; OPERATING §8.1 (every stop message)	No FMEA with severity or occurrence.
Acceptance criteria	BUILD_GUIDE §1.9, §1.10, §6.7, §7.8, §13.9	Some limits are judgments [Guessing].
Mains, fusing, E-stop	WIRING §1, §12, §14; OPERATING §8.1	See §6.
Guarding, pinch points	Nowhere	**Missing.**
Laser class, eye safety	WIRING §8	Class 2, labelled, no beam stop.
Structural loads	DESIGN §4 (beam sag, twist); BUILD_GUIDE §4.3 (joint plan)	**Missing:** joints, rail moments, drop arm, spine, PETG parts.
Tolerances; cables and tube	D-00 note 6, BUILD_GUIDE §4.1, §4.5; WIRING §11	Several PROPOSED, no stack-up; cables and tube not in the CAD.

8. Key assumptions

Input	Value used	Tag, source	Feeds
Slab mass	≤ 65 g (0.64 N)	`dims.py`	Cup SF, drop
Cup area, lip friction	78% of Ø25 effective; friction coefficient about 0.5	an estimate; a guess (DESIGN §4)	Hold 19.1 / 13.4 / 9.6 N; lateral SF
Buffer spring	SMC: 1.0 N at rest, 3.0 N at full stroke → about 1.2 / 2.4 / 3.0 N at 2 / 14 / 20 mm	[Likely], DESIGN §4	Trip, contact force
Moving masses	0.75 / 1.9 kg assumed (head / X carriage); CAD roll-up 0.40 / 1.62 kg, about 0.53 / 1.94 kg with allowances and a slab	DESIGN §4; allowances [Guessing]	Torque, twist
Screw, rotor	T8×8 at 40% efficiency (50–65% [Likely]); 2 N X friction; 60–80 g·cm² rotor	DESIGN §4; rotor [Guessing]	Torque
Motor torque	Z ≈ 0.15–0.2 N·m at 750 rpm at full current; X ≈ 0.15–0.3 N·m at 375 rpm at the 1.0 A run current	[Likely], DESIGN §4; no curve read	SF ≈ 2–4 (Z), ≈ 4–8 (X)
Detent, back-drive	≈ 0.015 vs ≈ 0.005 N·m	[Likely], DESIGN §4	Z hold unpowered
Beam torsion	J 1.0–2.5 cm⁴, G 26 GPa, both ends fixed	[Guessing on J], DESIGN §4	≈ 0.5 mm twist

Input	Value used	Tag, source	Feeds
Belt tension	15–25 N	[Guessing], BUILD_GUIDE §6.6	PETG loads
Load current	≈ 1.2 A typical, ≈ 3.4 A peak	[Guessing], WIRING §1.2	Fuses, E-stop
Bought parts	Catalog or listing envelopes; head camera 32 mm hole pattern and 20 mm board-to-lens	[Guessing] for the camera, BUILD_GUIDE	Clearances
Printed holes	0.1–0.3 small	[Likely], DESIGN §7	Fits
Label mix, card-window limit	The simulator's default mix; 2% on the holder	[Guessing]; a judgment (DESIGN §6, §7)	Sim results, face-down check

9. What we'd like you to review

A short answer with the question number is enough. Label and face-down risks are left to Phase 0 (DESIGN §7 has the full risk table).

1. **Frame joints.** The CAD has no brackets. BUILD_GUIDE §4.3 proposes [Guessing on the layout] two 4040 L brackets in each beam-to-upright corner, a back plate (none in front: the X rail ends 18 mm from each upright), and uprights held by L brackets bolted M5×30 through the 18 mm plywood into the back base rail. These joints react the head's overhang, and the twist estimate (Q2) assumes they fix the beam ends in torsion; nothing calculates their stiffness. Enough, or end-tap and bolt the uprights? (D-01, D-02)
2. **Beam twist.** ≈ 0.5 mm in Y at the cup at midspan (0.3–0.8 for J 2.5–1.0 cm^4) [Guessing on J]: $\theta = T \cdot L/(4GJ)$, ends fixed, T 1.46 N·m from the CAD centre of mass, G 26 GPa; brackets and rail-block play not included (DESIGN §4). The build check hangs 0.75 kg on the cup over pocket 6 (0.93 N·m [arithmetic: 0.75 kg × g × 127 mm], about 64% of the carriage's own 1.46 N·m), so its reading is not directly the estimate; a rear brace goes on above 1.0 mm (BUILD_GUIDE §6.7). Twist also tilts the head camera in the plane of the aim limits (Q11) and varies along X, while the aim check looks over one pocket (an inference; no doc relates them). Credible? Measure angle? (D-04)
3. **Rail loads.** Each axis runs on one MGN12H block (generic rails; the X listing states no preload). The X block carries the 1.62 kg carriage (CAD; about 1.94 kg with hardware and a slab), centre of mass 77 in front of the beam centre, Z motor top at Z 671. No moment-load, deflection or life check exists. One block enough? (D-04, D-05)
4. **Z torque at 0.8 A.** 0.026 N·m on the T8×8 for 0.75 kg at 1.2 m/s^2 without rotor inertia; with the CAD's 0.53 kg plus inertia, 0.025–0.027 N·m (DESIGN §4). At 0.8 A (40–55% of rating) SF ≈ 2–4 [Likely]; no torque curve. The test passes if a bare-pad probe agrees within 0.1 mm after 100 full-speed lifts; the fixes are 3000 mm/min, then about 60% of rated current (BUILD_GUIDE §7.8). More margin from the start? Is 100 lifts enough for a 330-slab run? (D-05)
5. **Z hold, power off.** Back-drive about 0.005 N·m (0.003–0.005 with the CAD mass) against about 0.015 N·m detent [Likely]. The test hangs 200 g on the cup, presses the E-stop and watches 1 minute; any creep means a brake or counterweight, whose mount is not in the CAD (BUILD_GUIDE §7.8). Design it in now? (D-05)
6. **Switch side load.** Each home switch is worked by a striker sliding across its hinge lever (a 45° lead-in, then a flat holding it 0.6 mm past the click, inside the 1.2 mm overtravel). The lead-in pushes the lever sideways about as hard as down; the datasheet rates only a straight push [Guessing that it holds]. Acceptable, or a roller on a cam? (BUILD_GUIDE §6.8; D-09, D-11)
7. **PETG parts.** The load-bearing parts (5–6 walls, 40–50% infill) carry the Z motor, the head and the belt (15–25 N [Guessing]) beside motors allowed about 50 °C. No doc gives a PETG temperature limit; nothing checks creep or stiffness. Which would you change? (D-08 `x_motor_mount`, D-09 `x_bracket`, D-10 `z_plate`, `z_motor_mount`, D-12 `head_foot`)
8. **Cup margin, marking, power-cut hold.** At the software's thresholds: 13.4 N, SF 21 at the −35 kPa seal; 9.6 N, SF 15 at the −25 kPa transit alarm; lateral SF 49 / 34 on a guessed friction coefficient of 0.5 [Certain,

arithmetic; 78% area an estimate] (DESIGN §4). Marking: 24 h on a hand pump, then 300 cycles + 24 h at mean ≤ −40 kPa (BUILD_GUIDE §1.10, §13.2). After a power cut the slab hangs on the closed cup line only, ≥ 60 s [Guessing target], with a cup-side reservoir as the fallback (BUILD_GUIDE §8.2). Is the marking plan enough for slab plastic, and the hold for unattended runs? (D-17)

9. **E-stop and mains.** The blocks are rated DC-13 1.5 A at 24 V [Certain, BOM.csv]; WIRING §1.3 asks for ≥ 5 A at 24 V DC, and the bus feeds capacitive inputs at up to ≈ 3.4 A [Guessing]. A weld is found only monthly; the pump restarts on release. The T3.15A fuse [Likely] is not checked against inrush; one earth bond crosses anodised joints [Likely]. Acceptable, or the E-stop in the mains feed (the WIRING §1.3 alternative) or a force-guided safety relay (linked contacts that reveal a weld)?
10. **Sensor fail-safety.** The carry sensor's "OK to move" state is dark, so a dead emitter or cut GND is caught only at the next grip [Certain] (WIRING §6a). The probe flag may chatter during the slow peel; the fix is a Schmitt stage [Guessing] (WIRING §6). Acceptable? (D-05, D-06, D-11)
11. **Head camera aim.** The learned check needs the camera within 1.34° toward the cup [Certain, CAD]; on a CGC slab in a pocket corner, about 0.01° toward the label puts over 2% of the card window on the holder [Likely]. The only adjustment is an M2 washer shim (about 0.5° [Likely]). What can the `camera_bracket` / drop-arm stack-up hold, and how would you measure it? (DESIGN §5; D-05, D-12)
12. **Guarding and remote start.** An optional lid switch that pauses between slabs and removes no power; a web page with no login that can start a run. What minimum guard and interlock before unattended runs, and should a start need someone at the machine? (§6)
13. **Tolerances.** D-00 note 6d marks these PROPOSED (no doc states them): printed parts ±0.3, plywood outline ±0.5, sheet-metal outlines ±0.2, dowel cut length ±0.5. The design's own fits: pocket play ±1.85 / ±1.75, rod holes drilled by the machine, Misumi cuts ±0.5 mm, printed holes 0.1–0.3 mm small [Likely]. Right, and holdable in a school shop? (D-00; D-03, D-06, D-07–D-17)
14. **Cables and tube.** Not in the CAD: the 1 m drag chain, the 350 mm Z loop, the vacuum tube (about 1 mm under the head ring at full stroke [Guessing]), about 4 m of head USB [Guessing]. Any routing or flex-life problem? (WIRING §11, §16; D-05)

10. How to read this package

10.1 Start here

Where	What it is
`package/SlabRail_Design_Review.pdf`	The review PDF. Start here.
`package/SlabRail_Complete_Documentation.pdf`	All the docs in one PDF.
`package/SlabRail_CAD.zip`	The CAD files in §10.4: exactly the list in `package/cad_manifest.txt`.
`web/slabrail.html`	One-page overview, as of 23 Oct 2026; where it disagrees with the docs, the docs win.
`web/slabrail_3d.html`	Interactive 3D model generated from the CAD: rotate, click a part to see what it is, hide groups.
`web/slabrail_replay.html`	Animated replay of a simulated 330-slab batch, run by the real machine software, with synthetic labels.
`docs/history/`	Design-round reports (rounds 2–5), the real-label evaluation and its independent check, sourcing data. **Historical: their numbers may be out of date.**

GitHub shows HTML as source: download a `web/` file and open it in a browser (online, for the fonts and the 3D library).

Documents. The README's "What's in this folder" table lists every doc. For review: DESIGN (architecture §2, crash layers §3, calculations §4, verification §6, risks §7), CONTRACT (the binding interface), WIRING,

FIRMWARE, BUILD_GUIDE (Phase 0 §1, acceptance tests §13), OPERATING (every stop message §8.1) and `cad/layout.py` (the master dimensions).

10.2 Drawing set

`cad/out/drawings/SlabRail_drawings.pdf`, 18 sheets, Rev A: a draft for design review, not released for fabrication. Previews are in `cad/out/drawings/png/`; `dims.json` beside them lists every dimension shown and its source, and `cad/drawings.py` re-measures each one on the model.

Sheet	Title	Sheet	Title
D-00	Index and general notes	D-09	Printed parts: X carriage
D-01	General arrangement	D-10	Printed parts: Z carriage plate, Z motor mount
D-02	Frame and extrusion cut list	D-11	Printed parts: Z end cap, Z switch mount, carry-height sensor
D-03	Bed, pockets and guide rods	D-12	Printed parts: head
D-04	X axis	D-13	Printed parts: station mount, drill template
D-05	Z axis and head stack-up	D-14	Printed parts: awl adapter, fill gauge
D-06	Flat parts (probe flag, carry flag)	D-15	Printed parts: Phase 0 jig (base, tray)
D-07	Printed parts: gate posts	D-16	Printed parts: Phase 0 jig (top, occluder mount)
D-08	Printed parts: X drive	D-17	Phase 0 test jig

10.3 Coordinates

`cad/README.md` and D-00 note 3: X left to right along the row (machine X is the cup axis), Y front (−) to back (+) with Y = 0 on the row centre line, Z up with **Z = 0 at the top of the pocket floor pads**; c is the cup-face height above it.

10.4 CAD files

The CAD zip holds the 64 files in `package/cad_manifest.txt`:

File	What it is
`cad/README.md`	Units, coordinates, poses, part names, how to open and regenerate the files
`cad/out/assembly/slabrail_<pose>.step` (4)	Full assembly: `home` (head over pocket 1 at c = 248, stacks as loaded), `pick_bottom` (cup at c = 6.6 on the last slab of pocket 11), `carry` (held slab at c = 248 between pockets 1 and 0), `station` (held slab at c ≈ 171.6). Measure on these.
`cad/out/assembly/slabrail_home.glb`, `slabrail_carry.glb`	The same assemblies as binary glTF meshes for any glTF viewer (metres, Y up, as glTF expects); for looking, not measuring
`cad/out/printed/`: 25 STEP, 25 STL, `print_list.json`	Printed parts in print orientation, with sizes and print notes. The `station_mount`, `phase0_base` and `phase0_tray` STLs are not clean meshes: check them in the slicer preview (`cad/README.md`).
`cad/out/cut_list.csv`, `probe_flag_blank.dxf`, `.svg`	Stock cut lengths (extrusions, lead screw, flags, 82 dowels); the probe flag blank at 1:1
`cad/out/checks.txt`, `cad/out/drawings/SlabRail_drawings.pdf`, `docs/BOM.csv`	All 177 check results; the drawing set; every bought part with price and link

In the repo only: the CadQuery source `cad/*.py`. Bought parts are envelopes, accurate where they meet other parts. Not in the CAD: frame brackets, cables, the vacuum tube, the electronics panel, the optional light roof and rear brace.

11. Glossary

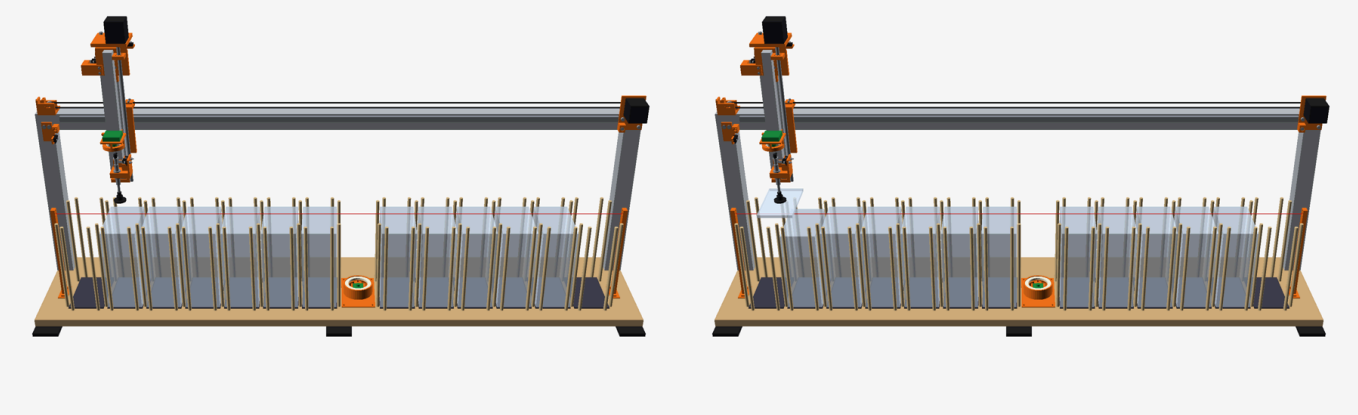
Term	Meaning
c, c_safe	Cup-face height above the pad top, the machine's Z; the free cup hangs 2 below it (`CUP_HANG`). Travel height c_safe = 248: rod tops 218 + two 11.5 holders + 5 + 2.
Cell, pocket, station	14 positions along the row: cell 7 is the station (upward camera), the rest are pockets 0–12; pocket 12 is the exception pocket.
Contact probe, probe window	Fail-safe optical sensor tripped by a flag on the spring buffer at 2.0 ± 0.5 of travel; a probing move goes at most 12 past the predicted contact.
Probe hard limit	FluidNC `probe_hard_limit`: a probe trip during a move that is not a probing move is a critical alarm.
Carry sensor	Carry-height sensor on the Z spine, blocked from c = 246.5 up.
Double pick	Two slabs lifted together, found by a placement step ≥ 11.6.
Back-label code	A code printed only on the back label (PSA back QR, graded CGC QR, CGC 2020–23 back barcode).
Face-down check, learned	Before the cup touches a slab, a head frame must prove it face-down: a back-label code, or (learned mode) its card window, a 55 × 5.5 mm strip of card back, matching references from two proven slabs. Active after a Phase 0 GO (threshold ≥ 0.55), 5 references and a clear camera-aim check over an empty pocket.
Statuses	`ok` (cert and grader confirmed), `needs_review` (a read the rules can't confirm), `conflict` (codes disagree), `no_read`, `other_grader` (a BGS, SGC or TAG cert URL).
QR, Code128, ITF	A 2D code and two 1D barcodes; ITF (interleaved 2 of 5) is on PSA labels.
J	The torsion constant of the beam section, used for the twist estimate.
Rodent, TMC2160, FluidNC	BTT controller board; its stepper drivers (StallGuard is sensorless stall detection, off here); the CNC firmware on it, driven by G-code over USB.
G38.3, `ALARM:4`	A probing move that stops on contact and does not alarm on a miss; FluidNC's alarm when the probe already reads tripped at the start (a cut fail-safe wire).
DC-13	IEC category for DC contacts switching electromagnets (the E-stop blocks' rating).

SlabRail: design rationale and engineering checks

Confidence tags: **[Certain]** = measured, computed, or checked by the CAD/simulator here. **[Likely]** = strong inference. **[Guessing]** = needs a real-slab test (see "Phase 0").

1. What it does

You load up to **330 slabs**: 11 stacks of up to 30, face-down. The machine reads every one, unattended, and writes a CSV of cert numbers with a photo of each label.



1. Load: 11 stacks of up to 30, face-down, label toward you. Pocket 0 (left) and pocket 12 (right) stay empty.
2. The cup lifts the TOP slab of stack 1 straight up, reads it, and sets it down in empty pocket 0.
3. Repeat until stack 1 is gone - then stack 2 moves into the now-empty pocket 1, and so on.
4. Done: every slab read, every stack one pocket to the left (order reversed). Rejects wait in pocket 12.

The run works like this:

1. The cup lifts the top slab of stack 1 straight up. A camera on the head reads the back of the label (QR code / barcode).
2. The cup puts the slab down in the empty pocket next to it.
3. The next slab is now on top, and the cycle repeats.
4. When stack 1 is gone, stack 2 moves into pocket 1, and so on down the row.

Slabs whose back has no readable code (pre-2016 labels, damage, glare) take a detour. They are held over an **upward camera** in the middle of the row, which reads the front label (barcode + printed digits). Anything still unreadable, read only by OCR, or from a grader other than PSA/CGC, goes to the **exception pocket** at the right end.

2. Why this architecture

Option	Verdict	Reason
Buy something	None exists	[Likely] No product feeds 200–300 graded slabs unattended. Raw-card sorters (Roca, PhyzBatch, SortSwift) exclude slabs. Sheet-fed scanners are rated "Slabs: No". See RESEARCH.md.
Hands-free 2D presentation scanner (~$350–650)	Best no-build option	[Likely] 20–30 slabs/min by hand, ~10 min per 250. The machine saves that time and adds a photo of every slab.
Gravity hopper / friction feeder	Rejected	[Certain, arithmetic] The bottom slab gets dragged out under the whole stack. A 300-slab column is about 2 m tall and 15–18 kg, so that's ~60 N of face-on-face drag across a glossy front. PSA slabs have no stacking ridges.
Slotted racks, lift-and-read in place (concept A)	Rejected	[Certain, panel geometry] A straight lift hits the neighbouring slab. It only works if every slab leans at 10°, which leaves 2.2 mm margins. Loading is one slot at a time.
Desktop robot arm (concept D)	Rejected	[Likely] Reach covers about 80–160 slabs. The arm that reaches 300 costs about twice the budget.
Short stacks + top pick by vacuum, one-pocket shuffle (concept B)	**Chosen**	Two research agents and the design panel arrived at it independently. No slab ever slides on another. The cup touches only the back. Loading is fast because you drop in handfuls.
B on a 2-D grid (4 motors) vs **one row (2 motors)**	**One row**	You asked for the fewest moving parts. One row removes the whole Y axis: 2 motors, 2 rails and the gantry squaring. Cost: a footprint of about 1.44 m × 0.29 m (CAD, feet included; the bed is 1416 × 289.5 mm). The grid version is in git (`grid-variant-v1`).

2a. What the independent design panel said (and what changed because of it)

Five agents each designed a different concept in depth. Three judges then scored all five, each through one lens: slab safety, reliability/buildability, and practical value.

Scores (1–10)

Concept	Safety	Buildability	Practical
A, slot racks	6.5	6	5.5
B, stacks + vacuum (the parent of this design)	5	6.5	4.5
C, shelf modules, pull and read	**8**	4	4
D, robot arm	3	3	2.5
E, "flip-through lanes" page-turner	6	**7**	**6.5**

Why this is still B (as SlabRail), not E.

- [Likely] E's slab physics is unproven. Its own author says so. It needs 300 hand-loaded slabs in a row to peel, pass a 2.6–4.3° balance point, and settle flush, every time.
- Pick-and-place of flat, smooth parts from stacks is what every SMT and packaging line does. Its failure modes are known, detectable and simulated here.
- The judges scored the original B, a 4-motor grid. Their main complaints have been fixed in SlabRail:

Judge's objection to B	Fix in SlabRail
Fingers can't reach slabs in tight corrals	Separate dowels, not walls. There is a 52 mm finger gap at the front and back of every pocket for lowering bundles of 5–10 when loading, and for taking slabs off the top in bundles of up to 5 when unloading. Never lift or flip a whole stack (OPERATING.md §6).
Printed posts can lean	Dowels sit in drilled holes, 15 mm deep, drilled through a printed vertical guide.

Judge's objection to B	Fix in SlabRail
Air-film double picks	The slow leg of the peel (2 mm/s) is placed where the cup really leaves the slab: at the probe trip the spring buffer is already squeezed about 2 mm (`buffer_trip_mm`, measured in BUILD_GUIDE §7.6), so the head first rises that far at 20 mm/s while the cup stays on the slab, then 2 mm/s through the release point and 1 mm past it, then a 0.5 s dwell low over the stack, so a slab clinging by an air film falls back a few mm onto its own stack. Squeeze-film force scales with speed, so it falls about 10× to ~0.1 N, below the slab's weight. Doubles are still detected at placement.
Klipper probe macro fragility	FluidNC native `G38.3` with an explicit bounded target.
Too many moving parts	2 motors instead of 4. No gantry squaring.
"Flipping and un-reversing costs more time than it saves"	Don't flip anything. Order doesn't matter, because the CSV records each slab's exact pocket and position, and unloading goes top-down in small bundles (OPERATING.md §6). Face-down loading is a hard rule: before the cup touches a slab, the head camera must see a back-label code (a PSA or CGC back QR, or the CGC 2020–23 back barcode) or, in the default learned mode, a card back it learned from proven slabs (§5); otherwise the run stops untouched for you to check. The head leaves the card window clear (the lens sits 5 mm toward the label) [Certain, CAD]; until the learned check is active, PSA 2016 and older labels, CGC Authentic, BGS and SGC stop, unless the run is started with `--all-face-down`; then a face-up slab would get the cup on its front.

- **What we keep from the other concepts:**
  - fail-safe sensor wiring;
  - a pre-run survey (every pocket probed while you are still there);
  - the Phase 0 tests before buying motion parts;
  - a photo of every slab;
  - CSV written incrementally;
  - count reconciliation.
- **Honest residuals:**
  - Every slab is lifted and carried. A vacuum loss drops it at most about 240 mm onto foam or a stack.
  - Hands-on time is still ~10 min per 300 for loading and unloading.

3. Everything that touches a slab

Contact	Motion	Force	Worst case
Suction cup on the **back**, centre of the card window	Static. The cup only moves vertically while touching.	Trips at ~1.2 N [Likely, SMC catalog spring force]. Vacuum clamp ~19 N over the 25 mm lip.	A faint cup ring or a grit micro-mark on the **back** (allowed). Mitigations: non-marking PU/HT1 cup, IPA wipe each run, 24 h marking test before use.
Slab front resting on the back of the slab below	Static. Pick = straight lift, 2 mm/s through the point where the cup leaves the slab, then 20 mm/s, with a 0.5 s pause just above the stack. Place = probed set-down at 20 mm/s (`G38.3`).	≤ 2 N at contact. Bottom slab of a 30-stack carries 19 N (1.8 kPa, same as stacking by hand).	Lateral slip at placement if a slab lands crooked. This is the most plausible front-scuff path. [Likely] Mitigated by ±1.75 mm pocket play, rods only at the edges, and slow probed set-down.
Bottom slab front on the pocket pad	Static	≤ 19 N	Grit on the pad. Wipe pads weekly. Microfiber face.
Slab edges on 8 mm wooden dowels	Only if the slab is off-centre by > 1.75 mm	< 1 N	Edge rub on wood. The dowels sit 12 mm from every corner, so corner tips never touch.

Contact	Motion	Force	Worst case
Head / camera / Z parts	None by design	—	[Certain, CAD] ≥ 5 mm at every checked pose, including a held slab at the Z top hard stop and a double pick of two thick holders; much more in normal travel (numbers in `cad/out/checks.txt`, §6). The lead screw is cut so its tip stays above anything the head carries.

Drops. [Certain, arithmetic]

- Losing vacuum at carry height means a fall of ≤ about 240 mm, about 0.15 J, onto a stack or pad. That is roughly a 25 cm desk drop. A flat landing is very unlikely to crack a slab [Likely]. A corner landing on a dowel top could chip one [Guessing].
- Exposure: about 2 s in the air per cycle for a slab whose back reads. A slab that needs the station is held longer: the front frames are taken over the lens, then decoding and OCR (up to 30 s on a Pi) happen with the slab over its planned output stack, not over the camera.
- Both solenoid valves are normally closed, so a power cut (E-stop, mains) seals the cup line. **The reservoir does not help then:** it is on the pump side of the closed vacuum valve. The slab hangs only as long as the cup line's own leak allows, seconds to minutes. [Certain, the plumbing in `hw.py` and BUILD_GUIDE §8.1] Acceptance test 13.4 measures it. While the Pi still has power (Ctrl-C twice, a USB cable out, a park that failed) the software keeps the pump holding a slab until you take it.

Crash prevention, layered.

1. **Height bookkeeping.** Every slab's thickness is measured when it is placed (the probe step).
2. **Every downward move is a probing move.** Each one is a `G38.3`, so an unexpected contact stops the head at once instead of driving on; probe windows never go more than 12 mm past the expected surface. A trip at full Z speed squeezes the 20 mm buffer about 6 mm (about 1.6 N [Likely]).
3. **Fail-safe contact sensor.** A cut wire reads as "touching", so the machine refuses to probe.
4. **Probe hard limit** (firmware). A contact-sensor trip during any move that is not a probe (a lift, a traverse) is an immediate critical alarm.
5. **Second contact sensor.** During the start-up survey, a cup that seals without the optical sensor tripping stops the run.
6. **Overfill laser beam.** It runs 4 mm above the fill line. If anything is too tall, the machine won't start. It cannot see anything above the 218 mm rod tops, such as a slab lying across the rods: the survey catches that, because its first probe window starts at travel height.
7. **Face-down check.** No back-label code seen from above = the cup does not touch the slab, unless the learned check (the default mode, §5) matches its card window with two card backs of proven slabs. A front-label code seen from above stops with a FACE-UP message; back and front codes together, two back labels or two certs in one frame mean a neighbour's label, and stop. Modes: `learned` (default), `stop` (back-label codes only), `off` (or `run --all-face-down` for one batch).
8. **Carry-height sensor (Z stall guard).** A slotted optical sensor on the Z spine and a flag on the Z carriage: blocked from c = 246.5 up to the top stop, clear below. Before every sideways move it must read blocked, and before every grip clear (a dead emitter or dirt reads blocked). A loaded head that stalled more than about 2.5 mm low therefore stops the run before it moves sideways; one that still reads blocked (c ≥ 245.5) carries even a double pick of two thick holders 2.5 mm over the rods. [Certain, CAD and simulator] After a stall the head never moves sideways with a slab: over its own pocket it probes straight down and vents, elsewhere it holds. Empty travel runs at 248 too, so the sensor's single edge sits just under both travel heights (about +0.2 s a slab). Not covered: `jog`, `calibrate` and `mark`.
9. **Plausibility checks.** A stack taller than predicted by > 4.5 mm, a vent that does not reach atmosphere, a vacuum sensor reading outside its range, or an exception pocket that's full stops the run. After any stop the

head re-reads its real position, lifts straight up, and puts a held slab back; where that is unsafe, nothing moves and the pump keeps holding it.

4. Calculations

Quantity	Value	Margin
Cup holding force, Ø25 mm at −50 kPa (78% effective area)	19.1 N	Slab weighs 0.64 N → **SF 30** vertical [Certain, arithmetic; the 78% effective area is an estimate]
Cup holding force at the software's seal level, −35 kPa (`vac_seal_kpa`)	13.4 N	**SF 21** on the 0.64 N slab, 19 with the 1.2 m/s² Z acceleration [Certain, arithmetic; same 78% area]. A grip counts as sealed only at this level or deeper.
Cup holding force at the transit alarm, −25 kPa (`vac_transit_alarm_kpa`)	9.6 N	**SF 15**, 13 with the Z acceleration [Certain, arithmetic; same 78% area]. A carry that has not raised the alarm holds at least this. The slab falls off at about −1.7 kPa (BUILD_GUIDE §13.4).
Lateral grip vs X acceleration 2 m/s², at −50 / −35 / −25 kPa	~9.3 / 6.4 / 4.5 N vs 0.13 N	**SF ≈ 70 / 49 / 34** [Certain, arithmetic: friction on the clamp force less the slab's weight; the friction coefficient of about 0.5 between cup lip and slab is a guess]
Z load (0.75 kg head, 1.2 m/s²)	8.3 N → 0.026 N·m on T8×8 at 40% efficiency (conservative: the screw's 20° lead angle gives about 50–65% at a friction coefficient of 0.15–0.25 [Likely])	A 1.5–2.0 A NEMA17 at 750 rpm on 24 V gives ≈ 0.15–0.2 N·m at full current, but Z runs at 0.8 A (40–55% of rating) → ≈ 0.06–0.1 N·m, **SF ≈ 2–4** [Likely]. Spinning up the rotor and screw adds about 0.006–0.009 N·m at 1.2 m/s² [Guessing: a 60–80 g·cm² rotor]; the CAD mass row below shows that the 0.75 kg covers it. The carry-height sensor catches a stall before the next sideways move but does not prevent it, so the 100-lift test (BUILD_GUIDE §7.8) must pass. A high-impedance 0.4–0.6 A "integrated" motor would stall: the BOM excludes it.
Z back-drive when unpowered	~0.005 N·m (0.75 kg at a back-drive efficiency of about 0.5; 0.4–0.7 for a friction coefficient of 0.2–0.1 gives 0.004–0.0065 N·m)	NEMA17 detent ≈ 0.015 N·m → holds [Likely]. Hold current is kept on anyway.
X drive (1.9 kg moving, 2 m/s² + 2 N friction)	5.8 N belt → 0.037 N·m on the 20T pulley (+ about 0.003 N·m for the rotor [Guessing])	The 17HS19-2004S1 (0.59 N·m holding at 2.0 A) runs at 1.0 A: about 0.3 N·m holding, less at 375 rpm, so ≈ 0.15–0.3 N·m → **SF ≈ 4–8** [Likely; no torque curve read]
X beam sag, 4040 resting on the uprights (1376 mm between their centres), 20 N at midspan	0.18 mm (simply supported; E 69 GPa; I = 8.89 cm⁴, Misumi's catalog value for HFS5-4040 [Likely, read from a catalog extract])	Vertical: every set-down is probed, so it uses none of the pocket play; BUILD_GUIDE §4.5 checks ≤ 0.3 mm. The beam's slope under the moving load tilts the head by under 0.02°, under 0.1 mm at the cup in X against ±1.85 mm of play. The beam's own weight (1.28 kg/m) adds a fixed 0.10 mm at midspan. [Certain, beam formula]
X beam twist from the head overhang (cup axis 127 mm in front of the beam centre)	≈ 0.5 mm in Y at the cup near the pad, worst at midspan; 0.3–0.8 mm for a torsion constant J of 2.5–1.0 cm⁴ [Guessing on J: no published value found for a slot-6 4040]	θ = T·L / (4·G·J) with both ends fixed in torsion: T = 1.93 kg × g × 77 mm = 1.46 N·m (the CAD puts the carriage's centre of mass 77 mm in front of the beam centre, not 127), L = 1.376 m, G = 26 GPa, times the 400 mm from the beam axis to the cup face. The earlier 0.8 mm put all 1.9 kg on the cup axis. The corner brackets and the rail block's play are not included. Within pocket play. Measure it during build (BUILD_GUIDE §6.7). Add a rear brace if > 1 mm.
Spring buffer force at trip (2 mm) / end of a probe window (14 mm) / bottomed (20 mm)	about 1.2 / 2.4 / 3.0 N [Likely: SMC lists 1.0 N at rest and 3.0 N at full stroke for ø10–32 buffers]	The software never probes more than 12 mm past contact; a trip at 6000 mm/min squeezes about 6 mm. The `ZPT25UUK20-B5-A10` has 20 mm of stroke: SMC sells no 25 mm stroke on a 25 mm pad. [Certain, SMC ZP catalog Table 2]
Stack height, 30 PSA slabs at 6.25–6.9 mm	188–207 mm	MAX line 210, overfill beam at 214, rod tops 218

Quantity	Value	Margin
Carry height (cup), loaded and empty	248 mm	A **double pick** of two 11.5 mm thick holders still clears the rods by 5 mm, counting the free cup's 2 mm hang below its calibrated height [Certain, CAD]. The carry sensor trips at 246.5: a head it still sees as "up" (≥ 245.5) clears them by 2.5 mm.
Moving mass, Z head (CAD: `build.assembly('carry')`, parts in group `head`)	0.40 kg from CAD volumes and catalog masses; ≈ 0.46 kg with the T8 nut and the head's share of cables and tube [Guessing]; ≈ 0.53 kg with a 65 g slab	Under the 0.75 kg used above. With 0.53 kg: 0.0185 N·m + 0.006–0.009 N·m of rotor and screw inertia = 0.025–0.027 N·m, so **SF ≈ 2–4** at 0.8 A still holds [Likely]. The 0.75 kg plus that inertia would need 0.033–0.035 N·m (SF ≈ 1.7–3). Back-drive with 0.53 kg: 0.003–0.005 N·m.
Moving mass, X carriage (groups `carriage` + `head`)	1.62 kg from CAD volumes and catalog masses; ≈ 1.87 kg with T-nuts, screw shanks, the belt and the drag chain's moving share [Guessing]; ≈ 1.94 kg with a slab	Matches the 1.9 kg used above (5.9 N, 0.037 N·m): the X margin is unchanged. Its centre of mass sits 77 mm in front of the beam centre (twist row).

Mass rollup inputs. Volumes are the solids of `build.assembly('carry')` (`cad/build.py`), summed by `Part.group`. Printed parts: PETG at 1.27 g/cm³ × 0.71, the solid fraction BOM.csv gives for the listed walls and infill (about 720 g of 1020 g solid) [Guessing for each part]; belt clamps at 100%. The aluminium flags, the steel buffer and lead screw (a solid Ø8 in the CAD) and the fastener heads (the only part of a screw that is modelled) use their CAD volume at 2.70 and 7.85 g/cm³. Extrusions, rails and carriage blocks are solid envelopes in the CAD, so catalog values replace them: the 2040 and 2020 at Misumi's 324 and 183 mm² sections × 2.70 g/cm³, the MGNR12 rail at 0.65 kg/m, each MGN12H block about 0.05 kg [Likely]. Z motor body 0.36 kg; head camera 20 g, LED ring 5 g, cup 9 g, each sensor board 4 g, M5 elbow 3 g [Guessing]. Heaviest items: Z motor 0.36 kg, Z spine 0.30, Z rail 0.22, drop arm 0.14, lead screw 0.13, printed parts 0.22 (0.15–0.31 kg for a solid fraction of 0.5–1.0).

5. Reading the label

- **Glare geometry.** The station camera sits on the slab's centre axis, the head camera 5 mm from it toward the label end, each with a small LED ring around the lens. **[Certain, mirror geometry]** The mirror reflection of a ring of radius r lands within r/2 of the lens axis, so within about 10 mm for the head's 40 mm ring (11 mm for a 42.5 mm one) and about 16 mm for the station's 63 mm ring. The label band sits about 40–65 mm off the lens axis, so it is glare-free without polarisers. That avoids the slab plastic's stress-birefringence "rainbow" risk that cross-polarisation has. [Likely] Room lights reflecting off the slab plastic are a minor effect. The LED ring is several times brighter on the label than a ceiling light's reflection, so contrast drops a little but nothing saturates. If Phase 0 shows glare anyway, add the optional black foam-board light roof.
- **Resolution.**
  - Head camera (8 MP IMX179, 3.6 mm lens, 140 mm away): 0.055 mm/px.
  - Station (Camera Module 3): 0.039 mm/px.
- **Decode sweep on synthetic labels in the real formats** (`software/tools/decode_sweep.py`):
  - **[Certain for the synthetic labels]** The back QR + barcode reads at ≤ 0.14 mm/px through blur and strong glare.
  - The front barcode needs ≤ 0.05–0.08 mm/px and low glare.
  - **No misreads occurred in any cell of the sweep.**
- **What real labels carry** [Certain, decoded from 64 real slabs' public studio scans; RESEARCH.md §4]. These formats are the rules in `slabscan/certs.py`:

Label	Back label	Front label
PSA current (9-digit certs, some 8)	QR `https://www.psacard.com/cert/<n>/` + Code128 `<n>`	Code128 `<n>`
PSA 2017+ "lighthouse" (8-digit)	QR `http://psacard.com/cert/<n>` (or the https form) + ITF	ITF
PSA 2016	ITF only	ITF
PSA before 2016	no code	ITF
CGC 2023+ (10-digit certs, a few 140 + 10)	QR `https://www.cgccards.com/certlookup/<n>/<grade>/`	Code128 `<n>`
CGC 2020–23 "OG"	QR `…cgccomics.com/certlookup/<n>/<grade>` + Code128 of 26–28 digits ending in `<n>`	no code
CGC Certified Authentic	no code	QR `…/certlookup/<n>/0_0/` + Code128 `<n>`
BGS, SGC	no code	no code (printed cert only)

- **Interpretation rules** (`slabscan/certs.py`, tested on the real payload strings):
  - **A QR names the cert and is `ok` on its own:** it carries its own error correction. PSA and CGC QRs are parsed with a right boundary, so a longer number is rejected, not cut. A CGC cert keeps all its digits: the old rule cut a real 13-digit cert to 12 and called it `ok` (a silent wrong cert, now a test).
  - **Bare digits are typed by symbology and length:** ITF of exactly 8 digits is a PSA cert (the cert zero-padded; any other ITF length is a misread, and a glare-cut `289412` was seen); Code128 of 26–28 digits is a CGC OG back (the cert is the last 10); Code128 of 10 digits not starting with 0, or 140 + 10, is a CGC front; Code128 of 8–9 digits is PSA (every real one). A Code128 in none of these formats (7 digits, or 10 starting with 0 like the BGS cert `0019865140`) is not typed: it is listed as unparsed and proves no label side, so a slab with nothing else ends in the exception pocket. [Certain, `certs.parse_payload`; 0 of the 119 distinct real payloads parse differently after this change] There is nothing left to learn about the ITF: the old padding-or-check-digit model could never learn on real labels and held every 2016 slab back. [Likely] An 8-digit Code128 with a leading zero still parses as a 7-digit PSA cert; the only real one seen is the phantom `09435764`, which the phantom guard needs to see.
  - **A barcode-only result (ITF or Code128, any grader) needs two printed symbols** (back + front label) **or OCR of the printed digits to agree.** One printed symbol is one source: the head camera's hover and held frames count once, and so do the station's. OCR that disagrees makes it `needs_review`.
  - **OCR also checks every barcode-only `ok`.** Both labels of a PSA 2016 slab print one ITF, and current labels one Code128 on both sides, so one fixed misread of a printed pattern can come back from both and agree on the wrong number. OCR that reads other digits makes it `needs_review`; OCR that reads nothing leaves it `ok`. The cost is one station visit and one OCR per slab without a QR read. [Certain, `runner.needs_ocr`; tested]
  - **An `ok` needs its grader confirmed** [Certain, `certs.consensus`, `certs.psa_itf_confirmed`]:

Evidence for the grader	Grader on an `ok`
A code that names it (a cert URL: PSA or CGC QR)	confirmed
Two different printed symbols in its format (back + front label barcode)	confirmed
An 8-digit ITF whose digits OCR of the printed number confirms	PSA (note "grader PSA from its 8-digit ITF")
One barcode in a grader's format, even with OCR agreeing (one Code128)	not confirmed: `needs_review`, grader kept ("grader … not confirmed")
An 8-digit ITF alone, or with OCR that read nothing	`needs_review` ("one printed barcode only")
An 8-digit ITF with OCR that read other digits	`needs_review` ("barcode N but OCR read M")
A code in no grader's format (bare digits in a QR, EAN-8, Codabar, Code39, a keyed "Cert #"), even beside an ITF	never confirms a grader

Evidence for the grader	Grader on an `ok`
OCR only	`needs_review`, grader `unknown`, exception pocket

The ITF rule makes a pre-2016 PSA slab (front ITF only) `ok`. It holds because in the real set ITF is printed only on PSA labels (CGC uses Code128; the 1 BGS and 1 SGC slab showed no code), and OCR reads the printed number on the slab in the cup, independently of the barcode's digits. [Certain, real set: 5 of 7 pre-2016 slabs `ok` native, 7 of 7 at machine scale; a stress run of 316 ITF-rule `ok` frames gave 0 wrong certs] Only a `needs_review` with grader PSA or CGC may stay in the output stack; `unknown`, `PSA/CGC` or no grader goes to the exception pocket. A grader guessed from the length still counts, so another grader's slab whose barcode has a PSA or CGC format can reach the output stack as `needs_review` [Certain, verifier probe: a 10-digit Code128 is typed CGC], and another grader's 8-digit ITF that OCR confirms would read `ok PSA` [Guessing that one exists]. Two settings switch the ITF rule off in effect: `custom_regex` (every ITF is then typed by it) and a `psa_cert_digits` without 7.

- **OCR finds a digit run, not a grader.** Tokens are 8–10 digits or 140 + 10. An OCR-only number is `needs_review` with grader `unknown` and always goes to the exception pocket (OCR also reads card numbers: `0079865140` for a BGS `0019865140`). OCR runs on a crop of the label band (3.2 barcode lengths along, 0.6 across) with `OMP_THREAD_LIMIT=1`: with Tesseract's default threads it timed out on every frame of a loaded 4-CPU computer [Certain, measured].
- **Phantom guard, twice.** `decode.py` drops a linear read lying inside another read of the same symbology and under a third of its thickness (the real `09435764` inside `27435764`). `consensus` then drops a phantom that got through, from every frame of that symbol, but only the thin read inside the other read's box (under half as thick, centre inside), and only when a different symbol (the QR, the other label side) or OCR confirms the other number ("dropped a phantom read …"). If the confirmed number is the thin read, nothing is dropped and the slab is a `conflict`; two reads whose boxes do not overlap are two labels, never a phantom. [Certain, `certs.drop_phantoms`; tests on the real payloads and boxes]
- **Decode passes:** on head-camera frames the CLAHE pass runs before the early exit when only a barcode was found (a QR the plain pass missed can decide the face-down check); station frames get an extra 0.75× / 0.6× downscale pass (CGC front Code128s with ink-spread bars read there).
- **Head frames are cropped to the slab under the cup** (`head_roi_half_mm` = 38.3 mm either side of its centre line), because the 3.6 mm lens also sees the neighbours' labels and, beside the slab, the lower neighbour stacks. The frame taken from above the stack alone never makes an `ok`, and a cert already recorded in the batch becomes `conflict` (exception pocket).
- **The back-label code that proved a slab face-down must be its own:** if the reads of the slab in the cup do not show that cert, or show another one, the slab is `conflict` ("… may have been FACE-UP …, check it by hand").
- **`psa_cert_digits`** (set from Phase 0): a PSA cert of a length Phase 0 never saw is `needs_review`.
- **Codes that disagree** produce `conflict` and the slab goes to the exception pocket. The machine never guesses.
- **CGC** is read and stacked like PSA. A **BGS/SGC/TAG** URL gives `other_grader` (exception pocket); real BGS and SGC labels carry no code, so they end as `no_read` or OCR-only.

Face-down: what a code seen from above proves

Only rules that held on every real label of their kind may prove a side [Certain, 64 real slabs; `certs.SIDE_PROOFS`]:

Code seen by the head camera	Proves	Evidence
PSA QR (`psacard.com/cert/`)	**face-down**	never on 35 PSA front images; on every 2017+ back
CGC QR with a cert and a grade other than `0_0` (cgccards or cgccomics)	**face-down**	0 of 26 graded CGC fronts have a QR; every 2023+ and OG back has one

Code seen by the head camera	Proves	Evidence
CGC 26–28-digit Code128	**face-down**	OG backs only; OG fronts carry no code
CGC QR with grade `0_0`	**face-UP** (Authentic front)	both Authentic fronts; their backs decode nothing. One official render of a 2023+ back also shows `0_0`: a wrong face-up call only stops the run
CGC 10- or 13-digit Code128	**face-UP** (CGC front)	0 of 26 2023+ backs carry a Code128
CGC QR without a cert	nothing (a hint in the message)	press images and mock-ups only [Likely]
PSA ITF or Code128	nothing	printed on both labels

- **Back and front codes together, two different back labels, or codes of two different certs in one frame** mean another slab's label is in view: the run stops.
- **Measured** [Certain, `tools/real_eval.py`]: 50 of 64 real backs proven face-down, 24 of 64 real fronts proven face-up, 0 false proofs either way (native scans); 48 and 23, 0 false, in the machine-equivalent frames. The 14 backs left unknown are exactly the slabs with no back-label code. These rules were written from the same 64 slabs, so this shows they fit them, not that they hold on new ones.
- **Every back-code proof teaches the learned check** (CGC included), once the slab is placed and its own read agrees (below).

The learned face-down check

Why. Slabs with no back-label code (PSA 2016 and older, CGC Authentic, BGS, SGC) stop the plain check every time: 43 of 330 in the simulator's default mix, 14 of 64 real slabs. Most of them show the same card back through the plastic, and a face-up slab shows a card front there instead.

The descriptor (`slabscan/facedown.py`) [Certain, code]:

- Only the card window, `face_down_window_mm` (x −27.5 … 27.5, y −18.5 … −13.0 mm from the head-camera axis = y −23.5 … −18.0 from the cup axis, −y = the label end; 55 × 5.5 mm = 302.5 mm², at least 300 required). It comes from the CAD (`layout.face_down_window_cam()`), placed from the card positions measured on 22 PSA and 27 CGC backs (`layout.HOLDERS`): 11.0 mm clear of the label band and clear of every head part at hover and held [Certain, `checks.py` frustum checks]. On the card plane (3.5 mm under the slab back, rounded corners) a PSA slab keeps it on the card anywhere in its play; a CGC slab pushed into a pocket corner has 1.52% of it on the holder (4.5% if the 3.5 mm were all air), checked against a 2% limit [Likely, `checks.py`]. The window shrank from 632.5 mm² because the round-3 one let 12 of 206 face-up CGC fronts through in an emulation of the real scans with every slab at the cup end of its play [Likely]: the strip of holder above a CGC card looks the same from both sides.
- Each rectangle is cropped with a 4 mm margin and box-resampled to 1 px/mm. That removes noise, focus and JPEG differences and about 1° of yaw, and costs about as much as a thumbnail.
- Glare (a 1 mm average at grey level 245 or more, grown by 2 mm) is masked everywhere.
- Pattern term: the window minus its own blur (σ 4 mm), so exposure and light fall-off drop out; normalised cross-correlation at every shift within ± 4 mm (the slab's play in its pocket).
- Colour term: a hue-saturation histogram (a grey-level histogram on grey frames).
- **Score = the lower of the two, and a slab needs two references** (from two different slabs of the matched back design) at the threshold, not only the best one. A false accept puts the cup on a front; a false reject only stops the run.
- Cost [Certain, measured on the development container]: about 2 ms to describe a full 3264 × 2448 frame and about 2 ms to compare it with 48 references. [Guessing] 10–20 ms on a Pi 5. It adds one head frame per run (the empty-pocket view), none per slab.

Safety layers [Certain, code; tested in the simulator]:

1. **References only from proven slabs,** and only once the slab is placed: a back-label code seen from above, if the slab's read is `ok` with the same cert and the read station saw no back-label code from below; your `--face-down-ok`, with the same placement checks and never with a front-label code in view; the Phase 0 seed. A learned acceptance never becomes one. A back design known only from your confirmations is used once 3 agree.
2. **Two references per match:** a new back design starts with one slab and matches nothing until a second slab joins it, so one odd reference (or one front taught by mistake) is never enough.
3. **`--face-down-ok` confirms one slab:** only after a face-down stop, only the top slab of that pocket. The web tick appears only then and is cleared after each resume.
4. **A texture floor**, and a match needs at least half the matched back design's own texture (a plain front with a glare spot has some texture, far less than a back).
5. **Glare:** a window with more than 2% glare never becomes a reference; more than 30% is not matched.
6. **The empty-pocket guard:** one head frame over an empty pocket per run. A window whose pattern looks like it (similarity ≥ 0.5) shows the head, not the slab: it neither matches nor teaches.
7. **The occlusion check:** on the newest 5 or more slabs proven on this machine, a pixel that tracks the empty-pocket view by the frame's exposure ratio moves with the camera. More than 10% of the window like that: the mode is not active.
8. **A measured threshold:** Phase 0 compares at least 10 usable backs with each other and with at least 15 face-up slabs, each face-up frame also with 4 reflections added in software, all by the same 2nd-best rule; a back that matches none of the others is left out. GO needs a gap of 0.10; the threshold is the midpoint, never below 0.55. It is stored with a tag of the window and descriptor it was measured for (`face_down_threshold_sig`) and refused without it.
9. **At least 5 references** (`face_down_min_refs`).
10. **The camera-aim check** (`facedown.aim_problem`), on the same view over an empty pocket: in the middle ±12 mm of the window along the row, over its full height and 1 mm beyond its cup-side edge, only the pad far below may show. More than 2 mm² of high-passed edges (4 grey levels or more) there means a head part is in the window, or about to be: the mode accepts and learns nothing that run, and the start line, every stop, the summary (`CAMERA AIM:`), `summary.json` (`camera_aim`), `selftest` and `focus head` say to re-aim. [Certain, sim: a square camera passes; with today's window it fires from 1.30°, the part entering at 1.33°] A verdict counts only for the run that took it (`CardBackRefs.stale`). A resume looks over an empty pocket at its survey (as before); if none is empty, it now accepts and learns nothing until it has emptied its first source pocket and looked over that, so a camera knocked while the run was stopped is seen [Certain, sim: a 1° knock caught with and without an empty pocket; the no-empty-pocket tests fail on the round-4 code]. The cost: with no empty pocket, every slab without a back-label code stops until then, so the operator should empty pocket 12 first. Its blind spots:
    - a head part as dark as the pad shows no edge, and layer 7 does not catch it either while it covers a strip of the window (a pad-dark foot over 9–26% of the window read 0.3–0.9% camera-locked) [Certain, sim];
    - only a tilt toward the cup is checked: a tilt away from it or a roll along the row moves the window toward the card top or off its side, and nothing sees that [Certain, code];
    - the outer window beyond ±12 mm also sees this pocket's rods and the neighbouring pockets, so it cannot be checked there [Certain, CAD];
    - the thresholds were set on the simulator's flat, noiseless pad: pad mottling, dust specks of about 1 mm, a tag or a sharp shadow made it fire on a square camera in a synthetic probe [Certain, synthetic]. That is safe but strict; a real empty-pocket frame from Phase 0 is needed to tune it.

Limits.

- [Certain, CAD aim check] **Camera aim:** a head-camera tilt of more than 1.34° toward the cup puts the cup pad into the window (1.31° with the software's own 147 mm mapping, by an independent ray cast). 1.5° is out of

reach: the cup pad sets the limit, and a window that clears it at 1.5° leaves under the software's 300 mm². Layer 10 catches a tilt toward the cup from about 1.3°; its blind spots are above. A head edge about 1 mm inside the window raised face-up scores in the simulator from 0.37 to 0.52 (hence the 0.55 floor). BUILD_GUIDE §12.3 checks the frame; [Likely] an M2 washer shim corrects about 0.5°.

- [Likely, CAD card-plane check] **CGC holders:** a CGC card top lies 93.8–96.2 mm from the holder's far end (27 real backs; PSA 97.2–100.0 on 22), so a CGC slab at the cup end of its play brings its card top about 3.4 mm nearer the cup than a PSA one. The window keeps about zero margin there: up to about 1.5% of it on the holder (4.5% at most), and a tilt of about 0.01° toward the label, or 0.10° of roll, takes it past 2%. The holder lengths and widths come from cropped scans and probably read short (CGC publishes about 80.0 × 134.6 mm) [Likely]; the card top to far end distance is the design input and is not affected.
- [Likely] **Ring glare:** the head ring's mirror image in the slab back covers 0.0% of the window for a slab square to the lens, 0.97 mm clear of the 2 mm glare halo. A slab tilted about 0.45° toward the label brings the halo in; from about 0.77° its frames stop becoming references (over 2% glare).
- [Certain, real images; Guessing for the machine] **Real separation:** on studio scans of 63 real slabs placed as the machine would see them, with two references 0 of 332 face-up frames and 223 of 252 face-down frames (glare included) were accepted at 0.55. With one reference, a Japanese blue Water-type front reached 0.62 under glare against one odd back; glare raised such scores. The real camera through slab plastic, sleeves, stickers and other games are untested.
- [Certain, sim] The simulator's face-up scores keep rising as the window shrinks: 0.24, then 0.43–0.46 with the round-3 window, now 0.57 in the Phase 0 simulation (threshold 0.69, lowest back 0.81; the simulator now places each card where real holders of its grader put it). The margin is synthetic. [Likely, CAD emulation of the real scans] With today's window the highest face-up score is 0.49–0.53 against the 0.55 floor, and about 101–103 of 126 face-down backs are accepted (the round-3 window: 113–116).

6. Verification done here (no real hardware)

Check	Tool	Result
Geometry: exact min-distance, all-pairs interference (including modelled fastener heads), held-slab sweep, hard-stop, rail-margin and line-of-sight checks at every pick, place, travel, station, hard-stop and Phase 0 pose	`cad/checks.py` (OpenCascade)	**177/177 pass** (re-run on 23 Oct 2026, about 12–15 min; the numbers are in `cad/out/checks.txt`) [Certain]. Picking the last slab next to full neighbour stacks clears them by 14.55 mm (the probe board's M3×16 tips); loaded travel clears the rods by 21.4 mm, a double pick of two thick holders by 5.33 mm. Both LED rings are modelled (40 × 30 × 5 and 63 × 47 × 5 mm, with their solder pads and leads): seated with zero overlap, the barb elbow 3.75 mm from the head ring at the full 20 mm stroke, nothing in either camera's view, a 60 mm ring fits the station seat too. The head ring checks run at 39.5, 40 and 42.5 mm (the ribbed 42.8 mm seat; a 42.5 mm ring has 0.15 mm of radial clearance). **Home switches** (Omron SS-5GL geometry): X clicks at cup X 19.00 and Z at c = 258.00; each striker holds the lever 0.6 mm past its click point (at the lever tip) to the hard stop, under the 1.2 mm overtravel; the levers are free in normal travel; the mounts, screws, finger and leads clear everything that moves (closest 1.49 mm, the Z carriage to `z_switch_mount`); the longest lever each mount takes is 27.24 mm (X) and 24.19 mm (Z), against a 23 mm limit. **Not checked:** belt slack, cables and the vacuum tube, the real shapes of bought parts (the buffer, elbow, rings and switches are modelled from catalog or listing dimensions), loads and deflection, the side load the strikers put on the switch levers, and the corner brackets and joining plates.

Check	Tool	Result
Face-down card window	`checks.py check_face_down_window`, `check_camera_aim`	[Certain] exact frustums from the lens (5 mm toward the label) at hover over stacks of 1, 15 and 30 and with a slab held: 0.0 mm³ of any part in view, nearest part 1.74 mm (1.94 held); 302.5 mm²; 11.0 mm clear of the label band. [Likely] On the card plane, a slab in a pocket corner: PSA 0.0% on the holder (top +3.39, side +0.22 mm); CGC 1.52% (top −0.03, side −0.23 mm; limit 2%). Aim tolerance before a head part enters: 1.34° toward the cup, over 4° every other way. Before more than 2% of the window is on the holder: 0.49° (PSA corner slab) and 0.01° (CGC corner slab); before it leaves a median card on a centred slab: 1.32°.
Carry-height sensor	`checks.py check_carry_sensor`	[Certain] trips at c = 246.50; beam clear from −14 to 242, fully blocked from 248 to the 262 top stop; flag 1.05 mm from the sensor, flag arm 4.6 mm from every part fixed to the X carriage, its lead 11.0 mm from anything moving; a head stalled just above where it reads clear (245.5) still carries a double thick pick 2.5 mm over the rods.
Probe flag, guide and sensor over the buffer's 20 mm stroke	`cad/checks.py check_probe`	[Certain] trip at 2.00 mm; beam clear for 1.5 mm, fully blocked from 3 to 20 mm; flag 1.05 mm from the sensor and 0.45 mm from its guide; 8 mm guide engagement at full stroke; a rotating (J) buffer stopped at 1.32°, 0.6 mm from the sensor; buffer ± 3 mm on its thread still covered by the ± 4 mm board slots; pad 1.5 mm clear of the buffer body at full stroke; nothing in the label view while reading; a ball-end key reaches the front board screw (1.63 mm clear).
Phase 0 jig vs machine	`cad/checks.py check_phase0`	[Certain] the occluder (foot, buffer, flag, elbow, board) sits 0.0 mm from the machine's relative to the lens and blocks the same share of the slab view; the face-up pass (mount 3 mm up) leaves the free cup 1.0 mm above the slab's front; every jig wall an M5×10 passes through is 4.0 mm.
Unit tests: cert parsing on the real payload strings, grader and side rules (including the 8-digit ITF rule, tried over every combination of reads), phantoms, decode at the design point, OCR fallback, the learned face-down descriptor and the camera-aim check (including a resume with and without an empty pocket), the carry-height sensor, config checks	`software/tests`	**`299 passed, 1 skipped in 1135.84s (0:18:55)`** (full suite on the final code, 23 Oct 2026) and the real-image test on its own: **`1 passed, 61 deselected in 137.29s (0:02:17)`** [Certain, the software fixer's runs]. On the code just before the last fix, the full suite with the real-image test gave `300 passed in 1202.90s (0:20:02)`. The skipped test runs `tools/real_eval.py` on the real image set when `SLABRAIL_REAL_IMAGES` points to it.
End-to-end batches in a simulator that **raises Crash on any unsafe move**	`software/tests/test_runner_sim.py`, `test_safety_fixes.py`, `test_bench_tests.py`, `test_face_down_learned.py`, `test_carry_sensor.py`	pass [Certain]. Scenarios: every slab moved and read; pre-2016 labels via the station; double pick detected; seal failure stops; overfill refused; CGC read, BGS to the exception pocket; face-down check, including FACE-UP and neighbour stops; an unexpected error with a slab in the cup; failed vent; vacuum sensor fault; Ctrl-C hard stop; a Z stall (40 mm and 10 mm losses) caught before the sideways move, a stuck or cut carry sensor, its selftest. Learned check: an old label passes once references exist; a face-up slab still stops; a plain face-up front with glare is refused; the round-2 head and a partly hidden window are refused; one wrong confirmation, or a neighbour's back QR in the frame, does not teach a front; a back code seen from below vetoes a reference; a Japanese back needs 3 confirmations; `--face-down-ok` is refused after another stop; a lower neighbour's back QR beside the slab is cropped out.

Check	Tool	Result
330-slab simulated batch (seed 7, default label mix, `--face-down-check`)	`sim --slabs 330 --face-down-check`	[Certain, sim, 23 Oct 2026, the round-5 code; the last fix touches only reads the simulator never makes, Likely] `needs_review 6, ok 324`: 5 BGS slabs OCR-only (pocket 12) and 1 pre-2016 PSA slab whose OCR read 9 digits (output stack). The other 20 pre-2016 PSA slabs are `ok` from their front ITF and OCR. No `ok` has a wrong cert or a wrong grader (checked against the simulated slabs). Face-down: 287 by back QR, 41 learned (lowest score 0.76), 2 confirmed (the first slabs, before 5 references existed); with back-label codes only, 43 would have stopped. `camera_aim: ok`. 14.2 s per slab, 79.4 min for the load (homing + survey 75 s), 991 head frames. With the head camera tilted 0.8° (`--sim-head-tilt 0.8`, round-3 window, 110 slabs) the aim check fired and every slab without a back code stopped (15 of 110). The mix (PSA current 170, 2017+ 56, 2016 12, older 21; CGC 2023+ 49, OG 12, Authentic 5; BGS 5) is [Guessing], and every simulated slab is face-down: the learned acceptances say nothing about face-up safety (the unit tests and Phase 0 cover that). Phase 0 on the simulator: GO, lowest back-to-back 0.81, highest face-up 0.57, threshold 0.69 (round-3 window: 0.82, 0.43, 0.62). The simulator places each card where real holders of its grader put it (`sim.HOLDERS`, the CAD's `layout.HOLDERS`).
Real labels: the machine's rules on public scans of 64 real slabs	`tools/real_eval.py`	[Certain for these images, 23 Oct 2026] Native (round-5 code): `ok` 60, `needs_review` 2 (pre-2016 PSA: OCR read other digits on `19802594`, nothing on `20908923`), `no_read` 2 (BGS, SGC: no code), **0 wrong certs in `ok`, 0 false face-down proofs**. Machine-equivalent (rescaled to the cameras, degraded, cropped as the runner does; 61 slabs; final code): 60 `ok`, 0 `needs_review`, 1 `no_read` (BGS), 0 wrong, 0 false; all 7 pre-2016 PSA slabs `ok` with "grader PSA from its 8-digit ITF". Before the ITF rule the 7 pre-2016 slabs were `needs_review` on both. With glare on the label (a run before the last fixes): 58 `ok`, 3 `needs_review`, 0 wrong. Before these rules the same images gave 1 silent wrong cert (the 13-digit CGC). **The rules were written from these 64 slabs**, and flat studio scans have no slab-plastic reflections: this shows the rules fit them, not that the machine will read yours.
Real slabs, real cameras, real cup	—	**Not done.** Phase 0 + acceptance tests in BUILD_GUIDE are mandatory before trusting it with your inventory.

7. Known risks and what retires them

Risk	Retired by
Your labels carry payloads unlike the 64 real slabs seen	Phase 0: scan 30–50 of your slabs with the real cameras, then `tools.real_eval --phase0` (0 wrong certs, 0 false face-down proofs). The report lists ITF lengths and payloads no rule reads. `custom_regex` in config handles a prefix or suffix.
Cup leaves visible marks	Phase 0B 24 h marking test with a hand pump, then the 300-cycle + 24 h test on the machine with the vacuum re-held all the time (BUILD_GUIDE §13.2). Swap cup material if needed (BOM lists two).
Slabs without a back-label code stop the run	The learned check, once Phase 0 says GO and 5 references exist: 2 stops in 330 in the simulator [Certain, sim; Guessing for real backs through plastic]. Until then use `run --all-face-down` only for stacks you checked yourself.
A real face-up front looks like a card back (blue Water-type fronts came closest)	Two references per match; a blue Water-type card in Phase 0's face-up pass; the back photo of every learned row audited in the first batches.
The head camera is aimed off, so a head part or the holder sits in the card window	The camera-aim check (layer 10) refuses a tilt toward the cup from about 1.3°. Not covered: a pad-dark head part, a tilt away from the cup, a roll (a CGC corner slab has about 0.01° toward the label): BUILD_GUIDE §12.3 look at `face_down_window.jpg`, a shim under the board. Software owner: measure aim in both axes from positive evidence (the back-QR box position, the card-top edge).

Risk	Retired by
The camera-aim check fires on a square camera (lint, dust, glue, a shadow on the pad)	Safe: the learned check stays off and slabs without a back code stop. Its thresholds need a real empty-pocket frame (Phase 0, `focus head`'s `focus_head_empty.jpg`). Software owner.
A CGC slab pushed into a pocket corner has up to about 1.5–4.5% of the window on the holder	[Likely] The 2% limit is a judgment, not a margin. Audit learned CGC rows. A +0.5 mm edge margin was tried and is out of reach: one rectangle with that margin leaves 266 mm² (the software needs 300), and the best 3-rectangle window (+0.53 mm at 300 mm²) moves y1 to −17.05, which cuts the aim margin to 0.97° against the 1.25° floor. The window stays as it is [Certain, CAD trade-off search; Likely that no shape does better: the search was a grid].
A neighbour's label in the head camera's crop	The crop to the slab's footprint; the two-cert rule; measuring `cam_head.hfov_deg`; a lens of F2.8 or faster (BOM).
A Z stall drags a slab sideways below the rod tops	The carry-height sensor: a loss over about 2.5 mm is caught before any sideways move (CAD and simulator). The §7.8 100-lift test still has to pass; jog, calibrate and mark do not read the sensor.
A phantom barcode confirms itself or drops the real read	[Certain, tested] Frames of one symbol count once, only the thin read inside the other read's box can be dropped, and a confirmed thin read makes a `conflict`. Two barcodes with the same misread are checked by OCR; if OCR reads nothing the slab stays `ok`.
Another grader's bare barcode is typed by its length (10 digits = CGC)	[Guessing that such labels exist] It is never `ok` (grader not confirmed), but as `needs_review` with a guessed PSA or CGC grader it stays in the output stack. The narrower length rules are applied now: a PSA Code128 is 8–9 digits and a 10-digit CGC one never starts with 0, so a 7-digit Code128 or a BGS-style `0019865140` is not typed at all (exception pocket) [Certain, `certs.py`; tested].
Another grader prints an 8-digit ITF, and OCR reads its printed number	[Guessing that one exists] The ITF rule would give `ok PSA`: the right digits, the wrong grader, in the output stack. In the real set ITF is only on PSA labels, but that set has 1 BGS and 1 SGC slab; BGS and SGC are reported to print a barcode or QR on some holders, type unknown [Likely, second-hand]. A sweep of 216 BGS, SGC and CGC frames found no ITF of any length [Certain, that set]. Put any non-PSA slabs you own through Phase 0 or `tools.real_eval` (OPERATING.md §5.2).
OCR confirms an ITF from a longer zero-padded number	[Guessing that it happens] The ITF rule compares certs with their leading zeros stripped, so OCR reading `0019865140` (a BGS-style number) would confirm an ITF `19865140` and give `ok PSA`. On every real ITF-rule `ok` frame the OCR digits equalled the ITF's exactly (316 of 316) [Certain, stress run]. Software owner: require the raw digits to match.
OCR picks one of two equally long digit runs at random	[Certain, verifier probe] `decode.ocr_digits` keeps the first longest token, and the order of a tie depends on the process's hash seed. Since OCR now decides `ok` versus `needs_review` for pre-2016 PSA slabs, the same slab can come out either way in two runs. Safe both ways (a non-matching token gives `needs_review`). Software owner: break ties by value, or return no read on a tie.
A homing switch lever is longer than the strikers allow, or wears under the sideways push	Measure each lever and cut it back past 23 mm (BUILD_GUIDE §6.8) [Certain, CAD: X takes 27.24 mm, Z 24.19 mm]. The 45° lead-in pushes the lever sideways as hard as down; the datasheet rates only a straight push [Guessing that it holds]: a 100-pass bench test before the build, and FIRMWARE.md step 6's homing repeat test after.
The 42.8 mm head ring bore prints small	[Likely] A 42.5 mm ring has 0.15 mm of radial clearance in the CAD, and printed holes come out 0.1–0.3 mm small: sand the bore for a ring over about 42.2 mm (BUILD_GUIDE §3.3).
The buffer and elbow differ from their catalog dimensions	Measure them on arrival (BUILD_GUIDE §3.3) before cutting the flag; the board slots give ± 4 mm of trip adjustment.
Beam twist larger than estimated	Dial-indicator check during the build. A rear diagonal brace is in the guide.
Double picks (slabs clinging)	Peel dwell. Detected at placement (step ≥ 11.6 mm) and flagged with the exact position.
Mixed weird holders (SGC 86 mm wide doesn't fit; BGS/TAG widths unmeasured; sleeves)	Loading rules in OPERATING.md. SGC is too wide for the pockets. Measure one BGS, TAG and CGC holder before loading them: anything wider than 82.6 mm is "do not load".

Interface contract (single source for docs, firmware, wiring and software)

Every document and file must agree with this page. If something here is wrong, fix it here first.

Machine: SlabRail

- **Layout:** one straight row of 14 cells, 92 mm pitch, centres at X = 42 + 92·cell.
  - Cell 7 (X = 686) is the read station: an upward camera, no stack.
  - The other 13 cells are pockets 0–12, left to right.
    - Pocket 0 starts empty.
    - Pockets 1–11 hold up to 30 slabs each (330 total).
    - Pocket 12 is the exception pocket.
- **Run order:** stack k is moved slab by slab onto pocket k−1, so every stack ends up one pocket to the left. Each slab is read on the way.
- **Loading rules:** slabs go in **face-down**, with the **label end toward the front** (toward the operator).
- **Sizes:** pocket interior 84 × 139 mm.
  - Guide rods are 8 mm hardwood dowels in the CAD (`ROD_D`); the BOM buys 5/16 in (7.94 mm). They stand 218 mm tall above the pad.
  - The fill line (MAX) is 210 mm.
  - An overfill laser beam runs at 214 mm along the row centre line.
- **Axes (only two):**
  - **X:** belt-driven, GT2 20T pulley, 40 mm/rev, 80 steps/mm at 1/16 microstepping, on an MGN12 1300 mm rail on a 4040 beam.
  - **Z:** NEMA17 (1.5–2.0 A/phase, ≤ 4 mH, ≥ 40 N·cm) with an integrated T8×8 lead screw (8 mm lead), 400 steps/mm at 1/16, on an MGN12 330 mm rail on a 2040 spine.
    - **The screw is cut to 340 mm below the motor face** (311–356 mm works). The screw tip is fixed to the X carriage right over the held slab (X +24, Y 33 from the cup axis). Cut to 340 it ends 283 mm above the pad, above everything the head carries. A 400 mm screw left uncut ends near 223 mm [Likely: integrated screws are usually measured from the motor face], so every lift would drive the held slab's back into its tip, and the tip would pass only 5 mm over the rod tops. [Certain, CAD `Z_SCREW_LEN`, `Z_SCREW_TIP`]
- **Work coordinates:**
  - X = the cup axis position in layout mm.
  - Z = the cup-face height "c" above the pad top.
  - Key heights: C_SAFE_EMPTY = C_SAFE_LOADED = 248, C_MAX = 250, STATION_READ_C ≈ 171.6, CARRY_SENSOR_C0 = 246.5 (the carry-height sensor's trip, read as ± 1 mm, `CARRY_BAND`). They include CUP_HANG = 2: the free cup hangs `buffer_trip_mm` below its trip-calibrated height. A measured `buffer_trip_mm` above 2 raises both travel heights by the difference (`config.load`), and one that would put loaded travel above C_MAX is refused. The carry sensor's trip must sit at least 1 mm under both travel heights (`config.check_heights`).
  - Hard stops and switches (CAD): Z bottom hard stop c = −14 (`z_end_cap`), Z top switch trips at c ≈ 258, Z top hard stop c = 262 (`z_plate` meets `z_motor_mount`). X hard stops at cup X 15 (`x_bumper_L`) and 1262 (left end of `x_motor_mount`); the X home switch trips at X 19 ± 2.
- **Homing:**

    - X homes to MIN (left end) with a microswitch and reads X = 22 there, so machine X = layout X (pocket 0 at X = 42). `calibrate x` stores the small residual as `offset_x`.
    - Z homes to MAX (top) with a microswitch and reads Z = 255. `calibrate z` stores the pad trip height as `pad_trigger_c`. The G54 work offset stays at zero; calibration lives in software.
- **Speeds:**
    - X: 15000 mm/min max, 2000 mm/s².
    - Z: 6000 mm/min, 1200 mm/s².
    - Probe: 1200 mm/min. **Every downward move is a guarded `G38.3`** (fast part at the Z speed, the probe window at 1200 mm/min, a held slab passes the rod tops at 1200 mm/min), so an unexpected contact stops it. A trip at 6000 mm/min squeezes the buffer about 6 mm before the head stops.
    - Halt (Ctrl-C twice): feed hold `!`, then soft reset Ctrl-X. Homing is required afterwards.
    - Motor current: X run 1.0 A, Z run 0.8 A (hold 0.5 A, to keep the head up).
- **Contact probe:** a slotted optical endstop (TCST2103) on the cup's spring buffer (M10, 20 mm stroke), wired **fail-safe** so a cut wire reads as TRIGGERED.
    - It is the FluidNC `probe` pin.
    - Probing uses `G38.3` (no alarm on a miss) with a bounded target.
    - `probe_hard_limit: true`: a probe trip during any move that is not a `G38` (a lift, a traverse, a jog, homing) is a critical alarm that needs a reset and `$H`. It must give zero alarms in the 300-cycle test (BUILD_GUIDE §13.2), or the sensor output gets hysteresis.
    - Buffer: SMC `ZPT25UUK20-B5-A10`, M10 spring buffer, **non-rotating (K)**, **20 mm stroke**. SMC sells no 25 mm stroke on a 25 mm pad. [Certain, SMC ZP catalog Table 2] The rotating `J20` also works, because `head_foot`'s guide holds the flag.
    - Flag: 1 mm aluminium, in the CAD (`probe_flag`), its tab clamped between the buffer's M5 port hex and an SMC `M-5ALU-4` barb elbow (no washer, thread sealant; tube toward +Y). It trips the beam after 2.0 ± 0.5 mm of buffer travel (`buffer_trip_mm`), set with the board's ± 4 mm slots.

CAD decisions after the build review (binding)

- **Checks:** `cad/checks.py` runs 177 checks; all 177 pass (re-run on 23 Oct 2026, about 12–15 min, numbers in `cad/out/checks.txt`). A CAD change is done only when they all pass again.
- **Rails on slots:**
    - The X rail centre is at Z = 410, on the 4040 beam's upper front slot. It is fixed with M3 screws into M3 T-nuts.
    - The Z spine (2040) sits 10 mm to the LEFT of the cup axis, so the Z rail is in its right slot, on the cup axis.
- **Z carriage:** its bottom is 295 mm above the cup face. The `z_end_cap` stops it at c = −14 (it would leave its rail at c = −19), and at the top hard stop (c = 262) 3.6 mm of carriage is still on the rail. The firmware soft limits (Z −8 … 255) sit inside both stops.
- **Z machinery above the held slab:** the Z spine bottom is 275 mm above the pad (the `z_end_cap` under it ends at 267) and the lead-screw tip is at 283, so even at the top hard stop a held slab clears every part fixed to the X carriage by 5 mm. [Certain, `cad/out/checks.txt`]
- **Bed:** the plywood has 41 × 41 mm notches at the two upright positions. The uprights pass through to the base rail.
- **Drop arm (2020 × 285 mm):** drill two 5.5 mm cross holes, 19 mm and 84 mm from the end that bottoms in `head_foot`'s socket, matching the head_foot and camera_bracket clamp holes.
- **Light roof:** OPTIONAL. Add it only if Phase 0 or the first runs show room-light glare on labels. If you do, it's black foam board on two 2020 posts bolted to the **outer end faces** of the uprights (X −68 and X 1348). Not to

their back faces: the beam-to-upright joining plates and brackets are there (BUILD_GUIDE §4.3). The CAD model leaves it out by default.

- **Guide rods:** 82 dowels at 12 mm from each pocket corner, which leaves a 52 mm finger gap front and back.
- **Head lens height:** 140 mm above the cup face at the 2 mm trip (`CAM_TO_SLAB`), so 142 mm above the free cup face. The buffer is set on its M10 thread to this (BUILD_GUIDE §7.5); the CAD allows ± 3 mm there.
- **Head lens position:** 5 mm from the cup axis toward the label end (`HEAD_CAM_DY = −5`), X on the cup axis, set by `camera_bracket` and, in Phase 0, by `phase0_top`. With the lens on the axis the M5 elbow 23 mm under it hid the card window.
- **Probe:** the flag, its guide (two fins on `head_foot`, a 1.9 mm slot) and the sensor post with ± 4 mm board slots are in the CAD, built on the SMC catalog heights of `ZPT25UUK20-B5-A10` (pad face to port top 108.5 mm) and Vishay's TCST2103 drawing. `checks.py` sweeps them over the 20 mm stroke.
- **Phase 0 occluder:** `phase0_occluder_mount` clamps to the jig's right post and holds `head_foot` on a 2020 × 66 mm stub, where the machine has it relative to the lens. For the face-up pass the mount slides 3 mm up the post.
- **Card window:** `FACE_DOWN_WINDOW_MM = [[-27.5, -23.5, 27.5, -18.0]]`, mm from the CUP axis (−y = the label end; 55 × 5.5 mm = 302.5 mm²). The software's `face_down_window_mm` is measured from the head-CAMERA axis: `layout.face_down_window_cam() = [[-27.5, -18.5, 27.5, -13.0]]`, which `gen_config` writes; the Phase 0 jig's flipped version is `[[-27.5, 13.0, 27.5, 18.5]]`. It is placed from `layout.HOLDERS`, the card position measured on 22 PSA and 27 CGC backs (card top to the holder's far end: PSA 97.18–99.95 mm, CGC 93.76–96.21 mm). `checks.py` proves it clear of every machine part at hover (stacks of 1, 15, 30) and with a slab held, and clear of the label band; on the card plane (card 3.5 mm under the back, rounded corners) a slab in a pocket corner has 0.0% of it on the holder for PSA and 1.52% for CGC (limit 2%). It also reports the camera-aim tolerance: 1.34° toward the cup (1.31° with the software's 147 mm mapping).
- **LED rings:** in the CAD. Head ring modelled 40 × 30 × 5 mm. Its seat on `camera_bracket` (and the pocket in `phase0_top`) is a 42.8 mm bore with three centring ribs, printed to fit a 40.0 mm ring, and a 33 mm lip: it takes OD 39.5–42.5 mm (shave each rib by half of OD − 40), and `checks.py` runs every head ring check at 39.5, 40 and 42.5 mm. Its LED face sits 1 mm under the lens front, the leads go out through a notch, and the barb elbow clears it by 3.75 mm at the full stroke. Station ring: a 64 mm bore on a 45° shelf (`station_mount`, `phase0_base`) that takes a 63 × 47 or a 60 mm ring, the leads down a slot into the ribbon slot. Each is fixed with three dots of glue.
- **Switch mounts:** printed, in the CAD, for a bare SPDT hinge-lever switch of the Omron SS-5GL size (2 holes 9.5 mm apart, M2×12 through ±1 mm slots). `x_switch_mount` sits under the beam's left end (2 × M5×10 button heads into the beam's lower front slot, at X −48.5 and −37) and is worked by a 17.4 mm finger on `x_bracket`'s left edge; `z_switch_mount` sits on the Z spine's +X face (2 × M5×10 button heads into that face's slot) and is worked by a 45° chamfer, 7.5 mm deep, along the `z_plate`'s top back edge from X 20 to its right end. Each striker slides across the lever: a 45° lead-in, then a flat that holds the lever 0.6 mm past its operating point (at the lever tip) up to the hard stop, under its 1.2 mm overtravel. Trips: cup X 19.00 and c = 258.00. Each mount slides about ±3 mm in its slot to set the trip. The lever must end within 23 mm of the switch's hinge end (`SW_LEVER_MAX`; a longer one is cut back to 20 mm); the X mount takes up to 27.24 mm and the Z mount 24.19 mm. Lead paths are modelled (X: under the beam into its lower rear slot; Z: behind the `z_motor_mount` to the spine top).
- **Carry-height sensor:** the spare TCST2103 board on the Z spine's −X face (`carry_sensor_mount`, 2 × M5×10 button heads into slot-6 T-nuts, beam at Z 561.5 above the pad top), and a flat 1 mm aluminium flag on `carry_flag_arm` (screwed to the `z_plate` front face). Its beam is clear at every cup height up to 242 and fully blocked from 248 to the 262 top stop; it trips at 246.5, set by sliding the mount.
- **Head crop:** `head_roi_half_mm = SLAB_W/2 − play = 38.3`, from the CAD: every real code lies within ± 33.5 mm of the slab's centre line.

Controller

- **Board:** BTT Rodent V1.x (V1.1 assumed) running FluidNC v4.1.1 or newer. BTT's official `rodent.yaml` is the base.
  - Set `r_sense_ohms` to the board's resistor marking: R075 → 0.075, R022 → 0.022. `config.yaml` ships with 0.022 on purpose: the wrong guess then gives weak motors, not 3.4× the current. Change all four lines to 0.075 only after reading R075 on your board.
- **Connection:** CH340 USB serial at 115200 baud to the Raspberry Pi: `/dev/ttyUSB0`, by-id `/dev/serial/by-id/usb-1a86_USB_Serial-if00-port0`.
- **Unused features:** no Y axis is defined, and no spindle or coolant pins are used. FluidNC still reports three axes (`MPos:x,0.000,z`).
- **Driver sockets:** X motor → socket X, Z motor → socket Z. Sockets Y and E are configured as ganged spares for X and Z, because the SPI chain needs all four. Leave them empty.
- **Inputs, all fail-safe (a cut wire reads as triggered):**
  - X home switch (NC) → X-MAX connector, gpio.35.
  - Z top switch (NC) → Z-MAX connector, gpio.33.
  - Optical contact probe → **Y-MAX connector, gpio.34** (not the board's Probe connector).
- **Soft limits (machine mm):** X 22…1253, Z −8…255.
  - After homing: X = 22, Z = 255.
  - Probe targets must stay above Z −8.
  - CAD must keep both carriages on their rails at every hard stop (X 15 / 1262, Z −14 / 262).
- **E-stop:** two NC contact blocks **in series** in the +24 V line. It cuts the Rodent, the pump, both valves and the LED buck, so FluidNC reboots and the USB serial device disappears. The software must treat a lost port as a stop, then reconnect and home again. A held slab then hangs only on the closed cup line until it leaks down: the operator takes it off.
- **Mains:** the IEC inlet sits in a closed ABS box with a cable gland to the LRS; no mains metal is exposed with the cord plugged in.
- **Firmware files:** `firmware/fluidnc/config.yaml`. Bring-up steps: `docs/FIRMWARE.md`.
- **Wiring:** `docs/WIRING.md` and `docs/img/wiring.svg`.

Raspberry Pi 5 (4 GB), Raspberry Pi OS Bookworm 64-bit, Python 3.11

- **GPIO (BCM numbering), each output driving a logic-level MOSFET module:**
  - 17 = vacuum pump (24 V)
  - 27 = vacuum valve, normally-closed 2/2 (24 V)
  - 22 = vent valve, normally-closed 2/2 (24 V)
  - 23 = head LED ring
  - 24 = station LED ring
- **GPIO inputs:**
  - 25 = optional lid/door switch to GND (internal pull-up)
  - 5 = overfill beam receiver output (HIGH = beam received). The laser is a ≤ 1 mW (Class 2) 650 nm module, fed from the Pi's 5 V, so it is on whenever the Pi is on.
  - 6 (pin 31) = carry-height sensor output through a 10 kΩ / 20 kΩ divider (HIGH = the flag is in the beam = the head is at carry height). A cut lead reads LOW, which refuses sideways moves; a dead emitter reads HIGH, which the runner catches by requiring LOW at every grip.
- **I2C1 (GPIO 2/3):** ADS1115 at 0x48.

    - Its A0 reads the XGZP6847A vacuum sensor (−100…0 kPa, 0.5–4.5 V at 5 V supply) through a 10k/20k divider.
- **Cameras:**
    - Head: USB UVC 8 MP IMX179 board with an M12 3.6 mm lens (F2.8 or faster), manual focus, looking down at the held slab's back from 140 mm, 5 mm toward the label from the cup axis. Its frames are cropped to the slab under the cup (`head_roi_half_mm`, `cam_head.hfov_deg`, `row_axis`, `lens_to_label_mm`: check them on the real camera).
    - Station: Raspberry Pi Camera Module 3 (standard) on CSI port 0, looking up at the held slab's front from 140 mm.
- **Lighting:** a 40 mm COB LED ring around the head lens (a 60 mm ring cuts into the drop arm; a 40 mm 12-SMD ring that fits the seat is the fallback) and a 60 or 63 mm ring around the station lens, both 12 V. The black foam-board "light roof" is optional (see the CAD decisions above).

Software (repo folder `psa-slab-scanner/software`)

- **Package:** `slabscan/`
    - `certs`, `decode`, `facedown`, `sim`, `runner`, `simworld`, `motion`, `hw`, `camera`, `config`, `results`, `psa_api`, `ops`, `factory`, `cli`, `web`
- **Config:**
    - `config/machine.toml` is generated by `python -m tools.gen_config` from `cad/layout.py`.
    - `config/calibration.toml` holds the values written by the calibrate commands (and `phase0 --save`).
    - `config/card_backs.npz` holds the learned face-down check's card-back references. Not in git.
- **CLI:** `python -m slabscan <command>`
    - `selftest [-i] [--motion]`: GPIO, vacuum sensor, pump and valves, both cameras, controller connection, beam gate, carry-height sensor (`--motion` measures where it trips).
    - `home`
    - `jog`: interactive.
    - `calibrate z`: probes the bare pad of pocket 0 and writes `pad_trigger_c`.
    - `calibrate x`: operator centres the awl over pocket 0 and pocket 12 and writes `offset_x` plus a scale check.
    - `focus head|station`: live sharpness and decode score to set focus. On the machine `focus head` first checks the camera's aim over the empty pocket 12 and saves `focus/focus_head_empty.jpg`.
    - `mark`: drives the awl to every cell for the drill template.
    - `run [--name NAME] [--front-photo always|fallback] [--all-face-down] [--yes]`
    - `resume BATCH_DIR [--face-down-ok] [--accept-heights] [--yes]`
    - `sim [--slabs N] [--mix MIX] [--eras E] [--face-down-check] [--jp-rate R] [--sim-head cad|cad_r2|open|none] [--sim-head-tilt DEG]`: full simulated batch, no hardware. `--sim-head-tilt` tilts the simulated head camera toward the cup (every command that takes `--sim` accepts it). `--mix` sets the label mix per grader and era (default: mostly PSA, some CGC, a few BGS); `--eras` is a PSA-only override. `--face-down-check` keeps the check on and counts the slabs it would stop at (a simulated operator confirms each). `cad` is today's head, `cad_r2` the round-2 head whose foot hid the card window.
    - Bench tests (not part of the CLI): `python -m tools.bench_tests cycles|rest|lifts|drop|powercut N` (BUILD_GUIDE §7.8, §13).
    - Real-label check (not part of the CLI): `python -m tools.real_eval INDEX.json [--machine] [--glare]` or `--phase0 DIR`: the run's reading order, consensus and face-down rule on photos of real slabs; exit 1 on a wrong cert in an `ok` verdict or a false face-down proof.

    - `web [--port 8080]`: phone-friendly start/stop/progress page and CSV download.
    - `phase0 [--sim N] [--out DIR] [--face-up N] [--save] [--force]`: the Phase 0 camera read-rate test (same as `python -m tools.phase0`). An `--out` folder that already holds a `phase0.csv` is refused (exit 2, nothing changed) unless `--force` is given. The read-rate report is written as soon as the operator types `q`. `--face-up` (default 0) adds the face-up pass for the learned face-down check; a GO needs 15. `--save` writes `face_down_threshold` and `face_down_threshold_sig` and merges the backs into `config/card_backs.npz`, only on a GO.
    - `refs [--drop CERT | --drop-operator]`: list or delete the learned face-down check's references.
- **Outputs:** `batches/<id>/certs.csv`, `certs.txt`, `report.html`, `summary.json`, `journal.jsonl`, `images/` (plus `run.json`, and `stop.json` after a stop).
    - `certs.csv`'s last column is `face_down`: `qr` (a PSA or CGC back QR), `code` (the CGC 2020–23 back barcode), `operator`, `learned(<score>)` or `off`. `report.html` has a Face-down column; `summary.json` a `face_down` entry, plus `itf_mode` (always `zero_pad`) and `itf_evidence`.
    - Learned mode adds `images/face_down_window.jpg` (the run's first slab with the card window drawn) and `images/face_down_empty.jpg` (the view over an empty pocket), and `summary.json`'s `face_down` gets `camera_aim` (`ok`, the failed check's message, "not checked (no view over an empty pocket)", or "not checked this run (…)" after a resume that never looked over an empty pocket); a failed aim check prints a `CAMERA AIM:` summary line.
- **Enrichment (optional, later):** `python -m tools.enrich batches/<id>` (PSA API, quota-metered).
- **Graders:** PSA and CGC are read and stacked normally. An `ok` needs its grader confirmed: a code that names it (a cert URL), two printed symbols in its format, or an 8-digit ITF whose digits OCR of the printed number confirms (`certs.psa_itf_confirmed`: grader PSA, note "grader PSA from its 8-digit ITF"; a pre-2016 PSA slab). Any other one-barcode slab is `needs_review` even when OCR agrees (OCR confirms its digits, not its grader), and an ITF alone or OCR alone is never `ok`. A code in no grader's format never confirms a grader. A PSA Code128 is 8–9 digits and a 10-digit CGC Code128 never starts with 0; any other Code128 is not typed. Every barcode-only `ok` also gets the front OCR. Other graders go to pocket 12, and so does every `needs_review` whose grader is not an accepted one (`unknown`: an OCR-only number or a code with no grader's format; `PSA/CGC`; none), whatever `needs_review_to` says. A grader guessed from a barcode's length counts as accepted.
- **Face-down check:** before the cup touches a slab, the frame from above must prove it is face-down. `face_down_check` sets how:
    - `"learned"` (the default): a back-label code (`certs.SIDE_PROOFS`: a PSA back QR, a graded CGC QR, the CGC 2020–23 back barcode), or a card window that matches two references (two different slabs of one back design) among the card backs of code- or operator-proven slabs (`slabscan/facedown.py`). It is active only with a Phase 0 threshold (≥ 0.55) and its matching `face_down_threshold_sig`, `face_down_min_refs` (5) references, the empty-pocket view, at most 10% of the window static on 5 machine-proven slabs, and a clear camera-aim check on that view (`facedown.aim_problem`: no edge in the middle of the window or within 1 mm of its cup side; a tilt toward the cup only), taken this run (`CardBackRefs.stale`: a resume whose survey finds no empty pocket takes the view over the first source pocket it empties, and accepts nothing before that). Until then it acts like `"stop"`. A code-proven or confirmed slab becomes a reference only after it is placed: never if a back-label code is seen from below, and a code proof only with an `ok` read of the same cert.
    - `"stop"`: only a back-label code.
    - `"off"`: no check. `run --all-face-down` (or the web start checkbox) does this for one batch; the choice is stored in `run.json` and kept on resume.
    - A front-label code seen from above (a CGC Authentic QR, a CGC front barcode) stops with a FACE-UP message. Back and front codes together, two back labels, or codes of two certs in one frame stop as "another slab's label is in view". Anything not proven stops untouched until the operator confirms with

`resume --face-down-ok` (or the web checkbox, shown only then). That is refused unless the batch's last stop was a face-down stop, and it confirms only that pocket's top slab.
    - Journal events: the stop line carries `face_down_stop` (pocket) and `why`; `face_down_confirmed`, `face_down_reference` (a confirmed slab learned after placement, with its cert and back photo), `face_down_learned` (score, ncc, hist, prototype, threshold, refs, texture, baseline, `position`), `face_down_baseline` (with `aim`), `face_down_ok_unused`; `stall_set_down` (the pocket a held slab was lowered onto after a Z stall). Events that are not transfers never use the `pos` key: `rebuild_state()` reads `pos` with `src` as a finished transfer.
- **Stops:** Ctrl-C once (or web Stop) = stop after the current slab. Ctrl-C twice = hard stop: the motors halt where they are, nothing moves afterwards, a held slab stays on the pump until the operator holds it and vents (CLI prompt or the web Vent button).
- **Carry-height sensor:** read before every sideways move (must be blocked), every Z or probe move with a slab in the cup, every grip (must be clear), and at a pause just above its trip on every loaded descent. A disagreement is a `CarryStop` ("Z stall suspected …" or "Carry-height sensor fault …"); after a stall a held slab is never carried sideways (lowered by one probe move over its own pocket, else held). Homing clears it. `jog`, `calibrate` and `mark` do not read it.
- **One process:** the machine is locked to one slabscan process (`/run/lock/slabrail.lock`); stop `slabrail-web` before running CLI machine commands.

Budget

The original figure was $2,000; on 22 Oct 2026 the user said $2,500–3,000 is fine as long as the machine works properly and is good quality. The user has school 3D printers and mechatronics instructors. The current BOM is about $2,182 with a 15% allowance, and the worst case about $2,380 (BOM.md has the breakdown; SHOPPING_LIST.md the orders with links). Upgrades are qty-0 BOM lines, each bought only when the test it names fails.

SlabRail build guide

Step by step, from the first camera test to the first batch of real slabs. It is written for a tech-school student working with mechatronics instructors. Interfaces (pins, axes, paths) come from CONTRACT.md. Wiring is in WIRING.md, firmware in FIRMWARE.md, parts in BOM.md, and the software in ../software/README.md. Day-to-day use is in OPERATING.md.

Confidence tags: **[Certain]** = read in the CAD, the code, a datasheet, or tested in the simulator. **[Likely]** = strong inference or several secondary sources. **[Guessing]** = a gap I filled. Every time estimate is **[Guessing]**.

Read this first

- **Phase 0 is two gates, and both come before any motion parts.** [Likely, BOM.md order plan]
  - **0A, camera test: ≈ $470**, or ≈ $281 if you borrow a school Pi 4 or Pi 5 (a Pi 5 4 GB costs $110 by itself in October 2026). Buy your own Pi only after 0A says GO.
  - **0B, cup test: ≈ $155**, with a spare SMC cup and buffer. A hand vacuum pump on the bench: no mains wiring, no electronics.
- **The full machine costs about $2,182.** [Likely ±10%] That is with a 15% allowance; it is about $1,862 if the school supplies its usual shop stock and about $1,787 with generic valves as well. A realistic worst case buying everything is about $2,380, inside the $2,500–3,000 you said is fine. Use the school's shop stock anyway: it is the cheapest headroom. Upgrades are in the BOM as qty-0 lines: buy one only when its test fails. Every order, with a link per line, is in SHOPPING_LIST.md.
- **Nothing here has moved a real slab yet.** [Certain] The CAD passes its clearance checks and the software passes its tests against a simulator. Real labels, the real cup and real wiring are untested. That is why Phase 0 comes first and section 13 has the tests to pass before you trust the machine with inventory.
- **Plan on 65–90 hours of hands-on work over 6–10 weeks**, plus about 45 hours of printer time. [Guessing] Drilling 82 rod holes, the wiring and the acceptance tests are most of it.
- **The CAD and BOM still have small gaps you fix during the build.** [Certain, from reading the CAD and BOM] Each one is handled where it comes up:

Gap	Where it is handled
The CAD has no brackets, and the bed covers the base-rail tops where the upright brackets would go.	§4.3
The SMC cup code `ZPT25UUK20-B5-A10` comes from SMC's catalog (25 mm pads come only in 10/20/30/40/50 mm stroke [Certain]). Proax lists the exact code, but back-ordered with no lead time [Certain, 22 Oct 2026]. `head_foot` has a flag guide, so the rotating "J" buffer works too.	§2, §7.6
The flag, guide and sensor post are built on bought-part numbers you must check on arrival: the buffer's catalog heights [Certain, SMC catalog], the M-5ALU-4 elbow's height (9.3 mm) [Likely], and the endstop board's header position and component heights (unknown; the listing does not even name the TCST2103). Measure them before you cut the flag; the board slots give ±4 mm. The carry-height sensor uses the same board.	§3.3, §7.4a, §7.6
The head camera's board-to-lens-front length (20 mm) and its 32 mm hole pattern are guesses in the CAD (the ELP board is 38 × 38 mm). The Phase 0 bridge takes any length from 10 mm up.	§1.5, §3.3
The head lens sits **5 mm toward the label from the cup axis**, set by `camera_bracket` (and by the jig's bridge in Phase 0): there is no adjustment. With the lens on the axis, the vacuum elbow 23 mm under it hid the card window the learned face-down check needs [Certain, CAD ray cast]. Print the current `camera_bracket`, `station_mount`, `phase0_top` and `phase0_base`: their LED ring seats changed too.	§1.5, §3.2, §7.7
The LED rings are sold by their outside diameter only. The CAD models a 40 × 30 × 5 mm head ring and a 63 × 47 × 5 mm station ring (a 60 mm one fits too) [Likely OD, Guessing ID and thickness]. The head seat takes any ring from 39.5 to 42.5 mm across: a 42.8 mm bore with three centring ribs that you shave for a ring over 40 mm. No 40 mm COB ring listing is live: a 12-SMD ring that fits the seat is the fallback.	§3.3

Gap	Where it is handled
The card window is placed from the card position measured on 22 PSA and 27 CGC backs [Likely ±0.5 mm]. PSA slabs keep it on the card. On a CGC slab pushed into a pocket corner it has about zero margin: about 1.5% of the window can lie on the holder (up to about 4.5% at most) [Likely, CAD card-plane check].	§1.2, §12.3
Camera aim: a tilt of more than about 1.3° toward the cup puts the cup pad into the card window (1.34° in the CAD check, 1.31° with the software's own mapping) [Certain]. Every learned run looks for that tilt over an empty pocket, and turns the learned check off for the run if it finds it [Certain, sim]. It does not see a head part as dark as the pad, a tilt away from the cup or a roll along the row, and a CGC slab in a pocket corner has almost no aim margin toward the label: for those, nothing catches it but your look at the frame.	§7.7, §12.3
The carry-height sensor's trip (c = 246.5) is set by sliding its mount along the spine's slot, then measured by `selftest --motion`.	§7.4a, §12.1
The only Class 2 laser found is 7 mm across: the left gate post now prints with a 7.2 mm hole. Drill only a post printed from an older STL.	§5.9
The X home switch and the Z top switch have printed mounts now (`x_switch_mount`, `z_switch_mount`), worked by a finger on `x_bracket` and a chamfer on `z_plate`. They are drawn for a short flat hinge lever: one that ends 23 mm or less from the switch's hinge end fits as it is, and a longer one is cut back to 20 mm first. Which lever the BOM's 10-pack has is not known [Guessing]. The strikers push the lever sideways as well as down: a 100-pass bench test on one switch comes first.	§6.8, §7.4
The Z lead screw must be cut to 340 mm below the motor face.	§7.3
The jig's M5×10 screws pass 4 mm printed walls with one washer each, and the probe board takes M3×16 with washers, as the CAD now models. Test the first jig screw anyway: the slot depth is a guess. [Guessing]	§1.5, §7.6

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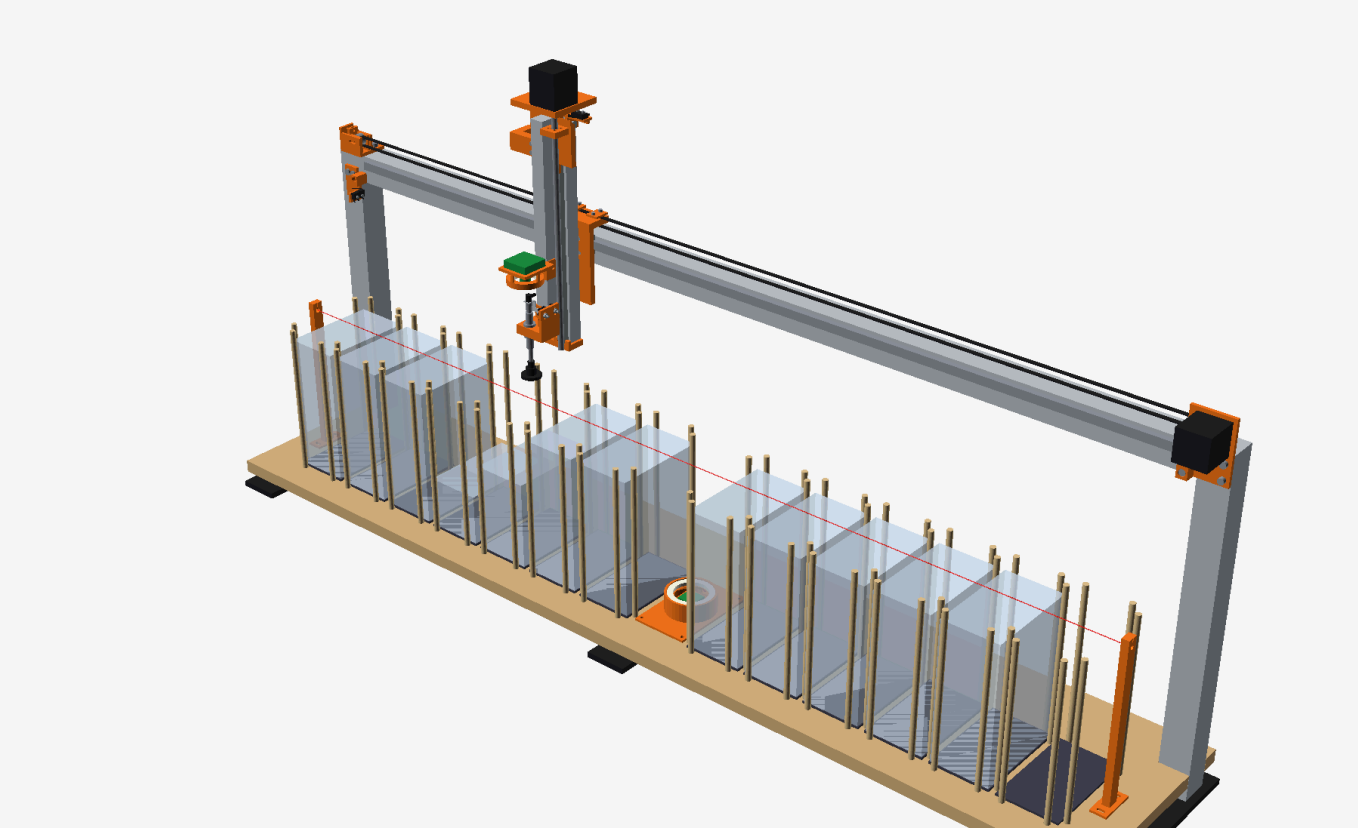
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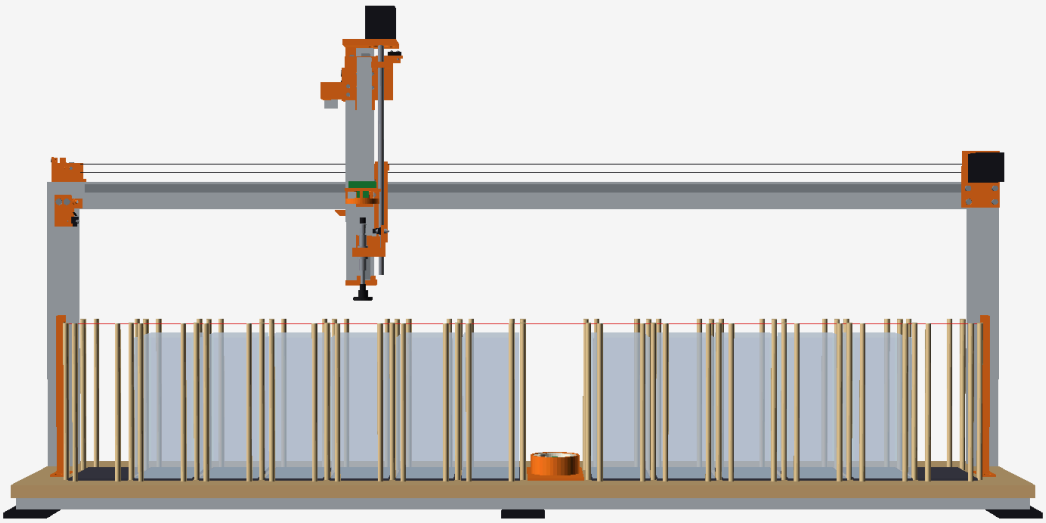
0. What you're building

SlabRail is one straight row of 14 cells under one gantry beam.

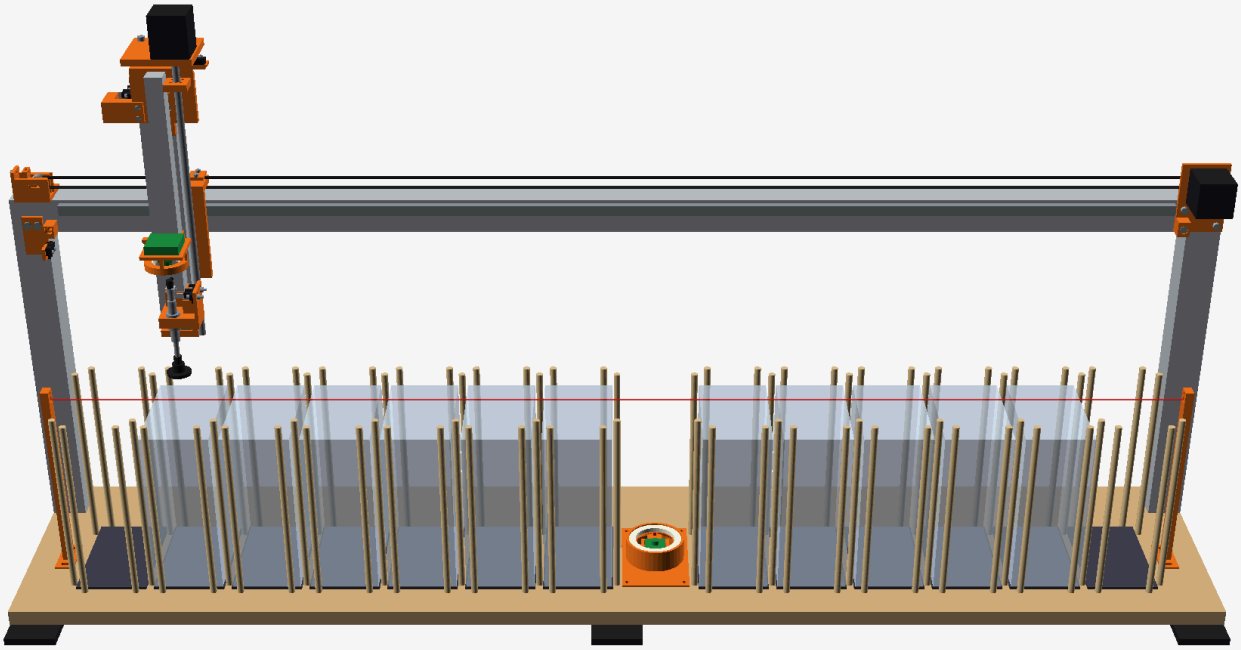
- **Cells:** 92 mm apart, centres at X = 42 + 92·cell.
  - Cell 7 (X = 686) is the read station, an upward camera.
  - The other 13 cells are pockets 0–12, left to right.

- **Loading:** pockets 1–11 hold up to 30 slabs each, face-down. Pocket 0 starts empty. Pocket 12 is the exception pocket.
- **Run:** a suction cup on a two-motor gantry moves every stack one pocket to the left, one slab at a time. It reads each slab's label on the way and writes `certs.csv`.

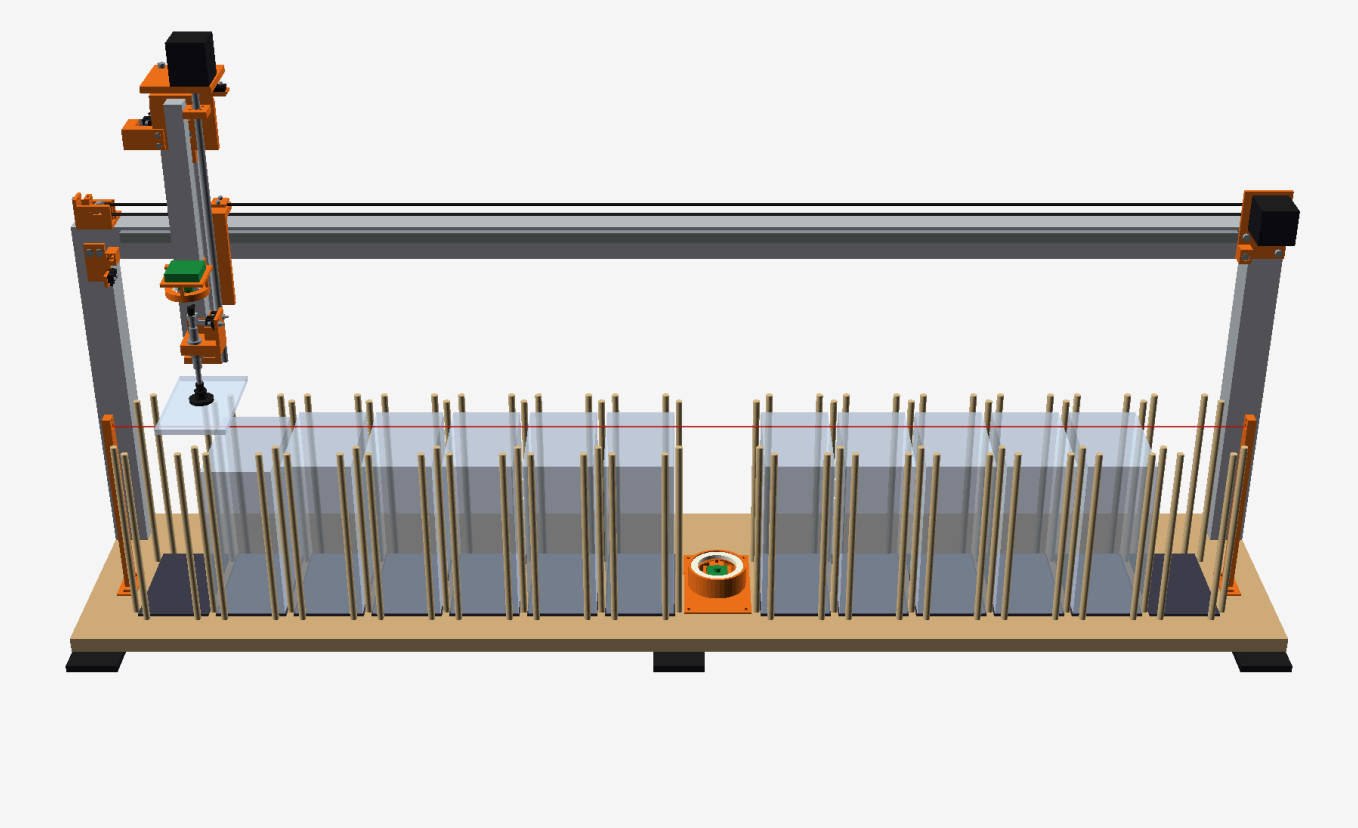




How a run looks: loaded, then the first transfer. The whole sequence, with all four frames, is in OPERATING.md §4.



1. Load: 11 stacks of up to 30, face-down, label toward you. Pocket 0 (left) and pocket 12 (right) stay empty.



2. The cup lifts the TOP slab of stack 1 straight up, reads it, and sets it down in empty pocket 0.

Item	Value	Source
Footprint	1416 × 290 mm bed; about 1.44 × 0.29 × 0.72 m overall, feet included	`cad/model.py` (assembly bounding box) [Certain]
Motors	X: NEMA17 on a GT2 belt. Z: NEMA17 with a T8×8 lead screw, cut to 340 mm.	CONTRACT [Certain]
Rails	X: MGN12 1300 mm on a 4040 beam. Z: MGN12 330 mm on a 2040 spine.	CONTRACT [Certain]
Pocket	84 × 139 mm inside, made by 5/16 in (7.94 mm) dowels 218 mm tall	`cad/layout.py` [Certain]
Capacity	11 × 30 = 330 PSA slabs	[Certain]
Cycle time	about 14 s per slab, so about 79 min for 330 (`sim --slabs 330`)	software [Guessing until measured]

0.1 Build order and time

The section numbers follow the parts of the machine, but the real order is different in one place. The rod holes are drilled *by the machine itself* (the head carries an awl that aligns the drill template), so §5 is split: 5A now, 5B after the machine moves (§12).

#	Step	Section	Hands-on	Printer / waiting	Instructor needed
1	Phase 0A: camera test (and the probe flag, if the SMC cup has arrived)	§1	6–7 h	about 17 h print	for power-tool rules (match-drilling, sawing the flag)
2	Phase 0B: cup bench test (hand pump)	§1.10	3–4 h	24 h test	no
3	Order the machine parts	§2	1 h	1–4 weeks delivery	no
4	Print the machine parts	§3	3 h	about 28 h	no
5	Frame	§4	4–6 h		**yes: squaring**

#	Step	Section	Hands-on	Printer / waiting	Instructor needed
6	Bed, part A (cut, notch, bolt down, cut pads)	§5A	3–4 h		for power-tool rules
7	X axis	§6	4–5 h		helpful for the dial indicator
8	Z axis, head and carry-height sensor	§7	5–7 h		no
9	Vacuum plumbing (finish)	§8	2–3 h		no
10	Electronics	§9	10–15 h		**yes: mains, crimp review, first power-up**
11	Firmware	§10	2–3 h		first homing: yes
12	Software	§11	1–2 h		no
13	Bed, part B: mark, drill, glue rods, pads	§5B	8–12 h	24 h glue cure	for power-tool rules
14	Calibration	§12	2–3 h		no
15	Acceptance tests	§13	6–8 h	about 3 days elapsed	first E-stop test: yes
16	First real batches: 30 slabs, then 100, then a full load	§14	2 h, then two more sessions of 2–3 h		no
	Total		**about 65–90 h**		

0.2 Cost

From BOM.md, with its 15% allowance; the orders line by line, with links, are in SHOPPING_LIST.md. $1,383 of the $1,897 parts subtotal is a price seen on the product page on 22 Oct 2026 [Certain]; the rest are search-text prices or estimates. None is a checkout price: shipping, tax and duty come on top, and the allowance is for them.

	Cost
Order 1a: Phase 0A (Pi, cameras, rings, USB extension, jig parts, endstop boards, flag sheet)	≈ $470 (≈ $281 with a borrowed Pi)
Order 1b: Phase 0B (two cups and buffers, pads, elbows, tube, PTFE tape, hand vacuum pump)	≈ $155
Order 2: the machine (frame, motion, bed, vacuum system, electronics, mains box, cables)	≈ $1,556
Full list	**≈ $2,182**
With school shop stock (filament, wire, glue, fuses, screws, flag sheet)	≈ $1,862
Lean (also generic valves)	≈ $1,787
Optional light roof, only if Phase 0 shows glare (conditional BOM lines)	+ ≈ $36
Worst case buying everything (tools, HT1 cups, roof, [Guessing] prices 20% high)	≈ $2,380: under $2,500
Upgrades, each only if its test fails (HIWIN rail, Z motor, constant-force spring, high-flex USB, branded pump, branded E-stop)	+ $20–420 each, about $920 if all of them
Junk slabs for §13, if they don't come from your stock	+ $150–400

Keep a tally of what you actually paid after each order (BOM.md).

---

1. Phase 0: camera and cup test (go / no-go)

Why: the two riskiest unknowns are (a) whether the cameras read *your* labels, and (b) whether the cup marks a slab. Both can be tested on a bench for a fraction of the machine's cost. If either fails, you have not bought any motion parts. **Order the machine only if both 0A and 0B pass.**

1.1 What Phase 0A needs

Rows marked "PHASE 0A" in BOM.csv. [Certain for the list; prices as in the BOM]

Part	From	Cost
Raspberry Pi 5 4 GB, 27 W PSU, 64 GB microSD, Active Cooler	BOM, or borrow a school Pi	$164.85 (or $0)
Camera Module 3 (standard) + 500 mm Pi 5 cable	BOM	$29.45
ELP 8 MP IMX179 USB camera, 3.6 mm M12 lens	BOM	$55
3 m USB 2.0 extension (the BOM line has two; Phase 0A needs one)	BOM	$13.90
LED rings: 40 mm for the head, 60 mm for the station (12 V COB)	BOM	$24
12 V DC wall adapter, 1 A or more, with a screw-terminal pigtail	BOM	$14.95
Two 2020 × 312 mm extrusion posts, cut square to ± 0.5 mm, and one 2020 × 66 mm arm stub	BOM	$16
M2 screw kit and M2 standoff kit	BOM	$17
6 × M5×10 + 6 M5 washers + 6 M5 slot-6 T-nuts (or plain M5 hex nuts, slid into the slot from the open end of the post or stub [Likely]); 1 × M5×30 + nylock; 2 × M3×16 + 4 washers + 2 nuts	school stock (their packs are order-2 lines), or pull the KB-3D M5 T-nuts and AFT M5×10 forward (about $19)	≈ $0–5
A strip of microfiber cloth, and spray adhesive or double-sided tape (the tray ledges, §1.5 step 4)	school stock, or pull the order-2 Pad facing and Spray adhesive lines forward ($35.96)	≈ $0
Optical endstop boards, two 2-packs (one for `head_foot`'s sensor post; the second becomes the carry-height sensor)	BOM	$5.08
1 mm soft aluminium sheet, at least 50 × 60 mm (the probe flag now, the carry flag later)	BOM / school stock	$11.99 (or $0)
Hookup wire, heat shrink, filament	BOM / school stock	$56.68 (or $0)
Printed: `phase0_base`, `phase0_tray`, `phase0_top`, `phase0_occluder_mount`, and `head_foot` (the machine's own part)	§3	
Total		**≈ $409 + 15% ≈ $470**, ≈ $281 with a borrowed Pi, ≈ $202 if the school also supplies the wire, filament, heat shrink and sheet

- **Why the 12 V adapter:** it powers the LED rings without any mains wiring, because the `phase0` tool leaves the lights on and never switches them. [Certain, `tools/phase0.py`] The rings and the cameras are the only powered parts in Phase 0A.
- **Why the USB extension and `head_foot` already:** in the machine the head camera reaches the Pi through about 4 m of USB, and the foot, buffer, flag, elbow, sensor and cup sit between the lens and the slab, inside the ring light. The jig reproduces both, so a GO means the machine's conditions. [Certain, `checks.txt`: the jig's occluder sits 0.0 mm from the machine's, relative to the lens, and blocks the same share of the camera's view of the slab]
- **The SMC cup, buffer and barb elbow are the real occluder.** They are in order 1b, but the cup's lead time can be 2–8 weeks [Likely], and Proax shows it back-ordered. **Order the two cups and the two `M-5ALU-4` elbows (and the PTFE tape) with 1a** if you accept about $77 at risk: the elbow is part of the occluder and also clamps the flag, so with cups but no elbow neither can go on the jig. Otherwise use an M10 bolt with a 25 mm washer as a dummy and leave the flag off; the label view does not depend on it [Certain, `checks.txt`: nothing on the head blocks the label].
- **With a Pi 4:** use the 15-pin camera cable that ships with the Camera Module 3, not the Pi 5 cable. [Likely]

1.2 Measure five of your slabs (15 min)

The CAD's slab size is in `cad/dims.py`: 80.3 × 135.5 × 6.6 mm nominal, design maximum 82.6 × 136.5 × 7.0. Use calipers to measure five slabs from your stock, including the newest and the oldest holders.

- **Width over 83.0 mm** or **length over 137.0 mm:** stop. The pockets (84 × 139) are too tight for them, so the CAD must change before anything is printed. [Certain, from `layout.py` play of ±1.85 / ±1.75 mm]
- **SGC holders are about 86 mm wide and do not fit at all.** [Likely, `layout.py` comment] Measure one if you deal SGC.
- **Also measure one CGC, one BGS and one TAG holder** if you deal them. [Guessing: no width for them has been measured; BGS may be wider than PSA] Anything wider than 82.6 mm is "do not load" (OPERATING.md §2). CGC matters most, because the machine reads CGC normally.
- **On two PSA slabs (and one CGC), measure from the card's top edge to the slab's FAR end** (the end away from the label, through the plastic). The learned face-down check looks at a window on the card back about 44 to 50 mm from the slab's label-end edge (`face_down_window_mm` = y −18.5 … −13.0 mm from the head-camera axis, which is −23.5 … −18.0 mm from the cup axis). A slab pushed to the cup end of its pocket play has its far end on the pocket wall, so this distance decides whether the window stays on the card. **It must be at least 93.5 mm** [Certain, CAD]. The real backs gave 93.8–96.2 mm on CGC and 97.2–100.0 mm on PSA [Likely ±0.5 mm, 27 CGC and 22 PSA backs]. If yours is under 93.5 mm, note it for §1.7: the window's label-side edge can then reach past the card's top edge onto the holder.

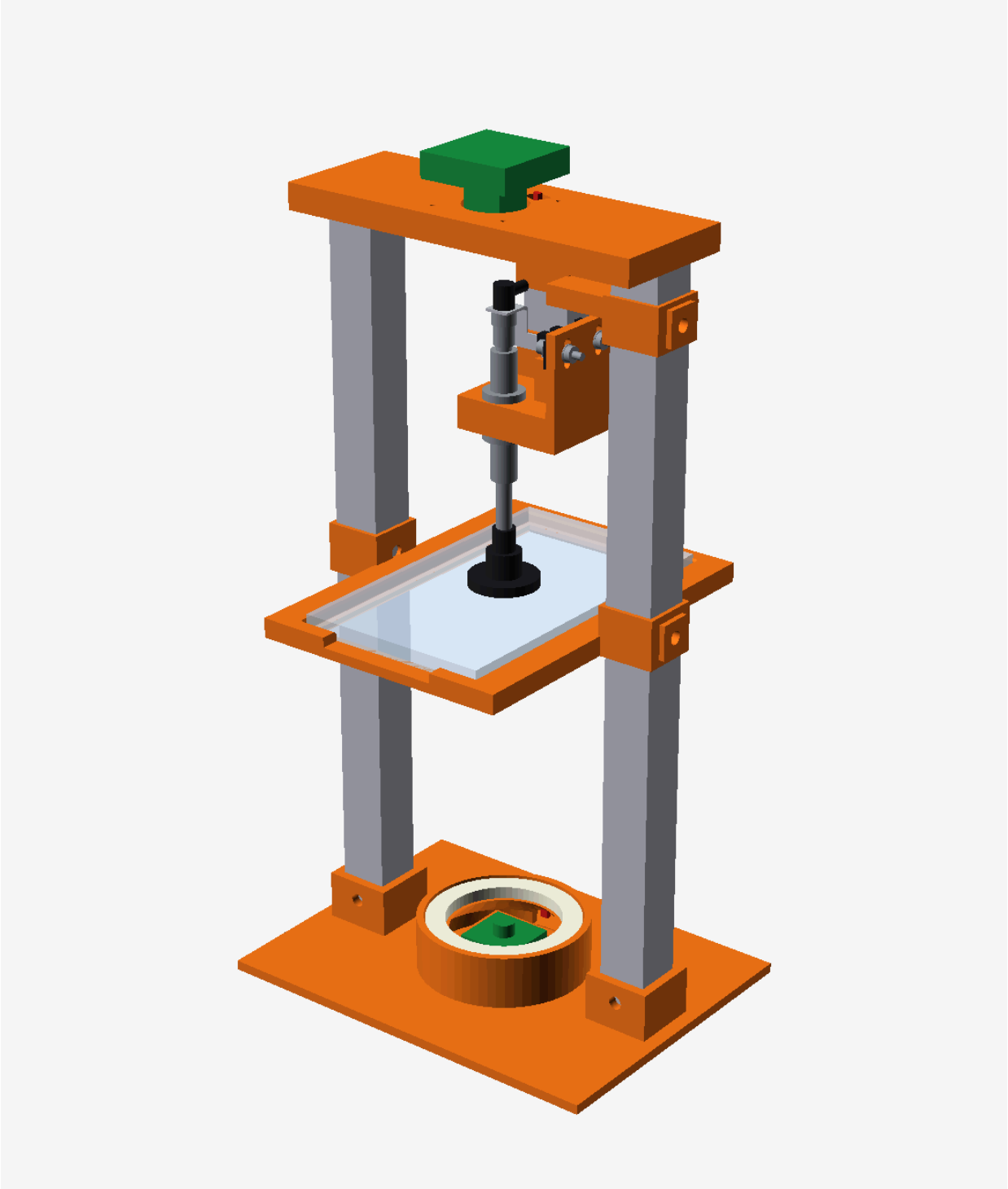
1.3 Print the jig (about 17 h printer time)

Print `phase0_base`, `phase0_tray`, `phase0_top` and `phase0_occluder_mount` in black PLA or PETG, and `head_foot` in PETG (settings in §3; it goes on the machine later). Use the current STLs: `phase0_top` (the bridge) and `head_foot` changed with the head-camera offset, and `phase0_top` and `phase0_base` got new LED ring seats. `phase0_top` changed again: it is now 10.6 mm thick, with a ribbed ring pocket. Black matters around the cameras, because it cuts stray light. Before printing `head_foot`, check the optical endstop board (§3.3).

1.4 Set up the Pi (1–1.5 h)

Do §11.1–11.4 now: the operating system, copying the repository, and `setup_pi.sh`. Phase 0 uses the same software as the machine.

1.5 Assemble the jig (1.5 h)



1. Screw the Camera Module 3 onto the four bosses in the base's camera mast with four M2×6 machine screws from the M2 kit, turned into the bosses' 1.8 mm holes (they cut their own thread). Lead the ribbon out through the slot in the mast and ring wall. Use the 500 mm cable to the Pi's CAM/DISP 0 port, fitted with the Pi off.
2. **Station LED ring:** lay the 60 or 63 mm ring LEDs up on the shelf inside the base's 64 mm ring seat, around the lens. Its solder pads and leads go down through the slot in the shelf, into the camera ribbon's slot. Centre it and fix it with three dots of glue: the bore alone does not hold it. [Certain, CAD `station_ring_seat`]
3. Push the two 2020 posts into the base sockets, down to the socket floors. Fix each with an M5×10 screw through the socket's cross hole into a T-nut in the post's slot.

- **Use ONE M5 washer under each M5×10 head** and **test the first screw:** it must pull the part tight before it bottoms in the slot. Every jig wall an M5×10 passes through is 4.0 mm, so with one washer the tip ends about 5 mm inside the profile face [Certain, CAD; Guessing that the slot is about 6 mm deep to the core]. If it bottoms, add a second washer. Do the same for every M5×10 in the jig.
- **No slot-6 T-nuts?** A plain M5 hex nut slid into the slot from the open end of the post works for the jig [Likely]. Slide the nuts in before the post goes into its socket.
- **Slide `phase0_occluder_mount`'s clamp onto the RIGHT post now**, with a T-nut in the post's outer slot, before the bridge goes on. Leave its screw loose.

4. **Face both tray ledges with a strip of microfiber cloth** (spray adhesive or double-sided tape; school stock, §1.1): the slab's front label rests on them. Slide the tray onto the posts. Set the **top of the cloth on the ledges 140 mm above the station lens front** (± 2 mm; if you measure to the bare ledge, set it 140 mm minus the cloth thickness), then fix both clamps with M5 screws and T-nuts. [Certain, `model.py`: slab front at lens + 140]
5. **Head camera on the bridge.** The bridge carries the board on four M2 standoffs.
   - **The bridge is not symmetric** (it runs 30 mm toward the label end and 25 mm toward the back from the slab centre): fit it with its wider side toward the tray's label end. The lens then sits 5 mm toward the label from the slab centre, as on the machine. [Certain, CAD]
   - Measure your ELP board's **board-to-lens-front length L**: from the board's underside to the front of the M12 lens holder, with the lens roughly in focus. The CAD guessed 20 mm. [Guessing in the CAD]
   - Standoff height = **L − 10.0 mm** (10.0 mm for L = 20). The bridge is 10.6 mm thick, so L must be at least 10 mm. Use brass M2 standoffs from the kit (stack two if needed); they screw into the bridge's 1.8 mm blind holes, 4 mm deep. [Certain, `checks.txt`]
   - The CAD assumes a 32 mm square hole pattern. Measure your board first; if it differs, drill new 1.8 mm holes in the bridge. [Guessing in the CAD]
   - About 2.0 mm of plastic separates each hole from the ring pocket [Certain, CAD]. Turn the standoffs in by hand; don't force them.
   - Fix the board with four M2 screws.
   - **Head LED ring:** push the ring LEDs down into the pocket on the bridge's underside, its pads and leads up through the slot. The pocket has three centring ribs, printed to fit a 40.0 mm ring: the ring presses in between them. For a bigger ring, shave each rib by half of (ring OD − 40), for example 1.0 mm for a 42 mm ring (§3.3). A ring under 40 mm sits a little loose: centre it by eye. Fix it with three dots of glue. The pocket holds it where the machine's seat does, relative to the lens. [Certain, `checks.txt`: 0.0 mm offset]
6. **The bridge sits on the post tops in its blind sockets,** so the 312 mm posts set its height. [Certain, CAD: post tops at 314, socket roof at 314] Put a slab in the tray and measure the **head lens front to the slab's top face (its back): 140 ± 2 mm**. If it is off, change the standoff height (washers on the standoffs, or a shorter or longer standoff). Don't pack the post tops unless you have to.
7. **Put the machine's occluders under the lens** on `phase0_occluder_mount`. The jig then holds them exactly where the machine does, relative to the lens. [Certain, `checks.txt`: 0.0 mm offset; the same share of the slab view blocked]
   1. **Arm stub:** push the 66 mm 2020 stub into `head_foot`'s arm socket until it bottoms. Clamp it square and match-drill the 5.5 mm cross hole through the printed hole and the stub together (19 mm from the stub's end). Fix it with an M5×30 and a nylock nut.
   2. **Occluder parts:** fit the buffer and cup, the flag, the elbow and the endstop board as in §7.5–7.6. With the M10-bolt dummy, leave the flag off.
   3. **Into the mount:** push the stub up into the mount's socket, the arm stub toward the back (away from the label end) and the sensor post on the right, exactly as in the machine. Fix it with an M5×10 and a T-nut in the stub's +X slot.

4. **Height:** put a slab in the tray. Slide the mount down the post until the cup rests on the slab's back and the **foot's underside is 62.0 ± 0.5 mm above the slab's back**. The buffer is then pushed in by its 2 mm trip travel, as in the machine. Tighten the clamp screw.
5. The cup lands on the slab centre, 5 mm behind the lens axis (away from the label), as on the machine, by construction. [Likely ± 1 mm, print tolerance] Check it by eye from above.
6. **To change slabs,** push the cup up by hand, only as far as the change needs.
7. **For the face-up pass (§1.7),** slide the whole mount 3 mm up the post. Never slide the stub in the mount instead: the sensor post meets the mount's beam after 2.5 mm. [Certain, CAD]

8. Plug the ELP camera into a black USB 2.0 port **through the 3 m USB extension**, as in the machine. Power both rings from the 12 V adapter (ring + to adapter +).

1.6 Focus both cameras (30 min)

Put one slab in the tray **face-down, label end toward you** (the finger-notch side), exactly as it will sit in the machine.

```
cd ~/psa-slab-scanner/software && source .venv/bin/activate
python -m slabscan focus head --no-motion             # turn the M12 lens; watch the score peak; lock the lens nut
python -m slabscan focus station --no-motion --save    # sweeps LensPosition, saves the best one to config/calibration.toml
```

- **Head:** with the occluders in place (§1.5 step 7), turn the lens slowly past the peak and come back to it. The line shows `READ …` when something decodes. Lock the lens nut once you are at the peak.
- **Station:** note the recommended LensPosition. The machine's station uses the same 140 mm distance, so the value carries over. [Certain, CAD]

1.7 Scan 30–50 of your own slabs (1–2 h)

```
python -m slabscan phase0 --out phase0_results --lens-position 7.1 --face-up 15 --save   # use the value focus station printed
```

Without the face-up pass, leave out `--face-up 15 --save`. Give each run a new `--out` folder: a folder that already holds a `phase0.csv` is refused, and nothing in it changes. Add `--force` only to overwrite an earlier run on purpose. [Certain, `tools/phase0.py`]

For each slab the tool asks you to type the printed cert number, then saves both images and what each camera decoded. **Type the cert every time:** the typed number is how it catches a *wrong* read, which is the failure that matters, and GO needs every slab typed. [Certain, `tools/phase0.py`] There is nothing to teach first: the ITF payload is always the cert zero-padded to 8 digits [Certain, real labels], so any order works.

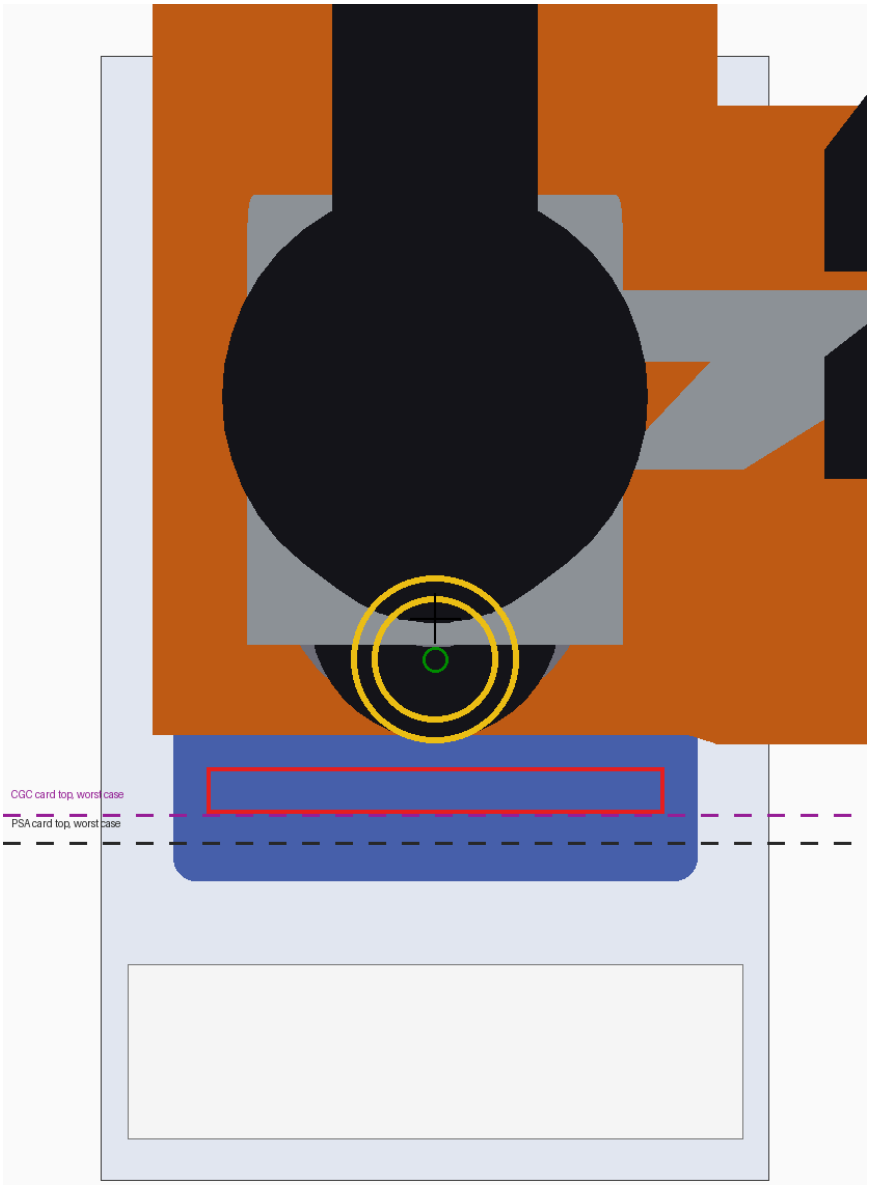
Include in the 30–50:

- current PSA labels (Code128, mostly 9-digit certs) and "lighthouse" labels (QR + ITF, 8-digit);
- at least 3 PSA labels from 2016 (back ITF, no QR) or older (no back code at all) [Certain, RESEARCH.md §4];
- the 2024+ holder and the older holder;
- CGC slabs if you deal them: 2023+ labels (back QR, front barcode), the 2020–23 blue "OG" labels (cgccomics.com QR and a long back barcode) and Certified Authentic (codes on the front only);
- scratched or scuffed labels and holders;
- one slab with a visible ceiling-light reflection, to check the light-roof question.

Type `q` when you are done. **The read-rate report (`phase0_report.txt`) is written at that point,** so stopping later loses nothing. [Certain, `tools/phase0.py`]

The face-up pass (recommended; for the learned face-down check). The machine can learn what a card *back* looks like through the slab and pick slabs without a back-label code when their card window matches (OPERATING.md §8.4). Phase 0 measures whether that can work: how well the backs match each other, and how well face-up slabs match the backs.

- **With today's head the card window is clear** [Certain, CAD frustum check], so this pass decides whether you can use the learned check. In the simulator it says GO (lowest back-to-back 0.81, highest face-up 0.57 with a reflection copy, threshold 0.69 [Certain, sim, 23 Oct 2026, today's smaller window and the card placed per grader; with the larger round-3 window it was 0.82, 0.43 and 0.62]). On studio scans of the 64 real slabs the same rule gave backs at 0.68 or more against face-up fronts at 0.45 or less, a GO at a threshold of about 0.56 [Certain for those images and the round-3 window; Likely optimistic, because they are flat scans, not the jig's camera]. With today's window the CAD's machine-scale emulation of the same scans put the highest face-up score at 0.49–0.53 [Likely].
- **Every similarity is the 2nd-best match:** the machine accepts a slab only when two card backs of two different slabs agree (one odd back, a card sitting low in its holder, matched a real blue Water-type front at 0.62). Each face-up frame is also scored with 4 reflections added in software, because on real fronts a reflection *raised* the score [Certain, real images; Guessing how the jig's reflections look].
- Start the tool with `--face-up 15`. After `q` it asks for 15 slabs **face-up**; its prompt repeats these steps.
  1. First slide the occluder mount 3 mm up the post (§1.5 step 7). The free cup then hangs 1.0 mm above the slab's front, so nothing touches it. [Certain, `checks.txt`] Use junk or low-value slabs anyway [Guessing: print tolerance could close a 1 mm gap].
  2. Change slabs with the cup pushed up no more than about 12 mm. [Certain, `checks.txt`: 3.16 mm clear at 12 mm; only 0.2 mm left at the full 20 mm with the mount lifted]
  3. Use 15 different cards, face-up, label end toward you, **including at least one plain card (one colour, little print), one full-art card and one blue Water-type card**: they are the hardest to tell from a back (a blue front came closest to the blue card back on real slabs).
  4. `s` (or Ctrl-C) ends the pass early; the slabs so far still count. Slide the mount back down to 62.0 mm afterwards.
- **Open `phase0_results/window_check.jpg`.** The red boxes are the card window (`face_down_window_mm`). They must lie on the card back, clear of the label and of the cup pad and foot (the CAD view is `img/face_down_view.png`). If a box reaches the label or the card's edge, note it (§1.2). If the label shows at the *bottom* of the image instead of the top, the camera is turned: put `face_down_window_mm = [[-27.5, 13.0, 27.5, 18.5]]` in `config/calibration.toml` and run Phase 0 again. [Guessing which way up the real camera is]



Head camera at hover: lens 147 mm above the slab, 5 mm toward the label (green circle; + = cup axis). Red: the face-down card window, 302 mm2, clear of every head part. Blue: the median PSA card, label at the bottom. Dashed: worst-case card tops (PSA, CGC). Yellow: where the camera sees its ring light mirrored in the slab (glare).

- `--save` writes `face_down_threshold` and `face_down_threshold_sig` to `config/calibration.toml` and the backs to `config/card_backs.npz`, **only on a GO**. Without `--save` it writes nothing to `config/`. [Certain, `tools/phase0.py`]

Then replay the machine's rules on your captures:

```
python -m tools.real_eval --phase0 phase0_results
```

It runs the run's own reading order, consensus and face-down rule on every Phase 0 frame, with the certs you typed as the truth. **It must print `WRONG CERT in an 'ok' verdict: 0` and `FALSE FACE-DOWN proofs (a face-up`**

`slab 'proven' face-down): 0`, and it exits 1 otherwise. [Certain, `tools/real_eval.py`] Phone photos of more slabs work too: list them in an `index.json` (the format is in `python -m tools.real_eval --help`).

1.8 Record what the labels actually encode

Open `phase0_results/phase0.csv` and `phase0_report.txt`. Fill this in and keep it with the machine: it settles the open questions in RESEARCH.md.

Question	Where to look	Your answer
Back QR payload format (`http://psacard.com/cert/<n>` or `https://www.psacard.com/cert/<n>/` expected)	`back_reads` column, `QRCode:` entries	
Front barcode symbology (ITF on lighthouse, 2016 and older labels; Code128 on current labels)	`front_reads`, the part before `:`	
Payload formats the rules recognised	report line `payload formats:`	
ITF payload lengths	report line `ITF payload lengths:` (all 8 expected: the cert zero-padded)	any other length: tell the software owner
Payloads no rule reads	report line `payloads no rule reads` (none expected)	if any: the label format differs from the real ones seen so far
Slabs whose back QR the head camera read	report line `slabs whose back QR the head camera read:`	low on 2017+ labels: focus, glare
CGC QR payload (`cgccards.com/certlookup/<n>/<grade>/`; `cgccomics.com` on OG labels)	`back_reads` for CGC slabs	
Any prefix or suffix around the cert	raw payloads	if yes, set `custom_regex` (§11.5)
Time per side (`back_ms`, `front_ms`)	CSV	replaces the [Guessing] frame times in `cli.py`
Cert lengths seen (7, 8, 9 digits …), counted without leading zeros	`typed` column	set `psa_cert_digits` in `calibration.toml` (§11.5). Include 7 if you have pre-2016 slabs: their certs are printed with a leading zero (`02074012` is a 7-digit cert). Leave it out, and every 7-digit cert is `needs_review`.
Widths of your CGC / BGS / TAG holders	your calipers (§1.2)	wider than 82.6 mm: do not load
Card top edge to the slab's far end	your calipers (§1.2)	at least 93.5 mm (real backs: CGC 93.8–96.2, PSA 97.2–100.0)
Card window on the card back?	`window_check.jpg`	boxes on the card / on the label / on the cup or foot
Learned check: how well backs match each other	report line `lowest back-to-back similarity:`	
Learned check: how well face-up slabs match a back	`highest face-up similarity:`	
Backs unlike the others (Japanese cards, other games, CGC)	lines `slab N matches none of the other backs`	
Learned check verdict	`GO for face_down_check = "learned"` with a threshold, or `HOLD for the learned check` and why	GO expected with today's head [Likely]
Machine rules on your captures	`python -m tools.real_eval --phase0 phase0_results`	`WRONG CERT …: 0` and `FALSE FACE-DOWN …: 0`

1.9 The gate

Result	Decision
`WRONG certs vs what you typed` is **0**, every slab typed, `confident read (ok)` is **≥ 95 %**, and `tools.real_eval --phase0` shows 0 wrong certs and 0 false face-down proofs	**GO** [Certain: the tools' own criteria]. The report also prints the rule-of-three bound (with 0 wrong in n slabs, the true wrong-read rate is probably below 3/n).
Any wrong cert at all	**NO-GO.** One silent misread is worse than 20 unread slabs. Follow the steps the report prints for the failing slabs and re-run; buy nothing else until a re-run gives 0 wrong.
ok below 95 % but no wrong certs	**Hold.** Look at the failing images. Typical fixes: refocus, move the ring, block room light. Then run again.
Back camera fails on most post-2017 labels	**Hold.** Every one of those would need the station detour (+3–5 s each). Fix the head image first. [Likely]
Glare spots over the label in the head images	Plan to build the optional light roof (CONTRACT).
Learned face-down check: **HOLD**, or no face-up pass	**Does not block the build.** The machine then stops at every slab without a back-label code (PSA 2016 and older, CGC Authentic, BGS, SGC) for you to check.
Learned face-down check: **GO** (with `--save`)	The threshold, its sig and the seed backs are saved. GO means: on your slabs, the backs matched each other (2nd-best match) at least 0.10 better than any face-up slab or its reflection copies matched a back; the threshold is the midpoint, never below 0.55. It is not a promise: whether 15 face-up slabs show the worst case is [Guessing]. On the machine the check still needs 5 references and a clear view of the window on 5 proven slabs before it accepts anything, a match needs two references from two different slabs, and you check the back photo of every learned acceptance in the first batches (§14).

1.10 Phase 0B: cup bench test (after 0A passes)

What it proves: the cup seals on your holders, the cup line holds, and the cup leaves no mark. **Parts:** order 1b, the rows marked "PHASE 0B": the SMC cup and buffer (and its spare), spare pads, the `M-5ALU-4` elbows, PU tube (its kit has the fittings), PTFE tape, IPA, and a hand vacuum pump with a gauge (brake-bleeder type). You also need shims (card, thin wood) or a small lab jack to raise the slab under the cup (BOM.md tools). **No mains wiring, no electronics, no instructor needed.** The machine's pump, valves, sensor and power supply come in order 2 and are tested in §8.3 and §13.

1. **Bench set-up** (30 min):
   - Clamp the buffer body in a bench vise by its M10 nuts (not by the rod), cup pointing down.
   - Screw the `M-5ALU-4` barb elbow into the buffer's top port, with one thin wrap of PTFE tape on its thread (§7.6 step 6; the flag is not needed here). Push about 1 m of 4 mm PU tube onto the barb and run it to the hand pump's hose (a stretch fit, or a fitting from the kit).
   - Wipe the cup lip with IPA.
2. **Seal** (on each slab): lay a junk slab face-down on a soft cloth on a block under the cup and raise the block with shims (or a lab jack) until the buffer is pushed in about 2 mm, the machine's trip point. Pump to −60 kPa (about 18 inHg). **Pass:** it gets there in a few strokes and holds. [Guessing target]
3. **Leak:** stop pumping and watch the gauge for 60 s. **Pass: it loses less than 5 kPa (about 1.5 inHg: brake-bleeder gauges read inHg) in a minute.** [Guessing target] Faster means the tube is not seated, the fitting leaks, or the lip has grit on it. Fix it and repeat. The rate also tells you how often step 5 needs re-pumping.
4. Do steps 2–3 on 5 different slabs, both holder generations, and a CGC if you deal CGC.
5. **24 h static mark test on a junk slab:**
   1. Photograph the junk slab's back under raking light: a phone torch held almost flat to the surface, about 10–20°, in a dim room.
   2. Set it up as in step 2, the cup on the back centre, and pump to −60 kPa.

3. **Keep the clamp on.** The closed line leaks, so re-pump to −60 kPa every time you check, at least every 30–60 min while you are around. Write down the time and the reading *before* each re-pump, and the reading next morning.
4. After 24 h, open the pump's release valve, then photograph the same spot the same way.

- **This is a screen, not the final test.** If most of your readings are above −40 kPa (overnight they will drift), most of the 24 h tested only the buffer spring's ~1.2 N, not the ~19 N vacuum clamp. The binding test is §13.2 on the machine, where the pump re-holds the line every few seconds for 24 h and logs it.

6. **Gate:**
   - **No ring, haze or residue that survives a dry microfiber wipe → GO.** Order the machine (order 2).
   - Anything that needs more than a dry wipe → **stop and ask.** The fallback is the Schmalz HT1 cup (BOM conditional line), but it needs an M10×1-to-G1/8 adapter nobody has sourced yet.
   - The cup won't seal on your holders → stop and find out why before ordering anything else.

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2. Ordering

Use BOM.md and BOM.csv. Order 2 is everything not marked "PHASE 0A" or "PHASE 0B". Get the Misumi extrusion quote first, because it moves the total most (BOM.md).

SHOPPING_LIST.md has the same lines as three orders, with a product link per line where one was found (22 Oct 2026) and the lines that still need one.

BOM.csv already has the parts this guide uses: the 285 mm drop arm, the Z spine with both ends tapped, the Adafruit #5648 MOSFET modules, bare SPDT microswitches, the 1 A / 3 A / 5 A branch fuses, 2.2 kΩ resistors and 100 nF capacitors, the I/O perfboard, mains and earth wire, the closed mains box, a Class 2 laser, an E-stop with two contact blocks, the 12 V adapter, the hand vacuum pump, the flag sheet, the `M-5ALU-4` elbows, a spare SMC cup, a spare rail block, and the carry-height sensor's parts (its board is the second endstop board). The M5×10 button-head line covers 6: 2 for the carry-height sensor's mount and 2 for each switch mount. [Certain, BOM.csv] The upgrade lines (HIWIN rail, Z motor, constant-force spring, high-flex USB cable, branded pump, branded E-stop) have qty 0: buy one only when the test its line names fails.

Check these lines when you order:

BOM line	What to do	Why
SMC cup + buffer `ZPT25UUK20-B5-A10` (×2, one spare)	Get the code confirmed in writing, or entered in SMC's configurator: a 25 mm flat urethane pad on a **non-rotating (K)** M10 buffer with **20 mm** stroke. The rotating `ZPT25UUJ20-B5-A10` works too. Download the STEP file while you are there and compare it with §3.3.	[Certain, SMC ZP catalog Table 2] 25 mm pads come only with 10/20/30/40/50 mm stroke, so the old 25 mm code does not exist. [Certain] Proax lists this exact code (back-ordered, no date on 22 Oct 2026). The flag, its guide and the sensor post are built on this buffer's catalog heights, so don't substitute another buffer.
`M-5ALU-4` barb elbow (×2)	SMC's own part, not a push-in elbow	A push-in elbow (15.9 mm tall) reaches past the head lens at full stroke. [Certain, SMC KQ2 catalog height placed in the CAD]
Z motor with lead screw	Buy one that lists 1.5–2.0 A per phase, ≤ 4 mH and an **8 mm lead** (T8×8). Not a 0.4–0.6 A "integrated" motor. The BOM's VXB 3-pack publishes all three (1.7 A, 3.1 mH).	The firmware runs Z at 0.8 A and 750 rpm; a high-impedance motor stalls there. The carry-height sensor stops the run after a stall, but it cannot stop the stalls.
Class 2 laser	The BOM's Quarton module is 7 mm across	The left gate post now prints with a 7.2 mm hole; a post printed from an older STL needs drilling to 7.2 mm (§5.9).
Dowels	The BOM's dowels are 5/16 in (7.94 mm)	Buy a 5/16 in brad-point bit and collar to match (BOM.md tools).

BOM line	What to do	Why
M5 screws	M5×25 for the 16 bed bolts, M5×30 for the 4 bracket-through-plywood bolts, M5×16/20, M5×12 button heads and 6 × M5×10 button heads as the CSV notes say	An M5×30 through the bed bottoms on the profile core before it clamps (§4.3).
Homing switches (10-pack)	A bare SPDT **hinge-lever** microswitch with a 19.8 × 10.2 × 6.4 mm body and two 2.35 mm holes 9.5 mm apart (Omron SS-5GL size). Prefer the short lever that ends at the body's end. Not a roller lever.	The printed mounts and strikers are drawn for that body and a flat lever ending at most 23 mm from the hinge end; a longer lever is cut back (§6.8). [Certain, Omron SS datasheet; Guessing which lever the BOM's pack has]
Overfill laser	The listing must say ≤ 1 mW or Class 2.	It is on whenever the Pi is on, at face height (WIRING.md §8).
Light roof (foam board, tape, 2 × 2020 posts, qty 0)	Buy only if Phase 0 showed glare (§1.9)	The roof is optional (CONTRACT).

Lead times [Guessing]: Amazon 2–5 days, Misumi about 1 week, AliExpress 2–4 weeks, the SMC cup 2–8 weeks [Likely, sibling listings; Proax shows it back-ordered with no date]: ask for the date in writing. LiMo ships the rails DDP by air in 8–14 working days. The Rodent was in stock at BIQU on 22 Oct 2026.

When the parts arrive:

- check that the 4040 profile is 20-series slot 6 with **two slots per face**;
- check that the Z motor's lead screw is **T8×8** (8 mm lead), not T8×2, and **measure the motor's phase resistance: about 1–2 Ω is right; 10 Ω or more is a high-impedance motor, don't use it**;
- measure the SMC buffer against §3.3 before you cut the flag (a K rod does not turn; a J rod does, and the flag guide handles it);
- measure the optical endstop boards (slot 3.1 mm, holes 19 mm apart, header position) before you print `head_foot` and `carry_sensor_mount`;
- measure both LED rings (OD, ID, thickness: §3.3) and the ELP board (hole pattern, board-to-lens length) before you print the Phase 0 parts, which come first;
- measure one homing switch: its body, its hole spacing and its lever, from the hinge end of the body to the lever tip (§3.3, §6.8);
- check that the laser receiver module fits the right gate post's 6.2 mm hole (never measured [Guessing]) and the laser the left post's 7.2 mm one;
- count the balls in the MGN12 carriages before you slide them anywhere (§6.1).

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3. Printing

3.1 Printer and material

- **Height and bed size** [Certain, `print_list.json` sizes]:
  - The gate posts are exported **standing upright, 229 mm tall: they need ≥ 230 mm of Z height.** A Prusa MK3S (210 mm) or MK4 (220 mm) can't do that: there, lay each post on its side with supports.
  - The fill gauge is exported lying flat and is 214 mm long: it needs 214 mm in one direction (or diagonally).
- **PETG**, 0.4 mm nozzle, 0.2 mm layers. Prusa's PETG settings are 230 °C nozzle on the first layer and 240 °C after, with the bed at 85 °C then 90 °C, on a powder-coated or textured sheet. [Certain, Prusa knowledge base as quoted in search results] Use your filament maker's numbers if they differ.
- **Don't print PETG on a bare smooth PEI sheet.** It can bond hard enough to tear the sheet. [Certain, Prusa] Use a textured sheet, or a glue-stick release layer.
- **Part cooling:** 30–50 % for PETG. [Likely]
- **Colour:**

- matte black for `camera_bracket`, `station_mount` and the Phase 0 jig, to cut stray light near the cameras;
- any colour for the rest.

3.2 Parts

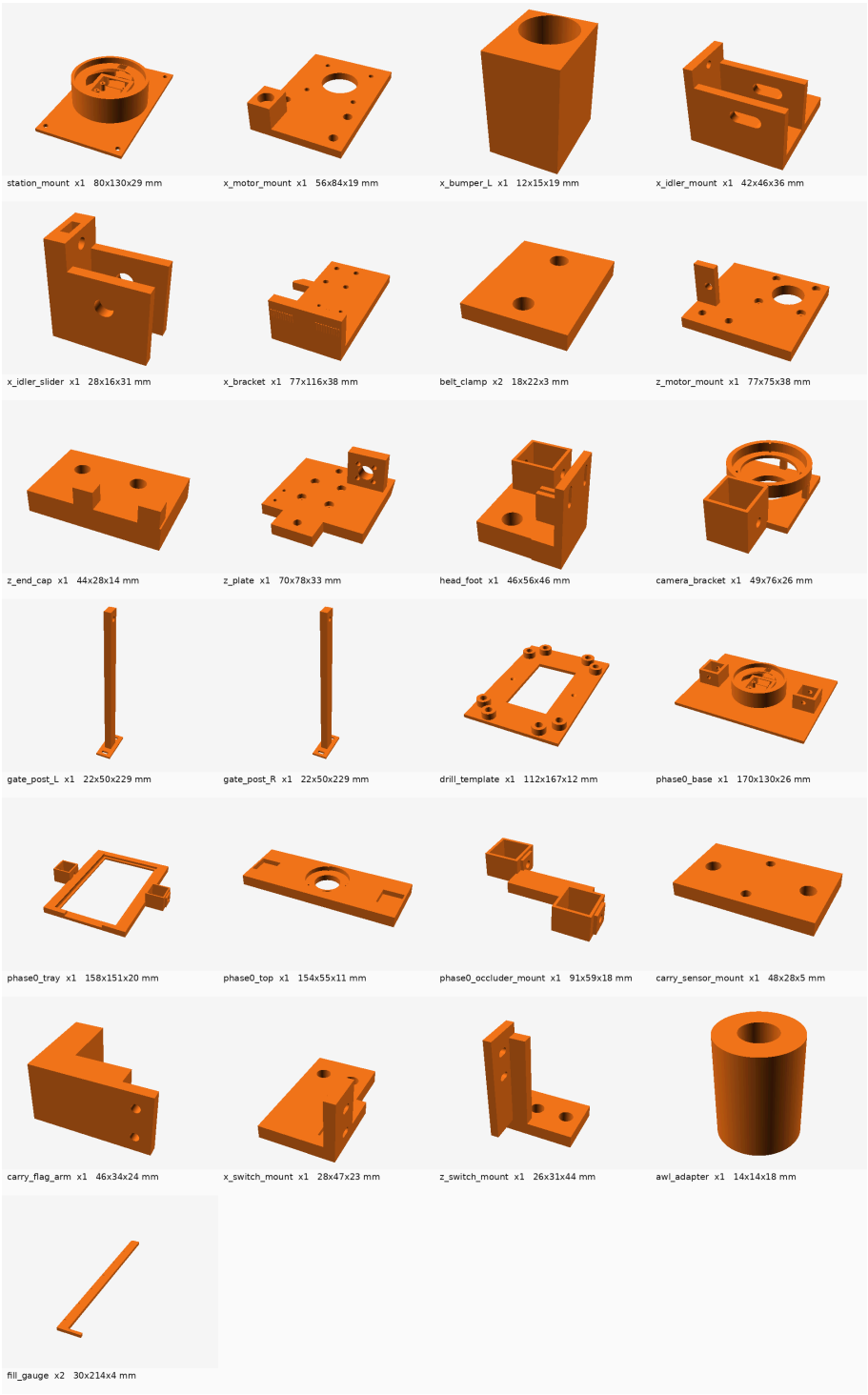
STL files are in `cad/out/printed/` (`print_list.json` has every size and note). Times are for a classic 0.2 mm printer (Prusa MK3-class, Ender 3) running about 5 mm³/s, estimated from each part's volume. Fast printers (Bambu, MK4) take roughly half. [Guessing]

STL	Qty	Material	Orientation	Walls / infill	Supports	Time
`phase0_base` (128.0 cm³; reprint an older one: new ring seat)	1	PLA/PETG black	as exported (plate down)	3 / 25 %	none	7.1 h
`phase0_tray`	1	PLA/PETG	as exported (ledges down)	3 / 25 %	none	2.8 h
`phase0_top` (75.7 cm³, 10.6 mm thick, ribbed ring pocket; reprint an older one)	1	PLA/PETG	as exported (ring pocket up)	3 / 25 %	none	4.2 h
`phase0_occluder_mount`	1	PLA/PETG	as exported (flat on its beam)	4 / 30 %	none	0.8 h
`head_foot` (31.0 cm³; Phase 0A, then the machine)	1	PETG	as exported (foot down)	5 / 40 %	none	1.7 h
`station_mount` (65.0 cm³, 29 mm tall; reprint an older one: new ring seat)	1	PETG black	as exported (base plate down)	3 / 25 %	none; the ribbon slot bridges	3.6 h
`x_motor_mount`	1	PETG	as exported: beam face down	5 / 40 %	none	1.3 h
`x_bumper_L`	1	PETG	as exported: beam face down	4 / 40 %	none	0.2 h
`x_idler_mount`	1	PETG	as exported (foot down)	5 / 40 %	none	1 h
`x_idler_slider`	1	PETG	as exported	5 / 50 %	none	0.3 h
`x_bracket` (70.7 cm³; reprint an older one: the finger off its left edge works the X home switch)	1	PETG	as exported: flat on its spine face	6 / 50 %	none	3.9 h
`belt_clamp`	2	PETG	flat	100 %	none	0.1 h each
`z_motor_mount`	1	PETG	as exported: motor face down, tab up	5 / 40 %	none	2.5 h
`z_end_cap`	1	PETG	as exported (lips up)	5 / 50 %	none	0.5 h
`z_plate` (40.9 cm³, with 2 × 2.5 mm pilots for the carry flag arm; reprint an older one: the 45° chamfer at its top right works the Z top switch, and it starts at the bed)	1	PETG	as exported: carriage face down, nut shelf up	6 / 50 %	none	2.3 h
`camera_bracket` (17.3 cm³; reprint an older one: the ring seat now has three centring ribs)	1	PETG matte black	as exported: shelf face down	4 / 40 %	tree supports under the ring seat if your printer sags (it bridges about 30 mm between its 4 legs)	1.0 h
`carry_sensor_mount` (6.2 cm³)	1	PETG	as exported: flat on its spine face	4 / 40 %	none	0.35 h
`carry_flag_arm` (17.2 cm³)	1	PETG	as exported: on its bottom face	4 / 40 %	none	0.95 h
`x_switch_mount` (6.8 cm³, new)	1	PETG	as exported: front plate down	4 / 40 %	none	0.4 h
`z_switch_mount` (4.9 cm³, new)	1	PETG	as exported: spine plate down	4 / 40 %	none (see the note)	0.3 h
`gate_post_L`, `gate_post_R`	1 + 1	PETG	as exported: upright on the foot, 10 mm brim (needs 230 mm Z)	4 / 30 %	none upright; on its side: supports	2.2 h each
`drill_template`	1	PETG (or PLA)	as exported (plate down)	4 walls, 50 %, bosses solid	none	3.6 h

STL	Qty	Material	Orientation	Walls / infill	Supports	Time
`awl_adapter`	1	PETG	as exported	100 %	none	0.2 h
`fill_gauge`	2	PLA	as exported (lying flat)	3 / 20 %	none	0.6 h each
Total (801 cm³ in 27 prints)						**≈ 44.5 h** (≈ 28 h after the Phase 0 prints)

Notes

- **Gate posts:** slow down the last 50 mm of the upright print; a tall thin part wobbles. If your printer is shorter than 230 mm, lay each post on its side with supports from the build plate.
- **`camera_bracket`:** print it shelf face down, as exported; the LED ring seat hangs between four legs and bridges about 30 mm. Use a brim.
- **`station_mount`, `camera_bracket`, `phase0_base` and `phase0_top` changed** (the lens sits 5 mm toward the label, and both LED ring seats are new), and so did **`x_bracket` and `z_plate`** (the switch finger and chamfer): print them from the current `cad/out/printed/`. [Certain, CAD export]
- **`z_switch_mount`:** where the shelf starts in front of the spine plate, a 3 × 11 mm strip of its first layers prints over air and may droop. File that edge flat before fitting: the Z carriage block passes 1.49 mm from the mount. Or add a support under that strip. [Certain, CAD: the unsupported strip and the 1.49 mm; Likely that it droops]
- **`head_foot`:** clear the 1.9 mm flag slot (in the two fins and through the foot) and the sensor post's board slots after printing. Use a junior hacksaw blade or a 1.5 mm needle file: a standard 12.7 mm hacksaw blade does not fit the slot in the foot. [Certain, CAD: that part of the slot is 9.3 mm long]



3.3 Check before you print the head and camera parts

Measure the real parts first. The CAD guessed some of these [Guessing in the CAD, BOM.md open gap 4] and took the others from datasheets you should check against your part:

Part	CAD value	Fits into
ELP camera board hole pattern	32 × 32 mm, M2 [Guessing]	`camera_bracket`, `phase0_top`

Part	CAD value	Fits into
ELP board-to-lens-front length (board underside to the lens holder front)	20 mm (`HEAD_LENS_PROJ`) [Guessing]	`camera_bracket` shelf height; the `phase0_top` standoffs (§1.5), which need 10 mm or more
ELP lens-holder footprint	16 mm square (a 25 mm window passes it) [Guessing]	`camera_bracket`, `phase0_top`
Optical endstop board	holes = the TCST2103's flange holes, 19 mm apart with the slot midway [Certain for the sensor, Vishay drawing]; header at the +Y end; **no component taller than 4 mm within 2 mm of the slot line** [Guessing]	`head_foot` sensor post (M3 slots, ± 4 mm)
Second optical endstop board (carry-height sensor)	the same: holes 19 mm apart, header assumed at the +Y end [Guessing]	`carry_sensor_mount` (M3×10 heads in pockets on the spine side)
Head LED ring OD, ID, thickness	modelled 40 × 30 × 5 mm [Likely OD, Guessing the rest]. The seat is a 42.8 mm bore with three centring ribs, printed to fit a 40.0 mm ring, and a 33 mm lip. It takes OD 39.5–42.5 mm: for a ring over 40, shave each rib by half of (OD − 40); for one over about 42.2 mm, also sand the bore, because printed holes come out a little small [Likely]. A ring under 40 sits up to 0.5 mm off-centre: centre it by eye (harmless [Certain, CAD]). ID about 24 mm (clear of the lens holder) up to OD − 3; 5 mm thick or less (a thicker ring sits lower, and each mm takes 1 mm off the 3.75 mm barb clearance at full stroke). [Certain, `checks.txt`: every head ring check passes at 39.5, 40 and 42.5 mm]	`camera_bracket`, `phase0_top` (measure before printing `phase0_top`)
Station LED ring OD, ID	modelled 63 × 47 × 5 mm: a 64 mm bore on a shelf, so OD 59–63.5 mm (a 60 or a 63 mm ring)	`station_mount`, `phase0_base` (measure before printing `phase0_base`)
Camera Module 3 holes	21 × 12.5 mm, M2	`station_mount`, `phase0_base`
Homing switch	Omron SS-5GL size: body 19.8 × 10.2 × 6.4 mm; two 2.35 mm holes 9.5 ± 0.1 mm apart, the first 5.1 mm from the hinge end; a flat hinge lever that ends at the body's end, 20 mm from the hinge end [Certain, Omron SS datasheet; Likely for the BOM's 10-pack]	`x_switch_mount`, `z_switch_mount` (M2 slots); a lever longer than 23 mm is cut back (§6.8)

Check on arrival, before you cut the flag [Certain, SMC ZP catalog p.13-11-12 for `ZPT25UUK20-B5-A10`, unless marked]:

Part	CAD value	What depends on it
SMC buffer: pad face to the top of the M5 port hex, at rest	108.5 mm	the flag's bend height
Pad face to the body's bottom end	44.5 mm	the pad-to-body gap
M10 × 1 body length; nuts	51 mm; 14 mm across flats, 3 mm thick	the foot hole, the guide fins
Free gap between the pad's hex and the body (must exceed the stroke)	21.5 mm vs 20 mm stroke	the probe window [Certain, `checks.txt`: 1.5 mm clear at full stroke]
Port top above the foot's top face, buffer at rest and fitted (§7.5)	31.5 mm	the trip point (the board slots give ± 4 mm)
`M-5ALU-4` elbow: height above the flag tab; thread length	at most 9.3 mm; at least 3.4 mm of M5 thread [Likely]	clearance under the head lens at full stroke (3.2 mm in the CAD)

If anything differs, fix the CAD and re-export (`cd cad && python build.py` on a PC with CadQuery), or drill new holes by hand.

3.4 After printing

- **Clearance holes:** run a 3.4 mm drill through every M3 clearance hole and a 5.5 mm drill through every M5 hole by hand. Printed holes come out undersize. [Likely]
- **`awl_adapter`:**
  - The insert pocket is drawn **6.4 mm**, the size for common M5 heat-set inserts (for example ruthex M5×9.5). [Certain, CAD; Likely for your insert] If your insert's maker gives another size, drill to it before you set the insert with the iron.
  - Press a 3 mm steel or stainless pin (hardened not needed; or a 3 mm drill-bit shank), ground to a point, into the 3.1 mm bottom hole with a drop of CA glue.
  - The point must be on the cup axis within 0.3 mm. Spin the adapter on the buffer and watch the tip: it should not wobble visibly.
  - **Check how your cup pad attaches to the buffer.** The adapter assumes an M5 male stud where the pad goes. [Guessing] If your buffer differs, make an equivalent: a sleeve that fits where the pad goes, carrying a 3 mm point on the axis.
- **Test-fit before assembly:**
  - each MGN12 carriage's four M3×8 screws through `x_bracket` and `z_plate` (counterbored);
  - the T8 nut flange on the `z_plate` shelf, and its four M3×16 + nuts;
  - the M10 buffer body through `head_foot`;
  - the probe flag sliding freely in `head_foot`'s 1.9 mm guide slot, with the board off (§7.6);
  - the second endstop board's M3×10 heads in `carry_sensor_mount`'s pockets, and an M3×10 turning into each of the `z_plate`'s two 2.5 mm pilots for the flag arm;
  - both LED rings in their seats: the head ring presses in between the three ribs of `camera_bracket`'s seat (shave them if it does not go in, §3.3);
  - each homing switch on its two M2×12 and nuts in its mount, sliding the full ±1 mm of the slots (`x_switch_mount`: heads on the mount's outer side, nuts on the switch; `z_switch_mount`: heads under the shelf, nuts on top);
  - the `x_idler_slider` sliding between the `x_idler_mount` cheeks, its M3 nut in the slot;
  - the M3 nuts in the `x_bracket` tab's nut traps, under the `belt_clamp` caps;
  - an M3 set screw in each gate post (2.5 mm hole: let it form its own thread, or tap M3).

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4. Frame

4.1 Cut list

From `cad/out/cut_list.csv` [Certain]. Misumi cuts to ±0.5 mm with square ends (BOM.md).

Profile	Use	Length	Qty
2040	base rail, lying flat under the bed	1416	2
4040	upright	401	2
4040	X beam, on top of the uprights	1416	1
2040	Z spine, **both ends tapped M5** (the two centre bores)	340	1
2020	head drop arm	285	1
T8×8 lead screw	Z screw, measured from the motor face: cut the motor's 400 mm screw (§7.3)	340 (311–356 works)	1
2020	Phase 0 jig posts	312	2
2020	Phase 0 jig arm stub (one 5.5 mm cross hole, match-drilled through `head_foot`, §1.5)	66	1

Profile	Use	Length	Qty
1 mm aluminium sheet	probe flag blank, cut to `cad/out/probe_flag_blank.svg` (§7.6)	24.4 × 55	1
1 mm aluminium sheet	carry flag, flat, two 3.4 mm holes 15.9 mm from the inner edge, 6 and 18 mm below the top edge (§7.4a)	19.9 × 36	1
2020	optional rear brace stock (§6.7)	600	1
5/16 in (7.94 mm) hardwood dowel	guide rod	236	82
18 mm (3/4 in, 17.9 mm actual) plywood	bed	1416 × 289.5	1

4.2 Where everything sits

Machine coordinates in mm, from `cad/model.py` [Certain]:

- X runs left to right along the row; Y runs front (−) to back (+); Z = 0 is the pad top surface.
- The bed top is at Z = −3, and the base rails' bottom is at Z = −41.

Part	X	Y	Z
Front base rail (2040 lying flat)	−68 … 1348	−87.5 … −47.5 (30 mm in from the bed's front edge)	−41 … −21
Back base rail	−68 … 1348	107 … 147	−41 … −21
Bed plywood	−68 … 1348	−117.5 … 172	−21 … −3
Left upright (4040)	−68 … −28	107 … 147 (on the back base rail)	−21 … 380
Right upright	1308 … 1348	107 … 147	−21 … 380
X beam (4040)	−68 … 1348	107 … 147	380 … 420
X rail (MGN12)	−10 … 1290	front face of the beam	centre 410, upper front slot

The uprights stand **on** the back base rail and pass through 41 × 41 mm notches in the bed. The beam sits **on top of** the uprights, flush with their outer faces.

4.3 Joints (the CAD has none; this is my recommendation)

[Guessing on the exact layout; Certain that the CAD and BOM leave it open]

Beam to upright (×2):

- two 4040 L brackets in the inside corner, on the upright's inner face and the beam's bottom face, one in each slot pair;
- one flat L joining plate on the back face, across the upright and the beam. **At the left end, put the plate's beam leg on the beam's upper rear slot:** the X home switch lead rises behind the beam into the lower rear slot just right of the upright, at X ≈ −24 (§6.8). [Guessing: the plate is not in the CAD]
- **No brackets on the front faces.** The X rail ends only 18 mm from each upright (BOM.md open gap 1), and at the right end the `x_motor_mount` plate covers the beam's front face (§6.4).

Upright to back base rail (×2):

- The bed covers the base-rail top all round the notch, so an L bracket can't reach the rail directly.
- Use two L brackets on the upright's **inner face**, sitting on top of the plywood.
- Bolt each one down with an M5×30 screw and a washer through its foot, the 18 mm plywood and a T-nut in the base rail's top slot (slots at Y = 117 and 137).
  - **Test one on a rail offcut first:** the screw must pull the bracket down hard before it bottoms in the slot. If it bottoms, add a second washer. Don't go to M5×25 here: with the bracket foot it leaves only 1–2 mm of thread in the T-nut. [Likely]
- The plywood then acts as part of the joint. Fit these after the bed (§5A).

- If it still racks, end-tap the upright bottom M5 (4 corner holes) and bolt up through drilled holes in the base rail.

Bed to base rails:

- **16 × M5×25** with washers into T-nuts, 8 per rail, in the rail's **outer** slot. [Likely] An M5×30 through 18 mm plywood leaves about 11 mm below the rail face, but the slot is only about 6 mm deep to the profile core, so it bottoms before it clamps the bed. **Test one on a rail offcut before you drill the bed.**
  - front rail at Y = −77.5, which is 40 mm from the front edge;
  - back rail at Y = 137, which is 35 mm from the back edge.
- Put the front-rail bolts on the cell wall lines (X = −4 + 92·k). That keeps them 16 mm clear of the nearest front rod hole and outside the pads. [Certain, CAD arithmetic]
- **Feet:** six rubber feet into the base rails' bottom slots, at both ends and at midspan (X ≈ 640). Shim every foot to the bench: a free midspan sags 1–3 mm under a full load. [Certain, CAD has the six; Guessing on the sag]

4.4 Assembly steps (4–6 h, with the instructor for squaring)

1. Slide T-nuts into every slot you will need **before** you close any joint. Drop-in T-nuts can be added later, but sliding ones can't. Count them per the BOM: about 126 M5 here (130 in all: the last 4 go into the Z spine in §7.1) and 65 M3.
2. Lay both base rails on a flat bench, 194.5 mm apart centre to centre (Y = −67.5 and 127).
3. Bed part A (§5A) goes on now, because the upright brackets bolt through it.
4. Stand the uprights in the notches on the back rail. Brackets snug, not tight.
5. Lift the beam onto the uprights. Brackets and back plates snug.
6. Square everything (below), then tighten from the bottom up and re-check.

4.5 Squareness checks (instructor)

Why these tolerances are loose: the machine drills its own rod holes (§5B), so it doesn't matter if the beam sits a little skewed to the bed. These checks keep the motion axes straight and the rods vertical. [Certain, by construction of the drill-template method]

Check	How	Target
Base rails parallel	centre-to-centre at both ends	194.5 ± 0.5 mm
Base rectangle square	the two diagonals between rail ends	equal within 1 mm
Bed flat	straightedge along the row centre line (Y = 0)	gap ≤ 0.5 mm per metre [Guessing]
Bed not twisted	level, or winding sticks, across the depth at both ends	≤ 0.5 mm across 290 mm [Guessing]
Uprights plumb, in X and in Y	machinist square on the bed plus feeler gauge, or a digital level	≤ 0.5 mm over 400 mm (≈ 0.07°) [Guessing]
Upright spacing	inner faces at the bottom and at the top	1336 ± 0.5 mm, equal top and bottom [Certain, CAD]
Gantry not racked	diagonals of the back face: beam top corner to the opposite base-rail bottom corner	equal within 1 mm
Beam height	bed top to beam bottom, at each upright	383 ± 0.5 mm, equal both ends [Certain, CAD]
Beam sag	same, at midspan	within 0.3 mm of the ends (DESIGN: 0.18 mm under 20 N)
Beam front face straight	string line or a 1.5 m straightedge along the rail face	≤ 0.5 mm bow [Guessing]

---

5. Bed and guide rods

5A. Now, before the X axis (3–4 h)

5.1 Cut the bed.

- Cut 1416 × 289.5 mm from the 3/4 in birch panel (610 × 2438 mm): one rip and one crosscut. Home Depot cuts it for free; a ≥ 1.42 m Baltic birch offcut from the school is better still.
- Birch plywood can have voids: drill a test hole away from the pockets first, because voids ruin rod holes. (BOM.md)

5.2 Cut the two upright notches at the back corners, one at each end.

- Each notch is 41 mm along X (open to the end) by 41 mm along Y.
- It runs from **24.5 mm to 65.5 mm in from the back edge** (Y 106.5 … 147.5). [Certain, CAD]
- Jigsaw or bandsaw, then file to fit. The upright must drop in without forcing.

5.3 Bolt the bed to the base rails (§4.3), with the frame square.

5.4 Prepare the rods (do this while the printers run):

1. Roll every dowel on a flat table. Reject bowed ones. Buy 100 so you can.
2. Cut 82 to **236 mm**: 15 mm in the plywood + 3 mm of pad + 218 mm above the pad. [Certain, CAD]
3. **Dome one end now.** Chuck the dowel in a cordless drill, spin it against 120 then 220 grit, and round the top to a 2–3 mm radius. This is the lead-in for slabs going in by hand. Thirty seconds each.
4. Lightly chamfer the other (bottom) end so the glue isn't scraped off on insertion.
5. **Test hole in scrap plywood:** drill 15 mm deep with the brad-point bit that matches your dowels (5/16 in for the BOM's 5/16 in dowels, 8 mm for 8 mm ones). A dowel must be a firm push fit.
   - Too loose: use a fresh bit of the matching size.
   - Too tight: sand the dowel end.

5.5 Cut the pads:

- 13 pads of **83 × 138 mm**: 2 mm black EVA, faced with tight-weave microfiber, bonded with spray adhesive. Total thickness about 3 mm. [Certain, CAD `PAD_T`]
- Cut with a sharp knife and a steel rule, corners square.
- **Don't lay them yet.** The template sits on bare plywood.

5B. After §11, once the machine homes and jogs

Order: §6–§11 first. You need X and Z moving, the probe working (FIRMWARE.md step 7), and the software installed. The bed must be bare: no rods, no pads.

5.6 Mark and drill, one cell at a time (4–6 h for 13 cells).

What the template does:

- `drill_template` is a one-cell jig with two cone dimples, 40 mm either side of the cell centre, on the row centre line, plus 8 drill-guide bosses. [Certain, `model.py`]
- The awl on the head drops into both dimples, so the template lines up with the machine's *own* X axis and cup path. Belt stretch, the switch position and beam skew are therefore all built into the holes.

Set-up:

- Fit the `awl_adapter` in place of the cup.
- Set its point level with where the cup face was, within 2 mm (`mark` asks for this).

- Make a **drill guide block**: drill a hole of the dowel size (5/16 in for the BOM's dowels) straight through a 30 mm hardwood block on the school's drill press.
  - The printed template sets each hole's *position*, not its *angle*. [Certain, BOM.md] The block sets the angle.
  - Why it matters: a rod leaning 0.5° puts its top 1.9 mm off at 218 mm, and the slab has only ±1.85 mm of play. [Certain, arithmetic]

```
python -m slabscan mark --dry-run     # prints the order and every dimple position
python -m slabscan mark               # the real thing
```

For each pocket the software:

1. Drives the awl over the left dimple and lowers it until it touches (a bounded probe).
2. Asks you to nudge the template until the point sits in the left dimple.
3. Moves to the right dimple. You swing the template on the left dimple until the right one is under the point, then **clamp the template** (two small clamps, or double-sided tape plus a clamp).
4. Parks the head three pockets away and prints **which holes to drill** for this pocket.
5. Waits until you type **`clear`** (drill and hands off the bed). Only then does the head move on. `s` = this pocket is drilled, stop here; `q` (or no keyboard) = stop without recording it. Every sideways move is slow (1500 mm/min), because the drill is right there. [Certain, `ops.mark`]

Drilling each hole:

1. Hold the guide block on top of the template boss, hole over hole.
2. Drill through boss and block with the brad-point bit (5/16 in for the BOM's dowels).
3. **Depth collar at 57 mm from the bit's cutting lips:** 30 mm block + 12 mm boss + 15 mm into the plywood. [Certain, CAD for the boss and the 15 mm] The collar stops on the top of the block.
   - The brad point's centre spur goes 1–2 mm deeper than the lips, and the plywood is only 18 mm. Check on the §5.4 scrap test hole that it doesn't break through the bottom. If it does, drill 14 mm: the rod tops end 1 mm higher, which is harmless. [Likely]
4. Vacuum the chips out of each hole.

Order and the end pockets:

- The order is pockets 1–11 first, then 0 and 12. [Certain, `mark --dry-run`]
- One dimple of each end pocket is past the X travel. For those, push two 5/16 in dowel offcuts through the template's shared-side holes into the neighbour's finished holes. The one dimple you can reach is then a check: **more than 1 mm off means stop.** Either the neighbour was drilled off, or X lost steps.
- The station cell (7) needs no holes of its own: its side rods belong to pockets 6 and 7. [Certain]
- Neighbouring cells share side holes. **Don't drill through the bosses over a neighbour's holes:** `mark` tells you which side to skip. Each template is placed by its own dimples, a few tenths of a mm off the neighbour's, so drilling those holes again would cut them oval and the dowels would lean. [Certain, `ops.drill_instruction`]
- Break: `python -m slabscan mark --start N` carries on **in the marking order** from pocket N, so pockets 0 and 12 still come last. `--only 0,12` does just those.

5.7 Glue the rods (2 h + 24 h cure).

1. Put a drop of Titebond II in each hole and spread it with a cotton swab.
2. Push the rod to the bottom with a twist, domed end up.
3. Wipe the squeeze-out with a damp cloth.
4. Hold a small square against the rod in X and in Y. Over 200 mm it must be within 1 mm. [Guessing target, from the play budget]
5. Do one pocket wall at a time, and check rods against their neighbours by eye along the row.

- Titebond II: about 3–5 min open time, clamp 30–60 min, no load for 24 h. [Likely, Franklin guidance via secondary sources] Longer if the shop is cold.

5.8 Rod check: the slab drop test. After 24 h, lower a junk slab flat between the rods of every pocket, from the rod tops to the plywood. Do it both ways round (rotated 180°). It must drop freely and never catch. If it catches, the rod at fault leans. Either pull it now, before the glue's full strength, or ream the hole and re-glue.

5.9 Lay the pads, the station mount and the gate posts.

- **Pads:** spray adhesive on the pad backs and the plywood, then press them in.
  - 0.5 mm clearance to the rods each side. [Certain, 83 vs 84 mm]
  - Use 3M Super 77 outdoors or in a ventilated room, and let it cure overnight before use. [Likely]
- **Station mount:** in cell 7, in place of a pad.
  - It has its own 3 mm base plate. Fix it with four #8 wood screws.
  - The station LED ring lies LEDs up on the shelf in its 64 mm seat, its leads down the shelf slot into the ribbon slot, fixed with three dots of glue (as in §1.5 step 2).
  - The camera ribbon exits toward the back.
  - Measure the station lens front height above the pad tops. It should be 23 ± 1 mm. If it isn't, set `station_read_c = <your height> + 148.6` in `calibration.toml`. [Certain: `STATION_READ_C = 23 + 140 + 6.6 + 2`, the last 2 mm being the free cup's hang below its calibrated height]
- **Gate posts:**
  - Left post at X = −20 (laser), right post at X = 1300 (receiver), both on Y = 0. [Certain, `model.py`]
  - **They stand on the bare plywood, with no shim.** The CAD puts the barrel 217 mm above the foot, so the beam lands at 214 mm above the pad top. [Certain, `model.gate_post`] A shim would lift the beam to 217 mm, and a stack up to about 216 mm would then pass the overfill check.
  - Fix each post with two wood screws through the slotted foot holes (they allow about ±9° of yaw for aiming).
  - **The BOM's Class 2 laser (Quarton VLM-650-03) is 7 mm across.** The left post prints with a 7.2 mm barrel hole for it; the right post keeps 6.2 mm for the receiver module. If your left post came from an older STL with a 6.2 mm hole, drill it out to 7.2 mm (or 9/32 in), square to the post. [Certain, the module's listed D7 × 21 mm; CAD `model.gate_post`]
  - Push the Class 2 laser into the left post's barrel hole and the receiver into the right one; lock each with an M3 set screw. Stick a "LASER - do not stare into the beam - Class 2" label on both posts.
  - Final beam alignment is in §12.5.

---

6. X axis

6.1 Rail prep (30 min). Generic MGN12 rails ship with storage oil and sometimes missing balls. (BOM.md)

- **Never slide the carriage off the rail** without a holder against its open end (a short scrap of rail, or a printed plug), or the balls fall out. [Likely]
- Clean the rail and carriage with IPA.
- Count the balls through the end seals if you can.
- Grease with lithium or PTFE grease.
- The carriage must roll the full length with an even, slightly sticky feel, and no notches.
- If balls fall out, fit the spare block from the BOM rather than refilling the old one. If the rail itself is notchy or binds after cleaning and greasing, that is the trigger for the BOM's genuine HIWIN line (replace rail and block together). [Likely]

6.2 Mount the rail on the beam's **upper front slot** (centre Z = 410). [Certain, CONTRACT]

- 52 × M3×10 screws into M3 T-nuts.
- Rail ends at X −10 and 1290, which is **58 mm in from each end of the beam**. [Certain, CAD]

1. Put all screws in finger tight.
2. **Align the rail to the beam's top face:** with a caliper depth rod at every screw, the beam top to the rail's top edge must read **4.00 ± 0.1 mm**. [Certain, CAD: rail top edge Z 416, beam top Z 420]
   - Better, if the shop has one: clamp a dial indicator to the carriage, put the tip on the beam's top face, and slide the carriage along. Adjust until the reading varies by ≤ 0.1 mm.
3. Tighten from the middle outward to about **1 N·m**: a firm squeeze on a short hex key, not a full-hand torque.
   - The M3 T-nuts are in aluminium. A third-party rail installation table gives 0.98 N·m for M3 into aluminium. [Likely]

6.3 Carriage and X bracket.

- Bolt `x_bracket` to the MGN12H carriage with **4 × M3×8** in the counterbores (20 × 20 mm pattern). Not M3×10: the carriage threads are only about 3.5 mm deep. [Likely]
- The belt-clamp tab reaches back over the beam top.
- The carriage must still roll freely end to end. If it binds near the ends, a screw head is touching.

6.4 Motor, idler and end bumpers. [Certain, CAD]

- **`x_motor_mount`** is a face plate on the beam's **front** face at the **right** end (X 1292 … 1348): 4 × M5 into T-nuts in both front slots (M5×10 through the 5 mm plate, and an M5×20 with its head sunk through the bumper block).
  - The motor sits **in front of the plate, its shaft pointing back** over the beam top. Fix it with 4 × M3×8 through the plate (heads on the beam side). A standard 24 mm shaft is long enough: the pulley needs 22.5 mm. [Likely]
  - The GT2 20T pulley goes on the shaft, teeth next to the plate, and the grub screw must sit on the shaft's flat. Use blue threadlocker.
  - The left end of this plate is the **right X hard stop** (the `x_bracket` lands on it at cup X 1262).
  - Because the plate covers the beam's front face here, the right upright gets back-face brackets only (§4.3).
- **`x_idler_mount`** goes on the beam top at the **left** end: 2 × M5×10 into T-nuts in the **rear** top slot, at X −52 and −26. The toothed idler sits in the `x_idler_slider` between the cheeks, on an M5×30 axle through the cheek slots and the slider. The M3×20 tension screw goes through the end wall into the nut in the slider's bridge.
- **`x_bumper_L`**, the left X hard stop: 1 × M5×20 (head sunk) into a T-nut in the beam's **lower front** slot, its face at X −15, so the `x_bracket` lands on it at cup X 15.

6.5 Belt.

1. Route the GT2 belt around the motor pulley and the idler in a vertical plane just above the beam top. Both ends come back to the `x_bracket` tab on the upper strand.
2. Lay each belt end **teeth down on the tab's GT2 rack** and press it with a printed `belt_clamp` cap: 2 × M3×12 + nut per cap (the nuts sit in the traps under the tab).
3. The lower strand runs **12.9 mm above the beam top** and must not rub it anywhere. [Certain, CAD]
4. Check pulley and idler alignment: the belt runs centred on both flanges, and the lower strand sits at the same height above the beam at both ends.

6.6 Belt tension. I found no maker's figure for 6 mm GT2. [Certain that the searches came up empty] Use this as a starting point and check it works:

- **Target 15–25 N.** [Guessing; one third-party calculator suggests 10–15 N, but this span is long]
- **Measure by deflection, sideways:** park the carriage at one end, out of the way. At mid-span, hook a small spring scale (BOM.md tools) onto the lower strand and pull it **sideways (in Y)** to 100 g (1 N).

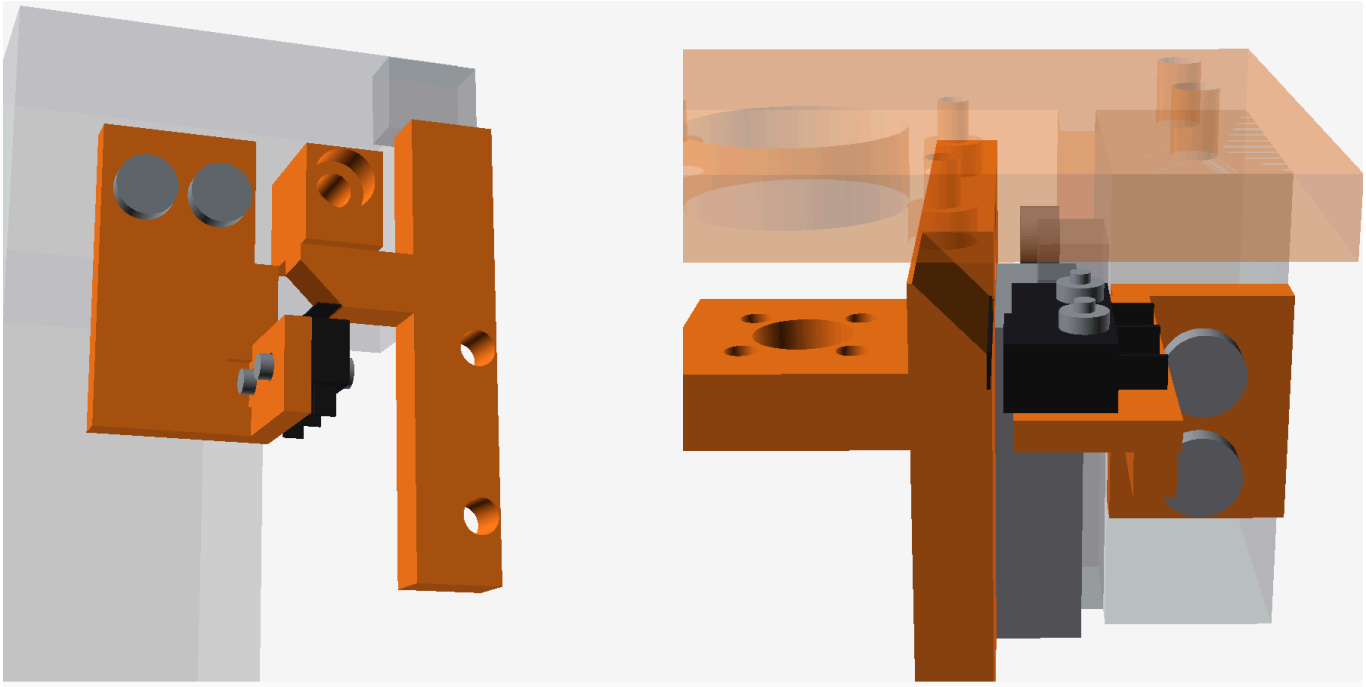
    - It should move **13–23 mm**, which corresponds to 15–25 N.
    - Not up or down: the lower strand runs only 12.9 mm above the beam top, and the upper strand about 12.7 mm above the lower one (a 20T pulley), so both block a vertical deflection of that size. [Likely, CAD geometry]
    - The formula is T = F·L / (4·d), with L = 1.35 m, the pulley-to-idler distance: the lower strand runs that whole span, because both belt ends clamp on the upper strand (§6.5). [Certain, statics]
    - Tighten by re-clamping one belt end one tooth further. Fine-tune with the M3 tension screw: it pulls the idler slider toward −X (its axle travels −6 … +2 mm in the cheek slots). Then tighten the M5 axle to lock the slider.
- **Functional checks** (after §10):
    - no tooth skips on full-speed moves end to end;
    - homing repeats within 0.2 mm (FIRMWARE.md step 6);
    - `calibrate x` reports a scale error under 0.5 mm across the row (§12.2).
- Too tight shows up as a motor that runs hot or a pulley that wears. Too loose shows as lost position.

6.7 Beam twist check, and the brace. DESIGN.md §4 estimates about 0.5 mm of twist at the cup at midspan (0.3–0.8 mm, depending on the profile's torsion constant) from the X carriage, whose centre of mass sits 77 mm in front of the beam centre. [Guessing in DESIGN] Measure it once the head is on (§7):

1. Park the head over pocket 6 (midspan) with the cup about 20 mm above the pad.
2. Put a dial indicator on the bed with its tip against the side of the buffer body, reading in Y (front–back).
3. Hang a 0.75 kg weight (a water bottle) from the cup and read the movement. At the cup's 127 mm lever the weight twists the beam with about 0.93 N·m, against about 1.46 N·m from the carriage itself (DESIGN.md §4) [Likely], so the carriage's own twist is about 1.6 × the reading.

- **Up to 0.6 mm read (1.0 mm scaled): fine.**
- **More than 0.6 mm read: fit the brace** (DESIGN.md's 1.0 mm threshold). [Guessing: no CAD for it]
    - Cut a vertical 2020 post (about 401 mm, from the 600 mm stock) and stand it behind the pockets at X ≈ 600, under the beam's rear bottom slot.
    - Fix it to the beam with two 2020 corner brackets, and to the bed with corner brackets bolted through into the back base rail's T-nuts.
    - Keep it clear of the drag chain and of the station camera ribbon (X ≈ 686).
    - Repeat the measurement.

6.8 X home switch (WIRING.md §7). [Certain, CAD `x_switch_mount` and `checks.txt`, unless tagged]



X home switch at cup X 19: the x_bracket's finger has just pressed the lever

Z top switch at c = 258: the z_plate's chamfer has just pressed the lever

How both switches work. A striker slides *across* the lever: first a 45° lead-in presses it down, then a flat holds it there however far the axis runs on. The hard stops are 4 mm past each click, and the switch's lever can only travel 1.2 mm past its click point (Omron SS-5GL, OT min 1.2 mm), so a striker that pushed the lever head-on would make the switch the stop and break it. The flat holds the lever 0.6 mm past its click point, measured at the lever tip. [Certain, Omron SS datasheet; CAD]

Before you fit either switch (both are from the BOM's 10-pack):

1. **Measure the lever** from the hinge end of the switch body to the lever tip. **Up to 23 mm:** fit it as it is. **Longer:** cut it flush with the far end of the body, 20 mm from the hinge end, with side cutters, then file the cut smooth. [Certain, `checks.txt`: the X mount takes a lever up to 27.24 mm, the Z mount up to 24.19 mm, against a 23 mm limit; Guessing which lever the pack has: long hinge levers often end 25–28 mm out] A longer lever reaches the carry flag (X) or the square part of the `z_plate` top (Z) and breaks on the first homing. A roller lever does not fit these strikers.
2. **Bench test one switch, about 100 passes.** The 45° lead-in pushes the lever sideways about as hard as it pushes it down, and the datasheet rates only a straight push. [Guessing that the lever takes it] Fit the switch in `x_switch_mount` on the bench and slide the `x_bracket`'s finger across the lever by hand about 100 times. The click must still come at the same place, and the lever must not wobble sideways. If it fails, stop and ask: the fallback (a roller lever on a cam that runs along the lever) is not in the CAD.

How the X switch sits. `x_switch_mount` goes under the beam's left end, on the front faces of the beam and the left upright, left of `x_bumper_L`. The switch hangs on its side plate, lever up, hinge end toward the back. A 17.4 mm finger on the `x_bracket`'s left edge presses it as the carriage moves left. It clicks at cup X 19.00, is held 0.6 mm past the click point from 18.7 down to the left hard stop at 15, and the lever is free again above cup X 24. Everything stays behind Y 84: in front of that, the carry flag and its arm sweep the whole Z travel at the left end.

Fit it (instructor: helpful):

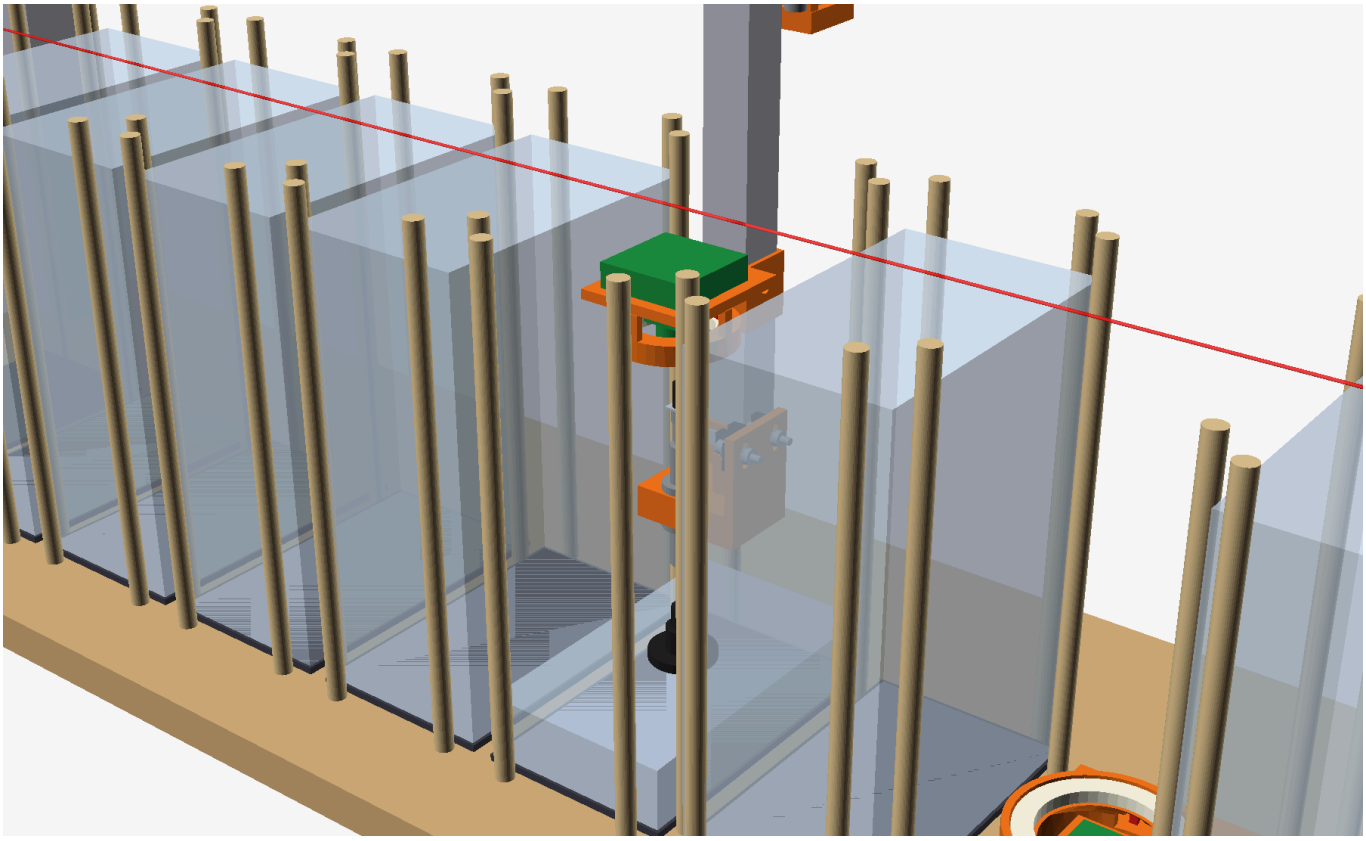
1. Put 2 slot-6 M5 T-nuts in the beam's **lower front** slot, left of `x_bumper_L`.
2. Fix `x_switch_mount` with **2 × M5×10 button heads** into them (at about X −48.5 and −37), its front plate flat on the beam and the upright. Snug, not tight.

3. Fit the switch to the side plate with **2 × M2×12** through the ±1 mm slots: heads on the mount's outer (left) side, nuts on the switch. Snug.
4. **Set the click depth, at the bumper.** Motors off, push the carriage by hand until the `x_bracket` rests on `x_bumper_L` (cup X 15). Slide the switch up in its slots until it clicks, then **about 0.3 mm more**, and tighten the M2s.
   - **Why 0.3, not 0.6:** the finger presses the lever about halfway along, so the lever tip moves about twice as far as the switch body (1.87 times). 0.3 mm at the body is the CAD's 0.6 mm at the tip. A 0.6 mm body slide would put the tip 1.1 mm past the click point, almost at its limit. [Certain, CAD lever geometry]
   - Set the 0.3 mm with calipers (the depth rod, from the switch body to an edge of the mount), not by eye.
   - **Check:** at the bumper the lever must still give a little when you press it with a finger.
5. **Set the click point along X.** Move the carriage right, then push it slowly left, and slide the mount along its slot until the click comes with a **4 ± 2 mm gap** between the `x_bracket` and `x_bumper_L` (cup X 19 ± 2, about 23 mm left of pocket 0's centre). The mount slides about ±3 mm. Tighten the M5s with the carriage at cup X 60 or more: at the bumper the carry flag blocks the hex key. [Certain, CAD]
6. **Lead (W52, WIRING.md §7):** from the terminals take the 2-core straight down to about 50 mm below the beam (below the corner brackets), back under the beam, then up behind it into the beam's **lower rear** slot. Keep the left joining plate off that spot (§4.3). [Certain, CAD lead path; Guessing on the brackets and plate, which are not in the CAD]
7. Check by hand over the whole X travel that the finger passes everything but the lever, and that nothing else (belt, cables, drag chain) touches the switch. [Certain, `checks.txt`: the finger clears every static part by 1.5 mm, and the moving parts clear the switch by 1.88 mm]

- It must click at least 2 mm before the `x_bracket` reaches `x_bumper_L` at X 15 (§6.4). The carriage still has 2.3 mm of rail at the bumper. [Certain, `checks.txt`]

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7. Z axis and head



All heights below are from `cad/layout.py` and `cad/model.py`. [Certain]

7.1 Z spine to X bracket.

- Bolt the 2040 × 340 spine to `x_bracket` with 4 × M5×12 into T-nuts in **both** of the spine's slots (bracket holes at Z 340 and 365).
- The spine sits **10 mm left of the cup axis**, so the Z rail in its right slot lands on the cup axis. (CONTRACT)
- **Set the spine's bottom end 275 ± 1 mm above the pad top.** That's 278 above the bare plywood, or 57 above a rod top if the rods are in.
- `z_end_cap` goes under the spine's bottom end: 2 × M5×12 **button heads** up into the tapped end bores (heads sunk). Its two lips sit under the Z carriage's skirts beside the rail: they are the **Z bottom hard stop** at c = −14. [Certain, CAD]
- The end cap's underside (267 mm above the pad) sweeps 49 mm over the rod tops, and everything fixed to the X carriage stays at least 5 mm above a held slab even at the Z top hard stop. [Certain, `checks.txt`]
- Drop two slot-6 M5 T-nuts into the spine's **left (−X) face slot** now, for the carry-height sensor's mount (§7.4a), and two into its **right (+X) face slot**, for the Z switch mount (§7.4). Those are the slots in the spine's narrow side faces, not the front slots: the front right slot holds the Z rail.

7.2 Z rail.

- MGN12 330 mm in the spine's right slot, 13 × M3×10 into T-nuts. Its bottom is 1 mm above the spine bottom, its top 9 mm below the spine top.
- Align it parallel to the spine edge with the caliper (as in §6.2).
- **Plumb:** ≤ 0.5 mm over the rail length, in X and in Y. Check with a square from the bed, or a digital level at 0.1°. [Guessing target: the cup's position must not drift across its 250 mm of travel]

7.3 Z motor, lead screw and nut.

1. **Cut the lead screw first: 340 mm below the motor's mounting face** (anything from 311 to 356 mm works). [Certain, CAD `Z_SCREW_LEN`] Left at 400 mm, its tip would end about 60 mm lower, inside the path of every held slab: each lift would drive the slab's back into it.
   - Wrap the cut point in tape, cut with a fine hacksaw, file the end square, and chamfer the first thread so the brass nut starts cleanly. Run the nut over the cut end by hand.
   - A screw sold as "350 mm" fits uncut if it measures 311–356 mm from the motor face.
2. `z_motor_mount` sits **on the spine's top end face**: 2 × M5×16 into the tapped end bores, plus 1 × M5×10 through its tab into a T-nut in the spine's **left** slot, 15 mm below the top. The tab stays clear of the rail and the carriage.
3. Motor **under** the mount, screw pointing down: 4 × M3×8 from below, heads in the counterbores (the `z_plate` meets this face at the top hard stop).
4. `z_plate` on the Z carriage: 4 × M3×8 in the counterbores (20 × 20 pattern). Not M3×10.
5. The brass T8 nut: **flange on top of the shelf**, 4 × M3×16 + nuts through the four holes at 45°. Leave those screws loose, run the plate up and down the full travel by turning the screw by hand, then tighten. The nut then centres itself. The screw runs 8 mm clear of the plate face.

- **The screw must not whip or push the carriage sideways at any height.**

7.4 Z top switch (WIRING.md §7). [Certain, CAD `z_switch_mount` and `checks.txt`, unless tagged] The picture is in §6.8. Measure (and if needed cut) the lever first, as in §6.8.

How it sits. `z_switch_mount` goes on the Z spine's right (+X) face near the top, fixed to the X carriage. The switch lies on its 3 mm shelf, to the right of the Z carriage: lever toward the front (the `z_plate`), hinge end to the right. The `z_plate` has a 45° chamfer, 7.5 mm deep, along its top back edge from X 20 to its right end (X 40 from the cup axis). As the head rises, the chamfer slides across the lever and the plate's back face holds it. It clicks at c = 258.00, is held 0.6 mm past the click point from 258.3 up to the top hard stop at 262, and the lever is free again below c = 253.

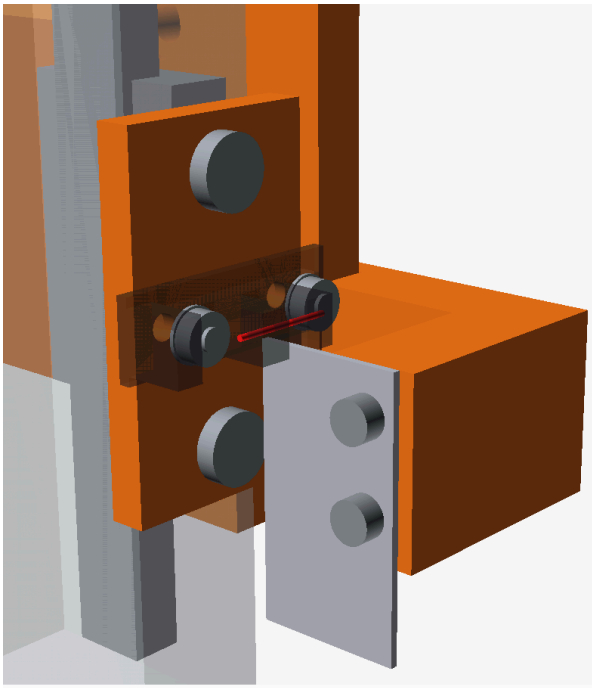
Fit it:

1. Use the two T-nuts in the spine's right-face slot (§7.1). Fix `z_switch_mount` with **2 × M5×10 button heads**, its plate flat on the spine. Snug, not tight.
2. Lay the switch on the shelf and fix it with **2 × M2×12** up from under the shelf through the ±1 mm slots, nuts on top. Snug.
3. **Set the click depth, at the top stop.** Motors off, turn the lead screw by hand until the `z_plate` meets the `z_motor_mount` (c = 262). Slide the switch toward the `z_plate` in its slots until it clicks, then **about 0.3 mm more** (calipers, as in §6.8 step 4), and tighten the M2s. At the top stop the lever must still give a little.
4. **Set the click height.** Turn the screw down about 10 mm, until the lever is free, then slowly up, and slide the mount in its slot until it clicks with the `z_plate` top **4 ± 2 mm below the `z_motor_mount`** (c = 258 ± 2). The mount slides about ±3 mm; upward, the M2 nuts and screw tips meet the `z_motor_mount` at about +3.5 mm [Certain, `checks.txt`: 3.58 mm clear]. Tighten the M5s.
5. **Lead:** cores 4 and 5 of the 8-core, back past the spine, then up behind the `z_motor_mount` to the spine top, where the 8-core's jacket is opened (WIRING.md §11).
6. Turn the lead screw by hand through the whole travel. Only the `z_plate`'s chamfer and back face may touch the lever, and the Z carriage block passes the mount (1.49 mm in the CAD, at c ≈ 260; file the strip under the shelf flat first, §3.2). [Certain, `checks.txt`]

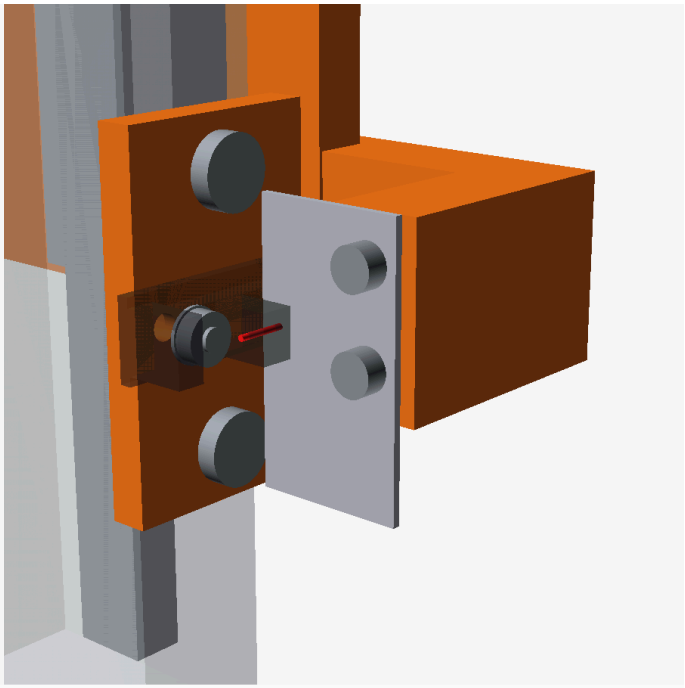
- It must click at c ≈ 258 mm: the cup face 258 above the pad top, ± 2 mm. At that point the `z_plate` top is still 4 mm below the `z_motor_mount`, which is the top hard stop at c = 262.

7.4a Carry-height sensor (the Z stall guard). [Certain, CAD `carry_sensor_mount`, `carry_flag_arm`, `checks.txt`]

The Z motor runs at about half its rated current and nothing in the controller notices a stall: a stalled lift leaves the head lower than the controller thinks. This sensor catches that before the head moves sideways. It is the second endstop board on the Z spine (fixed to the X carriage), with a flat aluminium flag on the `z_plate` (moving with Z). The flag's top edge enters the beam at cup height c = 246.5 mm and stays in it up to the top hard stop: the beam is clear at every height up to 242 and fully blocked from 248 (both travel heights) to 262. The software reads it on GPIO 6 before every sideways move (it must read blocked) and before every grip (it must read clear, which is how a dead emitter is caught). A head that still reads blocked is at least 245.5 high, where even a double pick of two thick holders clears the rod tops by 2.5 mm.



c = 244.5: flag 2 mm below the beam, slot clear



c = 262 (top stop): flag still in the beam

1. **Board to mount.** Put two M3×10 through `carry_sensor_mount` from its spine side, heads in the pockets, then the board on them, components away from the mount; a washer and nut on the sensor flange each. The arms point away from the spine and the slot runs vertically.
2. **Mount to the spine.** On the spine's left (−X) face, with 2 × M5×10 **button heads** into the two T-nuts from §7.1, the beam about 53.5 mm below the spine top (Z 561.5 above the pad top; the mount spans 537.5–585.5). Leave the screws just snug: you slide it in step 6. Button heads, because the flag arm passes 4.6 mm away.
3. **Flag arm.** Screw `carry_flag_arm` to the `z_plate` front face with 2 × M3×10 into the two 2.5 mm pilots (X −24, 2 and 14 mm above the Z carriage's bottom edge). Its leg runs beside the spine, outside the sensor's arm tips.
4. **Flag.** Cut it from the 1 mm sheet, flat, 19.9 × 36 mm (`cut_list.csv`), and file its **top edge straight**: that edge is the trip edge. Drill two 3.4 mm holes 4 mm from the outer edge, 6 and 18 mm below the top edge. Fix it to the leg's end with 2 × M3×8, inner edge toward the spine, top edge flush with the arm's top.
5. **Free-run test.** Turn the lead screw by hand through the full Z travel: the flag must pass through the slot without rubbing, about 1 mm clear each side (1.05 mm in the CAD).
6. **Set the trip, after §9 (wiring) and §12.1 (`calibrate z`):**
    1. Use a meter on the board's OUT (about 5 V blocked, about 0 V clear). `selftest` cannot run while `jog` has the machine, and restarting `jog` homes it, which loses the height. [Certain, `factory.open_hardware`, `cli.cmd_jog`]

2. `python -m slabscan jog`: move Z until the status line shows **height 246.5**. Slide the mount along its slot until OUT just turns blocked. Tighten both M5 screws.
3. Check: **clear at height 245.5, blocked at 247.5 and at 255.**
4. `python -m slabscan selftest --motion` must say `carry sensor … PASS … trips at about 246.x mm (set 246.5, allowed +/-1)`. It homes, steps the head down over pocket 0 from 249.5 to 243.5 in 0.25 mm probe moves and reports where the sensor turns clear. [Certain, `ops.carry_sweep`] Outside ±1 mm it fails and prints its readings: slide the mount and repeat.

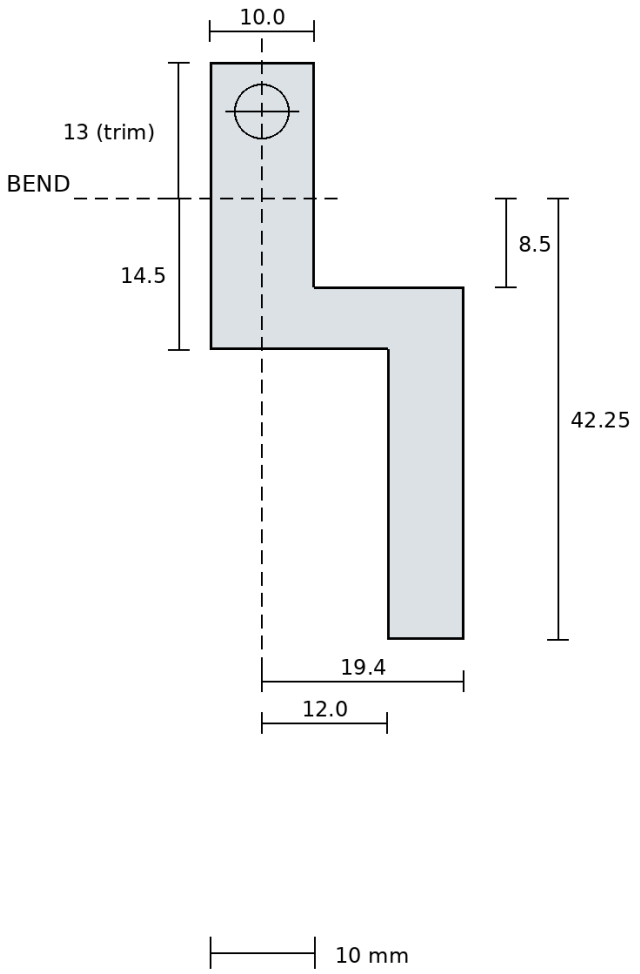
- **Not checked by the CAD:** the board's real component heights and header position, and the lead routing beyond the 0.2 m to the spine top (WIRING.md §6a).
- **`jog`, `calibrate` and `mark` do not read this sensor** [Certain, `ops.py`]: only runs, resumes, `focus`, the web page and `tools.bench_tests` do. Move carefully with those tools.

7.5 Drop arm, foot and cup.

1. **Drill the two 5.5 mm cross holes in the 2020 × 285 drop arm**, through the centres of two opposite faces, **19 mm** (for `head_foot`) and **84 mm** (for `camera_bracket`) from the end that goes into `head_foot`'s socket. The arm bottoms on the socket floor. [Certain, `model.py`: arm from 65 mm, foot hole at 84, camera hole at 149 above the cup face]
   - **Match-drill the foot hole:** push `head_foot` onto the arm until it bottoms, clamp it square, and drill through the printed hole and the arm together on the drill press.
   - Then mark the second hole **65 mm above the first**, centre to centre, and drill it.
   - Deburr both.
2. Fit `head_foot` with an M5×30 cross bolt and a nylock nut.
3. **Spring buffer:** the M10×1 body goes through the foot's 10.4 mm hole, with a nut each side. Fit the cup pad (SMC ZP25UU) on the buffer. If the foot comes from the Phase 0 jig, take it off the stub with the buffer, flag, elbow and board still on it.
4. Bolt the arm to `z_plate` with **2 × M5×12 button heads** into T-nuts in the arm's slot, heads in the plate's 2.5 mm recesses. A socket head doesn't fit: there are only 5 mm to the Z rail face. The two bolts sit below and above the carriage block. The arm can still slide.
5. **Set the height chain in this order** (buffer at rest, steel rule):
   1. With the camera bracket on (§7.7), move the buffer on its M10 nuts until the **head lens front is 142 ± 1 mm above the free cup face**. That is 140 mm above the cup face pushed up by the 2 mm trip travel, which is how the CAD measures it (`CAM_TO_SLAB`). [Certain, CAD]
      - The CAD allows the buffer ± 3 mm from its nominal place on the thread: the board slots still cover the trip, and the flag stays at least 5 mm deep in its guide. [Certain, `checks.txt`] More than that means the camera bracket's shelf height is off (§3.3).
   2. Then slide the arm in its T-nuts until **the Z carriage's bottom edge is 295 ± 2 mm above the cup face**. [Certain, CONTRACT: with this the end cap stops the carriage at c = −14 while it is still on its rail, and `calibrate z` lands in −4 … 0]
   - **Fit or re-set the flag and the trip (§7.6) after this step.** Moving the buffer moves the flag against the sensor.
6. **Vacuum tube** onto the `M-5ALU-4` barb elbow on the buffer's top port (§7.6 step 6: the elbow clamps the flag's tab, so the flag goes on first). The barb points to the back (+Y), away from the label. Leave **25 mm of slack**, because the rod rises up to 20 mm on contact. (WIRING.md §11)

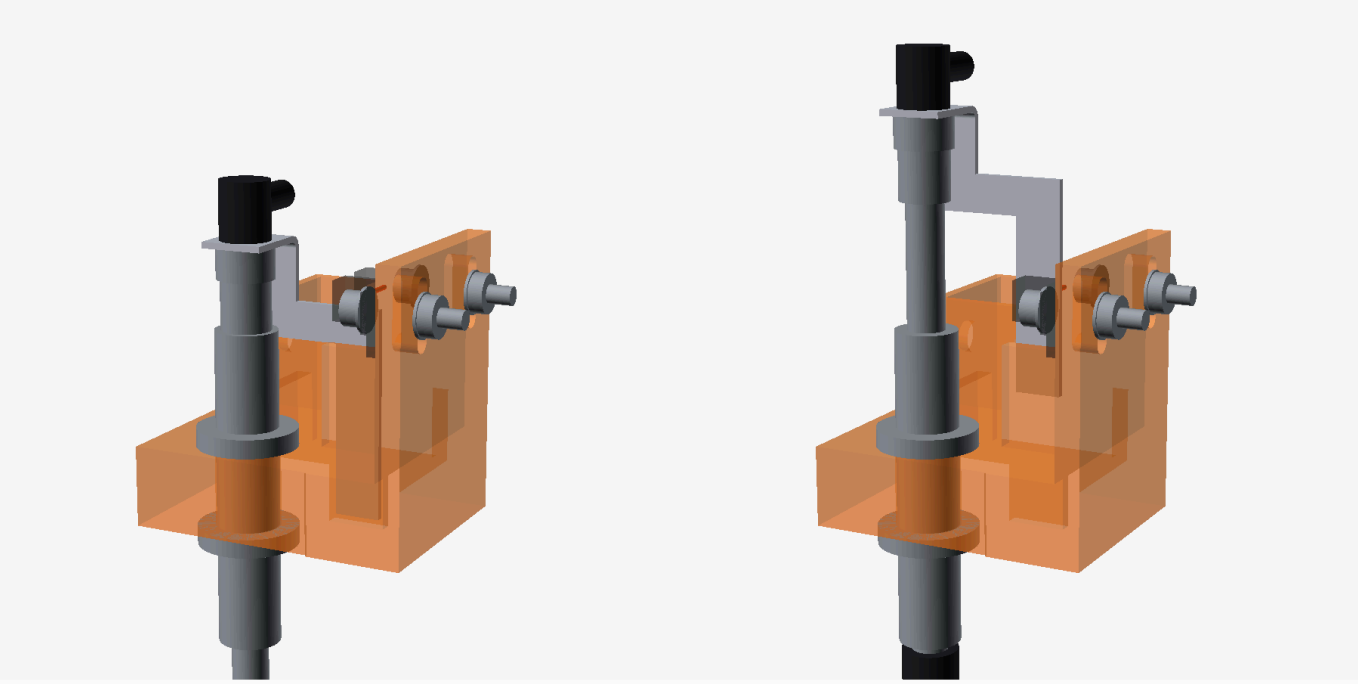
7.6 Contact sensor and flag (set it to trip at 2.0 ± 0.5 mm). The sensor is a slotted optical endstop on the foot's sensor post: a TCST2103 with a 3.1 mm slot, 7.7 mm deep, a 1 mm slit aperture, and its optical axis 2.6 mm below the arm tips. [Certain, Vishay doc 81147 drawing] The flag is a bent 1 mm aluminium plate clamped under the

vacuum elbow on top of the buffer. When the cup is pushed up, its blade rises through the slot and blocks the beam. [Certain, CAD `probe_flag`]



SlabRail probe flag, 1:1 (check the 10 mm bar)
1.0 mm aluminium sheet. Blank 24.4 x 55.25 mm.
Dimensions from the BEND line and the dashed
neck centre line (= hole centre line).

1. Cut and file to the outline. The blade's top
   edge is the trip edge: straight, square.
2. This view shows the face that faces the buffer.
   Neck in the vise, jaws at the BEND line; bend
   the tab 90 deg TOWARD YOU (r ~1 mm). Bent the
   other way the blade ends up on the wrong side.
   Plate stays flat; check 90 deg with a square.
3. THEN drill 5.2 mm on the centre line,
   7.0 mm from the plate's inner face
   (lands ~8.3 above the bend line flat).
4. Trim the tab to 2 mm past the hole.
5. Deburr. It rides in a 1.9 mm guide slot
   and a 3.1 mm sensor slot.



Cup free: flag 2 mm below the beam, slot clear

Pushed up 20 mm: flag through the slot, still in its guide

What the CAD checks [Certain, `cad/out/checks.txt`]:

- the beam is clear for the first 1.5 mm of buffer travel, the flag trips at 2.00 mm, and the beam stays fully blocked from 3 mm to the full 20 mm;
- the blade clears the sensor by 1.05 mm and its guide by 0.45 mm each side, over the whole stroke and with the board at either end of its slots;
- the blade stays 8 mm deep in the guide at full stroke (5 mm with the buffer 3 mm higher on its thread);
- a rotating (J) buffer turns at most 1.32° before the guide stops the blade, 0.6 mm from the sensor, so a J buffer works as well as a K;
- the flag, the buffer's port hex and the elbow stay out of the head camera's view of the label while it reads.
- **Not checked:** the 1.9 mm slot's print accuracy, the seal under the tab, the buffer's radial play, a twisted blade, and the tube on the barb.

Do this after §7.5 step 5, which moves the buffer. In Phase 0A, do steps 1–8 only.

1. **Measure first** (§3.3): the port top 31.5 mm above the foot's top face at rest; the board's holes are the sensor's flange holes; the elbow at most 9.3 mm above the tab.
2. **Cut.** Print `cad/out/probe_flag_blank.svg` at 100 % and check its 10 mm bar with calipers. Glue it to 1 mm **soft** aluminium (1050, 3003 or 5052; not 6061-T6, which can crack at a 1 mm bend radius [Likely]). Not thin black plastic: IR sees through it. Cut with a junior hacksaw or a piercing saw and file to the line; snips curl the edge. **The blade's top edge is the trip edge:** file it straight and square.
3. **Bend.** The drawing shows the face that will face the buffer. Clamp the neck in a vise with the jaws at the BEND line and bend the tab 90° toward you; bent the other way, the blade ends up on the wrong side. **Check 90 ± 0.5° with a square:** 1° of lean moves the blade about 0.5 mm over its length in the guide, and the guide leaves 0.45 mm each side. [Certain, arithmetic]
4. **Drill after bending:** 5.2 mm on the neck centre line, 7.0 mm from the plate's inner face. Trim the tab to 2 mm past the hole and deburr.
5. **Slide test:** with the board off, drop the blade into the 1.9 mm guide slot. It must slide freely over its whole length. If it rubs, straighten the bend or clear the slot with a 1.5 mm needle file.
6. **Fit.** The buffer is in the foot with its nuts (§7.5).

    - Put the tab on the port hex, **with no washer**: a washer would lift the flag and move the trip.
    - Put thread sealant on the elbow's M5 thread (a drop of pneumatic thread sealant, or one thin wrap of PTFE tape) and screw it in through the tab, at least 3 mm of thread engaged (about 4 turns). The elbow's own gasket seals on top of the tab. [Likely: 4 mm of thread]
    - Point the barb to the back (+Y), away from the label.
    - With a K buffer, align the blade with the slot before you tighten; with a J buffer the guide aligns it.
7. **Board.** Components toward the buffer, header toward the arm. Fix it with **2 × M3×16**, a washer under each head and another under each nut, centred in the post's slots. That is what the CAD models: the tip runs 3.9 mm past the nut, and a pick next to full neighbour stacks still clears them by 14.5 mm. [Certain, `checks.txt`]
8. **Free-run test.** Push the cup up by hand through the full stroke. The blade must slide with no rub and never touch the sensor.
9. **Set the trip** (machine only, after §9 has wired the sensor):
    1. Power the sensor from 5 V and put a meter on OUT: about 0 V with the slot clear, about 5 V blocked (WIRING.md §6).
    2. With a caliper depth rod, measure the buffer's port top above the foot's top face at rest.
    3. Push the cup up slowly by hand until OUT jumps. Measure again. The difference is the trip travel.
    4. Slide the board in its slots until it reads **2.0 ± 0.5 mm**. Board up = later trip.

    - The front (label-side) screw sits behind the buffer: reach it with a **2.5 mm ball-end key tilted 15–20° toward the label**. A straight key hits the buffer. [Certain, CAD: 1.63 mm clear at 20°] The rear screw takes a straight key (2.0 mm clear).
10. **Full-stroke test.** OUT must stay high from the trip to full compression, and the flag must never touch the sensor.
11. **Write the measured trip travel into `config/calibration.toml`** (after §11): `buffer_trip_mm = <your number>` and `buffer_trip_tol_mm = 0.1`. The software places the slow part of the peel around that point, and a trip later than 2 mm also raises both travel heights by the difference. [Certain, `config.py`, `runner._peel`] Re-measure it after any change to the flag, the sensor or the buffer's height.

Wiring: fail-safe (WIRING.md §6). **At rest the slot must be clear**; contact pushes the flag in. The simulator assumes a 2 mm trip. [Certain, `simworld.TRIGGER_COMPRESSION`]

Re-check after any knock. `calibrate z` (§12.1) also checks it in software: the probe must trigger when you push the cup up by hand, and release when you let go.

7.7 Camera bracket and LED ring.

- `camera_bracket` clamps the arm with the second cross bolt (M5×30 + nylock).
- **The head lens front is 142 ± 1 mm above the free cup face**, which is 140 mm above the cup face at the 2 mm trip. [Certain, CAD `CAM_TO_SLAB`] A held slab hangs on the free cup, so its back is about 142 mm from the lens; `focus head` (§12.3) focuses at that distance. Set it with the buffer nuts (§7.5 step 5).
- Fit the ELP board on the shelf (4 × M2 + nuts). The shelf height assumes a 20 mm board-to-lens-front length (§3.3).
- **Head LED ring:** push the ring up into the seat around the lens, between its three centring ribs, until it sits against the lip, LEDs down, its pads and leads through the notch in the lip. The ribs fit a 40.0 mm ring; for a bigger one (up to 42.5 mm), shave each rib by half of (OD − 40) first (§3.3). Centre a smaller one by eye. Fix it with three dots of glue: the bore alone does not hold it. Its LED face then sits 1 mm under the lens front, and the barb elbow clears it by 3.75 mm at the full 20 mm stroke. [Certain, `checks.txt`, for rings of 39.5, 40 and 42.5 mm]
- **The lens looks straight down, 5 mm toward the label from the cup axis.** The bracket sets that; there is no adjustment. [Certain, CAD `HEAD_CAM_DY`] With the lens on the cup axis, the vacuum elbow 23 mm under it

would hide the card window the learned face-down check needs; offset, every part on the axis throws its shadow away from the label side (`img/face_down_view.png`).

- **Keep the board square to the slab.** A tilt of more than about 1.3° toward the cup puts the cup pad into the card window [Certain, CAD aim check: 1.34°; 1.31° with the software's own mapping]. If it is tilted, shim the board: one 0.3 mm M2 washer (BOM.md tools) under one pair of its 32 mm-spaced holes tilts it about 0.54° [Likely].
- **What checks the aim** [Certain, `facedown.aim_problem`; tested in the simulator]: `focus head` on the machine (§12.3) looks over the empty pocket 12 first and says `camera aim: OK` or `FAIL`, and every run in learned mode looks again over the first empty pocket. A tilt toward the cup shows the cup pad or the foot there: the learned check then accepts nothing and learns nothing that run, and every slab without a back-label code stops for you. A resume whose survey finds no empty pocket accepts nothing until it has emptied its first source pocket and looked over that (OPERATING.md §8), so a camera knocked while the run was stopped is still seen.
- **What it does not see:** a head part as dark as the pad (no edge against it), a tilt away from the cup, and a roll along the row. On a CGC slab pushed into a pocket corner a tilt of about 0.01° toward the label already puts more than 2% of the window on the holder [Likely, CAD card-plane check]. Only your look at `face_down_window.jpg` (§12.3) covers those. Lint, dust or a tag on that pocket's pad can make the check fire when the camera is square: that is safe, but clean the pad first.
- Run the camera USB and the ring wires along the arm in clips every 50 mm (WIRING.md §11).

7.8 Z hold test and Z lift test (after §10; the lift test after §12.1)

Hold test (FIRMWARE.md open issue 4):

1. Put a 20–30 mm foam or wood block on the bed under the cup, and lower the cup to about 15 mm above it. The back-drive torque is the same at any height, so a low test checks the same thing and a fall costs nothing. Do the first trial before the camera is fitted.
2. With a 200 g weight hanging on the cup, press the E-stop.
3. Watch for 1 minute.

- **Any creep means a friction brake or counterweight is needed before unattended use** (the BOM's conditional "Z counterweight" line; its mount is not in the CAD). [Guessing in DESIGN: detent torque should hold]

Lift test (the Z motor runs at about half its rated current; the carry-height sensor stops a run after a stall, but only a strong enough motor prevents one):

1. One junk slab in pocket 1, pocket 0 empty, `calibrate z` done (§12.1).
2. `python -m tools.bench_tests lifts 100`: it probes pocket 0's bare pad, does 100 full-height lifts at full Z speed with the slab held, puts it back and probes the pad again without homing.
3. **Pass: the two pad readings agree within 0.1 mm.**

- **Fail:** Z lost steps. Try the free fixes first, one at a time, and repeat the test after each:
  1. Set Z `max_rate_mm_per_min` in `config.yaml` and `feed_z` in `calibration.toml` to 3000.
  2. Raise the Z motor's `run_amps` in `config.yaml` (the `motor0:` entry under `z:`, line 239; not the spare `motor1:` line below it) from 0.8 A to about 60 % of the motor's rating (1.2 A for a 2.0 A motor, 0.9 A for a 1.5 A one). The motor must stay under 50 °C after 10 min. [Likely]
  3. Check the motor is the specified 1.5–2.0 A, ≤ 4 mH type (BOM).
  - Only if it still fails, buy the BOM's conditional "Z motor upgrade" (one with a published torque curve).

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8. Vacuum plumbing

(2–3 h. Phase 0B used only the cup, buffer and tube; the pump, valves, reservoir and sensor are new here.)

8.1 Order of parts, from the pump to the cup [Certain, BOM + `hw.py` + WIRING.md §5]:

```
pump exhaust → room
pump inlet ◄── check valve (arrow toward pump) ◄── reservoir (2 in PVC, ~300 mL)
                                                    │
                                       vacuum valve Q2 (NC, GPIO 27)
                                                    │  cup side
                                       tee ── 2 mm silicone ── XGZP6847A sensor (on the I/O board)
                                                    │
                                       tee ── vent valve Q3 (NC, GPIO 22) ── room
                                                    │
                             4 mm PU tube through the drag chain and the Z loop
                                                    │
                                       M-5ALU-4 barb elbow ── buffer top port ── cup
```

8.2 Steps

1. **Reservoir:**
   - 2 in Sch 40 PVC, 6 in long, two caps solvent-welded on.
   - Drill (11/32 in tap drill) and tap one cap 1/8 NPT twice (in and out). Push fittings go in with PTFE tape.
   - Let the PVC cement cure for the time on its label before pulling vacuum on it. [Likely]
   - Don't use a PET bottle: it collapses at −80 kPa. (BOM.md)
2. **Check valve** between reservoir and pump inlet, arrow toward the pump.
3. **Valves (Clippard EV-2-24, 10-32 ports, 10-32 → 4 mm fittings).**
   - Put the **inlet** on the side that is at the higher pressure when the valve is closed:
     - vacuum valve: inlet on the **cup side**;
     - vent valve: inlet **open to the room**.
   - [Likely: the inlet pressure normally helps a direct-acting poppet seal. Check the inlet/outlet marks on Clippard's drawing; I could not open it from here. The leak test decides.]
4. **Sensor tee** on the cup side of the vacuum valve, so the Pi reads the cup line, not the reservoir. Keep the silicone tube short. (WIRING.md §5)
5. **Pump:**
   - on rubber feet, not hard-mounted;
   - a short silicone hose section at its inlet so vibration doesn't fatigue the fittings;
   - check its polarity (WIRING.md W23).
6. **Cup line:** cut the PU tube square with a knife and push it fully into every fitting until it bottoms, then tug it.

Why the reservoir sits on the pump side [Certain, `hw.py`]: opening the vacuum valve onto a pre-evacuated reservoir seals the cup within the software's 0.4 s seal window. The cost: when the power goes (E-stop, mains), the closed valve cuts the reservoir off, and a held slab hangs only on the small cup line until it leaks down. §13.4 measures how long. **Only if §13.4 fails even with a tight line**, the alternative is to put the reservoir on the cup side (pump → check valve → Q2 → reservoir → sensor/vent tee → cup): the slab then hangs much longer on a power cut, but every vent and every seal has to empty or fill the reservoir. Then raise `vent_s`, `vent_timeout_s` and `vac_seal_timeout_s` (to about 2–3 s [Guessing]) in `calibration.toml`, re-run `selftest -i`, and re-time the cycle.

8.3 Leak test [Certain for the selftest thresholds, Guessing for my tighter targets]

Run it once §9 has wired the pump, the valves and the sensor. Until the gate posts are on (§5.9), put `overfill_gate = false` in `config/calibration.toml`, or selftest's `overfill beam` line fails; §12.5 removes it

again.

Test	How	Pass
Seal	`python -m slabscan selftest -i`, press a slab on the cup	`seal` reaches −35 kPa
Cup line holds	same run: the `leak` line (valves shut, pump off)	< 1 kPa/s (selftest's own pass is < 2.5); the line also prints how long a slab would hang on the closed line
Vent	same run: the `vent` line	back **within 1 kPa of atmosphere** after venting (selftest's own pass): at −5 kPa the cup still holds about 3× a slab's weight
Reservoir holds	block the cup with a slab, pump down with the valve open, shut both valves, and watch `kpa`	holds −60 kPa for 60 s with the pump off (BOM.md valve test)

Finding a leak:

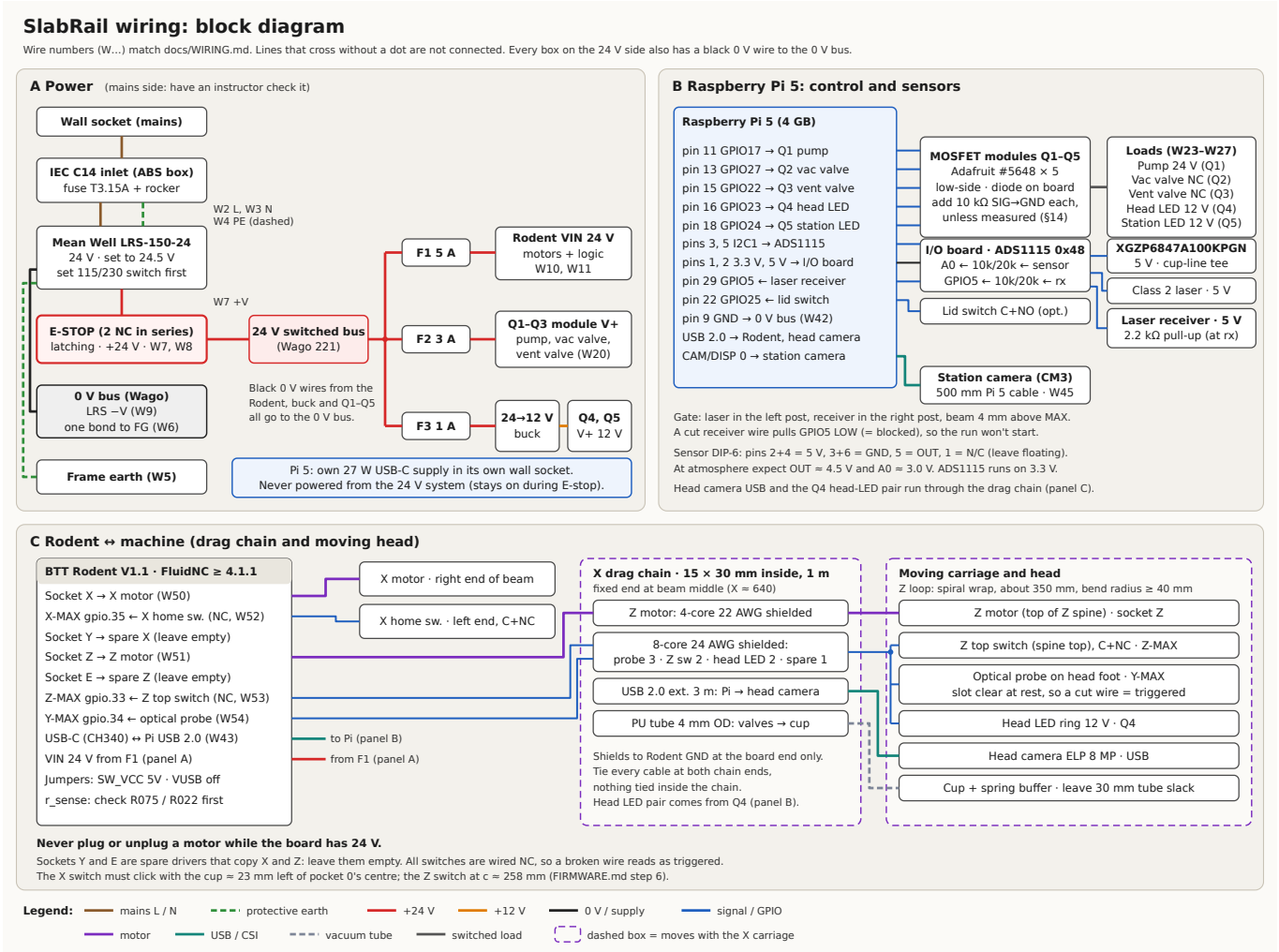
- Pinch the tube shut at points along the line, moving from the cup toward the valve. The leak is on the cup side of the first pinch point where the vacuum starts to hold.
- Usual culprits: a tube not fully seated on the barb, the elbow's thread on the buffer (sealant under the flag tab, §7.6 step 6), a cup lip with grit on it, the vent valve's seat.

---

9. Electronics

(10–15 h. **Instructor: mains wiring, crimp review, first power-up.**)

Follow WIRING.md section by section. The block diagram:



Order of work

1. **Panel** (1/4 in plywood behind the machine, Pi centred near X ≈ 690). It holds:
   - the PSU (terminal cover on) and the closed mains box with the IEC inlet;
   - the E-stop feed;
   - the Wago buses and fuse holders;
   - the Rodent;
   - the Pi;
   - the I/O board and the five MOSFET modules.
2. **Mains:** the IEC inlet snaps into the ABS mains box, never into the plywood; insulated receptacles on every tab; W2–W4 leave through the cable gland to the LRS; PE to the frame (WIRING.md §1.1, §12). **Instructor checks it before the first plug-in: box closed, no exposed mains metal anywhere with the cord plugged in.** Set the LRS-150's 115/230 V switch to your mains (115 V in the US). [Certain, WIRING.md]
3. **24 V side:**
   - E-stop in the +24 V line, **both NC contact blocks in series** (WIRING.md §1.3);
   - fuses F1 5 A (Rodent), F2 3 A (pump/valves), F3 1 A (buck);
   - the 12 V buck.
4. **Rodent:**
   - jumpers per WIRING.md §2;
   - **check the sense-resistor marking first** (FIRMWARE.md step 0: R075 or R022);

    - X motor to socket X, Z motor to socket Z. **Nothing in sockets Y or E.**
5. **Inputs:**
    - X home switch (NC) → X-MAX;
    - Z top switch (NC) → Z-MAX;
    - optical probe → **Y-MAX** (not the Probe connector).
    - All three are fail-safe: a cut wire reads as triggered. (CONTRACT)
    - the carry-height sensor's OUT → 8-core core 8 → the I/O board's 10 kΩ / 20 kΩ divider → Pi **GPIO 6** (pin 31) (WIRING.md §6a).
6. **Pi side:** MOSFET modules on GPIO 17/27/22/23/24, each with a 10 kΩ SIG-to-GND pull-down unless you measured one already there (WIRING.md §4), the I/O board (ADS1115 at 0x48 and the sensor divider), and the laser gate on GPIO 5 with its 2.2 kΩ pull-up **at the receiver**. The laser is the ≤ 1 mW (Class 2) module, not the kit's KY-008 (WIRING.md §8).
7. **Moving cables:** the drag chain (fixed end at the beam's top-back, X ≈ 640) and the Z service loop (WIRING.md §11). The 8-core cable's colours go in WIRING.md's table.
8. **Strain relief and labels on both ends of every cable** (WIRING.md §13).

Crimping: JST-XH and Dupont crimps with the proper tool (BOM.md tools). **Pull-test each one, and have the instructor look at the first ten.** Or buy pre-crimped leads.

Pre-power checklist: do every step of WIRING.md §14 with a meter, mains unplugged. The ones that bite:

- **Mains box closed, nothing at mains voltage exposed.**
- **24 V bus to the Pi 5 V pin: open.** 0 V bus to Pi GND: < 1 Ω.
- **E-stop pressed:** PSU +V to the 24 V switched bus open, and each of the two contact blocks opens it on its own.
- **MOSFET SIG to GND ≈ 10 kΩ on all five modules.** Open = add the pull-down now: without it a booting Pi can run the pump or open the vent.
- **Motor coils** 1–4 Ω each, coil to coil open, and **Y and E empty**.
- **Sense resistor value in `config.yaml` matches the board.** The file ships with 0.022 (the safe guess); change all four lines to 0.075 only if you read "R075". [Certain, FIRMWARE.md]

First power-up: WIRING.md §15, in its order:

1. Pi alone: I2C `48`, sensor ≈ 3.0 V at A0, gate ≈ 3 V clear and < 0.4 V blocked.
2. PSU set to 24.5 V with no load.
3. Release the E-stop and check the voltages, and that nothing warms up.
4. One load at a time with `pinctrl`.

---

10. Firmware

(2–3 h. Follow FIRMWARE.md steps 0–7.)

Step	What	The check that matters
0	Board set-up, power off	Resistor marking → `r_sense_ohms` (ships 0.022; R075 → all four lines 0.075). Jumpers 5V / 5V / NPN off / VUSB off / no DIAG.
1	Flash FluidNC v4.1.1+ (web installer)	The banner appears at 115200 baud
2	Load `firmware/fluidnc/config.yaml`	Then `$WiFi/Mode=Off`. **Never `$CD=config.yaml`.**
3	`$SS`, `$CD`, `?`, `$Limits`	No `ERR`; 4 drivers pass; Alarm until homed; nothing active at rest

Step	What	The check that matters
4	Motor direction, soft limits off for this test only	+X moves **right**, +Z moves **up**; 100 mm jog = 100 ± 0.5 mm
5	Switches and probe, fail-safe rows	**Unplugging any switch or the probe must read as triggered.** If not, stop and fix it.
6	First homing, hand on the E-stop (instructor present)	Z up first, then X left; `MPos:22.000,0.000,255.000`; repeat within 0.2 mm; cup over pocket 0 at X ≈ 42 ± 2; motors below about 50 °C after 10 min holding
7	`G38.3` probe test over pocket 0, and the probe hard limit	Trips at about −2; spread < 0.1 mm; unplugged probe gives `ALARM:4` and no motion; pushing the cup up during a slow upward `G1` gives the "Probe Hard Limit" alarm

Then do the Z hold test (§7.8) and the beam twist check (§6.7).

---

11. Software install

(1–2 h)

11.1 Operating system.

- Use Raspberry Pi Imager. **The contract says Bookworm, but Imager now offers Trixie (Debian 13) by default.** [Certain, raspberrytips.com]
- Pick **Raspberry Pi OS (other) → Raspberry Pi OS (Legacy, 64-bit)**, which is Bookworm with Python 3.11. [Certain, same source]
- In Imager's settings: hostname `slabrail`, your user, Wi-Fi, and **SSH on**.
- The software's tests also pass on Python 3.13 [Likely: the software agent's report], but `setup_pi.sh` has only been written for Bookworm's apt packages. [Certain]

11.2 Copy the project folder to the Pi so that it is `~/psa-slab-scanner`: every command in these docs starts with `cd ~/psa-slab-scanner/software`. The project is the `psa-slab-scanner` folder inside a larger git repository [Certain], so a plain `git clone` puts it one level deeper. Either:

```
git clone <repository URL> ~/codex                 # add -b <branch> if the project is on a branch
ln -s ~/codex/psa-slab-scanner ~/psa-slab-scanner
```

or copy only the `psa-slab-scanner` folder to `~/psa-slab-scanner` by USB stick. `gen_config` reads `cad/layout.py`, so the `cad/` folder must be there too. [Certain, `config.from_layout`]

11.3 Install:

```
cd ~/psa-slab-scanner/software
bash setup_pi.sh            # or: bash setup_pi.sh --service   (web page starts at boot)
sudo reboot
cd ~/psa-slab-scanner/software && source .venv/bin/activate
i2cdetect -y 1                      # 48 = ADS1115 (Phase 0: no 48 yet, that's fine)
rpicam-hello --list-cameras         # the Camera Module 3
v4l2-ctl --list-devices             # the USB head camera (not /dev/video0 on a Pi 5)
```

Don't `pip install` numpy or OpenCV into the venv. A pip numpy hides the apt one and breaks picamera2. [Certain, software README]

OCR threads. The `--service` unit sets `OMP_THREAD_LIMIT=1`, and `decode.py` sets it for command-line runs too. With Tesseract's default threads, OCR timed out on every frame on a loaded 4-CPU computer; single-threaded it read a station frame in about 3 s. [Certain, measured on the development computer; Guessing for the Pi]

11.4 Machine config. `setup_pi.sh` already ran `python -m tools.gen_config`. Run it again whenever `cad/layout.py` changes (`--check` says whether it is stale). It writes `config/machine.toml` and never touches `config/calibration.toml`. The face-down card window, the head crop (`head_roi_half_mm = 38.3`), both travel

heights (248), the buffer stroke (20) and the carry sensor's trip (`carry_sensor_c0 = 246.5`) all come from the CAD there; the window reads `face_down_window_mm = [[-27.5, -18.5, 27.5, -13.0]]` (mm from the head-camera axis; 55 × 5.5 mm, 302.5 mm²). [Certain, `config/machine.toml`]

11.5 Your own settings go in `config/calibration.toml`. It overrides `machine.toml`, and the calibrate commands write there too. A misspelt key is refused when the config loads.

```
serial_port = "/dev/serial/by-id/usb-1a86_USB_Serial-if00-port0"   # stable name for the Rodent's CH340 (CONTRACT)
psa_cert_digits = [7, 8, 9]  # the PSA cert lengths Phase 0 saw, without leading zeros (§1.8); another length becomes
                             # needs_review. Keep 7 if you have pre-2016 slabs (a cert printed 02074012 has 7 digits)
buffer_trip_mm = 2.0         # your measured trip travel (§7.6)
buffer_trip_tol_mm = 0.1     # once it is measured
# lid_switch = true          # only if you fitted the optional lid switch on GPIO 25
# custom_regex = "..."       # only if Phase 0 found a prefix/suffix around the cert (one capture group). It turns off
                             # the 8-digit ITF rule: every pre-2016 PSA slab is then needs_review
# cam_head: see §12.3 (the whole table must be given if you change any part of it)
# face_down_check = "stop"   # "learned" (default) | "stop" (back-label codes only) | "off" (no check, like --all-face-down)
# face_down_threshold = 0.56         # written by phase0 --face-up 15 --save on a GO (never below 0.55) ...
# face_down_threshold_sig = "..."    # ... with the card window it was measured for; a threshold without it is not used
# face_down_window_mm = [[-27.5, 13.0, 27.5, 18.5]]   # only if window_check.jpg showed the label at the bottom (§1.7)
# carry_sensor = false       # only to run without the carry-height sensor (then nothing sees a Z stall)
```

- **`face_down_check`:** leave it out to keep the default, `"learned"`. Until Phase 0 has saved a threshold, 5 references exist and 5 proven slabs showed the window clear, it acts exactly like `"stop"`: every slab without a back-label code stops for you (OPERATING.md §8.4). [Certain, `facedown.learned_ready`] Set `"stop"` if you want that stated, for example after a Phase 0 HOLD.
- **Don't type a threshold in by hand.** One without the `face_down_threshold_sig` that Phase 0 wrote with it is ignored, and one below 0.55 is refused. [Certain, `config.py`]

11.6 Check everything: `python -m slabscan selftest -i --motion`. It must end `SELFTEST PASSED` (exit code 0). With `-i` it offers to store the empty-cup vacuum reading as the sensor zero (`vac_baseline_kpa`): say yes with the cup empty. If you installed the boot service, stop it first (`sudo systemctl stop slabrail-web`): only one slabscan program may use the machine. The gate posts go on in §5.9, so keep `overfill_gate = false` (from §8.3) in `calibration.toml` until §12.5, or the `overfill beam` line fails. A `calibration` WARN is normal before §12, and so is a `carry sensor` FAIL from `--motion` until its trip is set (§7.4a step 6) and a `camera aim` WARN ("not checked yet") until `focus head` (§12.3) or the first learned run has looked over an empty pocket.

11.7 Learn the commands on the simulator before touching the machine: `python -m slabscan sim --slabs 330` and `python -m slabscan --sim web`.

---

12. Calibration

(2–3 h, after 5B: rods glued, pads down.)

Order: the awl is still on from §5B, so do **§12.2 (`calibrate x`) first**. Then take the awl off, fit the cup pad and do §12.1, §12.1a and the rest. If you did §12.1 first, re-run `calibrate z` after you refit the cup pad: a pad change moves the cup face. (Before `calibrate z`, `pad_trigger_c` is 0, which only makes `jog`'s safe height higher [Likely, `ops.py`, `config.py`].)

12.1 `calibrate z`. Cup fitted, pocket 0 empty.

```
python -m slabscan calibrate z
```

What it does:

1. Homes.
2. Asks you to push the cup up by hand (the probe must trigger) and let go (it must release).
3. Probes the bare pad of pocket 0 four times and writes `pad_trigger_c`, and stores the empty-cup vacuum zero.

- Probing never goes below Z −7.5. [Certain, `ops.z_floor`]

Pass:

- `pad_trigger_c` between −4 and 0 (FIRMWARE.md step 6);
- repeat spread < 0.1 mm (it warns above 0.3).

If it is off:

- **Outside −4 … 0:** re-check the 295 mm carriage-to-cup height (§7.5) before going on.
- **"No contact down to Z −7.0":** the head sits too high. (−7.0 = the Z after homing, 255, minus `c_max`, 250, minus the default 12 mm `--extra-depth`; it is never lower than −7.5.)

Re-run it whenever you change the cup pad.

12.1a Carry-height sensor trip. Now that heights are calibrated, set and measure it: §7.4a step 6. `selftest --motion` must report it within ±1 mm of 246.5.

12.2 `calibrate x`. Awl fitted.

First mark the two pocket centres. Nothing else marks them, and the pads now cover the plywood. Stick a small piece of masking tape on the pads of pockets 0 and 12 and draw a cross on each at the pocket centre: halfway between the left and right rod rows (along X), and halfway between the front and back rods (along Y). Measure from the rods with the calipers. Peel the tape off afterwards: ink on a pad can rub onto a slab's front label. [Likely]

```
python -m slabscan calibrate x
```

1. Centre the awl over pocket 0's centre: first by eye at the safe height, `ok`.
2. Then low, just above the pad: `z -10`, `z -1`, and so on. Downward moves stop on contact. Centre it exactly on the tape cross, `ok`.
3. Repeat for pocket 12.

- It writes `offset_x` and prints the distance error across the row.

Pass:

- **error under 0.5 mm** [Guessing target; the software only warns above 1.5 mm];
- `offset_x` within ± 1 mm. The holes came from `mark`, so the residual should be tiny.

Larger means the belt or a pulley grub screw slipped. Fix that, don't paper over it. [Certain, `ops.calibrate_x` message]

12.3 `focus head`. One slab face-down in pocket 0, label toward you. **Pocket 12 must be empty.**

```
python -m slabscan focus head
```

- **Camera aim first:** the head goes over pocket 12, probes it and looks at it from the empty travel height, as a learned run does. It prints `camera aim: OK` or `camera aim: FAIL - …` and saves the view, with the card window drawn in, as `focus/focus_head_empty.jpg`. On FAIL, look at the picture: lint or a tag on the pad gives the same message (clean it); otherwise shim the board (§7.7) and run `focus head` again. [Certain, `ops.aim_over_empty_pocket`] With a slab in pocket 12 it says "not checked".
- The cup comes down and rests on the slab: within about 2 mm of a held slab's distance.
- Turn the M12 lens to the peak score and lock the nut.
- The best frame is saved in `focus/`. Open it: the QR and barcode must look crisp, with no glare spot on the label.
- **Check the head-camera crop.** The run crops every head frame to the slab under the cup (`head_roi_half_mm` = 38.3 mm either side of its centre line: the slab's half-width less its play), because the 3.6 mm lens also sees the neighbours' labels and, beside the slab, the lower neighbour stacks. The crop is computed from `cam_head`'s

`hfov_deg` (full field of view across the image width), `row_axis` (`"x"` if the image width runs along the pocket row) and `lens_to_label_mm`. [Certain, `runner._head_roi`]

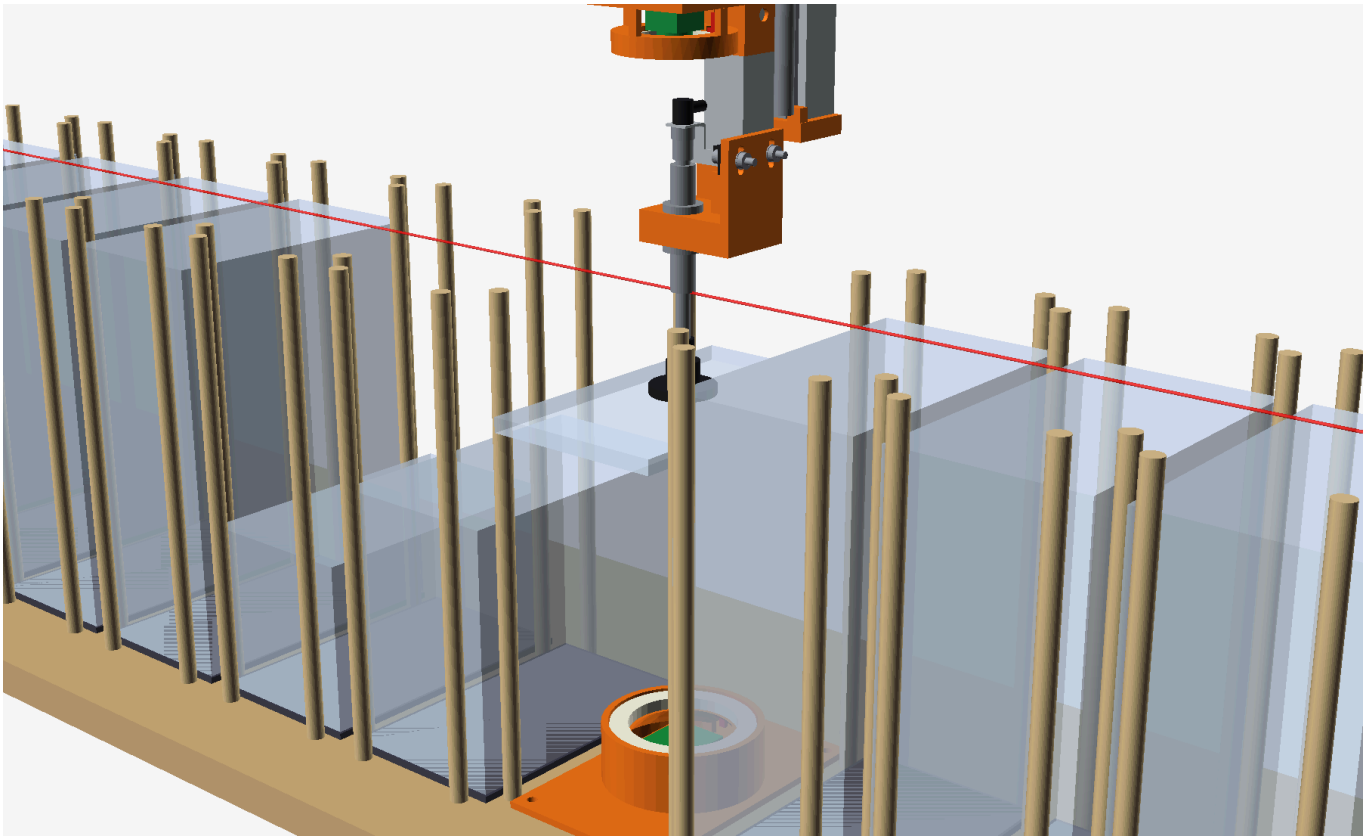
  - Lay a ruler across the slab along the row and read the width the saved frame shows: hfov = 2 × atan(half that width ÷ 140 mm).
  - If the image width does not run along the row, set `row_axis = "y"`.
  - Put the whole table in `calibration.toml`, e.g. `cam_head = { type = "usb", device = "auto", width = 3264, height = 2448, hfov_deg = 64.0, row_axis = "x", lens_to_label_mm = 140.0 }`.
  - Too large an `hfov_deg` makes the crop too narrow and cuts the label off: every slab then takes the slow station detour. Too small widens it past the slab (a real 70° lens configured as 64° crops about ±42.9 mm), and a lower neighbour's back label can come into view. That is not always visible as a `conflict`: on a slab with no back code of its own, a neighbour's QR could become its cert, or a face-up slab could look face-down. The rule "codes of two certs in one frame = another slab's label" stops most of it, so measure `hfov_deg` anyway. [Certain, real-image composites]

- **The learned check's card window** (after the first run, §13.1): open `batches/<id>/images/face_down_window.jpg` (the first slab, the window in red) and `face_down_empty.jpg` (the view over an empty pocket). **The red boxes must lie on the card back, about 3.4 mm (about 60 px) clear of the cup pad's edge** [Likely, CAD at hover: the window ends at y −18.0 mm from the cup axis, the cup's shadow at about −14.6; compare `img/face_down_view.png`]. The box is now a thin strip, 55 × 5.5 mm. If the pad or the foot reaches into a box, the camera is tilted toward the cup: shim the board (§7.7) and look again (the run's own aim check should have said so too: a `CAMERA AIM:` line). If a box reaches past the card's top edge onto the holder, or past its side, the camera is tilted away from the cup or rolled, which no software check sees: shim the other way, and note which holder (§1.2). The start line reads `face-down check: learned (…)` once Phase 0 has saved a threshold and 5 references exist.

12.4 `focus station --save`.

```
python -m slabscan focus station --save
```

The machine picks the slab out of pocket 0 and holds it over the station camera at c = 171.6 (the slab front hangs 140 mm above the lens). It sweeps LensPosition, writes the sharpest value that decodes, then puts the slab back.



12.5 Overfill beam.

1. Measure the beam height at both ends and in the middle (a card held in the beam, then a ruler): **214 ± 1 mm above the pad top.** That is above the 210 MAX line and below the 218 rod tops. [Certain, CONTRACT] The posts stand on the bare bed and need no shim (§5.9); only if it is more than 1 mm low, shim under the foot, and if it is high, file the foot.
2. Aim the laser dot at the centre of the receiver.
3. With the room lights on, turn the receiver's trimmer until a hand in the beam reads blocked. (WIRING.md §8: non-modulated receivers can be fooled by room light.)
4. If you added `overfill_gate = false` in §8.3, delete that line from `calibration.toml` now. Then `python -m slabscan selftest -i` → `overfill beam` PASS and `overfill beam blocked` PASS.
5. Stand a `fill_gauge` in a pocket's front finger gap, foot on the bed. The **underside of its top bar** is the MAX line (the gauge marks it 3 mm low, because the bed is 3 mm below the pad top); its top notch (198 mm) is not. The beam must pass about 3 mm above the gauge's top.

12.6 Check the whole set. `python -m slabscan selftest -i --motion` → PASSED. In learned mode it also has a `camera aim` line: the verdict the last run (or `focus head`) measured over an empty pocket. PASS, or WARN ("not checked yet") before the first one; FAIL means re-aim (§7.7). Keep a copy of `config/calibration.toml` somewhere off the Pi.

---

13. Acceptance tests before real slabs

Use junk slabs only: 15–20 cheap graded cards you can afford to scuff, including at least one CGC, one BGS or TAG (only if it measures ≤ 82.6 mm wide), one thick holder, and one old label if you can. **Take the lowest-value slabs from your own stock** and write off the ones the destructive tests use; buying 15–20 costs about $150–400 and is not in the BOM total (conditional line). **All tests must pass before your inventory goes in.**

Destructive on purpose: §13.3 drops two slabs (one onto the bare pad from about 240 mm). **May damage a slab:** §13.4 (a slab can fall from carry height when the vacuum leaks down) and §13.2 (that slab is the marking test). Use your least valuable slabs there.

13.0 The bench test tool

The CLI has no command to cycle the cup 300 times on one slab, drop a slab on purpose, or log the vacuum through an E-stop. `software/tools/bench_tests.py` does, using the runner's own pick, carry and place code, so each cycle is a real one. Run it from `software/` with the venv active:

```
python -m tools.bench_tests cycles 300     # 300 pick-read-place transfers between pocket 1 and pocket 0 (13.2)
python -m tools.bench_tests rest 24        # cup sealed on the top slab of pocket 1 for 24 h, re-held whenever above -40 kPa (13.2)
python -m tools.bench_tests lifts 100      # Z lift test (7.8)
python -m tools.bench_tests drop 2         # DESTRUCTIVE: carry the top slab of pocket 1 over pocket 2 and vent (13.3)
python -m tools.bench_tests powercut 300   # hold a slab at carry height and log the vacuum while you cut the power (13.4)
```

- Add `--sim` to try any test on the simulator first. `tests/test_bench_tests.py` runs every subcommand against the simulator. [Certain]
- Every test runs inside the runner's own guard, like a real run: a planned stop or an unexpected error (camera, controller ALARM) puts a held slab back or keeps the pump holding it and asks you to take it; Ctrl-C halts the motors at once. It never ends with a slab hanging on shut valves. [Certain, tested in the simulator]
- `cycles` reads the same slab again and again, so the run's rule "a cert already in the batch is a conflict" is switched off inside this tool only.
- It uses private runner methods, so re-run `pytest tests/test_bench_tests.py` after any software update. **Not yet run on hardware.**

13.1 Dry run: 100 cycles with 10 junk slabs

(about 1.5 h)

1. Put one junk slab in each of pockets 1–10, face-down, label toward you. Write down their certs.
2. `python -m slabscan run --name dry1`. Every slab moves one pocket left and gets read.
3. Move each slab back one pocket to the right by hand (pocket 9 → 10 first, then 8 → 9, … 0 → 1), and run again.
4. Ten runs = 100 transfers.

- Make one of the ten runs `--front-photo always`, so the station path gets exercised too.
- If you have them, swap in the BGS/TAG and the old-label slab for two of the runs. A slab with no back-label code (PSA 2016 and older, CGC Authentic, BGS) stops at the face-down check until the learned check is active: check it, then `python -m slabscan resume batches/<id> --face-down-ok` (the stop message prints the exact resume line with the batch folder), which confirms that one slab only. CGC 2023+ and OG slabs pass on their back QR. Note how often it stops.

Pass:

- 100/100 transfers with no unexplained stop;
- every PSA/CGC slab `ok` with the **right** cert (check `certs.txt` against your list). A pre-2016 PSA slab (its only code is the front ITF) is `ok` when OCR of its printed number agrees (note "… grader PSA from its 8-digit ITF"). When OCR reads nothing or other digits it is `needs_review` in the output stack: check its cert by hand;
- BGS/TAG lands in pocket 12 without rubbing on the rods: BGS labels carry no code, so as `no_read` or as an OCR-only `needs_review` (`other_grader` only if a BGS/TAG URL is read);
- `count check: every stack matches` in every summary;
- no visible marks.
- Note `mean cycle` from the summary: that's your real speed.

13.2 Cup marking test: 300 cycles + 24 h on one junk slab

(1.5 h of machine time + 24 h)

1. Photograph the junk slab's back under raking light (§1.10).
2. Put it in pocket 1, with pockets 0 and 12 empty.
3. `python -m tools.bench_tests cycles 300`. The slab ping-pongs between pockets 1 and 0, one full pick, read and place each time. The CSV also gives you 300 reads of one cert.
4. `python -m tools.bench_tests rest 24`. The cup sits sealed on the same slab for 24 h. The tool reads the line every 5 s and re-holds it for 2 s (pump on, valve open) whenever it rises above −40 kPa. At the end it prints the mean and minimum kPa, the % of time below −40 kPa, the re-holds per hour and the pump duty.
5. Inspect under the same raking light. Then wipe with a dry microfiber cloth and inspect again.

Pass:

- no ring, haze or residue after a dry wipe;
- 300/300 reads `ok` with the same cert;
- **zero "Probe Hard Limit" alarms in the 300 cycles.** Any alarm means the probe flag chatters at the slot edge during the slow lift: add hysteresis to the sensor output (WIRING.md §6), or set `probe_hard_limit: false` in `config.yaml`, and repeat;
- the rest test prints `vacuum: PASS` (**mean ≤ −40 kPa over the 24 h**); otherwise the clamp was not held and the marking result doesn't count: fix the leak (§8.3) and repeat;
- the cup lip is undamaged.

Fail: stop and ask. The fallback is the Schmalz HT1 cup (BOM conditional line), but it needs an M10×1-to-G1/8 adapter nobody has sourced yet; with one, re-run `calibrate z` and repeat.

13.3 Drop test from carry height

(30 min)

DESTRUCTIVE.

1. Pocket 1: two junk slabs. Pocket 2: a stack of about 10 junk slabs. Pocket 0: empty.
2. `python -m tools.bench_tests drop 2`. The tool says how far the slab will fall (about 170 mm here) and drops it only after you type `DROP`; anything else puts the slab back.
3. Put the dropped slab back in pocket 1. Then `python -m tools.bench_tests drop 0`, onto the bare pad: about 240 mm, the worst case. [Certain, DESIGN.md]

Inspect: corners, edges, both faces, and the top slab of the stack it hit.

Pass:

- no crack, chip or new scuff;
- the slab lands flat inside the rods, not hung on a rod top.

If a corner chips: you've found DESIGN.md's [Guessing] risk. Slow the X moves while loaded before going on: there is no config key for that yet, so it's a software change.

13.4 Power-cut test while holding a slab (valve fail-safe)

(30 min, **instructor present for the first E-stop**)

May drop a junk slab from carry height onto the stack in pocket 1.

1. Pocket 1: three junk slabs.
2. `python -m tools.bench_tests powercut 300`. The machine lifts the top slab to carry height.
3. **Press the E-stop.**

    - The tool logs the cup vacuum every second, using the Pi's own sensor (still powered).
    - The E-stop also cut the pump, and the closed vacuum valve cuts the reservoir off (§8.2): only the cup line holds the slab.
    - It stops logging when the vacuum rises above −3 kPa. The slab falls off at about −1.7 kPa. [Certain, arithmetic: 65 g over 78 % of a 25 mm cup]
    - Release the E-stop: the pump comes back and holds the slab; type `vent` at the prompt with a hand under it.
4. Repeat two more ways:
    - **pull the Pi's USB-C power** with the controller on: the Pi's GPIOs stop driving, so the modules' 10 kΩ pull-downs must switch the pump off and let both valves close;
    - **pull the machine's mains cord** with the Pi on its own supply.
    - Time these two with a stopwatch.

Pass:

- **the slab stays on the cup for at least 60 s in every case** [Guessing target: long enough to reach in and take it];
- nothing moves after the cut;
- after releasing the E-stop, `home` works.

Fail cases:

- the pump keeps running when the Pi is unplugged: the 10 kΩ SIG pull-downs are missing (WIRING.md §4);
- the slab drops in seconds: there's a leak in the cup line (§8.3);
- it drops in under 60 s with a tight line: see the cup-side reservoir option in §8.2.

13.5 Double-pick test

(20 min)

1. Make a clinging pair: two junk slabs with two small loops of painter's tape (sticky side out) between them, so they lift as one. The pair is about 13–14 mm thick.
2. Put the pair on top of a 5-slab junk stack in pocket 1.
3. `python -m slabscan run --name double`.
4. Then the realistic case, **untaped pairs, repeated:** wipe two slabs clean, press them together face to back, put the pair on top of pocket 1 and run again; do it 10 times. Count how many come up as a double. [Guessing target: 0; the slow peel is meant to stop air-film doubles]

Pass:

- The pair is carried over the rods without touching. Carry height 248 mm is set so even a double of two 11.5 mm thick holders clears the rods by 5 mm. [Certain, CAD]
- The pair is placed without a stop.
- The read slab's row shows `needs_review` with the note **`DOUBLE PICK: an unread slab is directly under this one in pocket 0`**, and the slab under it has its own `no_read` row whose note starts `UNREAD:` (same `position`, `out_index` one lower) (OPERATING.md §5.4).
- `count check` still matches.
- No untaped pair comes up as a double (step 4).

This is what the simulator does with `sim --clingy-rate 0.15`. [Certain]

13.6 Sensor-fault test: the machine must refuse to probe

(20 min)

A. Probe unplugged, machine idle. Unplug the optical sensor at the Rodent's Y-MAX connector, then try each command. [Certain for the messages, from `ops.py`, `runner.py` and FluidNC as documented in FIRMWARE.md]

Command	Expected
`python -m slabscan selftest`	`[FAIL] probe: reads TRIGGERED at rest …`
`python -m slabscan calibrate z`	refuses: "The probe reads TRIGGERED with the head at the top …"
`python -m slabscan jog`, then `probe`	"Refused: the probe already reads TRIGGERED …"
`python -m slabscan run`, with junk slabs loaded	It homes and goes to pocket 0 at the empty travel height, where the first probe window starts. The first `G38.3` is refused by FluidNC (`ALARM:4`) and the run stops with an `Unexpected error: MotionError … ALARM:4 …` message. **No probe move happens.** [Certain, `runner._find_top`]

B. Flag blocked from entering the slot (tape it back), probe plugged in:

- `run` → the survey of pocket 0 stops with **"Pocket 0: probe never touched anything (not even the pad)"**.
- Or, if the cup happens to seal on the pad: "the cup sealed on something but the contact sensor did not trip".
- The probe target is bounded, so the buffer is squeezed a few mm at most. [Certain, `runner.py`]

C. Cut the sensor's 5 V or cover its emitter: same as A (FIRMWARE.md step 5's fail-safe rows).

E. The carry-height sensor (junk slabs loaded) [Certain for the messages, `runner._carry_check`; tested in the simulator]:

1. Unplug its lead (or open core 8) and `run`. It must stop before the first sideways move: "Z stall suspected … the carry-height sensor (GPIO 6) reads CLEAR … Right after homing this is most likely the sensor itself …". Plug it back in.
2. Push a piece of card into its slot and `run`. It must stop at the first grip: the cup comes down onto the top slab of pocket 1 and stops there before it seals, with "Carry-height sensor fault …: it reads BLOCKED with the head at … mm, far below its trip height (246.5 mm) …". Take the card out.
3. `selftest --motion` → `carry sensor … PASS`.

Pass:

- no probe move in A or C: the cup never goes below the empty travel height;
- a clean stop in B;
- both stops in E, with nothing moved sideways and no slab gripped.

Plug back in and run `selftest -i`.

D. An unplanned error with a slab in the cup (junk slabs): start a run, and while a slab is held at carry height, pull the head camera's USB plug. Expect "Unexpected error: … The slab that was in the cup was put back on top of pocket N". [Certain, `runner._guarded`; tested in the simulator] If the controller is in ALARM instead, nothing moves: the pump keeps holding the slab and the prompt asks you to take it.

13.7 Stop, E-stop and resume

(30 min, extra but worth it)

1. 20 junk slabs across pockets 1–4. Start a run from the web page.
2. Press **Stop after this slab** mid-stack. Expect a park with nothing held, then **Resume**.
3. Press the **E-stop** while a slab is being carried.
   - The E-stop cut the pump too, so take the slab off **at once**, then put it face-down on top of the pocket the message names.
   - Release the E-stop and resume. The machine reconnects and homes first.

4. Press **Ctrl-C twice** (CLI run) while a slab is carried: the motors halt, nothing moves, the prompt asks you to hold the slab and type `vent`. Put it back as told and resume.
5. During a stop, take one slab off a stack and resume. It must refuse with "The stacks changed since the run stopped …". Put it back and resume.

Pass:

- every physical slab has **exactly one row** in `certs.csv`, and no cert repeats. (A double pick's `UNREAD` row shares the `position` of the slab above it: that is the one allowed repeat.)
- the count check matches.

13.8 Face-down check

(30 min; part B once the learned check is active)

A. Every build. [Certain for the expected messages, `runner._check_face_down`; tested in the simulator]

1. Junk slabs in pockets 1–4. Put one junk PSA slab **face-up** on top of pocket 2 (label end toward you), a face-down pre-2016 PSA label (no back-label code) on top of pocket 3, and, if you have one, a CGC 2023+ slab **face-up** on top of pocket 4.
2. `python -m slabscan run --name facedown`. Watch the start line: `face-down check: …`.
3. The run must stop at pocket 2 **before the cup touches the slab**: "Pocket 2: the top slab could not be confirmed FACE-DOWN before touching it …" (a PSA front's barcode is also on the back, so it proves nothing; if the learned check is active, its card window must not match). Turn the slab face-down and `resume batches/<id>` (no flag needed if it has a back QR).
4. It stops again at pocket 3. `resume batches/<id> --face-down-ok`. The slab is picked and its `face_down` column reads `operator`.
5. At pocket 4 it must stop with "Pocket 4: the top slab is FACE-UP (a CGC front-label barcode … is visible from above, and that code is printed on the FRONT label) …". Turn it face-down and `resume batches/<id>`: it is picked on its back QR.
6. Press **Stop after this slab** (web page) or Ctrl-C once, then try `resume batches/<id> --face-down-ok`. It must refuse: "--face-down-ok refused: the batch did not stop at the face-down check last time …". Resume without it.

B. Once the learned check is active (a Phase 0 GO with `--save`, 5 references, and the window clear on 5 proven slabs: the start line reads `face-down check: learned (threshold …)`):

1. If the start line still says "not active", it says why. If it says a part of the head is in the window, or the run printed a `CAMERA AIM:` line, compare `face_down_window.jpg` with `face_down_empty.jpg` (§12.3) and re-aim the camera (§7.7).
2. Put 3–5 junk slabs **face-up** on top of different stacks, including a plain card, a full-art card and a blue Water-type card. **Each must stop untouched.**
3. Put a pre-2016 face-down slab with an English Pokémon back on top of another stack. It must be picked without stopping, with `face_down = learned(0.xx)` in `certs.csv`, shown in bold in `report.html`. Its back photo must show the card back. (A match needs two references from two different slabs of that back design.)
4. A Japanese-back old label should stop until two back-code slabs with that back, or three of your confirmations, teach it.

Pass:

- A: every face-up slab stopped untouched (pocket 4 with the FACE-UP message); `--face-down-ok` worked after the face-down stop and was refused after the other stop.
- B: no face-up slab was touched, and every `learned(...)` row's back photo shows a card back.
- **Any face-up slab picked in B: set `face_down_check = "stop"` and don't use the learned check.**

13.9 Summary

Test	Pass criterion	Result
13.1 Dry run	100/100, all certs right, count check OK	
13.2 Marking	no mark after dry wipe; 300/300 reads; 0 probe hard-limit alarms; rest mean ≤ −40 kPa	
13.3 Drop (destructive)	no damage, lands in the pocket	
13.4 Power cut	holds ≥ 60 s in all three cases	
13.5 Double pick	flagged with an UNREAD row, carried clear, no crash; no untaped doubles	
13.6 Sensor fault	refuses to probe; clean stop; a slab is put back after an unplanned error; both carry-sensor faults stop before anything moves sideways or is gripped	
13.7 Resume	every slab exactly one row	
13.8 Face-down	face-up slabs stop untouched (a CGC front with the FACE-UP message); `--face-down-ok` refused after a non-face-down stop; with the learned check active, no face-up slab picked and every learned row shows a card back	
7.8 Z lift	bare pad within 0.1 mm after 100 lifts	

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14. First real batch, supervised

1. **Start small:** 30 low-value slabs in one pocket.
   - Before loading, record their certs another way: a phone app, a hand scanner, or by typing.
   - Then you can check the machine against something independent.
   - Slabs without a back-label code (PSA 2016 and older, CGC Authentic, BGS) stop the run at the face-down check until the learned check is active: check each one and `resume batches/<id> --face-down-ok`. Use `--all-face-down` only once you trust your own loading.
   - A pre-2016 PSA slab is `ok` only when OCR of its printed number agrees with its front ITF. When OCR reads nothing or other digits it is `needs_review` and stays in the output stack: type its cert in yourself after checking the label. A BGS or SGC slab that comes out `ok PSA` means the ITF rule is wrong for them (§15).
   - If the learned check is active (the start line says `face-down check: learned`), open the back photo of **every** `learned(...)` row in `report.html` (bold in the Face-down column) in the first batches, and the `face_down_learned` lines in `journal.jsonl`. A wrong one: `python -m slabscan refs` lists the references and `refs --drop CERT` deletes one (OPERATING.md §8.4).
2. **Watch the whole run.**
   - Hand near the E-stop for the first ten cycles.
   - Listen for scraping (a rod touched) and watch each set-down.
3. **Check every row:**
   - `certs.txt` against your list;
   - `report.html` photos against the slabs;
   - the slabs themselves under raking light for any mark.
   - **Any wrong cert at all → stop and investigate before the next batch.**
4. **Then 100 slabs** across 4 pockets, supervised.
5. **Then a full load**, staying in the room for the first hour.
6. **Don't leave it unattended** until you have three full runs with no unexplained stop and no wrong cert. [Guessing threshold]
   - Even then, keep it where nobody can reach into the bed.
   - Use the web page only on your own network: it has no login. [Certain, software README]

---

15. Troubleshooting

Symptom	Likely cause	Fix
`selftest`: probe reads TRIGGERED at rest	Cut wire, sensor unplugged, flag sitting in the slot, or a board with an LED circuit that holds the input	Check the flag is clear at rest; FIRMWARE.md step 5; WIRING.md §6 fallback (bare TCST2103)
Probe never triggers when the cup is pushed	Flag bent or loose under the elbow, the board slid in its slots, or the sensor mounted the wrong way round	§7.6 steps 5–10; re-set the trip
The flag rubs or sticks in the guide	Bend not square (more than 0.5° off), a twisted blade, or the printed slot undersize	§7.6 steps 3 and 5; clear the slot with a 1.5 mm needle file
"Probe Hard Limit" alarm during a run	The flag chatters at the slot edge during the slow lift, or something touched the cup during a lift or traverse	§13.2; hysteresis on the sensor output (WIRING.md §6); look for what touched the cup
`calibrate z`: no contact down to Z −7.0	Head sits too high: carriage-to-cup > 297 mm, or the Z switch is low	Re-set 295 ± 2 mm (§7.5), re-check the Z switch position (§7.4)
`calibrate z`: spread > 0.3 mm	Loose cup or buffer nuts; pad not flat; flag loose	Tighten the M10 nuts; re-glue pad 0; tighten the flag
`calibrate x`: error > 1.5 mm	Belt slipping, pulley grub screw loose, or the holes weren't drilled with `mark`	Grub screw on the flat with threadlocker; re-tension (§6.6)
Positions drift during a run	Lost steps: belt too loose, X current too low, or carriage binding	Tension; roll the carriage by hand power-off; check `run_amps`
Stop: "cup will not seal"	Dirty cup lip, slab face-up, warped holder, sleeve left on	Wipe the cup with IPA, remove sleeves, check orientation, then resume
Seal is slow or fails on every slab	Leak: tube not seated on the barb, the elbow's thread not sealed under the flag tab, vent valve seat	Leak test §8.3, pinch method; §7.6 step 6
Stop: "Vacuum lost in transit"	Leak, pump weak, slab knocked a rod	Leak test; watch the next carry; check rods near that pocket
Stop: "taller than expected" on a pocket you didn't touch	A slab sitting crooked on a rod; double-placed slab	Seat the slab flat; resume
Stop: "contact … above the stack - something is in the way"	Slab crooked in the output pocket, or a rod leaning in	Fix the slab; slab drop test on that pocket (§5.8)
Head camera vanishes or errors mid-run	USB extension through the drag chain (about 4 m)	Re-seat; check bend radius ≥ 40 mm; drag-chain-rated USB cable (BOM note). The run puts a held slab back ("Unexpected error: …").
Every slab goes to the station camera ("confirmed by OCR", slow runs)	The head crop cuts the label off: `cam_head.hfov_deg` or `row_axis` is wrong	§12.3 crop check
The run stops at every older PSA label (or CGC Authentic, BGS)	The face-down check: they have no back-label code, and the learned check is not active yet (the start line says why)	Expected until Phase 0 GO and 5 references. `resume batches/<id> --face-down-ok` per slab, or `run --all-face-down` for stacks you checked. OPERATING.md §8.4.
`CAMERA AIM: …` in the survey log and the summary, or "learned check not active: over an empty pocket the head camera sees edges where only the pad should show …"	A head part is in the card window over the empty pocket: the camera is tilted toward the cup, or lint, dust or a tag lies on that pocket's pad	Open `images/face_down_empty.jpg`. Clean the pad; otherwise shim the board (§7.7) and run `focus head` (§12.3). The learned check accepts nothing until a run's view is clear again.
Stop: "the top slab is FACE-UP"	A CGC front-label code (the Authentic QR or the front barcode) is visible from above	Turn the slab face-down, label end toward you, and resume.
Stop: "Z stall suspected …"	The carry-height sensor reads clear at travel height: Z lost steps on a lift, or its lead is cut (right after homing, the lead is likelier)	Power off and move the head up and down by hand: feel for a bind. Check the Z current (§7.8), `python -m tools.bench_tests lifts 20`, `selftest --motion`. Resume homes first. OPERATING.md §8.1 for the slab.

Symptom	Likely cause	Fix
Stop: "Carry-height sensor fault …"	It reads blocked far below its trip: dirt or a card in the slot, a dead emitter, a bent flag	Clean the slot, check the flag (§7.4a), `selftest --motion`, resume
`selftest --motion`: carry sensor trips outside 246.5 ± 1	The mount slid, or it was never set	§7.4a step 6
"--face-down-ok refused"	The batch's last stop was not a face-down stop	Resume without it. If a slab the check can't confirm comes next, the run stops there and you confirm that one.
"Another slabscan process is using the machine"	The boot web service, or a command in another window	`sudo systemctl stop slabrail-web`, or close the other window
Station camera not found	Ribbon backwards or not latched; wrong port	Power off, re-seat per the Raspberry Pi camera guide, CAM/DISP 0
Labels read on the bench but fail on the machine	Focus moved; ring not on; glare from room lights	`focus head` / `focus station --save`; fit the light roof
Many pre-2016 PSA labels `needs_review` "one printed barcode only (no QR) - confirm" or "barcode N but OCR read M"	Their front ITF is their only code, and it is `ok` only when OCR of the printed number agrees. OCR read nothing, or other digits	Check each one by hand. Then the `tesseract` line of `selftest`, `focus station --save`, the station ring. If you set `custom_regex` or `psa_cert_digits` (without 7), see §11.5. OPERATING.md §8.3.
A BGS, SGC or other non-PSA slab comes out `ok PSA`	The 8-digit ITF rule assumes only PSA prints an 8-digit ITF. BGS and SGC are reported to print some barcode or QR of unknown type [Likely]; none in the real set did [Certain, for that set]	Put your non-PSA slabs through Phase 0 (or `python -m tools.real_eval --phase0`) before trusting `ok PSA` on slabs with no QR. An `ok PSA` on a BGS or SGC slab means the rule is wrong for them: tell the software owner. OPERATING.md §5.2.
Overfill beam reads blocked with nothing there	Misaligned laser, receiver trimmer, a receiver wire cut (reads blocked by design)	§12.5; WIRING.md §8
Overfill beam reads clear when blocked	Room light swamps the receiver, or its GND wire is cut	Trimmer under room light; check the GND wire; light roof
Controller lost (E-stop) and resume says "still not answering"	The CH340 is slow to come back	Wait 10 s and press resume again; check the USB cable and the 24 V supply
Motors hot (> 50 °C)	`r_sense_ohms` doesn't match the board, or the current is above 70 % of the motor's rating	Power off. FIRMWARE.md step 0.
Z position drifts after many lifts	Z stalls: motor too weak at 0.8 A / 6000 mm/min, or a high-impedance motor	§7.8 lift test; lower Z speed to 3000
Z drops when the E-stop is pressed	Detent torque too low with the head's weight	Brake or counterweight before unattended use (§7.8)
Scraping as the head crosses the row	Spine bottom below 275 mm, an uncut lead screw, a tall rod, a loose cable loop	§7.1 height; §7.3 screw length; trim that rod; re-tie the Z loop (WIRING.md §11)
`run` says calibration missing	`calibration.toml` lacks `pad_trigger_c` / `offset_x`	§12.1–12.2

Sources

- **Repository files (primary for every [Certain] tag above):**
  - `cad/layout.py`, `cad/model.py`, `cad/out/cut_list.csv`, `cad/out/checks.txt` (177/177 checks passed, re-run on 23 Oct 2026), `cad/out/printed/print_list.json`;
  - `software/slabscan/*.py`, `software/tools/phase0.py`, `software/tools/real_eval.py`, `software/config/machine.toml`;
  - CONTRACT.md, WIRING.md, FIRMWARE.md, BOM.md, DESIGN.md.

- **Homing switch body, holes, lever and travel (SS-5GL: OP 8.8 ± 0.8 mm, FP max 13.6 mm, OT min 1.2 mm, holes 2.35 mm at 9.5 ± 0.1 mm):** Omron SS subminiature basic switch datasheet (`en-ss.pdf`)
- **Prusa PETG settings and the smooth-PEI warning:** help.prusa3d.com/article/petg_2059
- **TCST2103 dimensions (slot 3.1 × 7.7 mm, axis 2.6 mm, flange holes 19 mm):** Vishay doc 81147, drawing 6.550-5040.01-4; Farnell listing
- **SMC buffer strokes and dimensions (Table 2; ø20/25 at 20 mm stroke: E 44.5, F 108.5, G 51):** SMC ZP catalog, RS copy, pages 13-11-9 and 13-11-12
- **SMC buffer letters (J rotating, K non-rotating; B5 = M5 female, A10 = M10×1):** Proax ZPT25BNJ10-B5-A10, Proax ZPT25BSK20-04-A10
- **SMC M-5ALU-4 barb elbow:** automationdistribution
- **M3 screw torque into aluminium (0.98 N·m):** AL Motion installation & lubrication PDF
- **GT2 tension, third-party range only:** everycalculators GT2 calculator, printpal belt tension wiki
- **M5 heat-set insert hole (generic 6.4 mm):** heat-set insert spec sheet
- **Titebond II times:** woodworkingadvisor glue chart, Dictum Titebond II Extend
- **Raspberry Pi OS Trixie default, Bookworm as Legacy:** raspberrytips: Trixie released, raspberrytips: which version
- **Clippard EV-2 (10-32 in-line, 28 inHg vacuum in the 2005 catalogue):** EV-2-24 datasheet; I could not open it from here.

Wiring

Every wire in SlabRail, plus how power is laid out and how to check it before power-up. The block diagram is img/wiring.svg. Firmware set-up is in FIRMWARE.md, and pin numbers come from CONTRACT.md.

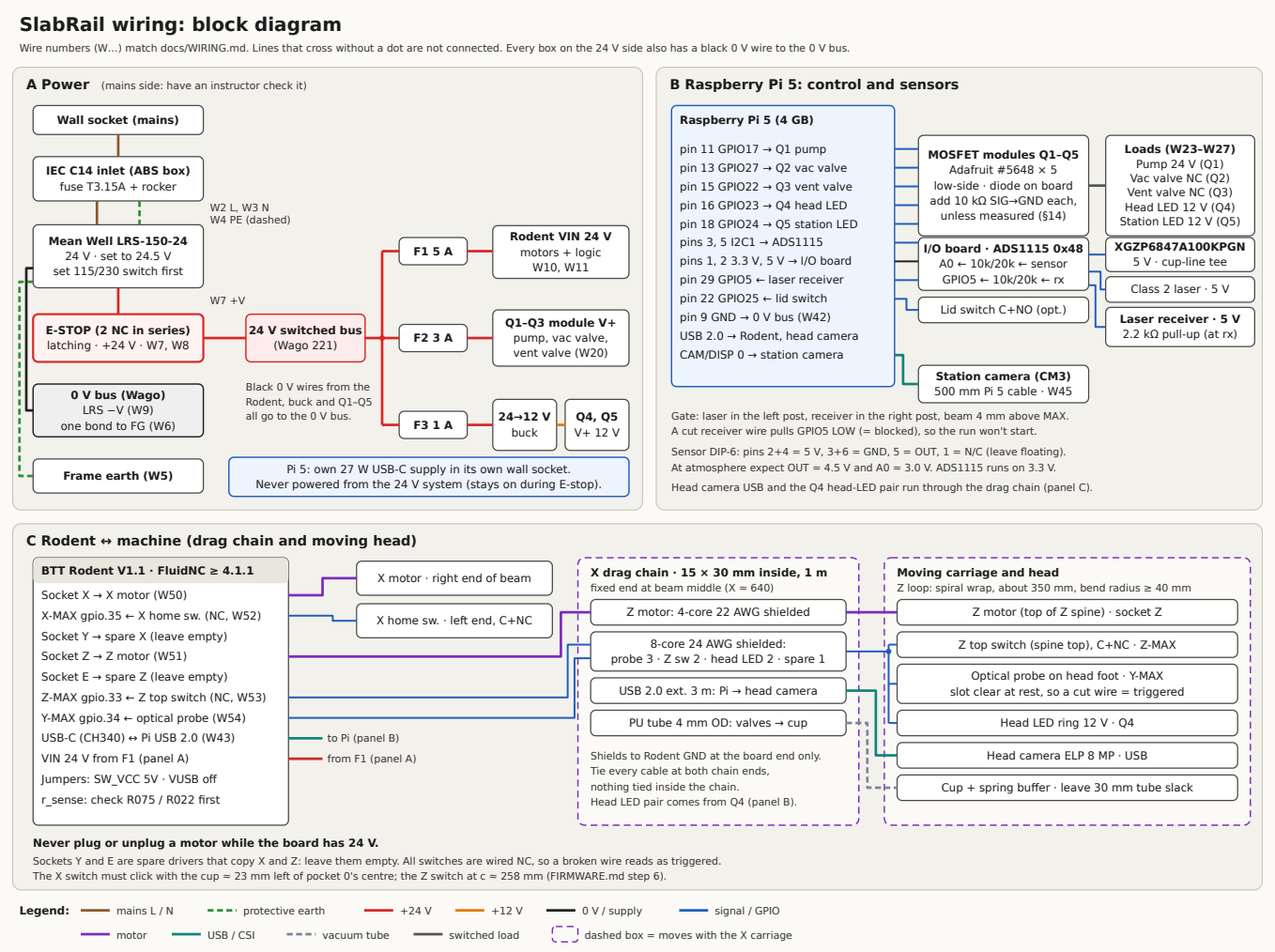


Figure: docs/img/wiring.svg

Confidence tags: **[Certain]** = primary source (datasheet, schematic, FluidNC source), **[Likely]** = strong inference or several secondary sources, **[Guessing]** = a gap I filled. Vendor product pages could not be opened from this session. The module facts come from search-engine extracts of the listings, and every "measure" step below exists because of that.

Safety rules.

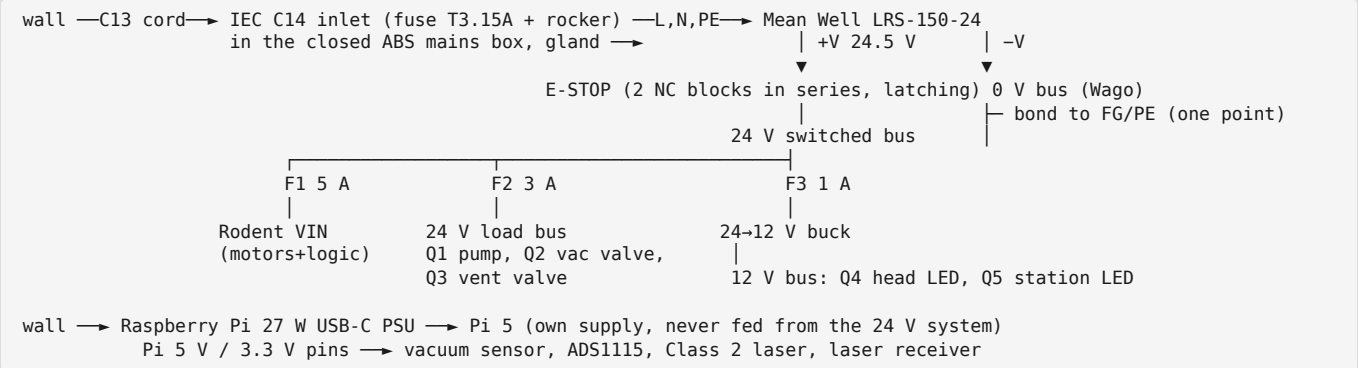
- Have a mechatronics instructor check the mains side (§1.1) before the first power-up.
- Never plug or unplug a stepper motor while the Rodent has 24 V.
- Everything after the 24 V supply is SELV (safe extra-low voltage). Most of the work is there.

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1. Power architecture



1.1 Mains side (instructor check)

- **Inlet: an IEC C14 inlet with a built-in fuse holder and rocker switch** (BOM "Mains inlet"). One part gives you the inlet, fuse and power switch.
- **It lives in the mains box, not on the plywood panel** (BOM "Mains box": a small ABS box, a cable gland and fully insulated 6.3 mm receptacles or tab boots). [Likely: snap-in IEC modules are made for 1–3 mm panel walls, so on 6 mm plywood they end up glued or hanging, with live tabs open.]
    - Cut the inlet's rectangular hole in the box wall and snap the inlet in.
    - Put fully insulated receptacles (or boots) on every inlet tab, including the fuse-holder side.
    - W2–W4 leave through the cable gland, tightened on the cable jacket, to the LRS terminal strip.
    - Close the lid with its screws. **With the cord plugged in, no metal at mains voltage may be exposed anywhere** (inlet back, fuse holder, LRS terminals). The instructor checks this (§14).
    - An ABS box needs no earth bond. A metal box (Hammond 1590 type) must be bonded to PE.
    - Fuse **T3.15A** slow-blow (5 × 20 mm), as in the BOM. [Likely: suits a 156 W supply on 115 V. Check the LRS-150 datasheet's input current and inrush before final choice]
- **Mean Well LRS-150-24: 24 V, 6.5 A, 156 W, output adjustable 21.6–28.8 V.** [Certain from reseller copies of the datasheet]
    - **It has a 115/230 VAC selector switch.** [Certain, reseller listings] Set it to your mains (115 in the US) **before** the first power-up. On 115 V mains, the 230 position won't run properly. On 230 V mains, the 115 position destroys the supply.

- Wire L, N and PE with 18 AWG (0.75 mm²) mains-rated wire, using the colours in the wire table.
  - **Colour mapping** (the BOM line is the EU set; US 18 AWG MTW is easier to buy, and the instructor decides which): line = brown (EU) or black (US); neutral = blue (EU) or white (US); earth = green/yellow (EU) or green (US). W2–W6 below accept either set. [Certain, IEC 60445 and US NEC conventions]
  - Keep the mains wire inside the mains box and the LRS terminals. On the low-voltage side black means 0 V and white +5 V (the colour table in §3), so never reuse leftover mains wire there.
  - The terminal labels are printed on the supply's terminal strip. Follow them.
  - Fit the supply's terminal cover.
- The Pi's 27 W USB-C supply plugs into its own wall socket, or into the same power strip as the machine cord. **It is never fed from the 24 V system.** That is what keeps the Pi alive when the machine power is cut.

1.2 Why one 24 V supply, and why this one

- **Every motor-side load is a 24 V part.** The Rodent's driver input, the BOM's pump and both Clippard EV-2-24 valves all run on 24 V. Only the LED rings need 12 V, from one small buck.
- **The Rodent's input range starts at 24 V.** The manual says "DC24V to DC56V". [Certain] So 24 V is the bottom of its range. **Set the LRS to 24.5 V** with its V.ADJ trimmer (no load). That leaves margin for wire drop while staying inside the pump and valve ratings (24 V ± 10 % is normal for solenoids [Likely]).
- **The load budget is under 4 A peak against a 6.5 A rating.** [Guessing, before parts are measured]

Load	Typical	Peak	Note
X motor (1.0 A phase, through the TMC2160)	0.3 A	1.0 A	supply current is lower than phase current
Z motor (0.8 A phase)	0.2 A	0.8 A	
Rodent logic	0.1 A	0.1 A	
Vacuum pump (24 V diaphragm)	0.4 A	1.2 A at start	measure yours
2 × valve	0.05 A	0.1 A	about 1 W coils [Guessing]
12 V buck for 2 LED rings	0.1 A	0.2 A	at 24 V
Total	**≈1.2 A**	**≈3.4 A**	LRS-150-24 = 6.5 A

The smaller LRS-100-24 (4.5 A) would also carry it. The 150 W model costs a few dollars more and copes better with the capacitor inrush every time the E-stop is released. [Likely]

1.3 E-stop

What it is: a 22 mm latching mushroom E-stop with NC contacts, in its own box (BOM item). It sits **in the +24 V line straight after the supply**. Pressing it cuts:

- the Rodent (both motors and the controller's logic);
- the pump and both valves;
- the LED buck.

What it doesn't cut: the Pi, which stays alive on its own supply, and the sensors it powers.

What happens on E-stop

- **Motors go limp.** DESIGN.md estimates the Z head holds on detent torque. The test is in FIRMWARE.md's open issues. [Likely]
- **Both valves are normally closed, so both close.** The cup line stays sealed, and a held slab stays on the cup for a while. Acceptance test 13.4 in the build guide checks this.
- **The Rodent switches off**, and its USB-serial chip with it. The Pi sees `/dev/ttyUSB0` vanish.
  - The software must stop and tell you "controller lost power".
  - After you twist the E-stop to release it, run `home` before anything else.

Contact rating and the two contact blocks (required)

- **Use two NC contact blocks wired in series** (BOM: a box with two NC blocks, or add a clip-on 22 mm NC block, about $3). Why: the contact switches 24 V DC straight into the Rodent's and the buck's input capacitors. Every release charges them through nothing but wire and contact resistance, and that inrush can weld a small contact closed. [Likely] A welded NC E-stop is invisible: the button is down and the machine is still live. With two blocks in series, one welded block still leaves the other to open.
- The target is blocks rated for **at least 5 A at 24 V DC**. The BOM's ECX1030 blocks are rated **DC-13 1.5 A at 24 V** [Certain, AutomationDirect], below that target and below the bus's ≈3.4 A peak (§1.2) [Guessing on the peak]; no 22 mm block rated ≥ 5 A at 24 V DC was found. DC-13 is the inductive-load category, so the resistive rating is probably higher [Guessing], but it is not listed. Two blocks in series and the monthly E-stop check are the mitigation for now. **Open issue for review:** the mains-side E-stop below, or a force-guided safety relay, removes it.
- **Do not "fix" this with an ordinary 24 V relay** switched by the E-stop: its contacts see the same inrush and can weld the same way. Only a force-guided safety relay would help.
- The alternative is to put the E-stop in the supply's mains feed (AC-rated contacts, and the LRS's own soft start removes the DC inrush). That means mains wiring in the E-stop box: instructor only.
- **Check it monthly** (OPERATING.md §7): with the machine idle, press the E-stop. The software must report the controller lost, and the 24 V switched bus must read 0 V on a meter.
- Only if F3 actually blows when you release the E-stop, change it to 2 A (the ring wiring takes it).

Mounting: at the front of the machine, where you can hit it without reaching over the bed.

1.4 Branch fuses (ATC blade, inline holders)

Fuse	Feeds	Rating	Why
F1	Rodent VIN	5 A	Above 3.4 A peak, below the 18 AWG wire limit
F2	Pump + both valves (24 V load bus)	3 A	Pump start current. Protects the 22 AWG load wiring.
F3	24→12 V buck (LED rings)	1 A	Protects the 22/24 AWG ring wiring

The BOM's "Branch fuses" line has three inline holders with 1 A, 3 A and 5 A fuses. [Certain, BOM.csv]

1.5 12 V buck for the LED rings

- Use the BOM's 24→12 V 3 A buck module. The Pololu D36V28F12 (12 V, 2.4 A, 12.9–50 V input) is a documented alternative. [Certain, Pololu specs]
- Set or check 12.0 V before connecting the rings.
- Its ground goes to the 0 V bus.

2. Rodent jumpers

Jumper	Setting	Why
SW_VCC (limit-input voltage)	**5V**	The optical probe is powered from V-Lim, and its output stage only works if V-Lim matches its 5 V supply (§6).
V-Probe	**5V**	The Probe connector is unused. 5 V is the safe choice.
NPN EN	**off**	Not needed
VUSB	**off**	BTT: "Do not insert the jumper when the device is in use." [Certain]
DIAG X/Y/Z/E	**none**	They connect driver stall outputs to the limit pins.

3. Wire tables

Wire gauge

- 18 AWG for mains and the 24 V trunk.
- 22 AWG for motors and 24 V/12 V loads.
- 24–26 AWG for signals.

Colours

Colour	Use
red	+24 V
orange or yellow	+12 V
white	+5 V (Pi side)
black	0 V
blue	switched minus (load return through a MOSFET)
green/yellow	protective earth only
other	signals

Labelling: write the W-number on both ends of every cable.

Connector types on the Rodent are 2.54 mm pitch 3-pin (limits, probe) and 4-pin (motors) headers. They look like JST-XH in BTT's renders. [Likely] **Check against your board before you crimp.** Power is on screw terminals. [Certain]

3.1 Mains and 24 V / 12 V power

W	From	To	Wire	Ends	Notes
W1	Wall socket	IEC C14 inlet	C13 cord	moulded	BOM power cord
W2	Inlet L (after fuse and switch)	LRS L	18 AWG brown/black, 300 V	fork crimp or ferrule	**mains**
W3	Inlet N	LRS N	18 AWG blue/white, 300 V	fork or ferrule	**mains**
W4	Inlet PE	LRS FG (⏚)	18 AWG green/yellow (US: green)	fork or ferrule	earth
W5	LRS FG	Frame bond point on the base 2040	18 AWG green/yellow (US: green)	M5 ring + serrated washer into a T-nut	§12
W6	LRS −V	LRS FG	18 AWG green/yellow (US: green), short	ferrules	the **only** 0 V-to-earth bond
W7	LRS +V	E-stop box contact IN	18 AWG red (one core of a 2-core cable, about 1.5 m)	ferrule / contact screw	
W8	E-stop contact OUT	24 V switched bus (Wago 221-415)	18 AWG red (other core of W7's cable)	ferrule / lever	
W9	LRS −V	0 V bus (Wago 221-415, two joined if needed)	18 AWG black	ferrule / lever	
W10	24 V switched bus	F1 5 A holder → Rodent VIN "VCC"	18 AWG red	lever / ferrule into the screw terminal	
W11	0 V bus	Rodent VIN "GND"	18 AWG black	ferrule	
W12	24 V switched bus	F2 3 A holder → 24 V load bus (Wago)	20–22 AWG red	lever	
W13	24 V switched bus	F3 1 A holder → buck VIN+	22 AWG red	lever / solder or screw	
W14	0 V bus	buck GND (input side)	22 AWG black		
W15	Buck VOUT 12 V	12 V bus (Wago 221-413)	22 AWG orange		

3.2 MOSFET modules and loads (Q1–Q5, Adafruit #5648, see §4)

Each module has an input header (load supply **V+**, **GND**, **SIG**) and two output terminals (**+**, which is tied to V+, and **−**, which the MOSFET switches to GND). Use the silkscreen names on your board.

W	From	To	Wire	Ends	Notes
W20	24 V load bus	Q1, Q2, Q3 V+	22 AWG red	JST PH or 0.1" pad	The module V+ **must** be the load's own supply, or the flyback diode can't clamp.
W21	12 V bus	Q4, Q5 V+	22 AWG orange	JST PH or pad	
W22	0 V bus	Q1–Q5 GND	22 AWG black	JST PH or pad	
W23	Q1 out + / −	Pump + / −	22 AWG red / blue, ≤ 0.5 m	screw terminal, ferrules	Check pump polarity: DC diaphragm pumps are polarised.
W24	Q2 out + / −	Vacuum valve coil (EV-2-24 leads)	22–24 AWG	screw terminal; solder + heat-shrink at the valve	Coil polarity doesn't matter for an EV-2 [Likely]
W25	Q3 out + / −	Vent valve coil	22–24 AWG	as W24	
W26	Q4 out + / −	Head LED ring (through the 8-core, cores 6 and 7, §11)	24 AWG	screw terminal / solder at the ring	
W27	Q5 out + / −	Station LED ring	22 AWG 2-core, about 0.8 m	screw terminal / solder	Runs beside the CSI ribbon through the bed slot

3.3 Raspberry Pi 5 header and the I/O board

What the I/O board is: a small perfboard, about 70 × 50 mm, next to the Pi. It carries the ADS1115, the XGZP6847A, the three dividers (vacuum sensor, laser gate, carry-height sensor) and screw terminals. It connects to the Pi header with female Dupont leads.

Physical pin numbers are the standard 40-pin layout. [Certain]

W	Pi pin	Signal	To	Wire	Notes
W30	1	3.3 V	ADS1115 VDD	26 AWG Dupont	Gives the ADS 3.3 V I2C pull-ups
W31	2	5 V	I/O board 5 V rail: sensor VDD, laser, receiver	24 AWG	about 50 mA total
W32	6	GND	I/O board GND rail	24 AWG	
W33	3	GPIO2 SDA	ADS1115 SDA	26 AWG, < 20 cm	
W34	5	GPIO3 SCL	ADS1115 SCL	26 AWG, < 20 cm	ADS ADDR → GND gives 0x48 [Certain, TI datasheet]
W35	11	GPIO17	Q1 SIG (pump)	24 AWG	
W36	13	GPIO27	Q2 SIG (vacuum valve)	24 AWG	
W37	15	GPIO22	Q3 SIG (vent valve)	24 AWG	
W38	16	GPIO23	Q4 SIG (head LED)	24 AWG	
W39	18	GPIO24	Q5 SIG (station LED)	24 AWG	
W40	22	GPIO25	Lid switch C (switch NO → GND, pin 20)	24 AWG 2-core	Optional. Closed = lid shut.
W41	29	GPIO5	Gate divider midpoint (§8)	24 AWG	
W48	31	GPIO6	Carry-height sensor divider midpoint (§6a)	24 AWG	GPIO 6 is used by nothing else [Certain, docs, software and firmware searched]
W42	9	GND	**0 V bus** (one wire)	22 AWG black	Common reference for the MOSFET inputs. Keep it even though USB also joins the grounds.
W43	USB 2.0 (black) port		Rodent USB-C	USB-A → USB-C **data** cable, 1 m	Serial to FluidNC
W44	USB 2.0 (black) port		Head-camera extension (§10)		

W	Pi pin	Signal	To	Wire	Notes
W45	CAM/DISP 0		Station camera	500 mm Pi 5 camera cable	§10

3.4 Rodent inputs and motors

W	From	To (Rodent)	Wire	Ends	Notes
W50	X motor (right end of beam)	Socket **X**	22 AWG 4-core, about 1.3 m, static	4-pin 2.54 mm housing	Along the beam's back slot to the middle, then down with the bundle. The BOM's extension is needed if the motor lead is 1 m.
W51	Z motor (top of Z spine)	Socket **Z**	4-core 22 AWG shielded high-flex, about 2.5 m, through the chain	4-pin housing; shield → Rodent GND at board end only	
W52	X home switch (left end of beam): **C** and **NC** tabs	**X-MAX**: signal and GND	24 AWG 2-core, about 1.3 m, static	3-pin housing (V-Lim pin empty)	§7. Not "X-MIN": the board's names don't matter, the config does.
W53	Z top switch on the spine: **C** and **NC**	**Z-MAX**: signal and GND	8-core cores 4 and 5, through the chain	3-pin housing	§7
W54	Optical probe board: VCC, GND, OUT	**Y-MAX**: V-Lim, GND, signal	8-core cores 1, 2 and 3, through the chain	3-pin housing	§6. Yes, Y-MAX: there is no Y axis.
W55	Carry-height sensor board (Z spine): VCC, GND, OUT	VCC and GND: spliced into cores 1 and 2 at the spine top (the probe's 5 V and GND). OUT: core 8 → **I/O board divider** → **Pi GPIO 6** (not the Rodent)	the board's 3-pin cable, about 0.2 m, then the 8-core through the chain	solder + heat-shrink at the splice; screw terminal at the I/O board	§6a
—	nothing	Sockets **Y** and **E**	—	—	Spare drivers (FIRMWARE.md). Leave them empty.
—	nothing	Probe, E0-MAX, E1-MAX, V-MOS, RS485, spindle	—	—	unused

3.5 Overfill gate (Pi side)

W	From	To	Wire	Ends	Notes
W46	I/O board 5 V and GND	Class 2 (≤ 1 mW) laser module + and − (left gate post)	24 AWG 2-core, about 1.2 m	Dupont / solder + heat-shrink	§8. Along the bed's front edge in clips.
W47	I/O board 5 V, GND, divider input	Receiver VCC, GND, OUT (right gate post)	24 AWG 3-core, about 1.2 m	Dupont / solder	2.2 kΩ OUT-to-VCC pull-up soldered **at the receiver** (§8)

4. MOSFET modules

Pick: Adafruit MOSFET Driver, product #5648, ×5. It switches the pump, the vacuum valve, the vent valve and both LED rings, all low-side. [Certain for the specs below, from Adafruit's listing as quoted by several resellers]

Why this module

- **It is specified for 3 V logic.** The listing says "signal (3V to 20V logic level)". It uses an AO3406 N-MOSFET (30 V, 70 mΩ). A 3.3 V Pi pin turns it fully on, with no guessing about gate threshold.
- **The flyback diode is on the board** (1N4007). The pump and valve coils need one, and an on-board diode can't be soldered in backwards or forgotten.
- **Its current rating covers every load here:** 1.5 A continuous, 3 A peak. The JST PH input is rated 2 A.
- **The 30 V rating is enough on a 24.5 V bus**, because the diode clamps coil spikes to about 25.5 V. [Likely]

If you substitute "D4184" (AOD4184) modules (the BOM keeps 1N5819 diodes as a qty-0 conditional line for this case): those modules switch more current, but they have no flyback diode. Their makers also give no 3.3 V drive figure, so whether they switch fully at 3.3 V is unconfirmed. [Likely] If you use them:

- solder a diode across each pump and valve coil, cathode (stripe) to +;
- with the load on, check that the voltage across the module's output is under 0.2 V.

Don't use IRF520 modules: they aren't logic-level.

Rules for all five modules

- **Module V+ must be the load's supply:** 24 V for Q1–Q3, 12 V for Q4–Q5. The diode returns the coil current to V+.
- **Add a 10 kΩ resistor from SIG to GND on each module** unless you measured one already there. Power off and measure SIG to GND **before the first power-up** (§14 step 13). [Guessing whether #5648 has one; Adafruit's listing does not mention one] The Pi's GPIOs float while it boots or is off, and the pull-down keeps the pump and valves off. Without it, a floating vent gate can open the vent and drop a held slab.
- **If your pump draws more than 1.2 A running or 3 A at start, use a bigger module for Q1 only.** Measure it on the bench with a meter in series.

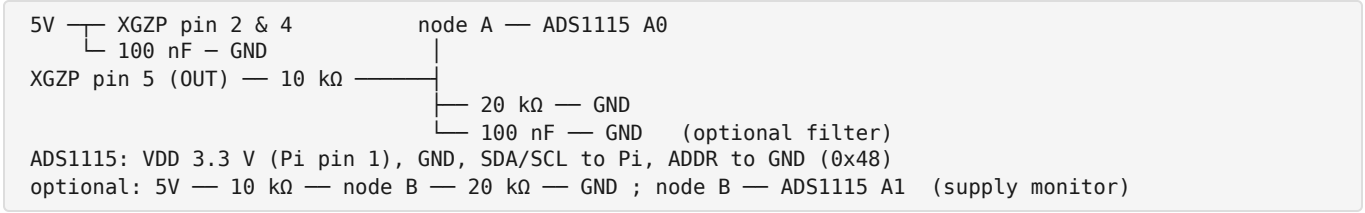
5. Vacuum sensor (XGZP6847A + ADS1115)

Part: CFSensor XGZP6847A100KPGN. Analog, 5 V supply, −100…0 kPa ("GN" = negative gauge), 0.5–4.5 V out. [Certain, CFSensor ordering code via distributor extracts] **A part number ending in "33" is the 3.3 V version. Don't buy that one.**

Pins (DIP-6) [Certain, CFSensor datasheet as quoted]:

Pin	Name	Connect to
1	N/C	nothing. The datasheet says leave it floating.
2	VDD	I/O board 5 V rail
3	GND	I/O board GND
4	VDD	5 V (duplicate of pin 2; connect it too)
5	OUT	10 kΩ → node A
6	GND	GND (duplicate of pin 3; connect it too)

Circuit



Why the divider

- The ADS1115 runs on 3.3 V, so that its I2C pull-ups match the Pi. Its inputs must stay within GND − 0.3 V to VDD + 0.3 V = 3.6 V. [Certain, TI datasheet]
- The sensor's 4.5 V output × 20/(10+20) = **3.0 V**. Even a sensor stuck at its 5.1 V rail gives 3.4 V.
- `hw.py` already uses `DIVIDER = 2/3` and the ±4.096 V range. [Certain]

Expected readings

- **At atmosphere: about 4.5 V at pin 5 and about 3.0 V at A0, so hw.py shows about 0 kPa.**

    - This assumes the standard direction: 0.5 V at −100 kPa, 4.5 V at 0 kPa, as `hw.py` expects. [Likely; CFSensor can calibrate either direction]
    - If you read about 0.5 V at atmosphere, your sensor is reversed. Tell the software owner to flip the formula. Don't rewire.
- **Optional supply monitor on A1:** the output is ratiometric to the 5 V supply, and the Pi's 5 V can sit anywhere from 4.9 to 5.2 V. That error is ±3 kPa, which is fine for the −35 kPa seal threshold. Wiring A1 lets the software correct it later. [Likely]

Tubing: the sensor has a 3 mm barb. Use 2 mm ID silicone tube (BOM) from the barb to a tee on the **cup side** of the vacuum valve, so it reads the cup line, not the reservoir. Mount the I/O board near the valves so this tube stays short.

6. Contact probe (fail-safe optical endstop)

Part: a slotted optical endstop board with a TCST2103 sensor (BOM), on the head foot's sensor post. A 1 mm aluminium flag, clamped under the vacuum elbow on top of the spring buffer, rises through the slot as the cup is pushed up (BUILD_GUIDE §7.6; it is in the CAD as `probe_flag`).

Wiring (W54):

- board **VCC** → Rodent Y-MAX **V-Lim**, which is 5 V with the SW_VCC jumper on 5 V;
- board **GND** → Y-MAX **GND**;
- board **OUT/SIG** → Y-MAX **signal**.

How the Rodent's limit input works [Certain, V1.0.2 schematic]

- The signal pin is the cathode of an opto LED, fed from V-Lim through 1 kΩ.
- **When the sensor output pulls low (sinks about 4 mA), the input reads LOW = "not touching".**
- When the output is high or disconnected, the opto is off and the input reads HIGH = **triggered**. The config uses `pin: gpio.34`, active high.

Mechanical rule (this is what makes it fail-safe)

- **At rest, the slot must be clear**, so the light reaches the detector and the output is LOW.
- **Contact pushes the flag into the slot.** The output goes HIGH, which reads as triggered.
- Set the board in its slots so the flag blocks the beam after 2.0 ± 0.5 mm of buffer travel (BUILD_GUIDE §7.6). The sim assumes 2 mm.

Why this way round: every failure lands on "triggered", and FluidNC then refuses to probe (ALARM:4).

Failure	Input reads	Result
Signal wire cut	open = HIGH	triggered → safe
Sensor GND or VCC cut	output can't sink = HIGH	triggered → safe
Emitter LED dead, or dirt in the slot	dark = output HIGH	triggered → safe
Rodent loses 24 V	everything off	machine stopped anyway

Check your board's polarity first. Most of these boards output HIGH when the slot is blocked, which is what you want. [Likely]

1. Power it from 5 V on the bench.
2. Measure OUT to GND: about 0–0.4 V with the slot clear, about 5 V with a card in the slot.
3. If yours is the other way round, don't use it. Inverting the logic would turn a cut wire into "not touching".

Hysteresis (only if needed): the firmware's `probe_hard_limit` alarms on any probe edge outside a `G38` move. If the flag chatters at the slot edge during the slow peel, BUILD_GUIDE §13.2's 300 cycles will show "Probe Hard

Limit" alarms. Then add hysteresis to the sensor output, keeping the fail-safe rule (a cut wire must still read triggered): for example a 74HC14 Schmitt inverter pair powered from V-Lim, with a 10 kΩ pull-up on its input [Guessing; have the instructor check it], and repeat FIRMWARE.md step 5's fail-safe rows. Or set `probe_hard_limit: false` and live without it.

Indicator LED caution: some boards have an indicator LED arranged so that it holds OUT at mid voltage when "high". The Rodent's opto then stays half on, and the input never reads triggered. FIRMWARE.md Step 5 catches this.

- **Fallback:** wire a bare TCST2103 directly. Detector collector to signal, detector emitter to GND. Emitter LED anode to V-Lim through 220 Ω, cathode to GND. [Likely, standard TCST2103 use]

Why not the board's Probe connector: its opto input is the other way round. The sensor would need to source current at rest, and an emitter failure would read "not touching". (Explained in config.yaml.)

6a. Carry-height sensor (Z stall guard)

Part: the second optical endstop board from the BOM's packs (the same TCST2103 board as the probe), on `carry_sensor_mount` on the Z spine's left face. A flat 1 mm aluminium flag on `carry_flag_arm` (on the `z_plate`) enters its slot from below once the cup is at least 246.5 mm above the pad, and stays in it up to the top stop (BUILD_GUIDE §7.4a). [Certain, CAD]

What reads it: the Pi, not the Rodent: FluidNC knows nothing about it. The runner reads it before every sideways move (it must read blocked), before every grip (it must read clear), and on every Z move with a slab in the cup. [Certain, `runner._carry_check`, `hw.PiCarry`: majority of three reads 1 ms apart]

Wiring (W55, W48):

- At the top of the Z spine, where the 8-core's jacket is already opened for the Z switch (§11): board **VCC → core 1** (+5 V, shared with the probe), **GND → core 2**, **OUT → core 8**. Solder and heat-shrink each splice. The board is fixed to the X carriage, so its lead runs only in the X drag chain and never enters the Z service loop. At the head end, core 8 is cut back and insulated.
- At the panel end: core 8 to the I/O board:



- 5 V × 20/(10 + 20) = 3.3 V at GPIO 6 with a push-pull output, and the 20 kΩ pulls GPIO 6 LOW when the lead is cut. [Certain, arithmetic]
- The board's 5 V comes from the Rodent's V-Lim (core 1). The Pi and the Rodent share a ground through W42 and the USB cable. [Certain, §3.3]

Polarity: HIGH = the flag is in the beam = the head is at carry height. Check it on the bench first, as for the probe (§6): OUT about 0–0.4 V with the slot clear, about 5 V with a card in it. Don't use a board that is the other way round.

Check the level at GPIO 6 once it is wired [Likely: many of these boards pull OUT up through about 10 kΩ, and the 30 kΩ divider then gives only about 2.5 V]: with a card in the slot, GPIO 6 must read at least 2.5 V. If it is lower, solder a 2.2 kΩ pull-up from OUT to VCC **at the board**, as on the laser receiver (§8). At the board end a cut wire still reads LOW.

Failure	GPIO 6 reads	Result
OUT wire (core 8) cut	LOW (the 20 kΩ)	"clear" at travel height: sideways moves are refused ("Z stall suspected … right after homing this is most likely the sensor itself") → safe
Board VCC cut, or the Rodent unpowered	LOW	the same → safe

Failure	GPIO 6 reads	Result
Board GND cut	HIGH (OUT floats up to VCC) [Likely]	"blocked" everywhere: **not safe by itself**; caught at the next grip, where it must read clear ("Carry-height sensor fault …")
Emitter LED dead, or dirt or a card in the slot	HIGH	the same: **not safe by itself**, caught at the next grip
Flag bent away, or the mount slid down	wrong trip height	`selftest --motion` measures the trip; outside 246.5 ± 1 mm it fails

Why a dead emitter is not simply fail-safe here: the flag blocks the beam at carry height, so "dark" is the "OK to move" reading. The grip rule (the cup is far below the trip at every pick, so the sensor must read clear there) is what turns a dark sensor into a stop. [Certain, the CAD report's own warning, implemented in `runner._carry_check`]

7. Limit switches (NC)

Use bare SPDT hinge-lever microswitches with C / NO / NC solder tabs (the Omron SS-5GL size: 19.8 × 10.2 × 6.4 mm body, holes 9.5 mm apart). The printed mounts take a flat lever that ends at most 23 mm from the hinge end; cut a longer one back to 20 mm (BUILD_GUIDE §6.8).

- Solder the cable to **C** and **NC**: closed at rest, open when pressed.
- C goes to the connector's GND pin and NC to its signal pin. Leave V-Lim empty.

Don't use the 3D-printer "mechanical endstop" boards as they come. (The BOM lists bare lever microswitches for this reason.) Their LED-and-resistor circuit differs between makers. On some, it feeds the Rodent's opto LED and hides a pressed switch. [Likely: forum reports]

- If you only have those boards, desolder the switch, or cut the traces and wire straight to its C and NC tabs. Check C, NO and NC with a meter.

How it reads [Certain, schematic + config]

- At rest: switch closed → opto on → LOW → OK.
- Pressed, or a wire broken: open → HIGH → triggered. Outside homing that is ALARM:1, so a broken switch wire stops the machine instead of letting it crash.

Where the switches go [Certain, CAD `x_switch_mount`, `z_switch_mount`, `checks.txt`]

- **X home switch:** on the printed `x_switch_mount`, under the beam's left end, on the front faces of the beam and the left upright (2 × M5×10 button heads into T-nuts in the beam's lower front slot, left of `x_bumper_L`). Lever up, hinge end toward the back. A finger on the `x_bracket`'s left edge slides across the lever (a 45° lead-in, then a flat). It must click when the cup axis is at layout X ≈ 19 mm, about 23 mm left of pocket 0's centre, **± 2 mm**.
  - The printed `x_bumper_L` is the hard stop at cup X 15, so the switch must trip at least 2 mm before the `x_bracket` reaches it. The carriage still has 2.3 mm of rail there. The finger's flat holds the lever 0.6 mm past its click point (at the lever tip) all the way to the bumper.
  - **Lead (W52):** from the terminals straight down to about 50 mm below the beam (below the corner brackets), back under the beam, then up behind it into the beam's **lower rear** slot, and along that slot to the middle (§11). At the left joint, keep the flat joining plate's beam leg on the upper rear slot so it does not cover the spot where the lead enters (BUILD_GUIDE §4.3). [Certain, CAD lead path; Guessing on the brackets and plate, which are not in the CAD]
- **Z top switch:** on the printed `z_switch_mount`, on the Z spine's right (+X) face near the top (2 × M5×10 button heads into T-nuts in that face's slot), fixed to the X carriage. The switch lies on the mount's shelf, lever toward the `z_plate`. A 45° chamfer along the `z_plate`'s top back edge slides across the lever, and the plate's back

face then holds it. It must click at c ≈ 258 mm (the cup face 258 mm above the pad top), ± 2 mm. That is 4 mm before the `z_plate` meets the `z_motor_mount` (the top hard stop at c = 262).
    - **Lead:** cores 4 and 5 of the 8-core (§11), back past the spine, then up behind the `z_motor_mount` to the spine top, where the 8-core's jacket is opened.
- Setting each click point (the switch in its M2 slots, the mount in its M5 slot) is in BUILD_GUIDE §6.8 and §7.4. Exact positions are checked in FIRMWARE.md Step 6.

Lid switch (optional, GPIO25): use **C and NO**, closed when the lid presses it, so a shut lid reads LOW (`Button.is_pressed`). A cut wire then reads "lid open", which is safe.

8. Overfill laser gate

Parts (BOM): a ≤ 1 mW (Class 2) 650 nm laser module with a 6 mm barrel, and a non-modulated laser receiver module. Both are 5 V parts.

- **Do not use the KY-008 laser that comes in the receiver kit.** KY-008 modules are usually sold as 5 mW (Class 3R). [Likely] The laser is on whenever the Pi is on (W46 is the Pi's 5 V), and the beam runs along the open row at about 214 mm, the height of a face bent over the stacks while loading. A Class 2 beam is protected by the blink reflex; a 3R beam is not. The listing must say "≤ 1 mW" or "Class 2 / Class II". Keep the kit's receiver.
- The laser sits in the left gate post and the receiver in the right post, with the beam 4 mm above the MAX line. An M3 set screw in each post locks the barrel.
- Stick a "LASER - do not stare into the beam - Class 2" label on both posts.
- **Never look into the beam.** Aim it so it always ends on the receiver.
- [Guessing] If you want the laser off between runs as well, switch its 5 V with a small low-side MOSFET from a spare GPIO, on only during the start-up check. The software does not do this yet. With the laser off the gate reads "blocked", so it stays fail-safe.

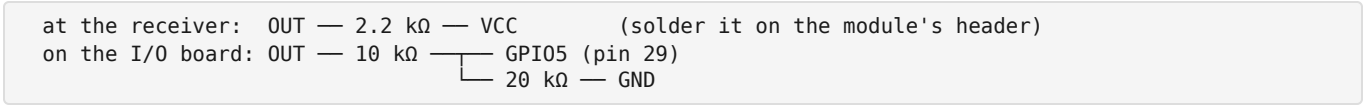
Laser wiring (W46):

- 5 V → the module's + (red) lead; GND → − (black). Confirm with a meter that it lights.
- 24 AWG 2-core, about 1.2 m.

Receiver wiring (W47):

- VCC → 5 V, GND → GND, OUT → I/O board.
- 24 AWG 3-core, about 1.2 m.

Level shift and fail-safe



- Most of these receivers are LM393 boards with an open-collector output and a 10 kΩ pull-up. [Likely] A bare 10k/20k divider on a 10 kΩ pull-up gives only about 2.5 V, which is marginal for a 3.3 V input.
- The extra 2.2 kΩ pull-up raises "high" to about 2.7–3.1 V, and a push-pull output gives 3.3 V. Both are inside the Pi's safe range. [Certain, arithmetic]
- The 2.2 kΩ **must sit at the receiver end**. Then a cut OUT wire leaves only the 20 kΩ, which pulls GPIO5 LOW = "blocked", and the run won't start.

Condition	GPIO5	Meaning
Beam received	about 3 V, HIGH	clear

Condition	GPIO5	Meaning
Beam blocked (stack too tall, or the head is down in the beam plane)	LOW	blocked
OUT wire cut, receiver unpowered, laser dead	LOW	blocked → safe
Receiver GND wire cut	HIGH	**false "clear"**: the selftest must block the beam and see LOW

Check before wiring: measure the receiver's OUT with the beam on and with it blocked. It must be **high when the beam is received**. If your module is the other way round, swap it. Don't invert in software, because a cut wire would then read "clear".

Room light: non-modulated receivers can be held HIGH by bright room light. Adjust the module's trimmer (under the light roof, if you fitted the optional roof) until a blocked beam reads LOW with the room lights on.

9. LED rings

Parts (BOM): two 12 V COB "angel eye" white rings: **40 mm** for the head (a 60 mm ring would cut into the drop arm; the seat in `camera_bracket` and `phase0_top` is a 42.8 mm bore with three centring ribs and a 33 mm lip: it takes a ring of 39.5–42.5 mm OD, up to 5 mm thick, and the ribs are shaved for a ring over 40 mm) and 60 or 63 mm for the station (a 64 mm bore on a shelf in `station_mount` and `phase0_base`). Measure OD, ID and thickness before printing the ring seats (BUILD_GUIDE §3.3). If no 40 mm COB ring can be found, a 40 mm 12-SMD ring that fits the seat is the fallback. **Leads:** the head ring's pads and leads leave through the notch in its seat's lip; the station ring's drop through the shelf slot into the camera ribbon's slot and leave with it. [Certain, CAD]

Switching

- Low side, from GPIO23 (head ring, Q4) and GPIO24 (station ring, Q5).
- Ring + to the 12 V bus through the module's "+" terminal, ring − to the module's "−" terminal.
- If the ring came with a small inline driver box, keep it. Wire the 12 V and module side to its input. [Likely: many angel-eye rings must use their driver]

No dimming by PWM

- Pi 5 GPIO23 and 24 only have software PWM. The rolling-shutter cameras would show dark bands. [Likely]
- Switch the rings fully on and set brightness with camera exposure and gain.
- If a ring is still too bright at minimum exposure, measure its current at 12 V. Then add a series resistor that halves it, rated for at least 4 × the power it burns.

Turn-on time: the ring drivers may soft-start, and camera auto exposure needs time to adapt. `hw.py` already waits 0.3 s after switching a ring on (`PiLights.settle_s`) before the first frame. [Certain, code] If the first frame of each slab still looks dark, raise it.

10. Cameras

Station camera: Camera Module 3 on CAM/DISP 0

- Use the 500 mm Pi 5 camera cable (22-pin 0.5 mm to 15-pin 1 mm, BOM: Raspberry Pi SC1130). **Don't go longer.**
- Mount the Pi on the electronics panel **behind the bed, centred near X ≈ 690** (the station is at X = 686). The ribbon then runs from the station mount's rear slot along the bed and down to the Pi in about 350 mm.
- Fit the ribbon with the power off. Lift the connector latch, insert, and close it. Follow the Raspberry Pi camera guide for which way the contacts face.
- Keep the ribbon away from the motor cables. Cross them at right angles if you have to.

Head camera: ELP 8 MP USB, IMX179

- Its own USB lead joins the BOM's 3 m USB 2.0 extension, which runs through the chain and the Z loop to a **black USB 2.0 port** on the Pi.
- Total length is about 4 m. USB 2.0 allows 5 m [Certain], so add nothing more. If the camera drops out, the first fix is a true drag-chain-rated USB cable (BOM note).

11. Cable routing: drag chain and Z service loop

X drag chain

Part (BOM): an openable chain, 15 × 30 mm inside, 1 m long. Openable links let the USB plug go in. Buy the R38 bend radius or larger if you have the choice. Don't go below R28. [Guessing on the sizes sold]

Fixed end: on the top-back of the X beam, at the **middle of travel, X ≈ 640**.

Moving end: on the X bracket's top tab, with a printed adapter.

Length check [Certain, chain geometry]

- With the fixed end at mid-travel, the chain needs travel/2 + π·R + margin = 1231/2 + π·38 + 100 ≈ 0.84 m, so 1 m is enough.
- With the fixed end at a beam end, it needs travel + π·R ≈ 1.35 m. Buy 1.5 m in that case.

What goes inside

#	Item	About	Carries
1	4-core 22 AWG shielded high-flex	Ø 5.5 mm	Z motor (W51)
2	8-core 24 AWG shielded high-flex	Ø 6 mm	see below
3	USB 2.0 extension	Ø 5 mm	head camera
4	PU tube, 4 mm OD	Ø 4 mm	cup vacuum line

That fills about 30 % of the chain. Lay the items side by side, untwisted, with nothing crossing. Don't tie anything inside the chain.

8-core assignment. Record your cable's real colours here.

Core	Use	Rodent end	Head end
1	+5 V probe and carry sensor	Y-MAX V-Lim	optical board VCC; carry board VCC (spliced at the spine top)
2	probe and carry sensor GND	Y-MAX GND	optical board GND; carry board GND (spliced at the spine top)
3	probe signal	Y-MAX signal	optical board OUT
4	Z switch signal	Z-MAX signal	Z switch NC
5	Z switch GND	Z-MAX GND	Z switch C
6	+12 V head ring	Q4 "+"	ring +
7	head ring return	Q4 "−"	ring −
8	carry sensor OUT	I/O board: 10 kΩ / 20 kΩ divider → Pi GPIO 6 (§6a)	carry board OUT, spliced at the spine top; insulated past the splice
shield	drain	Rodent GND, board end only	cut back, heat-shrink

Off the fixed end:

- The bundle drops behind the beam to the electronics panel.
- Tie it every 100 mm to the rear brace or to printed clips.

- Keep it more than 100 mm behind the head's path.
- The static X motor cable (W50) and X switch cable (W52) run in the beam's back slot from their ends to the middle, then join this bundle. The X switch lead comes up into the lower back slot just right of the left upright (§7).

Z service loop (head moves 263 mm relative to the carriage)

What's in it:

- the 8-core carries on past the carriage to the head;
- at the top of the Z spine, open the jacket over about 40 mm and take cores 4 and 5 out to the Z switch;
- in the same opening, splice the carry-height sensor's lead to cores 1, 2 and 8 (§6a); core 8 ends there;
- heat-shrink over the opening.

The 8-core, the USB extension and the vacuum tube then form one loop wrapped in the BOM's 10 mm spiral wrap.

Loop path: from a tie point at the top of the spine down to the top of the drop arm. Then tie the bundle along the drop arm every 50 mm down to the head.

Loop size:

- Make it about 350 mm long, with no bend tighter than 40 mm radius.
- Check by hand at c = 0 and c = 255. The loop must never hang lower than the top of the drop arm. At c = 0, that point is 350 mm above the pads (`model.ARM_Z`: the Z carriage 295 mm above the cup face, plus 55), well above the 218 mm rod tops. [Certain, layout numbers]

Tube slack: leave 25 mm at the cup's `M-5ALU-4` barb elbow, which points to the back (+Y). The buffer rod rises up to 20 mm on contact (SMC `ZPT25UUK20-B5-A10`, 20 mm stroke). Keep the tube off the head lens and its LED ring: at full stroke it runs about 1 mm under the ring [Guessing; the tube is not in the CAD].

12. Grounding and shielding

Protective earth

- Inlet PE → LRS FG (W4).
- LRS FG → frame (W5): an M5 ring terminal with a serrated washer into a T-nut on the base 2040. Scrape the anodising under it.
- Check that each extrusion joint conducts. Anodising insulates, so a joint may need a serrated washer. [Likely]

0 V bond: one short wire from LRS −V to FG (W6). Nowhere else.

- The Pi's ground joins the 0 V bus through W42 and the USB cable, so the whole low-voltage side has one ground.
- The Pi's own USB-C supply is floating (2-pin), so this creates no mains ground loop. [Likely]

Heavy current returns: pump, valve and motor currents return to the 0 V bus through their own black wires, never through the Pi.

Shields: connect them at the Rodent end only (its VIN GND terminal or any GND pin). Cut back and insulate the far end.

Separation:

- Keep motor cables at least 50 mm from the CSI ribbon, the ADS1115 and the sensor leads.
- Keep the I2C leads under 20 cm.

13. Strain relief and connectors

- **Screw terminals:** use a ferrule on every stranded wire (PSU, Rodent VIN, MOSFET outputs, fuse holders, E-stop). **Never put solder-tinned wire in a screw terminal.** It creeps and goes loose.
- **Wago 221 lever connectors** take stranded wire directly. Use them for the 24 V, 12 V and 0 V buses.
- **Tie every cable to the panel within 50 mm of its connector.** No connector should carry cable weight.
- **Drag chain:** tie each cable to the tie-bar at both chain ends.
- **Head:** use P-clips or tie mounts on the drop arm.
- **Crimps:** crimp JST-XH and Dupont housings with the proper tool and pull-test each one. Or buy pre-crimped leads.
- **Solder joints:** heat-shrink every one (dividers, cable breakouts, LED ring leads).
- **Moving parts:** put the pump on rubber feet and don't hard-mount it to the panel. Fit a short silicone hose section at the pump so vibration doesn't fatigue the fittings.

14. Pre-power checklist (multimeter)

Do this with everything wired, **mains unplugged**, and the E-stop released.

Mains (instructor)

1. Inlet PE pin to the X beam, Z spine and base: **< 0.5 Ω**.
2. Inlet L to PE and N to PE, switch on: **open (> 1 MΩ)**.
3. Inlet L to N, switch on: **not a short (> 10 Ω)**. The supply's input circuit shows some reading.
4. Fuse fitted, T3.15A. **LRS 115/230 switch on your mains voltage.** Terminal cover on. 4a. **Mains box closed, gland tight, insulated receptacles on every inlet tab: no exposed mains metal anywhere with the cord plugged in** (§1.1). Not on the plywood panel.

24 V side

5. 24 V switched bus to 0 V bus: **> 1 kΩ**. It rises as the meter charges the capacitors. Under 10 Ω is a short: find it.
6. E-stop pressed: LRS +V to 24 V switched bus **open**. Released: **< 0.5 Ω**. Check both NC blocks: each one alone must open the circuit when pressed (meter across each block).
7. F1 = 5 A, F2 = 3 A, F3 = 1 A fitted.
8. Rodent VIN polarity: red to "VCC", black to "GND". [Certain, terminal labels]
9. **24 V bus to Pi 5 V pin: open. 0 V bus to Pi GND: < 1 Ω.**

Motors, switches, probe

10. Motor coils, at the motor plug:
    - each coil pair **1–4 Ω** (the 17HS19-2004S1 is about 1.4 Ω [Likely]);
    - coil to coil **open**.
    - X plugged into socket **X** and Z into **Z**. **Nothing in Y or E.**
11. Limit switches, at the unplugged board-end housing: signal to GND **≈ 0 Ω at rest, open when pressed**.

Loads

12. Each MOSFET output: "+" to "−" with the load connected:
    - pump: a few tens of ohms [Guessing];
    - valve coil: hundreds of ohms [Guessing];
    - **never close to 0 Ω**.

13. Divider resistors: measure each 10 kΩ, 20 kΩ and 2.2 kΩ (10k and 100k are easy to mix up), including the carry sensor's pair (§6a): its GPIO 6 node to GND, with the Pi lead and core 8 unplugged, about 20 kΩ. **MOSFET SIG pull-downs:** SIG to GND on each of Q1–Q5, Pi lead unplugged: **about 10 kΩ**. Open means the pull-down is missing: add it (§4) before any power-up.

Board set-up

14. Rodent jumpers as §2. `r_sense_ohms` matches the resistor marking (FIRMWARE.md Step 0).
15. Sensor: pin 1 not connected; pins 2 and 4 to 5 V; pins 3 and 6 to GND.

15. First power-up

1. **Pi only (machine cord unplugged).** Boot the Pi, then check:
   - `i2cdetect -y 1` shows **48**;
   - the sensor gives about 3.0 V at A0 (about 0 kPa);
   - the gate divider gives about 3 V with the beam received and under 0.4 V when blocked. The laser and receiver run from the Pi's 5 V, so this works without 24 V.
2. **Machine cord in, E-stop pressed, rocker on.** Measure the LRS output with no load and set **24.5 V** with V.ADJ.
3. **Release the E-stop.**
   - Rodent VIN ≈ 24.4 V and the Rodent LEDs light;
   - buck output 12.0 V;
   - nothing gets warm in the first minute: Rodent driver area, buck, modules.
4. **Test each load from the Pi** with Raspberry Pi OS's `pinctrl`: `pinctrl set 17 op dh` turns the pump on and `pinctrl set 17 op dl` turns it off. [Likely, Bookworm tool] Do the same for 27, 22, 23 and 24, one at a time. Each output must drive only its own load. 4a. **Carry-height sensor** (it is powered from the Rodent, so the E-stop must be released): `pinctrl get 6` must read low with the slot clear and high with a card in it, and GPIO 6 must measure at least 2.5 V high (§6a). `python -m slabscan selftest` shows the same as its `carry sensor` line.
5. Now follow FIRMWARE.md from Step 1.
6. **After 10 minutes of motors holding:** the motor bodies may be warm but **below about 50 °C** (comfortable to hold a finger on). Hotter than that means the sense-resistor setting is wrong: power off (FIRMWARE.md step 0).

16. Open risks

1. **[Likely] Laser receiver output varies by maker.** The 2.2 kΩ pull-up and divider make both common output types safe, but measure before wiring.
2. **[Likely] Sensor output direction.** It is assumed to be 0.5 V at −100 kPa. Check about 4.5 V at atmosphere on first power-up.
3. **[Certain] E-stop DC rating is below target.** The BOM's blocks are rated DC-13 1.5 A at 24 V, against the ≥ 5 A target and the ≈3.4 A bus peak (§1.2, §1.3). Two NC blocks in series and the monthly E-stop check stay required; a mains-side E-stop or a force-guided safety relay is the fix to decide before unattended use.
4. **[Guessing] Head USB at about 4 m.** It works on paper. A cheap extension in a moving chain is the weakest cable in the machine. Watch for camera dropouts in the soak test.
5. **[Guessing] Whether Adafruit #5648 has a SIG pull-down.** Measured in §14 step 13 before the first power-up; add 10 kΩ if it reads open (§4). BUILD_GUIDE §13.4 ("unplug the Pi, the pump must stop") checks it again.

6. **[Guessing] Probe flag chatter** with `probe_hard_limit: true`: see §6 "Hysteresis".
7. **[Likely] The carry sensor's high level through the divider** may be only about 2.5 V with a 10 kΩ on-board pull-up: measure it (§6a), add 2.2 kΩ at the board if low.
8. **[Certain] A dark carry sensor reads "OK to move".** A dead emitter, dirt or a cut GND is caught only at the next grip (§6a); until then it cannot see a stall. The run checks at every grip, so that window is at most one slab's moves.

Resolved: the BOM now lists Adafruit #5648 modules, bare microswitches, the 1/3/5 A fuses, the mains box, a Class 2 laser and a two-block E-stop used on this page, and the 8-core's spare core now carries the carry-height sensor (§6a). Both limit switches have printed mounts with their lead paths in the CAD (§7). The CAD has printed hard stops that keep both carriages on their rails (CONTRACT), so the old Z-rail margin issue is closed.

Firmware: FluidNC on the BTT Rodent

This page covers flashing FluidNC, loading `firmware/fluidnc/config.yaml`, checking it, and testing every motor, switch and the contact probe before the Pi software takes over. Wiring is in WIRING.md. The interface rules are in CONTRACT.md.

Confidence tags: **[Certain]** = read in a primary source (BTT's files, the FluidNC v4.1.1 source code), **[Likely]** = strong inference or several secondary sources, **[Guessing]** = a gap I filled.

Read this first

1. **Check the current-sense resistors before you plug in any motor.** BTT's documents disagree, and a wrong value overheats the motors.
   - The Rodent V1.1 manual says 75 mΩ, and a V1.1 owner reports parts marked "R075". [Certain that the V1.1 manual says so; Likely for real V1.1 boards]
   - The V1.0 schematic and BTT's `rodent.yaml` say 22 mΩ. [Certain]
   - **`config.yaml` ships with `r_sense_ohms: 0.022`, on purpose.** Change all four lines to `0.075` only after you have **read "R075"** on your board's resistors.
   - Why this default: on a TMC2160 the motor current scales with 1/R_sense. Setting 0.075 on a 22 mΩ board gives 3.4 × the set current (3.4 A on X, 1.7 A held all the time on both motors in PETG mounts). Setting 0.022 on a 75 mΩ board gives 0.29 ×: weak motors that stall at the first homing, with no slab in the machine. The wrong guess should fail cold, not hot. [Certain, from the driver current formula]
2. **Use FluidNC v4.1.1 or newer.** [Certain] FluidNC puts itself in a config alarm on any key it doesn't know, so an older build may reject this file. BTT has shipped Rodents with old firmware, so reflash even a new board. [Likely]
3. **Never plug or unplug a motor while the board has 24 V.** It can destroy the driver. [Certain, standard for TMC drivers]
4. **The Z soft limits (−8 … 255) sit inside printed hard stops.** [Certain, `cad/out/checks.txt`] The Z carriage sits 295 mm above the cup face. The `z_end_cap` stops it at c = −14 (it would leave its rail at c = −19); at the top the `z_plate` meets the `z_motor_mount` at c = 262, 4 mm past the switch, with 3.6 mm of carriage still on the rail.
5. **`probe_hard_limit: true`** turns a contact-sensor trip during any non-probe move into an immediate critical alarm (Step 7 tests it). The 300-cycle test must show zero such alarms (BUILD_GUIDE §13.2).

Why this board

The Rodent fits the job: [Certain, BTT manual and schematic]

- four TMC2160 drivers on SPI;
- 24–56 V motor input;
- optocoupled limit inputs with on-board 10 kΩ pull-ups;
- a USB-C port with a CH340K USB-serial chip;
- an official FluidNC config from BTT.

Two axes only need two drivers, so the extra drivers become plug-in spares (see below). I found no reason to switch boards.

The cheaper MKS DLC32 would also run FluidNC, but its plug-in stepstick drivers and 3.3 V GPIO endstops would need more parts for the same fail-safe wiring. [Likely]

Board facts used by the config

Item	Value	Source
MCU	ESP32 (V1.0: bare ESP32-D0WD-V3, V1.1: ESP32-WROOM-32U module with external antenna)	BTT manual [Certain]
Motor power input	**24–56 V DC**, screw terminal "GND / VCC"	BTT manual [Certain]
Driver sockets	Labelled X, Y, Z, E. STEP/DIR/EN go through an I2S shift register.	BTT manual table, `rodent.yaml` [Certain]
X socket	STEP I2SO.2, DIR I2SO.1, EN I2SO.0	[Certain]
Y socket	STEP I2SO.5, DIR I2SO.4, EN I2SO.7	[Certain]
Z socket	STEP I2SO.10, DIR I2SO.9, EN I2SO.8	[Certain]
E socket	STEP I2SO.13, DIR I2SO.12, EN I2SO.15	[Certain]
SPI chain	One chip-select (gpio.5). MOSI gpio.23, MISO gpio.19, SCK gpio.18. Chain order from MOSI: X, Y, Z, E.	`rodent.yaml` [Certain]; order also stated in a Marlin port PR [Likely]
Limit inputs (3-pin: signal, GND, V-Lim)	X-MAX gpio.35, Y-MAX gpio.34, Z-MAX gpio.33, E0-MAX gpio.32, **E1-MAX gpio.37 on V1.0 but gpio.39 on V1.1**	manual images V1.0 and V1.1, V1.0.2 schematic [Certain]
Probe input	gpio.36, "V-probe / GND / P-36"	manual [Certain]
Pull-ups	10 kΩ to 3.3 V on GPIO 32–37 (V1.0.2 schematic). GPIO 34–39 have no internal pull resistor.	schematic [Certain]; ESP32 [Certain]
Limit input circuit	1 kΩ from V-Lim to an EL357N opto LED, whose cathode is the signal pin. **Pulling the signal to GND turns the LED on, and the GPIO reads LOW.** Open = HIGH.	schematic [Certain]
Probe input circuit	The other way round. The signal pin drives the opto LED through 1 kΩ to GND, so the signal must be HIGH to read LOW. Optional 10 kΩ pull-up ("NPN" jumper).	schematic [Certain]
V-Lim / V-Probe voltage	Jumper: 5 V, 12 V or VCC. VCC is banned above 24 V.	manual [Certain]
VUSB jumper	Ties USB 5 V to the board's 5 V rail. **"Do not insert the jumper when the device is in use."** Only for flashing with no 24 V.	manual V1.0 image [Certain]
USB serial	CH340K, powered from the board's 3.3 V. It appears as `/dev/ttyUSB0`, by-id `usb-1a86_USB_Serial-if00-port0`.	schematic [Certain]; by-id name from a V1.1 Klipper port [Likely]
V-MOS outputs	gpio.12, gpio.2, gpio.4 (12–36 V, own input). **Not used.** The Pi switches the vacuum and lights.	manual [Certain]

What `config.yaml` sets

	X	Z
Driver socket	**X** (motor0, chain 1)	**Z** (motor0, chain 3)
Spare socket, ganged	Y (motor1, chain 2)	E (motor1, chain 4)
steps/mm	80 (GT2 20T, 1/16)	400 (T8×8, 1/16)
Max rate / accel	15000 mm/min, 2000 mm/s²	6000 mm/min, 1200 mm/s²
Run / hold current	1.0 A / 0.5 A	0.8 A / 0.5 A
Homing	cycle 2, toward min, `$HX` alone blocked	cycle 1 (first), toward max
Position after homing	X = 22 (switch trips at about 19)	Z = 255 (switch trips at about 258)
Soft limits (machine mm)	**22 … 1253**	**−8 … 255**
Home / limit switch	X-MAX connector, gpio.35, NC, hard limit on	Z-MAX connector, gpio.33, NC, hard limit on
Contact probe	—	Y-MAX connector, gpio.34, active high, fail-safe

Other choices:

- No Y axis is defined. FluidNC still reports three numbers (`MPos:x,0.000,z`), because it pads the axis count to 3 "since senders might assume 3 axes". [Certain, `Axes.cpp`] The software's status and `[PRB:]` parsers rely on that.
- No spindle, coolant, user outputs, OLED or SD card section.
- `idle_ms: 255` means the motors are never switched off, so Z cannot drift down.
- `must_home: true` keeps the machine in alarm until `$H` has run.

Why the spare sockets are configured. [Certain, FluidNC source comment on `spi_index`: "every physical position in the chain must be represented by a motor entry, even unused ones, or the chain's data alignment breaks"]

- FluidNC reads a TMC chip by clocking data through every chip behind it. It works out how many chips there are from the highest `spi_index` it knows.
- So the Y and E chips are given entries, as second motors (`motor1`) on X and Z. They step together with X and Z.
- **Leave them empty.** If the X or Z driver dies, move that motor's plug to Y or E and nothing else changes.
- **Never plug a second motor into Y or E.** It would move with X or Z.

Where the travel numbers come from (cad/layout.py and cad/model.py, October 2026):

- Pocket centres run from X = 42 to 1238.
- Printed end bumpers stop the `x_bracket` at cup X 15 (`x_bumper_L`) and X 1262 (the left end of `x_motor_mount`). The MGN12H carriage still has 2.3 / 5.3 mm on its rail there. The soft range 22 … 1253 stays 7 / 9 mm inside the bumpers. [Certain, `checks.txt`]
- The X home switch must trip at X 19 ± 2: at least 2 mm before the left bumper.
- Z's top of travel is 250 (C_MAX). The Z switch trips at c ≈ 258, the top hard stop is at 262 (3.6 mm of carriage left on the rail), and the bottom hard stop is at −14 (5 mm left; the carriage would leave the rail at −19). [Certain, `checks.txt`]
- If the rail lengths or pocket count change, recompute `max_travel_mm` and `mpos_mm` with the same margins.

Step 0: board set-up (power off)

1. Read the board version on the silkscreen (V1.0 or V1.1). Read the marking on the big resistors next to each driver (R075 or R022), then set `r_sense_ohms` to match.
2. Set the jumpers:
   - SW_VCC (limit voltage): **5V**.
   - V-Probe: **5V**. The Probe connector is unused, but 5 V is the safe setting.
   - **NPN EN: off.**
   - **VUSB: off.**
   - **No DIAG jumpers.** They tie driver stall outputs onto the limit pins.
3. Fit the Wi-Fi antenna (V1.1 has a u.FL connector) only if you plan to use the web UI.

Step 1: flash FluidNC

You need a laptop with Chrome or Edge, which have Web Serial. [Likely]

1. Power the Rodent from 24 V, or fit the VUSB jumper with 24 V **off**. Remove the jumper again after flashing.
2. Connect USB-C to the laptop. Windows may need the CH340 driver. [Likely]
3. Open the FluidNC web installer at `https://installer.fluidnc.com`. Pick the latest stable release (v4.1.1 or newer) and the **ESP32 "wifi"** build. Choose a fresh install, which erases the flash. [Likely]

4. If the installer can't connect, hold **BOOT**, tap **RESET**, release BOOT, then retry. Both buttons are on the board. [Certain]
5. Open a serial terminal at 115200 baud and tap RESET. You should see the FluidNC banner and startup messages. [Certain, FluidNC default baud]

Step 2: load `config.yaml`

Use any one of these. All of them store the file in the ESP32's internal flash as `config.yaml`, which is the default config name (`$Config/Filename`). [Certain]

A. FluidTerm, on the laptop (easiest). FluidTerm comes in the FluidNC release zip.

1. Run it and pick the Rodent's COM port.
2. Press **Ctrl+U**, pick `firmware/fluidnc/config.yaml`, and keep the destination name `config.yaml`. Ctrl+U sends the file over XMODEM. [Certain, `fluidterm.py`]
3. Press **Ctrl+R** to reset the board.

B. Web UI over Wi-Fi.

1. Join the board's access point: SSID `FluidNC`, password `12345678`. [Certain, defaults in the v4.1.1 source]
2. Browse to `http://192.168.0.1`.
3. Upload `config.yaml` with the flash file manager, then restart.

C. From the Pi, headless (later updates, without a laptop).

1. Install the tool into the software's venv (system `pip` on Bookworm refuses with "externally-managed-environment"; pyserial is already in the venv): `~/psa-slab-scanner/software/.venv/bin/pip install xmodem`
2. Stop the slabscan software (`sudo systemctl stop slabrail-web` if the boot service runs), then run the script with the venv's Python: `~/psa-slab-scanner/software/.venv/bin/python upload_config.py /dev/ttyUSB0 firmware/fluidnc/config.yaml`

```
# upload_config.py  -  run on the Pi with the venv's python (step C.2)
import sys, time, serial
from xmodem import XMODEM
port, path = sys.argv[1], sys.argv[2]
s = serial.Serial(port, 115200, timeout=1)
time.sleep(3)                       # the ESP32 resets when the port opens; let it boot
s.reset_input_buffer()
s.write(b"$Xmodem/Receive=config.yaml\n")
time.sleep(0.5)
getc = lambda n, timeout=1: s.read(n) or None
putc = lambda data, timeout=1: s.write(data)
with open(path, "rb") as f:
    print("sent OK" if XMODEM(getc, putc).send(f) else "XMODEM FAILED")
s.write(b"$Bye\n")                  # restart FluidNC so it loads the new file
```

The command `$Xmodem/Receive` exists in v4.1.1 and is what FluidTerm sends. [Certain] The script itself has not been run against a board. [Guessing]

Once the config is in:

- Turn the radio off: `$WiFi/Mode=Off`. [Certain, setting name] The default mode falls back to an open-to-anyone access point with a published password. Anyone in range could jog the machine.
- **Do not run `$CD=config.yaml`.** It overwrites the commented file with a bare dump. Plain `$CD` only prints the config.
- Runtime edits like `$/axes/x/soft_limits=false` last until the next restart. [Certain]

Step 3: check that the config loaded

Send each command and compare with the expected result.

Command	Expect	If not
`$SS` (startup log)	No `[MSG:ERR:` lines. An "Axis count 3" line. Four driver lines, each followed by "driver test passed". The I2SO and SPI bus lines.	"Ignored key …" means the key is unknown to your FluidNC version: update FluidNC. "TMC driver not detected" on all four means an SPI or 24 V problem. On one only, that driver is damaged.
`$CD`	Your values come back, e.g. `steps_per_mm: 80.000` under `x`.	Wrong file loaded: check `$Config/Filename` and `$LocalFS/List`.
`?`	`<Alarm|MPos:0.000,0.000,0.000|...>` before homing. Alarm is normal: `must_home`.	`<Idle` before homing means `must_home` was not read.
`$Limits` (send `!` to stop it)	A live line of active limit pins and `P` for the probe. **Nothing active at rest.**	See Step 5.

The exact wording of startup lines varies by version, so check the meaning, not the exact text. [Likely]

Step 4: motor direction (one axis at a time)

Before homing, FluidNC's position is a made-up 0, which is outside the X soft range (22…1253). With soft limits on, it would stretch or clip your test jogs. [Certain, `Cartesian::constrain_jog`] Turn them off just for this test.

1. Power off. Move the carriage to the middle of the beam by hand, slowly. Turn the Z lead screw by hand until the cup is about 100 mm above an empty pad.
2. Power on. Keep a hand on the E-stop. Send:

   ```
   $/axes/x/soft_limits=false
   $/axes/z/soft_limits=false
   $X
   $J=G91 G21 X10 F600
   ```

   The carriage must move 10 mm to the RIGHT, toward the X motor end.
3. Send `$J=G91 G21 Z10 F300`. **The head must move UP 10 mm.**
4. If an axis runs backwards, power off and fix it one of two ways:
   - swap the two wires of ONE coil (A1↔A2) at the motor plug; or
   - add `:low` to that axis's `direction_pin` on **both** of its motors (for example `I2SO.1:low` and `I2SO.4:low` for X). Upload the file again.
5. Check distance: jog X +100 and measure. It must be 100 mm ± 0.5. If it's off by 2× or 0.5×, the pulley or microstep setting is wrong.
6. Restart (`$Bye` or the RESET button). The soft limits come back on.

Step 5: switches and probe

Run `$Limits` and watch the live line while you do each step.

Action	Expected	Why
Nothing touched	No X, no Z, no P	NC switches closed, probe slot clear
Press the X home switch	`X` appears in NegLimitPins	Switch opens, input goes high
Unplug the X switch at the board	`X` stays on	**Fail-safe check:** a broken wire must look like "pressed"
Press / unplug the Z top switch	`Z` appears in PosLimitPins	Same
Push the cup up about 3 mm by hand	`P` appears	Flag enters the slot
Unplug the probe at the board	`P` stays on	**Fail-safe check**

Action	Expected	Why
Cover the sensor's emitter side, or cut its 5 V	`P` stays on	**Fail-safe check**

If any fail-safe row shows the pin as not active, **stop**. The switch or sensor is wired the wrong way, or the module's LED circuit is feeding the input. Fix the wiring before you home. WIRING.md, sections 6 and 7, shows the right way. The `?` report also lists active pins as `|Pn:XZP` (standard Grbl). [Likely]

Step 6: first homing

1. Put the head about halfway up, over an empty area. Hand on the E-stop.
2. Send `$H`.
   - **Z must go up first**, touch the switch, back off 3 mm and touch again slowly.
   - **Then X goes left** and does the same.
   - Then `?` shows `<Idle|MPos:22.000,0.000,255.000|...>`.
3. Repeat `$H` five times. After each one, jog to the same spot (`G90 G0 X42 Z240`) and check the cup lands on the same point (a pencil mark on the pad). Spread must be under 0.2 mm.
4. **Check that the switch positions match the numbers in the config.**
   - Jog until the cup is centred over pocket 0 at about Z 240. `?` should show X ≈ 42 ± 2. (The X switch must trip within ±2 mm of X 19, because the left bumper is at X 15.)
   - If not, move the X switch, not the config. `calibrate x` corrects only the small remainder, and a large offset eats into the soft-limit margins.
   - Do the same for Z after `calibrate z`: `pad_trigger_c` must be between −4 and 0.
5. **Motor temperature:** leave the motors holding for 10 minutes, then touch both motor cases. **They must stay below about 50 °C** (warm, comfortable to hold a finger on). Hotter means the `r_sense_ohms` value does not match the board, or the current is set above 70 % of the motor's rating: power off and check Step 0.

Homing takes up to about 40 s in the worst case. The software allows 180 s for `$H` and 60 s for other commands. [Certain, arithmetic: 1354 mm at 50 mm/s plus 289 mm at 25 mm/s; `motion.home()`]

Step 7: probe test with `G38.3`

Over the empty pocket 0, after homing:

```
G90 G21
G0 X42 Z30
G38.3 Z-6 F1200
```

Expected:

- the cup touches the pad and stops;
- `[PRB:42.000,0.000,<z>:1]`, then `ok`;
- `<z>` is about −2. That is the buffer squeezed 2 mm at the trip point.
- Repeat ten times. The spread of `<z>` should be under 0.1 mm. [Guessing on the spread; it depends on the flag and the pad]

Other checks:

- **Miss test:** `G0 Z30`, then `G38.3 Z25 F1200`. This ends with `[PRB:…:0]` and `ok`. G38.3 does not alarm on a miss. [Certain, FluidNC `mc_probe_cycle`]
- **Fail-safe test:** unplug the probe and send `G38.3 Z20 F600`. You must get `ALARM:4` (probe already tripped at start), and nothing moves. [Certain, FluidNC `mc_probe_cycle`]

- **Soft-limit test:** `G38.3 Z-20 F600` must be refused, because −20 is below the −8 soft limit. [Likely: soft-limit alarm or error, depending on state]
- **Probe hard limit test** (`probe_hard_limit: true`): `G0 Z100`, then send `G1 Z130 F100` (a slow move UP, away from everything) and push the cup up by hand while it moves. FluidNC must stop at once with a critical alarm (`ALARM` with "Probe Hard Limit" in the message). Clear it with a reset (Ctrl-X, or the RESET button) and `$H`. [Certain: FluidNC v4.1.1 `Probe.cpp` raises it when the probe trips in a cycle, jog or homing that is not a `G38` move] If nothing happens, the key was not read: check `$CD` for `probe_hard_limit: true`.

How the software talks to FluidNC

`software/slabscan/motion.py` (`GcodeMachine`) is a plain line-by-line Grbl sender.

Connection

- Opens the CH340 port at 115200 baud. Prefer `/dev/serial/by-id/usb-1a86_USB_Serial-if00-port0` as `serial_port` in `config/calibration.toml`, because `ttyUSB0` can renumber. If the configured port is missing, the software tries `/dev/serial/by-id/*`, `/dev/ttyUSB*`, then `/dev/ttyACM*`.
- Opening the port resets the ESP32, so it waits up to 8 s for the FluidNC start-up banner.
- Sends `$X` (up to 5 tries). The software still refuses every move and probe until it has sent `$H` on this connection, because `$X` marks the axes homed without homing them.

Commands

Purpose	Sent	FluidNC answers
Home	`$H`, then `G90`, `G21`, `G54`, `G92.1`. If `$#` shows a G54 offset, it also sends `G10 L2 P1 X0 Z0`.	`ok` after both homing cycles
Move	`G1 X… F…`, or `G1 Z… F…` **up only**. Never Y. Moves are machine-frame numbers because G54 is zero.	`ok` when queued
Move down	`G38.3 Z… F…` (the Z speed for the fast part, 1200 for the probe window and the rod band): every downward move stops on an unexpected contact	`[PRB:…:0]` (no contact) or `:1` (contact: the run stops)
Halt (Ctrl-C twice)	`!` (feed hold), wait for `Hold:0`, then Ctrl-X (soft reset)	the banner; homing is required again
Wait for done	`G4 P0`, then `?` polled until `<Idle|MPos:x,y,z…>`	`Idle`. `Alarm` or `Door` raises MotionError.
Probe	`G38.3 Z<target> F1200`	`[PRB:x,y,z:1]` (hit) or `:0` (miss), then `ok`
Work offset	`$#` (reads `[G54:…]`)	All zeros after `home()`

The probe report prints machine coordinates. [Certain, `report_probe_parameters`]

What the software does not use

- The vacuum, valves and lights are on the Pi's GPIO, not FluidNC. There are no `user_outputs`, so the `M64`/`M65` method in motion.py must not be called.
- Any `error:` or `ALARM:` reply raises `MotionError`. The runner then re-reads the real position and parks: a held slab is put back on its stack, lifting straight up first. If the controller is in Alarm (for example a probe hard limit), nothing moves: the pump keeps holding a slab on the cup and the operator takes it (CLI prompt or the web Vent button).
- The serial port is opened exclusively. A second slabscan process gets `ControllerBusy` (and the machine lock refuses it first): stop `slabrail-web` before running CLI machine commands.

How the software handles power loss and limits [Certain, `motion.py`, `runner.py`]:

- **E-stop = controller power off.** The E-stop cuts the Rodent's 24 V (WIRING.md §1), and the CH340 is powered from the board. So the serial device **disappears**, and pyserial raises `SerialException`.
  - `GcodeMachine` turns that into `ControllerLost`. The runner stops without moving. If a slab is on the cup, the Pi keeps the pump and valve switched on; after an E-stop they have no 24 V, so the slab hangs only on the closed cup line until it leaks down. The message says "The controller lost power or its USB link (E-stop pressed, or the cable came out) …" and where the slab came from.
  - The next command reconnects (it waits up to 15 s for the port), waits for the banner, and must home again before anything moves.
- **Probe targets** are clamped to machine Z −7.5 (`z_min + 0.5`), inside the −8 soft limit. `calibrate z` probes no lower than −7.0 by default.
- `[PRB:…]` has three numbers before `:1`/`:0`, so the regex matches. [Certain]

Open issues

1. **Resolved: Z rail margins.** [Certain, CAD] `Z_CARRIAGE_ABOVE_CUP` is now 295 mm (the 285 mm drop arm), with printed hard stops at c = −14 and c = 262 (see "Where the travel numbers come from"). `checks.txt` shows 5 mm of carriage on the rail at the bottom stop and 3.6 mm at the top stop.
2. **[Likely] Sense resistor value by board version.** See "Read this first". If BTT publishes a V1.1 config, compare it with this file.
3. **[Likely] Optical endstop boards differ.** Some have an indicator LED circuit that can hold the Rodent's opto on. The Step 5 fail-safe rows catch this, so never skip them.
4. **[Guessing] Z hold when unpowered.** DESIGN.md estimates the head stays put on E-stop (detent torque > back-drive). Test it (BUILD_GUIDE §7.8): E-stop with the cup about 15 mm above a foam block and 200 g on the cup, then watch for 1 minute. If it creeps, add a friction brake or counterweight before unattended use (the BOM's conditional "Z counterweight" line).
5. **[Likely] Z torque margin at 0.8 A.** The Z motor runs at about half its rated current, at 750 rpm, and FluidNC has no stall detection. A stall while lifting a slab leaves the head lower than the software thinks. The carry-height sensor (on the Pi's GPIO 6, not on the Rodent: WIRING.md §6a, BUILD_GUIDE §7.4a) now stops a run before any sideways move once a stall has left the head more than about 2.5 mm low [Certain, CAD and simulator]; it does not prevent the stall. BUILD_GUIDE §7.8 has a 100-lift test (`python -m tools.bench_tests lifts 100`): the bare pad must re-probe within 0.1 mm. If not, lower Z `max_rate_mm_per_min` here and `feed_z` in `calibration.toml` to 3000, then raise Z `run_amps` to about 60% of the motor's rating (motor under 50 °C). Only if it still fails, buy the BOM's conditional Z motor. A closed-loop motor is not a drop-in here: the Rodent has no step/dir header for one. [Likely]

Operating SlabRail

Weekly use: loading, running, reading the results, unloading, maintenance, and what to do when it stops. The build and the acceptance tests are in BUILD_GUIDE.md. Commands are run from `~/psa-slab-scanner/software` (BUILD_GUIDE §11.2 puts the project there) with the venv active (`source .venv/bin/activate`).

Confidence tags: **[Certain]** = read in the code (`software/slabscan/`) or the CAD. **[Likely]** = strong inference. **[Guessing]** = not yet measured on a real machine.

Read this first

- **Only use it on inventory after every BUILD_GUIDE §13 test has passed.** [Certain that none of this has run on real hardware yet] Until then, every batch is a test batch.
- **`certs.csv` has one row per slab you loaded.** [Certain, `runner.py`] When two slabs come up together (DOUBLE PICK), the read one gets its row and the one underneath gets an extra `no_read` row whose note starts `UNREAD:`. That slab is in the **output pocket**, directly under the read one, not in pocket 12 (§5.4). Always check the count line in the summary.
- **Slabs without a back-label code stop the run until the learned check is active.** [Certain, `runner._check_face_down`] Before the cup touches a slab, the head camera must prove it is face-down: a code printed only on the back label (a PSA back QR, a CGC back QR, or the CGC 2020–23 back barcode), or, in the default `learned` mode, a card window that matches the card backs of slabs proven earlier (§8.4). The learned part works only after a Phase 0 GO, 5 references and a clear view of the card window on 5 proven slabs; today's head leaves that window clear [Certain, CAD]. Until then, PSA 2016 and older labels, CGC Authentic, BGS and SGC slabs stop the run until you check each one (§8.1): 14 of 64 real slabs in a public-scan sample [Certain, RESEARCH.md §4]. CGC 2023+ and 2020–23 slabs no longer stop. For stacks you checked yourself, start with `--all-face-down` (§3).
- **A slab whose front code is visible from above is FACE-UP, and the run says so.** A CGC front-label barcode or a CGC Authentic QR seen from above stops the run untouched with a "FACE-UP" message. A PSA front proves nothing either way (its barcode is printed on both labels).
- **A Z stall stops the run before the head moves sideways.** [Certain, `runner._carry_check`; tested in the simulator] The carry-height sensor on the Z spine must read "at carry height" before every sideways move, so a stall that left the head more than about 2.5 mm low is caught (a smaller one still clears the rods). The run then stops, and a held slab is lowered straight onto its own pocket, or held on the pump for you (§8.1).
- **Only the E-stop removes motor power.** Ctrl-C twice halts the motors at once and nothing moves afterwards, but they stay powered. [Certain, `runner.hard_stop`]
- **A wrong cert is worse than no cert.** The software refuses to guess: two codes that disagree give `conflict`, and a barcode-only read (ITF or Code128, any grader) with nothing to confirm it gives `needs_review`. Even two barcodes that agree get their printed digits checked by OCR. An `ok` also needs its grader confirmed: a code that names it (a cert URL), two printed codes in its format, or an 8-digit ITF whose digits OCR of the printed number confirms (that names PSA). So **a pre-2016 PSA slab is `ok` only when OCR agrees with its front ITF**, its only code; when OCR reads nothing or other digits it is `needs_review`. Any other single barcode stays `needs_review` even when OCR agrees. A number read only by OCR has no grader and always goes to pocket 12. Respect those statuses. Don't paste `needs_review` certs into your inventory unchecked.
- **The web page has no login.** [Certain, `web.py`] Use it only on your own home or shop network. Never port-forward it.

Contents

1. Before each run

2. Loading rules
3. Starting a run
4. What the machine does
5. Reading the results
6. Unloading
7. Maintenance
8. When it stops
9. Quick card

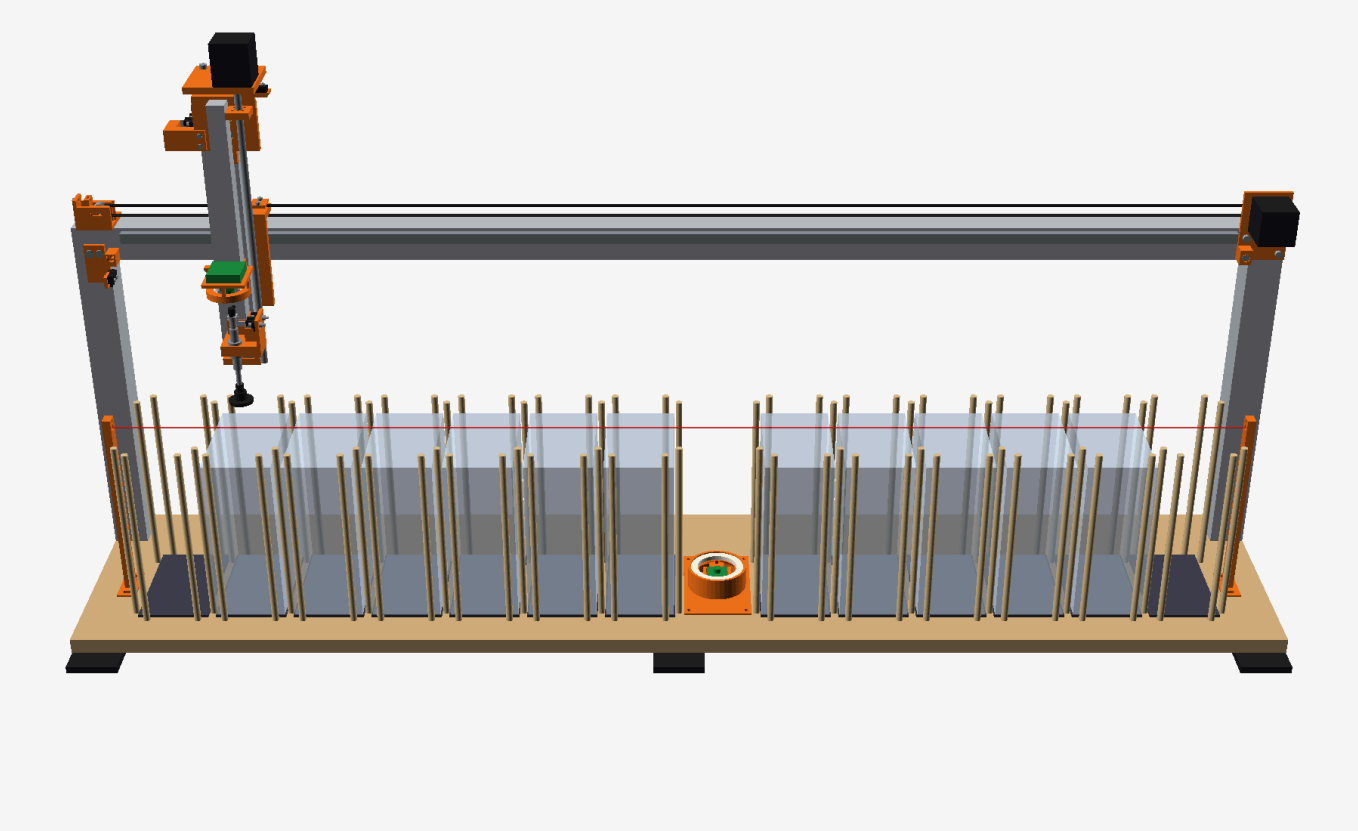
---

1. Before each run

(2 minutes)

- ☐ **Wipe the cup lip with isopropyl alcohol** on a lint-free wipe, then let it dry for a minute. [Certain, DESIGN.md] The cup is the only part that touches your slabs.
- ☐ Pads clean: no grit, no lint. Use a lint roller.
- ☐ **Pocket 0 and pocket 12 are empty.** The run refuses to start otherwise. [Certain, `runner.survey`]
- ☐ Nothing is lying on top of the stacks, and nothing is above the fill line.
- ☐ Light roof on, if you fitted one. Lid closed, if you fitted the lid switch.
- ☐ Nothing in the carry-height sensor's slot on the Z spine (a card or dust there stops the run at the first pick).
- ☐ E-stop released. Hands out of the bed.
- ☐ Using the command line? Stop the web service first (`sudo systemctl stop slabrail-web`): only one slabscan program may use the machine at a time, and the second one is refused.

2. Loading rules



1. Load: 11 stacks of up to 30, face-down, label toward you. Pocket 0 (left) and pocket 12 (right) stay empty.

Rule	Why
Face-down (label and card facing the pad, the back facing up). A hard rule.	The cup only touches the back. Before the cup touches a slab, the head camera must see a back-label code, or (learned mode, once active) a card back it knows. If it can't (a face-up slab; a slab with no back-label code: PSA 2016 and older, CGC Authentic, BGS, SGC) the run stops with nothing touched until you check it (§8.1). With `--all-face-down` that check is off and a face-up slab gets the cup on its front. [Certain, `runner.py`]
Label end toward the front (toward you)	It keeps the arm out of the camera's view of the label. [Certain, CONTRACT]
Remove all sleeves and bags	A smooth sleeve may well seal under the cup [Guessing], but the slab can then slide out of the sleeve during the 2 m/s² moves, and nothing detects a sleeve. A sleeve also slides on the stack.
Pockets 1–11 only. Pocket 0 and pocket 12 start empty.	Stack 1 moves into pocket 0. Pocket 12 collects exceptions. [Certain]
Fill to the gauge, never above it. Stand a `fill_gauge` in the front finger gap, its foot on the bed, against the stack. **The underside of its top bar is the MAX line.** The notches mark 0, 5, 10 … 30 slabs of 6.6 mm; the top notch (198 mm) is *not* the MAX line.	The bar underside is 210 mm above the bed, so 207 mm above the pad: the gauge errs 3 mm on the safe side because the bed sits 3 mm below the pad top. About 30 standard PSA slabs, or about 27 CGC. [Certain, `model.fill_gauge`, `layout.py`] Higher, and the overfill beam blocks the start.
Each slab flat and centred between the rods. Lower bundles of 5–10 with your fingertips in the front and back gaps (52 mm wide), following the bundle down; let go only when it rests on the stack. Never drop a bundle in.	A slab sitting on a rod stops the run ("taller than expected"). [Certain]

Rule	Why
PSA and CGC can be mixed, in any order and any holder thickness	Both are read and stacked normally. [Certain, `certs.py`] Measure one CGC holder before the first CGC batch: it must be ≤ 82.6 mm wide. CGC 2023+ and 2020–23 "OG" slabs are proven face-down by their back QR. CGC Authentic slabs carry their codes on the front only, so they stop at the face-down check like old PSA labels until the learned check is active. [Likely, CAD card-plane check] The card window has about zero margin on CGC: a CGC slab pushed into a pocket corner can have about 1.5% of the window on the holder (up to about 4.5% at most); PSA slabs stay on the card. Audit learned CGC rows (§8.4).
BGS and TAG: measure one first. Load them only if the holder is ≤ 82.6 mm wide (the pocket is 84 mm). BGS labels carry no code, so a BGS slab stops at the face-down check (until the learned check is active) and ends in pocket 12 as `no_read` or as an OCR-only `needs_review`; `other_grader` appears only if a BGS, SGC or TAG URL is read.	[Guessing] BGS holders may be wider than PSA's, and no width here has been measured. A wider holder rubs or jams on the dowels at every pick and placement.
SGC: don't load. Its holder is about 86 mm wide and the pocket is 84 mm.	[Likely, `layout.py`; measure one of yours] It won't go between the rods. Scan SGC by hand.
Thick PSA holders are OK. They are read and flagged "thick holder".	Fewer fit under the line, so trust the gauge, not a count.

You don't need to sort, flip or order anything. The CSV records exactly where every slab ends up (§6).

3. Starting a run

Command line:

```
python -m slabscan run --name "show-stock-oct"                # asks one question, then runs
python -m slabscan run --name "high-value" --front-photo always  # also photographs every front (+~3 s/slab)
```

- `--front-photo always` is worth using on expensive slabs. A photo of both faces is your record against swapped cards and fake labels. [Likely, RESEARCH.md]
- `--all-face-down` turns the face-down check off for this batch (it is remembered on resume). Use it only when you loaded every stack yourself and checked it, for example a batch of older labels that would otherwise stop at every slab. A slab that is face-up after all then gets the cup on its front. The web page has the same choice as a checkbox.
- **The run refuses to start** until `calibrate z` and `calibrate x` have been done. [Certain, `cli.py`]
- **The start line `face-down check: …` says what the check does this run** [Certain, `runner.face_down_status`]:
  - `back-label codes only - every slab without one (a PSA back QR, a CGC back QR or the CGC 2020-23 back barcode) stops for you` (`face_down_check = "stop"`);
  - `learned check not active yet: <why>. Until then every slab without a back-label code (…) stops for you, as in 'stop' mode` (the default mode, not ready yet: normal in the first batches after Phase 0);
  - `learned check not active: over an empty pocket the head camera sees edges where only the pad should show … Re-aim the head camera (BUILD_GUIDE 7.7) …` (the camera-aim check failed, §8.4; the same in every stop of that run);
  - `learned check not active: the head camera's aim is not checked this run yet …` (a resume whose survey found no empty pocket: §8);
  - `learned (threshold 0.56, 23 references in 1 prototype(s) …): a slab without a back-label code is picked only if its card window matches a known card back` (§8.4);

- `OFF - a slab loaded face-up gets the cup on its front` (`--all-face-down`, or `face_down_check = "off"`).

Web page:

1. On the Pi, run `python -m slabscan web`, or install the boot service with `bash setup_pi.sh --service`.
2. On your phone, open `http://<pi-address>:8080/`. The command prints the address.
3. Type a batch name.
4. Choose **front photo: only when needed** or **every slab (+3 s)**.
5. Press **Start run** and confirm.

The page shows:

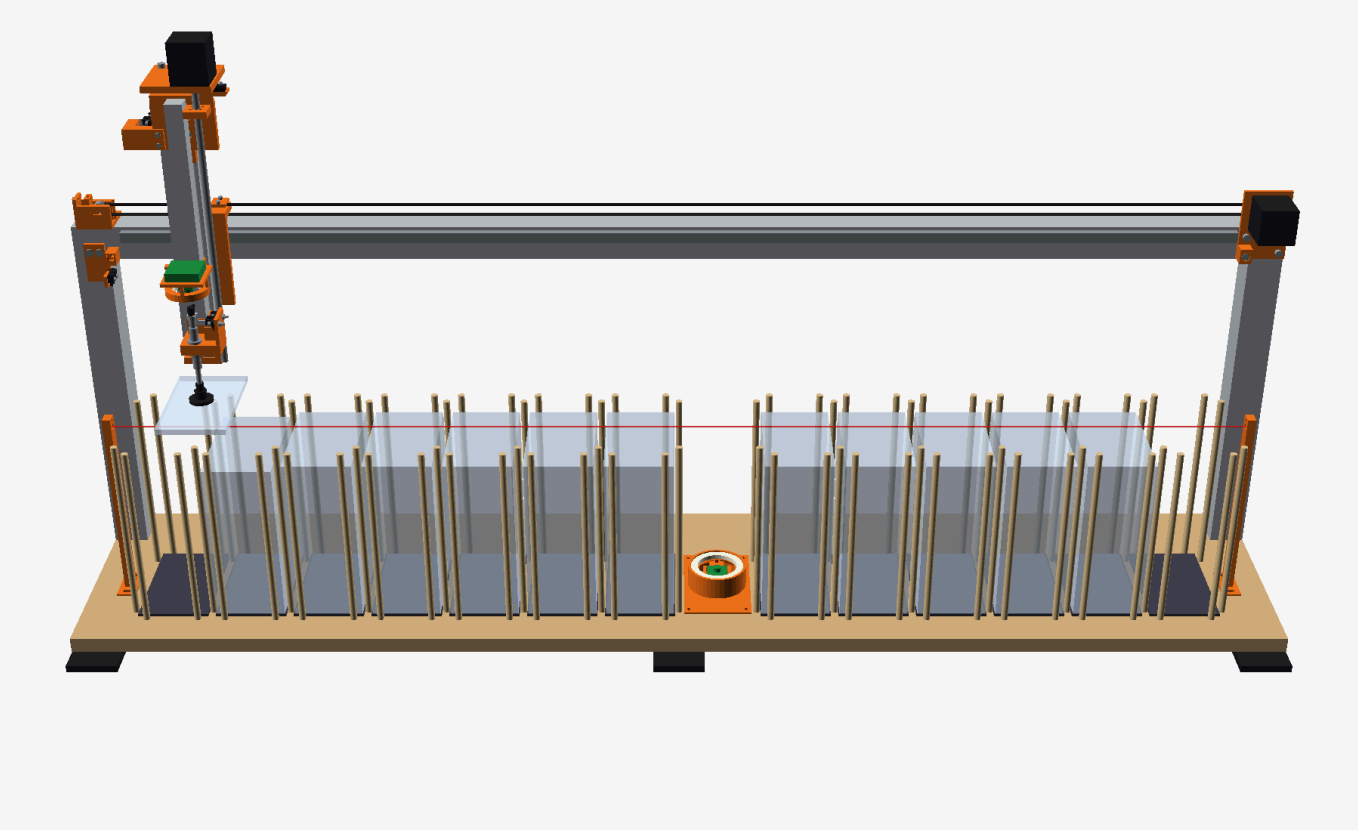
- the counts, the current pocket and the last label photo;
- the last 10 slabs;
- the stop message;
- downloads of `certs.csv` and `certs.txt`.

Stopping on purpose

You do	The machine does
Stop after this slab (web), or Ctrl-C once	It finishes the slab in the cup, parks with nothing held, and halts. Resume later. [Certain]
Ctrl-C twice	**Hard stop.** The motors halt where they are (feed hold, then a controller reset) and **nothing moves after that**. They stay powered. A slab in the cup stays on it, held by the pump; the program asks you to hold it and type `vent` (web: the Vent button), and you put it face-down on top of the pocket it names. The next run or resume homes first. [Certain, `runner.hard_stop`, `cli.hand_off_slab`]
E-stop	It cuts the 24 V: motors, pump and valves all stop. **The only stop that also removes motor power**: use it for anything dangerous. A held slab then hangs on the closed cup line for seconds to minutes only: take it off right away. §8.1.

4. What the machine does

1. **Homes:** Z straight up, then X to the left switch.
2. **Survey** (1–3 min): it probes the top of every pocket. [Certain, `runner.survey`; time from the simulator]
   - It checks that pockets 0 and 12 are empty and that nothing is above the MAX line.
   - It measures every stack height.
3. **Shuffle one pocket left.** Stack 1 is moved slab by slab onto pocket 0, stack 2 onto pocket 1, and so on to stack 11 onto pocket 10.



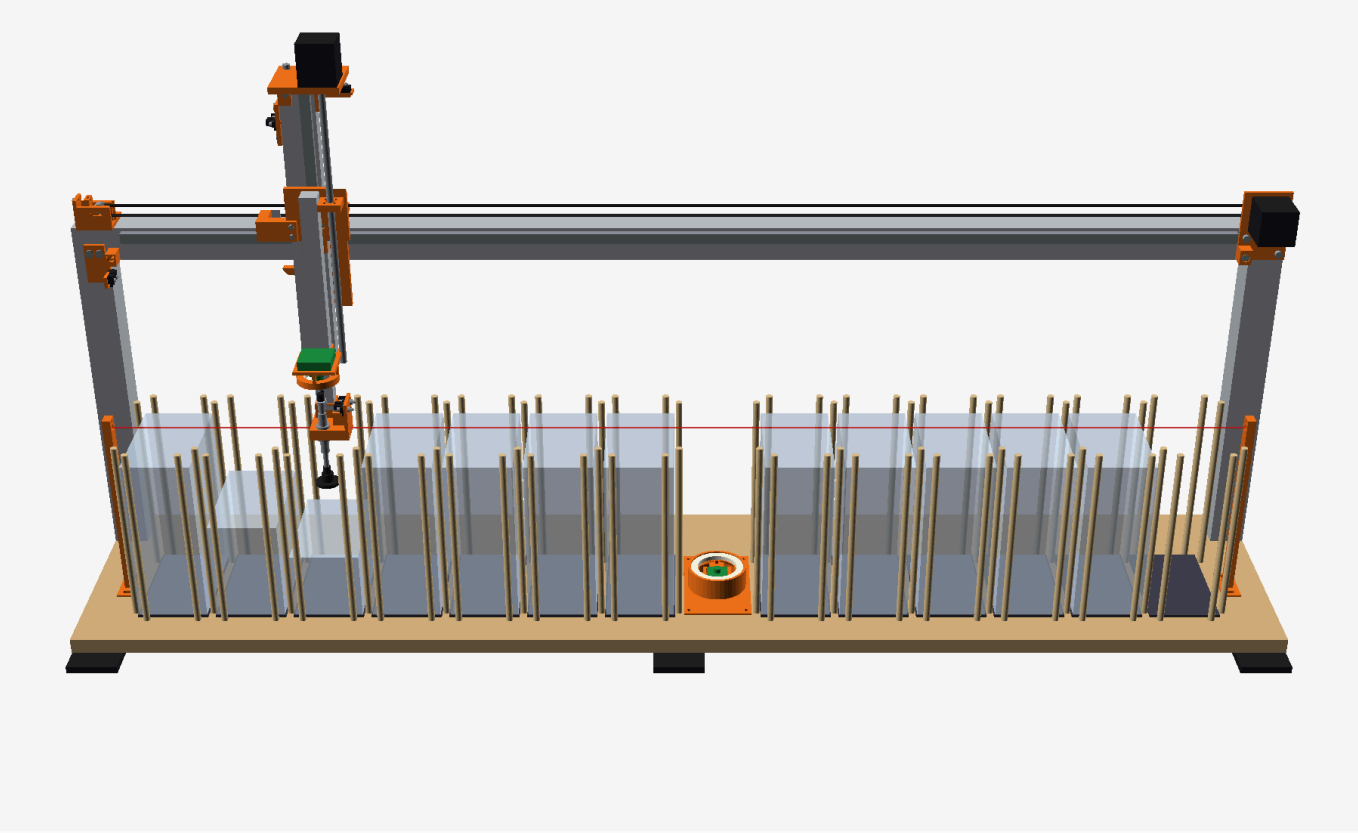
2. The cup lifts the TOP slab of stack 1 straight up, reads it, and sets it down in empty pocket 0.

For every slab [Certain, `runner.transfer`]:

1. A look from just above the top slab: the face-down check (a back-label code must be visible, or in learned mode a card window that matches a known card back) and a first read of the back. In learned mode the survey also takes one head frame over the first empty pocket (`images/face_down_empty.jpg`): every card window is compared with it, so a head part in the way is noticed, and the camera's aim is checked on it. A failed aim check logs a `CAMERA AIM (pocket N, …)` line and turns the learned check off for that run. [Certain, `runner._take_baseline`] A resume whose survey finds no empty pocket takes that view over the first source pocket it empties instead (§8).
2. A slow probe down to it. Every downward move is a probing move, so anything unexpected in the way stops the head.
3. The cup line must read atmosphere, then the cup seals.
4. The lift is slow (2 mm/s) through the point where the cup leaves the slab, then pauses 0.5 s just above the stack, so the slab underneath stays behind.
5. It lifts to carry height, 248 mm, and takes two more photos of the back. Before it moves sideways, the carry-height sensor must confirm the head really is up there (a stalled Z motor would leave it low). [Certain, `runner._go`]
6. **If the back can't be read** (old label, glare), or the read is barcode-only (no QR), it carries the slab over the upward camera in the middle of the row and photographs the front, then decodes it and runs OCR on the printed digits, with the slab over its output stack. OCR that reads other digits makes the slab `needs_review`. For a pre-2016 PSA slab (front ITF only), OCR that agrees is what makes it `ok`. [Certain, `runner.needs_ocr`, `certs.psa_itf_confirmed`]
7. It probes down onto the destination stack (slowly past the rod tops), measures the slab's thickness from the contact height, vents the cup and checks the vent worked before it lifts.

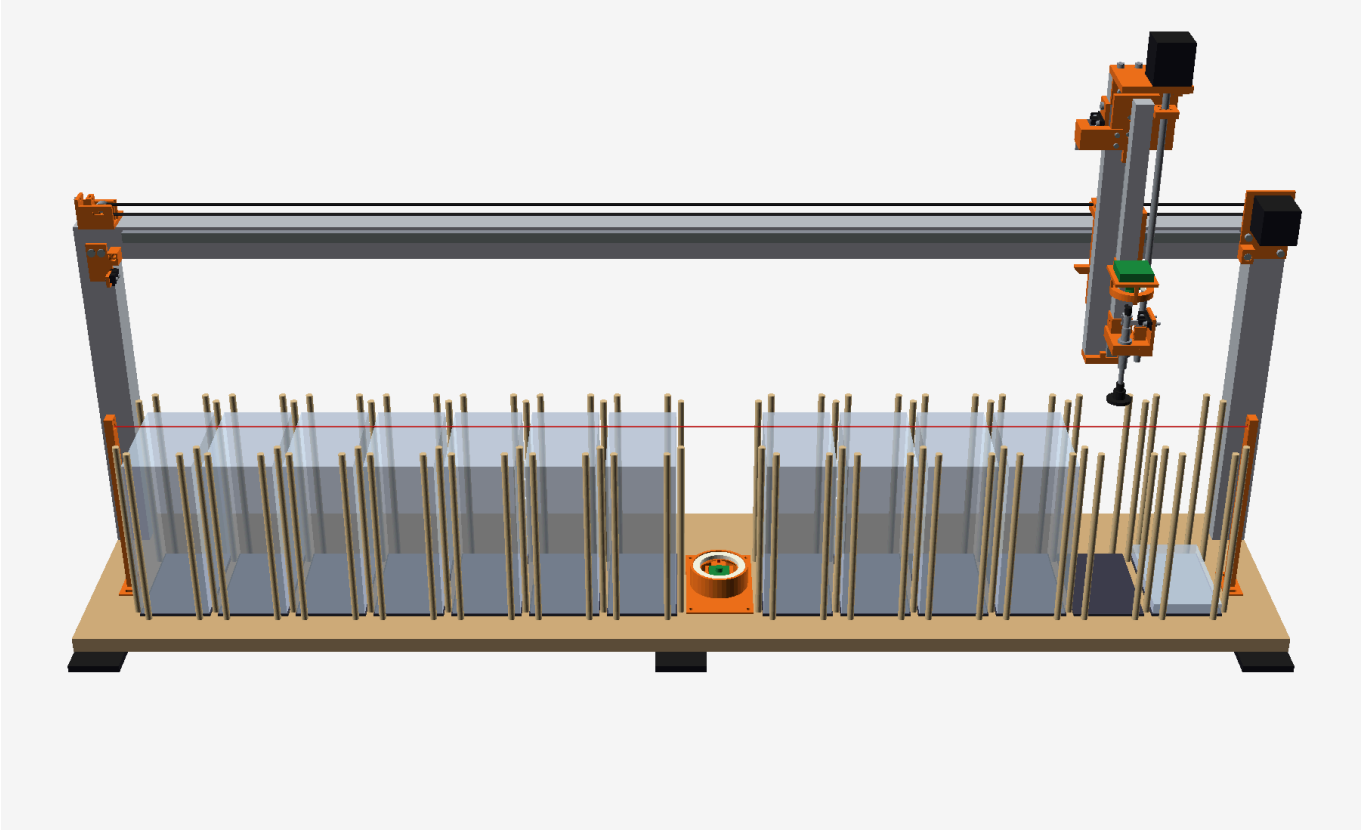
Where each slab goes [Certain, `runner.transfer`]:

- **`ok`, and `needs_review` with grader PSA or CGC,** go onto the next output stack.
- **`needs_review` with any other grader** (`unknown`: OCR only, or a number with no grader's format; two graders, `PSA/CGC`; none) goes to **pocket 12**, whatever `needs_review_to` says: it may not even be PSA or CGC.
- **A grader guessed from a barcode's length** (note "grader … not confirmed") counts as PSA or CGC, so that slab stays in the output stack. Another grader's slab whose barcode has a PSA or CGC format can therefore end up there too. [Certain, `runner.transfer`] Check every `needs_review`.
- **The same goes for an `ok` from the 8-digit ITF rule.** Another grader's slab with an 8-digit ITF whose printed number OCR reads would come out `ok PSA`, with the right digits and the wrong grader, in the output stack. [Guessing that such a slab exists: none of the 64 real slabs had one] See §5.2.
- **A code in no grader's format is not read as a cert:** a 7-digit Code128, a 10-digit one starting with 0 (a BGS-style number), or an ITF of any length but 8. It is listed as unparsed, so a slab with nothing else readable ends `no_read` (or OCR-only) in pocket 12. [Certain, `certs.parse_payload`]
- **`no_read`, `conflict` and `other_grader`** go to **pocket 12**, except the `UNREAD` row of a double pick: that slab came along under a read one and sits in the output pocket (§5.4).



3. Repeat until stack 1 is gone - then stack 2 moves into the now-empty pocket 1, and so on.

When a stack is empty, the next stack moves into the pocket it just left.



4. Done: every slab read, every stack one pocket to the left (order reversed). Rejects wait in pocket 12.

At the end:

- stacks 1–11 sit in pockets 0–10, each one pocket left of where you loaded it;
- pocket 11 is empty;
- pocket 12 holds the slabs to check by hand;
- the head parks over pocket 0.

Speed: about 14 s per slab, so about 79 minutes for a full 330-slab load in the timing simulation (`sim --slabs 330`). [Guessing until your first runs] Slabs that need the front camera and OCR take up to 30 s longer, and every face-down-check stop waits for you. BUILD_GUIDE §13.1 measures the real figure, which appears as `mean cycle` in the summary.

5. Reading the results

Everything is in `batches/<batch id>/`:

File	What it holds
`certs.csv`	One row per slab, written row by row as it runs, so a crash loses nothing.
`certs.txt`	Only the `ok` certs, one per line, in machine order. Paste it into a spreadsheet or Card Dealer Pro.
`report.html`	Every row with its label photo(s). Opens on a phone.
`summary.json`	Counts per status, duplicates, the per-stack count check, `needs_attention`, `itf_mode` (always `zero_pad`: the ITF is the cert padded to 8 digits) and `itf_evidence` (slabs where QR and ITF agreed), mean cycle time, and `face_down` (§5.3), which in learned mode holds `camera_aim`.
`images/`	`NNNN_back.jpg` for every slab, plus `NNNN_front.jpg` when the front camera was used. NNNN = `position`. In learned mode also `face_down_window.jpg` (the run's first slab with the card window drawn in red) and `face_down_empty.jpg` (the view over an empty pocket).

File	What it holds
`journal.jsonl`, `run.json`, `stop.json`	Machine log for `resume` (including the face-down events, §8.4), the run options, and why it last stopped.

5.1 `certs.csv` columns

[Certain, `results.COLUMNS`]

Column	Meaning
`batch_id`	Folder name: date-time plus your `--name`
`position`	1, 2, 3 … the order the machine handled the slabs
`source_pocket`	The pocket the slab was loaded in (1–11)
`source_index`	Its place in that stack when loaded: **1 = top**
`out_pocket`	The pocket it is in now (0–10, or 12 for exceptions)
`out_index`	Its place in that pocket now: **1 = bottom**, so the top slab has the highest number
`cert`	Cert number, leading zeros stripped
`grader`	`PSA`, `CGC`, `BGS`, `SGC`, `TAG` or `unknown` (an OCR-only number, or a code with no grader's format)
`status`	See 5.2
`sources`	Which codes agreed, joined with `+`: `back_qr`, `back_barcode`, `front_barcode`, `front_qr`, `ocr`. A QR's tag follows the label side its payload proves, whichever camera read it: `back_qr` is a PSA or graded CGC QR, `front_qr` the CGC Authentic QR.
`raw_payload`	Exactly what the code said, e.g. the QR's URL
`note`	Why it isn't plain `ok`, plus flags like "thick holder", "thin/odd step" and "DOUBLE PICK"
`image_front`, `image_back`	Photo paths, relative to the batch folder
`scanned_at`	Local time
`face_down`	How the slab was proven face-down before the cup touched it: `qr` (a PSA or CGC back QR seen from above), `code` (the CGC 2020–23 back barcode), `operator` (your `--face-down-ok`), `learned(0.93)` (card-window match, with its score: check its photo, §8.4) or `off` (check off). Empty on a double pick's `UNREAD` row.

5.2 Statuses

[Certain, `certs.consensus` and `runner.transfer`]

Status	Meaning	Where the slab is	What you do
`ok`	A machine-readable code read with confirmation, and its grader confirmed: a QR, or a barcode on two printed symbols (back and front label) or confirmed by OCR of the printed digits, plus a code that names the grader (a cert URL), two printed codes in its format, or an 8-digit ITF whose digits OCR confirms (grader PSA; note "barcode confirmed by OCR; grader PSA from its 8-digit ITF": a pre-2016 PSA slab). A two-barcode `ok` has had its digits checked by OCR too; if OCR read nothing, it stays `ok`.	output stack	Nothing. It's in `certs.txt`.

Status	Meaning	Where the slab is	What you do
`needs_review`	A cert was found but isn't trusted. The note says why: "grader PSA not confirmed: no code names it and only one printed barcode has its format (OCR confirms its digits, not its grader) - check it is a PSA slab" (one Code128 only, often after "barcode confirmed by OCR"), "one printed barcode only (no QR) - confirm" (for example a pre-2016 PSA slab whose OCR read nothing), "barcode N but OCR read M - verify by eye" (OCR read other digits), "OCR only (no barcode or QR read), grader unknown - check by hand", "grader not confirmed (a code with no grader's format)", "read only in the frame from above the stack (a neighbour's label?)", "N-digit PSA cert, but Phase 0 saw only … digits", or **DOUBLE PICK**. Any note can start "dropped a phantom read Code128:…": a thin false read inside a real barcode, dropped only when a different code or OCR confirms the other number. If the confirmed number is the thin read itself, the slab is a `conflict` instead.	**output stack** (the default `needs_review_to = "output"`) if the grader is PSA or CGC; **pocket 12** otherwise	Check the photo against the slab, and type the cert in yourself.
`no_read`	Nothing decoded on either face	pocket 12. **Exception:** a row whose note starts `UNREAD:` is the hidden slab of a double pick; it is in the output pocket the row names (§5.4)	Scan or type it by hand. Look at the photo: glare? damaged label?
`conflict`	Two codes on the slab disagree (e.g. QR ≠ barcode), the cert was already recorded earlier in this batch, or the back-label code that proved it face-down before the pick was another cert ("… this slab may have been FACE-UP and the cup may have touched its front - check it by hand")	pocket 12	**Check it carefully.** It can be a misread or a neighbour's label in view, but it's also what a doctored label or a reused cert number looks like. For the last kind, also look at the slab's front for a cup mark.
`other_grader`	A BGS, SGC or TAG URL was read	pocket 12	Handle outside SlabRail.

`needs_review` slabs with grader PSA or CGC stay in the output stack by default. To send them to pocket 12 too, put `needs_review_to = "exceptions"` in `config/calibration.toml`. [Certain, `config.py`, `runner.transfer`]

What `needs_review` means. The machine found a number, but one piece of the evidence it needs for an `ok` is missing or disagrees: a second printed code or OCR for the digits, or a code that confirms the grader. The note names which. On the 64 real slabs only 2 pre-2016 PSA slabs ended there on the images as they are (OCR read nothing on one and other digits on the other), and none at the machine's resolution. [Certain for that set; `tools.real_eval`]

Two settings change what can be `ok` (BUILD_GUIDE §11.5) [Certain, `runner._decide`, `certs.parse_payload`]:

- `psa_cert_digits`: a PSA cert of a length not in the list is `needs_review`. Lengths count without leading zeros, so **include 7** if you have pre-2016 slabs: a cert printed `02074012` has 7 digits. Leave it out, and every 7-digit pre-2016 cert becomes `needs_review`.
- `custom_regex`: with it set, every ITF is typed by your pattern, so the 8-digit ITF rule never fires and every pre-2016 PSA slab is `needs_review` ("grader PSA not confirmed"). Safe, but set it only if Phase 0 found a prefix or suffix around the cert.

The 8-digit ITF rule rests on one assumption [Guessing]: that only PSA prints an 8-digit ITF. In the real set only PSA labels carry ITF, but that set has just 1 BGS and 1 SGC slab, and neither showed a code. BGS and SGC are reported to print a barcode or a QR on some holders, of unknown type [Likely, second-hand]. If you deal other

graders, put a few of their slabs through Phase 0 (or photograph them and check the photos with `python -m tools.real_eval`, BUILD_GUIDE §1.7) before you trust `ok PSA` on slabs with no QR: an `ok PSA` on a BGS or SGC slab means the rule is wrong for them. Tell the software owner.

5.3 The summary lines

Shown at the end of the run and in `summary.json`:

- **Repeated certs:** the second slab with a cert already in the batch becomes a `conflict` row (note "cert … was already recorded at #N in this batch") and goes to pocket 12. Either you loaded the same slab twice, the camera saw a neighbour's label, or a fake reuses a real cert number. [Likely, RESEARCH.md] Pull both slabs and compare them. The `duplicates` line counts what is left after that rule and is normally "none".
- **`count check:`** for each stack, the slabs counted against its surveyed height ÷ the measured slab thickness. "Every stack matches" means within one slab. Anything else says "check these stacks by hand". [Certain, `runner.count_check`]
- **`needs attention:`** every non-`ok` row with its pocket and position counted from the bottom. That is your to-do list.
- **`face-down:`** how the batch's slabs were proven face-down, e.g. `face-down: 287 by back QR, 41 by the learned card-back match, 2 confirmed by you` (a CGC 2020–23 back barcode adds "N by a back-label barcode"). If any were learned, it adds the lowest learned score and its position: check the back photos of the learned ones. In `report.html` the Face-down column shows the same per slab, with learned rows in bold. `summary.json`'s `face_down` holds the mode, the counts, every learned row (position, pocket, score, back photo), the references, this run's stops and `camera_aim`: `ok`, what was wrong, "not checked" when no view over an empty pocket exists yet, or "not checked this run (the resume survey found no empty pocket, and no source pocket was emptied after it)". [Certain, `cli.face_down_line`, `runner._face_down_summary`]
- **`CAMERA AIM: …`** (only when the aim check failed): "The learned check accepted nothing this run. Re-aim the head camera (BUILD_GUIDE 7.7)". [Certain, `cli.summary_lines`]

5.4 Double picks

Sometimes two slabs cling together and come up as one. The machine can't see the lower one, but it notices at set-down: the contact step is ≥ 11.6 mm instead of about 6.6. [Certain, `t_double`]

- The read slab's row gets `needs_review` and the note **"DOUBLE PICK: an unread slab is directly under this one in pocket N"**.
- **The slab underneath gets its own row:** status `no_read`, the note starting **`UNREAD: lifted together with #position`**, the same `position` number as the read slab, the same `out_pocket`, and `out_index` one lower. [Certain, `runner.transfer`]
- **It is in the output pocket, not in pocket 12**, directly under the read slab.
- **To do:** take that slab out and scan it by hand (or put it in the next batch).
- The count check still matches, because the machine counts a double as two. [Certain, simulator]

5.5 The exception pocket

- **What lands there:** every `no_read`, `conflict` and `other_grader` slab (except the `UNREAD` rows of double picks), in the order they arrived. Its rows have `out_pocket = 12`; `out_index` 1 is the bottom.
- **When it fills:** it holds about 28 before the machine stops with "Exception pocket 12 is full". [Certain, arithmetic: it stops when the next two thick holders would cross the MAX line] Empty it and resume (§8). Counting there restarts from what is left.

6. Unloading and matching rows to slabs

Where the stacks are: stack *k* is now in pocket *k − 1* (stack 1 in pocket 0 … stack 11 in pocket 10). Pocket 11 is empty and pocket 12 holds the exceptions. **Each stack's order is reversed:** the slab that was on top is now at the bottom (`out_index` 1), and the **top slab of every pocket has that pocket's highest `out_index`**. [Certain, by construction]

Never lift a whole stack out or turn it over. A full pocket is about 200 mm and 2 kg of slick slabs, and fingertips in the gaps only reach their top ends: slabs slip out of the middle, and their fronts slide across each other's backs.

How to unload one pocket:

1. Have a tray or box ready next to the machine.
2. Take slabs off the **top**, **at most 5 at a time**, fingertips in the front and back gaps.
3. Put each bundle in the tray **face-down, as it came out**, each new bundle behind (or under) the previous one, so the order is kept.
4. The first slab you take is that pocket's highest `out_index`; count down from there. In `certs.csv`, sort that pocket's rows by `out_index` from high to low: that is the order you unload them in.
5. Do pocket 12 the same way.

To find one slab (for example a `needs_review` row):

- Look up its `out_pocket` and `out_index`.
- Count up from the bottom of that pocket (`out_index` 1), or down from the top (the highest `out_index`). A DOUBLE PICK's `UNREAD` row has its own `out_index` (one below the read slab), so there are no gaps.
- Compare the slab with its `image_back` photo in `report.html` before you act on it.

Next batch: unload every pocket. The next run needs pocket 0 and pocket 12 empty again.

7. Maintenance schedule

When	Task	Why / how
Every run	Wipe the cup lip with IPA (§1)	Grit on the lip is the main scuff risk. [Certain, DESIGN.md]
Every run	Look at the cup lip	Replace it if it's nicked, flattened or hardened
Every run	Lint roller over the pads	Grit under a stack's bottom slab
Weekly, or every ~1,000 slabs	Wipe the pads' microfiber with a slightly damp microfiber cloth, then let them dry	DESIGN.md: pads weekly
Weekly	`python -m slabscan selftest -i`	Watch the trends: seal time and the `leak` kPa/s. A creeping leak means a fitting or the cup is going.
Weekly	Check the vacuum tube's slack at the cup elbow, and the Z cable loop	The buffer rod rises up to 20 mm on contact (WIRING.md §11)
Monthly, or every ~5,000 slabs	**Belt:** deflection check, a 100 g sideways pull at mid-span moves the lower strand 13–23 mm (BUILD_GUIDE §6.6). Check the pulley grub screw too.	If you re-tension, re-run `calibrate x`.
Monthly	Rails: wipe, then a little lithium/PTFE grease on both	Dry rails get notchy. [Likely]
Monthly	Spot-check frame bolts, the gate post screws, the camera bracket and the lens lock nut	Vibration. Re-run `focus head` if the lens moved.
Monthly	Contact-sensor check: `selftest -i` probe trigger; push the cup up by hand and feel for the flag rubbing in its guide	The flag can bend in a knock (BUILD_GUIDE §7.6)

When	Task	Why / how
Monthly	Carry-height sensor: blow its slot clear, then `selftest --motion` → `carry sensor … trips at about 246.x mm`	Dust in the slot reads "blocked" and stops the run; a mount that slid moves the trip (BUILD_GUIDE §7.4a)
Monthly	**E-stop check:** machine idle, press the E-stop. The software must report the controller lost, and a meter on the 24 V switched bus must read 0 V. Release, `home`.	The contact switches the 24 V capacitor inrush and can weld closed without any sign (WIRING.md §1.3). If the bus stays live, stop using the machine and replace the contact blocks.
Replace cup pads	When the lip is damaged, the seal gets slow, the leak grows, or after any slab shows a mark. Spare SMC ZP25UU pads are in the BOM.	**Re-run `calibrate z` after every pad change**: the new pad's face height differs. [Likely]
Replace pad facing	When the microfiber is worn shiny or stained	Re-glue a new facing (BUILD_GUIDE §5.5)
After any crash or a knock to the head	`calibrate z`, `calibrate x`, `focus head`, `focus station --save`, `selftest -i --motion`	Positions and focus may have moved

Keep a copy of `config/calibration.toml` off the Pi.

8. When it stops

When the machine stops on its own:

- the message says what happened and what to do;
- it's saved in `batches/<id>/stop.json` and shown on the web page.

What happens to a slab in the cup [Certain, `runner.secure`, `cli.hand_off_slab`]:

- **After a planned stop or an unexpected error** (a camera dropping off USB, a decoder bug, a controller error), the head first re-reads its real position, lifts straight up, and puts the slab back on top of the stack it came from, so it gets read again. If the vacuum is low, it carries the slab back slowly and probes it down.
- **After a Z stall** (the carry-height sensor read clear at travel height), the head may be lower than the controller thinks, so the slab is never carried sideways. Over its own pocket the head goes straight down in one probe move (it stops on contact, whatever the real height) and vents the slab there. Anywhere else nothing moves, and the pump holds the slab for you. [Certain, `runner._set_down_after_stall`]
- **When moving is unsafe**, nothing moves at all: Ctrl-C twice, the controller lost (E-stop or USB), a controller ALARM, a failed vent, a vacuum sensor fault, a slab already hanging on the cup. The pump keeps holding the slab while the Pi has power. The command line then asks you to hold the slab and type `vent` (the web page shows a **Vent** button); put it face-down on top of the pocket the message names. Never just close the program with a slab on the cup: the valves then shut and it hangs only until the line leaks down.

After fixing the cause: `python -m slabscan resume batches/<id>`, or **Resume** on the web page. Resume asks "hands clear?" first (skip it with `--yes`). Resume:

1. reconnects and homes;
2. probes every pocket again and compares it with what the journal expects;
3. continues.

In learned mode, empty pocket 12 before you resume. The camera's aim verdict counts only for the run that took it, because the camera could have been knocked while the run was stopped. The resume survey checks the aim over an empty pocket. If no pocket is empty (normal mid-run, unless you emptied pocket 12), the learned check accepts nothing and learns nothing until the run has emptied its next source pocket and looked over it: until then every slab without a back-label code stops for you ("… the head camera's aim is not checked this run yet …"). [Certain, `runner.survey`, `runner._work`; tested in the simulator]

Every slab still appears exactly once. [Certain, in the simulator: `tests/test_resume.py`, 8 passed when I ran it on 2026-10-22]

8.1 The table

Messages are quoted from the code, shortened where marked "…". [Certain]

Message (start of it)	What happened	What you do
Before the start		
"… has no pad_trigger_c / offset_x yet. Run `calibrate z` and `calibrate x` first"	Not calibrated	BUILD_GUIDE §12.1–12.2
"Close the lid to start."	Lid switch open	Close it
"The overfill light beam is blocked with the head at the top: a stack is above the MAX line, something is lying across the stacks, or the laser and its receiver are out of line …"	Something is taller than 214 mm, or the beam is out of line	Check every stack with the fill gauge and look for a slab lying on top. Then `selftest -i` for the beam.
"The cup line holds vacuum … (… kPa; an empty cup reads …): a slab may be stuck to the cup. Nothing will move, and the pump is switched on …"	A slab is left on the cup, or the sensor zero drifted	Hold the slab, vent (the prompt, or the web Vent button), put it face-down on top of the stack it came from, then start again. If the cup is empty: `selftest -i` stores a new zero.
"Another slabscan process is using the machine … (stop it with: sudo systemctl stop slabrail-web) …"	The web service or a second window has the machine	Stop the other one first
During the survey		
"Pocket N (empty_start / exceptions) must start empty but is … mm tall."	Something is in pocket 0 or 12	Empty it and restart
"Pocket N is filled above the MAX line …"	Overfilled	Take slabs off and restart
"Pocket N: probe never touched anything (not even the pad)."	The contact sensor didn't trip	Check the flag and the sensor (`selftest -i`). **Don't resume until the probe test passes.**
"Pocket N: the cup sealed on something but the contact sensor did not trip. Contact sensor fault …"	The second, vacuum-based contact check caught a dead sensor	Same as above
"Pocket N: the cup sealed (… kPa) before the head moved down - a slab is hanging on the cup. Nothing will move …"	A slab stuck to the cup from before	Hold it, vent (prompt / Vent button), put it face-down back on the stack it came from, start again
While moving slabs		
"Pocket N: the top slab could not be confirmed FACE-DOWN before touching it (no back-label code was seen from above (a PSA QR, a CGC QR or the CGC 2020-23 back barcode)[; a CGC QR without a cert number …][; learned check: …]). … If it is face-down (older PSA labels, CGC Authentic, BGS and SGC slabs have no back-label code), resume with --face-down-ok …"	A face-up PSA slab (its front barcode proves nothing), or a slab with no back-label code that the learned check could not match (the reason after "learned check:" is in the rows below)	Look at the top slab of pocket N. Face-up: turn it face-down, then `resume`. Face-down: `resume --face-down-ok` (web: tick "the top slab of pocket N, where the face-down check stopped, is face-down"; the tick only shows after a face-down stop). It confirms that one slab only, and only right after this stop. For a batch of older labels you checked, start it with `--all-face-down`.
"Pocket N: the top slab is FACE-UP (a CGC front-label barcode (…) / a CGC Authentic front-label QR (grade 0_0) (…) is visible from above, and that code is printed on the FRONT label). … Turn the top slab of pocket N face-down, label end toward you, then resume. Only if it already is face-down, resume with --face-down-ok …"	A CGC slab lies face-up	Turn it face-down and resume. If it really is face-down, confirm it: it is picked, but never learned from.

Message (start of it)	What happened	What you do
"… a … back-label QR … and a … front-label … are both in view: another slab's label is in view", "… two different back labels are in view (A, B): another slab's label is in view" or "… codes of two different certs are in view …"	A neighbour's label is inside the head camera's crop	Look at the top slab of pocket N and its neighbours. If it is face-down, confirm it. If this happens often, check `cam_head.hfov_deg` (BUILD_GUIDE §12.3).
"… learned check not active yet: face_down_threshold is not measured (Phase 0 measures it)"	No Phase 0 GO saved	Phase 0 with `--face-up 15 --save` (BUILD_GUIDE §1.7), or set `face_down_check = "stop"`.
"… learned check not active: face_down_threshold was not measured for this card window (face_down_threshold_sig …)"	The card window changed since Phase 0, or the threshold was typed in by hand	Run Phase 0 again with `--save`
"… learned check not active yet: N of 5 card-back references" or "… checked on 5 slabs proven face-down on this machine …, N so far"	The first batches: not enough code-proven (or confirmed) slabs placed yet	Normal. Every back-code slab whose read is `ok` adds one once it is placed.
"… learned check not active: over an empty pocket the head camera sees edges where only the pad should show, in the card window or within 1 mm of its cup side (… mm2 …): a part of the head is in the window, or about to be. The head camera is tilted toward the cup, or something lies on the pad there (lint, dust, a tag): look at the picture first. Re-aim the head camera (BUILD_GUIDE 7.7), then compare images/face_down_empty.jpg (pocket N) with images/face_down_window.jpg. Dirt on that pocket's pad looks the same: clean it"	The camera-aim check failed on this run's view over an empty pocket: the camera is tilted toward the cup (with today's head a part enters the window at about 1.3°), or something lies on that pad. The learned check accepts nothing and learns nothing this run. [Certain, `facedown.aim_problem`]	Confirm the slab as above. Open `images/face_down_empty.jpg`: clean the pad if it shows lint or a tag; otherwise shim the camera board (BUILD_GUIDE §7.7) and run `focus head`. The next run checks again.
"… learned check not active: the head camera's aim is not checked this run yet: its last check is from before (a resume whose survey found no empty pocket), and a knock since then would not be seen. It is checked over the first source pocket this run empties; until then every slab without a back-label code stops for you"	A resume found no empty pocket at its survey, so it could not check the camera's aim yet. The learned check accepts nothing and learns nothing until the run has emptied its next source pocket and looked over it (journal `face_down_baseline`). [Certain, `runner.AIM_STALE`]	Confirm the slab as above. To avoid more of these stops, empty pocket 12 before that resume (§5.5): its survey then checks the aim at once.
"… N% of the card window shows the same thing on every proven slab as over an empty pocket (allowed 10%): a part of the head is in the way" or "… the card window looks like the view over an empty pocket (…): the head hides it"	A head part is in the camera's view of the card window: a camera tilted toward the cup (with today's head a part enters at about 1.3°), or a tube or cable hanging in the view [Certain, CAD aim check]	Confirm the slab as above. Compare `images/face_down_window.jpg` with `face_down_empty.jpg`; shim the camera board (BUILD_GUIDE §7.7, §12.3).
"… only N known card back(s) of one design match it (0.xx); a match needs 2 slabs of that design at the threshold 0.yy (one odd reference, or a card front that looks like one, is not enough)"	One reference matched, a second did not: one odd back (a card sitting low in its holder), or a front that looks like one	Look at it. Face-down: confirm it.
"… best match with a known card back 0.xx < threshold 0.yy"	A face-up slab, a card back the machine has not learned (Japanese, another game), or a sleeve or sticker over the window	Look at it. Face-down: confirm it.
"… it matches a card back (0.xx) known only from N of your confirmations; it is used once 3 confirm it"	A back design taught only by your confirmations	Confirm it if it is face-down; after 3 agreeing confirmations that back is used
"… the card window shows far less pattern (texture …) than that card back …: a plain card front, or glare"	Very likely a face-up plain card, or a glare spot	Look at it carefully before confirming

Message (start of it)	What happened	What you do
"… nothing visible in the card window (blank, dark or washed out)", "… glare covers N% of the card window" or "… the card window is not inside the frame"	Head ring off or glare on the window; or `cam_head` / `face_down_window_mm` wrong	Check the head ring and `face_down_window.jpg` (BUILD_GUIDE §12.3)
"Z stall suspected <where>: the carry-height sensor (GPIO 6) reads CLEAR, but the controller has the head at … mm, where the flag on the Z carriage must block its beam (from 246.5 mm up). Nothing moved sideways. …" plus "The slab that was in the cup was lowered straight down onto pocket N, where it came from …" or "The slab from pocket N is still on the cup … it was not carried anywhere: nothing moves, and the pump keeps holding it …"	The Z motor lost steps while lifting (a bind, a heavy pull, too little current). Right after homing, the message blames the sensor lead instead: a cut or unplugged lead, a bent or missing flag, a mount that slid. [Certain, `runner._carry_check`]	If a slab is still on the cup: hold it, vent (prompt / Vent button), put it face-down on top of pocket N. Then, power off, move the head up and down by hand and feel for a bind; check the Z current (BUILD_GUIDE §7.8); `python -m tools.bench_tests lifts 20`; `selftest --motion`. Resume homes first.
"Carry-height sensor fault <where>: it reads BLOCKED with the head at … mm, far below its trip height (246.5 mm). …"	Dirt or a card in the sensor's slot, a dead emitter LED or a bent flag: then a Z stall could not be seen	Clean the slot, check the flag (BUILD_GUIDE §7.4a), `selftest --motion` (clear below 245.5, blocked above 247.5), resume.
"Unexpected contact over pocket N at … mm while moving down to … mm [with a slab in the cup]: something is higher than it should be …"	A crooked, added or dropped slab, or a slab lying on the rods	Look before you touch anything. Fix it, then resume.
"Pocket N is … mm taller than expected - was a slab added or is one sitting crooked?"	A slab is on a rod, or someone added one	Seat it flat (or take the added slab off), then resume
"Pocket N: no contact where the stack should be (predicted … mm). Check the contact sensor, then resume."	The stack is lower than expected: a slab was removed, or the sensor is weak	Count the stack against the CSV. `selftest -i`. Resume.
"Pocket N: cup will not seal on the top slab (dirty cup, warped slab?)."	No seal, even after a second try	Wipe the cup with IPA. Check the slab is face-down, flat and unsleeved. Resume.
"Vacuum lost after peel / in transit / at the read station (… kPa). A slab may have dropped." plus one of: "… It was carried back slowly and set down on top of pocket N …", "… the slab was NOT on the cup any more: it fell somewhere between pocket N and where the head was …", or "… nothing was found on top of pocket N. The head waits at the safe height with the pump on."	Vacuum dropped while holding. The machine never lets go in mid-air: it carries the slab back slowly over its own pocket and probes it down. [Certain, `runner._set_down_weak`]	Do what the second part says. Find and check any dropped slab. Run `selftest -i` (leak) before resuming.
"Vent failed …: the cup line still reads … kPa …" or "A slab came up with the vented cup …"	The vent path (GPIO 22 module, coil or wire) did not let go. The slab rests on the stack, still sealed to the cup; nothing moves.	Hold the slab and pull the vacuum tube off the cup fitting (or unscrew the cup). Never slide the slab sideways. Fix the vent, `selftest -i`, resume.
"Vacuum sensor fault: … Without it the machine cannot tell whether a slab is held or let go, so nothing moves."	The sensor reads outside its range (an open wire, a lost 5 V)	If a slab is on the cup, hold it and vent. Fix the sensor wiring (WIRING.md §5), `selftest -i`, resume.
"Exception pocket 12 is full. Empty it …, then resume."	About 28 exceptions	Unload pocket 12 (§6) and resume. It recounts from what's left.
"Placing onto pocket N: no contact - output stack lower than expected."	The output stack is lower than recorded	Check that stack for a fallen or removed slab, then resume
"Placing onto pocket N: contact … above the stack - something is in the way."	An object, or a crooked slab, on the output stack	Clear it, then resume
"Placing onto pocket N: the cup touched … mm above the stack, as if nothing is in the cup. The slab from pocket N may have fallen off on the way …"	The slab fell off, or the vacuum sensor reports a seal that isn't there	Look at the cup and the bed (rods, stacks, read station). Find the slab, then resume.

Message (start of it)	What happened	What you do
"Lid opened - run paused with nothing held."	Lid opened	Close it and resume
"Stopped on request - nothing held. Resume to continue."	You pressed Stop	Resume when ready
Controller and power		
"The controller lost power or its USB link (E-stop pressed, or the cable came out). The head stopped where it was. If you did not press the E-stop, the motors may still be powered: press it now before reaching in. [A slab from pocket N is still on the cup …]"	E-stop, or the USB came out. The Pi keeps the pump and valve switched on; after an E-stop they have no power, so the slab hangs on the closed cup line only for a short while. [Certain, `runner.lost_note`]	Press the E-stop if you didn't. **Take the slab now** (after an E-stop the vent has no power either: hold it and pull it off gently), or vent at the prompt / Vent button if the pump still runs. Put it face-down on top of pocket N. Release the E-stop, then resume. It reconnects and homes first: Z straight up, then X.
"The controller is still not answering … check the USB cable and the 24 V supply"	The CH340 didn't come back within 15 s	Wait 10 s and resume again. Check the E-stop is released, the 24 V supply, and the USB cable.
Anything else		
"Hard stop (Ctrl-C twice): the motors were halted where they were and nothing moved after that. … [A slab from pocket N is on the cup; the pump keeps holding it …]"	You stopped it hard	Hold the slab, type `vent` at the prompt (web: Vent), put it face-down on top of pocket N. Resume homes first.
"Interrupted (Ctrl-C) during the stop itself. The head may have stopped anywhere and a slab may be in the cup …"	Ctrl-C while the machine was putting a slab back	Check the head. If a slab is on it, the prompt asks you to vent. Then resume.
"Unexpected error: <ErrorType>: …. The slab that was in the cup was put back on top of pocket N; it will be read again on resume."	An unplanned error, e.g. a camera dropping off USB or a decoder bug. The slab was parked.	Fix the cause (§8.3), then resume.
"… Could not park the machine automatically (…). A SLAB from pocket N is still on the cup; the pump keeps holding it …"	The put-back failed, or the controller is in ALARM (for example "Probe Hard Limit")	Hold the slab, vent (prompt / Vent button), put it face-down on top of pocket N. Check the head, fix the cause, resume. A probe hard-limit alarm during a lift means the flag chatters or something touched the cup: BUILD_GUIDE §13.2.
"<ErrorType>: … The machine stopped without parking …"	Rare now: an error before the run's own guard started (for example while opening the hardware)	Check the head anyway, then resume.
On resume		
"The stacks changed since the run stopped: pocket N: … mm now, … mm when the run stopped (±n slabs) …"	A slab was added, removed or moved while stopped	Put things back as they were. If it says "It looks like one slab went from pocket A to pocket B …", move that top slab back as told. Or use `resume --accept-heights` if you changed things on purpose: the count check will flag it.
"exception pocket 12 was emptied to … mm; counting continues from …"	Info: you emptied pocket 12	Nothing
"resume: the run stopped before the first survey finished - starting it from the beginning"	Info	Nothing
"--face-down-ok refused: the batch did not stop at the face-down check last time, so there is no slab for it to confirm. Nothing moved. …"	The last stop was not a face-down stop	Resume without it. If a slab the check can't confirm comes next, the run stops there and you confirm that one.
"--face-down-ok was for the top slab of pocket N, but the next slab is in pocket M: not used" (log line)	Info: the confirmation is only for the slab it stopped on	Nothing; if pocket M's slab can't be confirmed either, the run stops there

8.2 Venting the cup by hand

If the software is running and asks you to type `vent` (or the web page shows **Vent**), use that. Only if a slab is stuck on the cup and the software isn't running, hold the slab with one hand and open the vent valve for a second:

```
pinctrl set 22 op dh; sleep 1; pinctrl set 22 op dl
```

GPIO 22 is the vent valve. [Certain, CONTRACT] `pinctrl` is the Raspberry Pi OS GPIO tool WIRING.md §15 uses. [Likely] Don't do this while `slabscan` is still running, because both would drive the same pin. With the E-stop pressed the vent valve has no power: release the E-stop first, or hold the slab and pull the vacuum tube off the cup fitting.

8.3 Repeated stops of one kind

Pattern	Look at
"cup will not seal" on many slabs	A cup lip worn or dirty; a leak (`selftest -i` → `leak`)
"Vacuum lost in transit" more than once	A leak in the cup line; the pump; whether the slab is knocking a rod (watch a carry)
"taller than expected" on random pockets	Slabs placed crooked: X calibration (`calibrate x`), belt tension, a leaning rod
Camera errors	The head camera's USB extension in the drag chain (WIRING.md §16 risk 4)
Many `no_read`	Focus (`focus head`, `focus station --save`), a ring not lighting, glare. Fit the light roof.
Many pre-2016 PSA labels `needs_review`: "one printed barcode only (no QR) - confirm" or "barcode N but OCR read M"	Their front ITF is their only code, and it is `ok` only when OCR of the printed number agrees. OCR read nothing or other digits: the `tesseract` line of `selftest`, `focus station --save`, the station ring. [Certain, `certs.consensus`]
Every pre-2016 PSA label `needs_review` "… grader PSA not confirmed …", or "N-digit PSA cert, but Phase 0 saw only …"	`custom_regex` is set (it turns off the 8-digit ITF rule), or `psa_cert_digits` leaves out 7 (§5.2)
After a resume, every slab without a back-label code stops ("… aim is not checked this run yet …")	No pocket was empty at the resume survey. Empty pocket 12 before resuming (§8).
A BGS, SGC or other non-PSA slab reads `ok PSA`	The 8-digit ITF rule is wrong for that grader (§5.2). Treat its `ok PSA` rows as `needs_review` and tell the software owner.
`CAMERA AIM:` on every run	The head camera is tilted toward the cup, or a pocket pad has lint or a tag on it: BUILD_GUIDE §7.7, §12.3
A face-down-check stop at every older PSA label (and CGC Authentic, BGS)	Expected until the learned check is active (the start line says why). Check the stacks yourself and start such batches with `--all-face-down`.
"Z stall suspected" more than once	The Z axis: a bind (lead screw, nut, rail), the motor current, the motor itself (BUILD_GUIDE §7.8 lift test). Not the sensor, if `selftest --motion` passes.
Learned acceptances with scores just above the threshold	Open their back photos (§8.4). If one shows a card front, set `face_down_check = "stop"` and check that slab: the cup touched its front.
Every slab goes to the front camera (slow runs, many "confirmed by OCR")	The head crop is wrong: `cam_head.hfov_deg` / `row_axis` don't match the real camera, so the label is cut off (BUILD_GUIDE §12.3)
"Probe Hard Limit" alarms	The probe flag chatters at the slot edge during the slow lift, or something touched the cup: BUILD_GUIDE §13.2, WIRING.md §6

8.4 The learned face-down check

[Certain for how it works, from `slabscan/facedown.py` and `runner.py`; Guessing for how well it works on real slabs: it has only run in the simulator]

What it does. Most slabs without a back-label code show the same card back through the plastic (every English Pokémon card has the same back). The machine compares the card window (`face_down_window_mm`, a 55 × 5.5 mm strip on the card next to the label) with the card backs of slabs it has seen proven face-down. A match lets the cup pick the slab without stopping. A face-up slab shows its card front there instead.

The three modes (`face_down_check` in `config/calibration.toml`):

- `"learned"` (the default): a back-label code, or a card-window match. Until the check is active it acts exactly like `"stop"`.
- `"stop"`: only a back-label code proves it. PSA 2016 and older labels, CGC Authentic, BGS and SGC slabs stop every time.
- `"off"`, or `run --all-face-down` for one batch: no check. A face-up slab gets the cup on its front.

When it is active. All of these must hold; the start line says which one is missing. [Certain, `facedown.learned_ready`]

1. Phase 0 measured a threshold on your slabs and said GO, and `--save` wrote it with its `face_down_threshold_sig` (BUILD_GUIDE §1.7). No threshold below 0.55 is used.
2. At least 5 card-back references exist.
3. The view over an empty pocket was taken this run: during the survey, or, on a resume whose survey found no empty pocket, over the first source pocket it empties.
4. On the newest 5 or more slabs proven on this machine, at most 10% of the card window looked the same as over the empty pocket. More means a part of the head is in the view.
5. The camera-aim check found no edge in the middle of the card window, or within 1 mm of its cup side, on this run's view over an empty pocket (the verdict is stored with that view in `card_backs.npz`). [Certain, `facedown.aim_problem`]

- **Today's head leaves the window clear** [Certain, CAD frustum check: nothing in view at hover over stacks of 1, 15 and 30 or with a slab held]. In the 330-slab simulation the check became active after the first slabs and stopped for 2 of 330 (43 without it) [Certain, sim].
- [Certain, CAD aim check] Camera aim: a tilt of more than about 1.3° toward the cup puts the cup pad into the window (1.34° in the CAD check, 1.31° with the software's own mapping). Condition 5 catches that tilt from about 1.3° in the simulator [Certain, sim]. It does **not** see:
  - a head part as dark as the pad (no edge against it; condition 4 missed such a part over 9–26% of the window in the simulator);
  - a tilt away from the cup, or a roll along the row: they move the window toward the card top or off its side. On a CGC slab pushed into a pocket corner about 0.01° toward the label already puts over 2% of the window on the holder [Likely, CAD card-plane check]. Its thresholds were set on the simulator's flat pad, so lint or dust on a real pad can make it fire on a square camera: that is safe (the check stays off). A head edge about 1 mm inside the window raised face-up scores in the simulator from 0.37 to 0.52 (hence the 0.55 floor). Check `face_down_window.jpg` on your first run (BUILD_GUIDE §12.3).

A match needs two references. A slab is accepted only when two references from two different slabs of the same back design reach the threshold, not just the best one. [Certain, real slabs: one odd back, a card sitting low in its holder, matched a real blue Water-type front at up to 0.62 under glare; with the second reference no face-up front came above 0.44, at the cost of about 6% more face-down slabs stopping]

Where the references come from.

- Slabs proven by a back-label code seen from above, but only after the slab is placed, only if its read is `ok` with the same cert as that code (a neighbour's QR in view gives another cert), and never if the read station saw a back-label code from below. Each refusal is logged: "not used as a reference: the read station saw <a PSA back-label QR | a CGC back-label QR | a CGC 2020-23 back-label barcode> from below, so this slab was FACE-UP", "… its read is '…', not 'ok' …" or "… the back-label code seen from above (cert …) is not this slab's cert …".
- Slabs you confirmed with `--face-down-ok`, after the same placement checks, and never if a front-label code was in view from above.

- A back design known only from your confirmations is used once 3 of them agree. One wrong confirmation alone teaches nothing.
- The Phase 0 backs, if Phase 0 said GO and you used `--save`.
- **Never from a learned acceptance**, so the check cannot teach itself a wrong appearance. Never from a window with glare on more than 2% of it, with too little pattern, or that looks like the empty-pocket view.
- A new back design starts with one slab, so it matches nothing until a second slab joins it: one odd reference can only ever be one of the two votes.
- Up to 6 back designs (prototypes) with the newest 8 references each are kept.

Where they live: `config/card_backs.npz`, next to `calibration.toml`. Not in git.

```
python -m slabscan refs                    # list: prototype, source, batch and position, cert, back photo
python -m slabscan refs --drop 12345678    # delete the references taken from the slab with this cert
python -m slabscan refs --drop-operator    # delete every reference you confirmed (keeps the back-QR ones)
```

Deleting the file starts over, including the empty-pocket view. A file written for another card window is not used. [Certain, `facedown.CardBackRefs`]

Auditing learned acceptances. Do it for every learned row in the first batches.

- `certs.csv`: `face_down` = `learned(0.93)`; `report.html` shows those rows in bold in the Face-down column. Open each back photo: it must show a card **back**.
- The summary line names the lowest learned score.
- `journal.jsonl`, one line per event:
  - `face_down_learned`: score, its pattern (`ncc`) and colour (`hist`) parts, prototype, threshold, references, texture, and similarity to the empty-pocket view;
  - `face_down_reference`: each reference taken from a slab you confirmed (position, cert, back photo, result);
  - `face_down_confirmed`: your confirmation and what became of it;
  - the stop line carries `face_down_stop` (the pocket) and why;
  - `face_down_baseline`: the empty-pocket view, with its camera-aim verdict (`aim`). After a resume with no empty pocket it comes mid-run, when the first source pocket is emptied; `face_down_ok_unused`: a confirmation that did not apply.
- **After a wrong `--face-down-ok`** (you confirmed a slab that was face-up): the cup touched its front, so check that slab. Then delete what it taught: `refs --drop <its cert>` (the cert is in its `face_down_reference` line).

Limits. [Certain, round-3 window] On studio scans of 63 real slabs, placed where the machine's camera would see them, clean frames gave 0 face-up fronts accepted and all 14 no-code backs accepted (with one reference); glare raised the scores of some fronts, which is why a match now needs two. With two, 223 of 252 face-down frames (glare included) and 0 of 332 face-up frames were accepted at 0.55. [Likely, CAD emulation of the same scans] Today's smaller window accepts about 101–103 of 126 face-down backs and 0 of 206 face-up slabs (highest face-up 0.49–0.53, under the 0.55 floor). The larger round-3 window let 12 of 206 face-up CGC fronts through with every slab at the cup end of its play: the strip of holder above a CGC card looks the same from both sides. [Guessing] The real camera, through slab plastic, under your lights, has not been measured; nor have sleeves and stickers, and 15 face-up slabs in Phase 0 say little about the worst face-up slab. Japanese backs need their own references (two back-code slabs with that back, or 3 of your confirmations).

9. Quick card

Print this and tape it to the machine.

```
BEFORE   cup wiped (IPA) - pockets 0 and 12 empty - nothing above the gauge's top bar - sleeves off
LOAD     face-DOWN (hard rule), label toward you, pockets 1-11, bundles of 5-10 lowered by hand
         PSA+CGC mixed OK.  BGS/TAG only if <= 82.6 mm wide (go to 12).  NO SGC.
```

```
START    python -m slabscan run --name NAME   (web: Start run)
         start line "face-down check: ..." = what the check does (learned, or back-label codes only)
         slabs without a back-label code (PSA 2016 and older, CGC Authentic, BGS) stop the run until
         the learned check is active: check, then resume --face-down-ok (only right after that stop;
         that one slab), or start a checked batch with --all-face-down
         "FACE-UP" stop = a CGC front code seen from above: turn that slab face-down, resume
         "Z stall suspected" = head low: take a held slab as told, check Z, selftest --motion
         "CAMERA AIM" / "sees edges where only the pad should show" = re-aim the head camera or clean that pad
STOP     web "Stop after this slab" / Ctrl-C once = clean stop.
         Ctrl-C twice = motors halt NOW, nothing moves; a held slab stays on the pump -> hold it, type vent.
         E-STOP = the only stop that cuts motor power. Use it for anything dangerous.
RESUME   python -m slabscan resume batches/<id>    (or web: Resume)
         learned mode: empty pocket 12 first, so the camera aim is checked at once
AFTER    stack k is now in pocket k-1, order reversed: top slab = highest out_index.
         Unload from the TOP, max 5 at a time, face-down into a tray. Never lift or flip a stack.
         Pocket 12 = check by hand (incl. OCR-only rows).  needs_review = check by hand
         (a pre-2016 PSA slab is ok only when OCR confirms its front ITF).
         DOUBLE PICK = an extra UNREAD row: that slab is in the OUTPUT pocket, under the read one.
         face_down = learned(0.xx) -> open its back photo.  refs / refs --drop CERT = references.
WEEKLY   pads wiped, selftest -i.   MONTHLY  belt, rails, bolts, E-stop check, carry sensor (selftest --motion)
NEW CUP PAD -> calibrate z
```

SlabRail bill of materials (October 2026)

The line-by-line list with links is in `BOM.csv`. The same parts, grouped into the three orders you actually place, are in SHOPPING_LIST.md. This page has the totals, the order plan, where to save and where not to, the tools, and the design gaps found while pricing.

Confidence tags: **[Certain]** = primary source (for a price: the product page itself, opened on 22 Oct 2026), **[Likely]** = several secondary sources or a dated listing, **[Guessing]** = estimate.

Bottom line

- **The full list costs about $2,182 with a 15% allowance.** [Likely ±10%] You have said $2,500–3,000 is fine if the machine works properly, so there is about $300–800 of headroom.
- **The total did not change this round.** The two homing switches now have printed mounts. They need 4 more M5×10 button heads, 4 more M5 T-nuts and 4 × M2×12 with nuts, all from packs already in the list, and about 33 cm³ more filament. The switch line now says to buy the short hinge lever. In round 4 two lines moved between orders and the confidence tags were aligned; in round 3 the list went up $209 ($182 of parts plus the allowance), almost all of it real product-page prices replacing estimates (below).
- **$1,383 of the $1,897 subtotal is a price seen on the product page** [Certain, 22 Oct 2026], $175 is [Likely] and $339 is [Guessing] (mostly the Misumi extrusions, which block scripts: get that quote first, and Amazon lines whose price was not visible). The allowance covers shipping already seen but not in the line prices (LiMo DDP $39, Digi-Key Marketplace $28, MakerStock $13.49, the endstop boards $5.70) and the ones not shown (ELP express and duty, Proax shipping and brokerage, AFT's small-order fee, Misumi).
- **The worst case fits too.** [Guessing] Add the tools you will probably have to buy ($45–50), the conditional lines Phase 0 is designed to trigger (HT1 cups and the light roof, about $77 with the allowance), and the [Guessing] lines coming in 20% high (about +$78): about **$2,380**, still under $2,500.
- **About $1,862 if the school supplies its usual shop stock** (filament, wire, glue, fuses, common screws, the flag sheet). [Likely] It is the cheapest headroom: the same worst case then comes to about $2,050.
- **About $1,787 lean:** school stock, plus generic direct-acting valves instead of Clippard. [Likely] (The bed is already the Home Depot birch panel: the Baltic birch blank is no longer listed online.)
- **Keep a running tally** (a spreadsheet: line, quoted, paid). Order 1 (Phase 0) is quoted at ≈ $626 and order 2 at ≈ $1,556. If order 1 comes in more than about 10% over its quote, re-quote order 2 before buying it.
- **Spend the headroom only when a test asks for it.** The conditional lines are there for that: HT1 cups if Phase 0B leaves a mark, the light roof if Phase 0A shows glare, and one upgrade per acceptance test that can fail (below). Adding parts before the tests doesn't make the machine work better. If every conditional line triggered at once, the total would reach about $3,330; that many failed tests would mean the design needs rethinking, not more parts. [Guessing, arithmetic]
- **Not in any total: junk graded slabs for the acceptance tests** (15–20, some deliberately dropped). Take your lowest-value slabs from stock and write off 2–3 of them, or budget $150–400 to buy them (BUILD_GUIDE §13).
- **Things that add the most by themselves:**
  - buying every tool new (about +$460, total about $2,640: over $2,500, inside $3,000);
  - the genuine HIWIN rail tier (about +$420, conditional line);
  - a branded pump with a flow curve (about +$200 [Guessing], conditional line).
- **Six lines in the total still have only a search hint, no product link:** the 40 mm COB head ring (a 12-SMD ring that fits the seat is the fallback), the branch fuse holders, the drag chain, the M5×10 button heads and the two shielded moving cables (the igus cable in their notes is unshielded, 4-core and sold by the foot). Their estimates stay in the total, and a few linked prices were not visible (Amazon blocks scripts): SHOPPING_LIST.md marks each one. Look them up before the order they belong to.

Order plan

The design says to pass two Phase 0 gates before buying any motion parts. Each order, with a link per line, is in SHOPPING_LIST.md.

Order	What	Cost incl. 15%
1a — Phase 0A, camera test	Pi 5 with PSU, microSD and cooler, both cameras, the 3 m USB extension, both LED rings, a 12 V wall adapter, M2 screws and standoffs, two 2020 jig posts and the 66 mm arm stub, the optical endstop boards, the flag sheet, hookup wire, heat shrink, filament. Rows marked "PHASE 0A" in the CSV. **Plus, from school stock:** a handful of M5/M3 screws (6 × M5×10, 6 M5 washers, 6 M5 T-nuts or plain M5 nuts, 1 × M5×30 + nylock, 2 × M3×16 + 4 washers + 2 nuts) and a strip of microfiber with spray adhesive or double-sided tape for the tray ledges: their packs are order-2 lines.	**≈ $470**; ≈ $281 with a borrowed school Pi; ≈ $202 if the school also supplies filament, wire, heat shrink and the flag sheet
1b — Phase 0B, cup bench test	Two SMC cups and buffers (one is the spare), spare pads, the M5 barb elbows, PU tube (with its fittings kit), PTFE tape, IPA, a hand vacuum pump with gauge. No mains wiring, no electronics. Rows marked "PHASE 0B". Buy after 0A passes, or order the two cups and the two elbows with 0A: Proax shows the cups back-ordered with no lead time, and cup, buffer and elbow are the jig's real occluder.	**≈ $155**
2 — Machine	Extrusions, rails and a spare block, motors, belt, brackets, fasteners, bed, dowels, Rodent, valves, pump, reservoir and its push fittings, vacuum sensor and ADC, 24 V PSU, mains inlet and mains box, MOSFET modules, fuses, I/O board, limit switches (short hinge lever), laser gate, drag chain, flex cables, E-stop, and the screws for the carry-height sensor and the two switch mounts (the sensor's board is already in the 0A endstop packs; the switches' M2 screws in the 0A M2 kit). Add the light-roof lines only if Phase 0 showed glare, the HT1 cups only if the urethane cup marked, and an upgrade line only when its test failed.	**≈ $1,556** (+ ≈ $190 for the Pi set if you borrowed one for Phase 0)

If Phase 0A fails, you have spent at most ≈ $470 (≈ $281 with a borrowed Pi). If Phase 0B fails, ≈ $626. The first fix to try would be the HT1 cup (+ ≈ $41), but it needs an M10×1-to-G1/8 adapter nobody has sourced yet: until one is found, a cup that marks means stop and ask. Nothing from order 2 has been bought at that point. [Certain, by construction of the tags]

Totals by category

Category	Total	Usually at school	Biggest line
Frame (extrusion)	$158.00	—	2 × 2040 × 1416 base rails, $48
Motion	$191.69	$10.99	VXB Z motor 3-pack, $99.99
Fasteners & brackets	$268.99	$49.13	CNC Kitchen heat-set insert kit, $26.99
Bed, rods & light roof	$158.38	$35.74	PureBond birch panel, $60.17
Controller & compute	$225.80	—	Raspberry Pi 5 4 GB, $110
Cameras & lighting	$137.30	—	ELP 8 MP head camera, $55
Vacuum & gripper	$294.61	$12.00	2 × Clippard EV-2-24 valves, $77.90
Sensors	$92.42	$11.99	5 × XGZP6847A, $36
Power & switching	$197.28	$47.12	AutomationDirect E-stop set, $36.75
Wiring & cable management	$146.09	$84.13	22 AWG hookup wire set, $19.95
Printed parts (filament)	$26.78	$26.78	2 kg PETG
Acceptance tests	$0.00	—	junk slabs, conditional (qty 0)
Subtotal	**$1,897.34**	**$277.88**	
Spares, shipping, duties (15%)	$284.60		
Grand total	**$2,181.94**		
Minus school stock (+ its 15%)	−$319.56		**≈ $1,862**
Lean swap (generic valves), with school stock			**≈ $1,787**

What changed in this round ($0.00):

- **Homing switches:** the CAD now has printed mounts (`x_switch_mount`, `z_switch_mount`), so the 2020 corner brackets no longer carry the switches. The line now asks for the short hinge lever (Omron SS-5GL size, a lever ending at the body's end): the mounts take a lever up to 23 mm from the hinge end, and a longer one is cut back to 20 mm (BUILD_GUIDE §6.8).
- **M5×10 button heads:** 6 needed, not 2 (2 per switch mount). Same pack, same price [Guessing, a small pack of 10 or more].
- **M5 T-nuts:** about 130 needed, not 126 (2 per switch mount). The 150-pack still covers them and the 6 for the Phase 0 jig.
- **M2 kit:** 4 × M2×12 and 4 nuts more, for the switches. No price change.
- **M2 standoffs:** the jig's bridge is now 10.6 mm thick, so a standoff is the board-to-lens length minus 10.0 mm, with 2.0 mm of plastic around each hole.
- **Head LED ring:** the seat is now a 42.8 mm bore with three centring ribs: it takes a ring of 39.5–42.5 mm OD.
- **Filament:** 801 cm³ in 27 prints (was 768 in 25). The 2 kg still cover it.

What changed in round 4 ($0.00):

- **Push fittings kit** ($11.99) from order 1b to order 2: its 1/8 NPT fittings go only into the reservoir cap. Phase 0B uses the tube and the barb elbow.
- **PTFE tape** ($2.36) split out of the reservoir set ($19.93 → $17.57) into its own order-1b line: Phase 0B seals the elbow's M5 thread with it.
- **Confidence tags** now follow each line's price evidence, in the CSV and the shopping list alike. In the CSV, six lines whose price was never seen went from [Likely] to [Guessing] (the two M2 kits, the M5×12 button heads, the M2.5/M3 standoff kit, the resistor kit, the awl pin) and three whose price was seen went to [Certain] (the M-5ALU-4 elbows, the vacuum sensors, the MOSFET modules). In the shopping list, the mains inlet and the mains wire went from [Likely] to [Guessing], as in the CSV.
- **Specs and notes:** the awl pin is one spec now (steel or stainless, hardened not needed); the station ring seat takes a 60 or a 63 mm ring; the head ring line has a fallback; the moving cables link to a search, not to the per-foot igus page; "measure before printing" names the Phase 0 part, which is printed first.

What changed in round 3 (+$182.08, +$209.39 with the allowance):

- **Links and prices from a sourcing run** (22 Oct 2026), spot-checked line by line. Every line in the total now carries a product link except the six above. Where the link is [Likely] or [Guessing], the old search hint stays after it. Price changes of $10 or more: up, the Z motor +$69.99, vacuum sensors +$20.00, E-stop +$19.75, mains and earth wire +$15.00 (still an estimate), microSD +$13.95, pad cloth +$13.48, pad foam +$10.99, heat-set inserts +$10.99, check valves +$10.60; down, the LiMo rails and block −$23.90, the hand pump −$14.01, the two SMC cups −$13.76 (Proax), the head camera −$11.00 (the ELP store), the dowels −$10.00. Every other line moved by less than $10.
- **Spec corrections from the sourcing run, applied because they are plainly right:**
  - **Bed:** the Woodcraft 12 × 60 in Baltic birch blank is no longer listed; the line is now a Home Depot 3/4 in PureBond birch 2 × 8 ft panel (17.9 mm), one rip and one crosscut.
  - **Dowels:** 5/16 in (7.94 mm) × 12 in, so the brad-point bit and collar are 5/16 in.
  - **Class 2 laser:** the only module found (Quarton VLM-650-03) has a 7 mm body. The left gate post now prints with a 7.2 mm barrel hole for it (BUILD_GUIDE §5.9).
  - **Pump "upgrade":** the VXB 10 L/min pump is probably the main pump itself (VXB's SKU is the Amazon ASIN) [Likely]. The upgrade line is now an unsourced branded pump, about $200 [Guessing].
  - **Optical endstop boards:** the 3-pack is out of stock; two 2-packs give 4 boards (probe, carry-height sensor, 2 spares). The listing does not name the TCST2103: measure it.
  - **Flag sheet:** at least 50 × 60 mm, for the probe flag and the new carry flag.

- **Not applied:** the sourcing run's igus chainflex cable for both moving cables. It is unshielded and 4-core, so the head cable would become two runs with renumbered cores. The shielded 8-core line stays [Guessing, price not seen]; the igus cable is listed as a verified alternative in the CSV.
- **New lines for the carry-height sensor** (BUILD_GUIDE §7.4a, WIRING.md §6a): the board (the second one in the endstop packs, $0), the carry flag (cut from the flag sheet, $0), 2 × M5×10 button heads ($6, new), the M3 screws, washers, nuts and T-nuts (from the existing packs, $0), its 0.2 m lead (the board's own 3-pin cable, $0) and the two printed parts ($0). A 10 kΩ + 20 kΩ divider for GPIO 6 comes from the resistor kit.
- **Phase 0 jig:** six M5 washers, not twelve (one under each M5×10: the jig walls are now 4 mm).

What changed in the earlier rounds (+$266 in all): the second SMC cup and buffer, the M5 barb elbows, a spare rail block and the jig's arm stub (+$60), and +$112 of parts the first review found the design needs (a closed mains box, a Class 2 laser, a second E-stop contact block, a hand vacuum pump, a 12 V adapter, separate head and station LED rings, more screw sizes, M2 standoffs, set screws), plus the allowance going from 10% to 15% (+$94).

Conditional lines are in the CSV with qty 0 and are not in the total. Buy them only if a test fails:

- Schmalz HT1 cups, 2 × $18 [Guessing: the price needs a Schmalz login]. Buy if the urethane cup marks slabs. The 25 mm HT1 cups are G1/8-threaded and need an M10×1-to-G1/8 adapter nobody has sourced yet: find it before Phase 0B. Until then, a cup that marks means stop and ask.
- Stepper extension cable, $8. Buy (or make) it if the X motor's lead is too short.
- Light roof (4 foam boards $5, gaffer tape about $12, two 2020 × ~500 mm posts $14). The roof is optional (CONTRACT): buy it only if Phase 0 or the first runs show room-light glare.
- 1N5819 flyback diodes, $6. Only if you swap the Adafruit #5648 modules for ones without an on-board diode (for example D4184 modules).
- Junk graded slabs for the acceptance tests (above).

Upgrades, also qty 0: buy one only when its test fails (about $920 if all of them). [Likely for the failure each one removes; prices as tagged]

Upgrade	About	Buy only if
Genuine HIWIN MGNR12 rail + block (replace both)	$420 [Certain in EUR, Guessing on the exchange rate]	The generic X rail fails the §6.1 roll test (notches or binding after cleaning and greasing), or binds in §13.1
Z motor with a published torque curve	$55 [Guessing]	The §7.8 lift test still fails after the free fixes: Z at 3000 mm/min and Z current raised to about 60% of the motor's rating, motor under 50 °C. No off-the-shelf motor with an 8 mm lead and a 311–356 mm screw was found: ask for a custom screw length.
Constant-force spring for Z	$20.36 [Likely]	The §7.8 hold test creeps. Its mount is not in the CAD.
High-flex USB cable (L-com CAUAAHFX-3M + coupler)	$50 [Likely]	Head-camera dropouts in §13.1, §13.2 or later runs that the spare cable does not cure. It is rated 10,000 flex cycles, not continuous drag-chain duty.
Branded 24 V vacuum pump with a flow curve	$200 [Guessing]	The generic pump fails the §8.3 seal or reservoir test although the line holds
Schneider Harmony XB5AS8444 E-stop (2 NC) + box	$175 [Likely]	The monthly E-stop check ever finds the 24 V bus live. [Likely] Even these blocks are rated below the capacitor inrush, so keep two in series and the check.

Not bought, on purpose: a closed-loop Z motor (the Rodent has no step/dir header for one, and FluidNC uses StallGuard only for homing [Likely]); a spare Rodent (a dead board is a safe stop, and its Y and E driver sockets are already spares [Certain, CONTRACT]); a better head camera (it gives 0.055 mm/px against the 0.14 mm/px the back QR needs [Certain, synthetic labels]; buy one only if Phase 0A fails on sharpness, not on glare). **The Z stall detector is now in the design** at about $6: the spare TCST2103 board on the Z spine with a flag on `z_plate`

(BUILD_GUIDE §7.4a). It sees a loaded head that ended more than about 2.5 mm low before any sideways move; it does not make a weak Z motor strong, so the §7.8 lift test still has to pass.

The allowance is 15%. [Likely] The US removed the de minimis duty exemption in 2025, so parcels from abroad (ELP from China, LiMo, Proax from Canada) carry duty, and several shops add shipping the line prices leave out (above).

The 5 biggest cost items

1. **Raspberry Pi 5 4 GB, $110** [Certain, CanaKit, board only]. DRAM costs pushed the list price from $60 to $70 (Dec 2025), then $85 (Feb 2026), then about $110 (Apr 2026). [Likely]
2. **VXB NEMA17 + TR8×8 400 mm, 3-pack, $99.99** [Certain, VXB]. 1.7 A, 1.7 Ω, 3.1 mH, 0.44 N·m: every number the Z motor rule asks for is on the page. You need one; the other two are spares. A single motor with the same published specs would save about $70.
3. **Two Clippard EV-2-24 direct-acting NC valves, $77.90** [Certain, Clippard; AHE listed $25.90 each on sale].
4. **Two SMC `ZPT25UUK20-B5-A10` cups and buffers, $70.24** [Certain, Proax: CAD 49.34 each, back-ordered]. Canadian shipping and brokerage are extra.
5. **The bed panel, $60.17** [Certain, Home Depot], then the Rodent ($56, in stock at BIQU) and the ELP head camera ($55, ELP's own store).

The 15% allowance ($285) is bigger than any single part.

Where to save

- **Valves.** Generic 24 V direct-acting 2/2 NC micro valves cost about $6 each instead of $39, saving about $66.
  - They work if they are truly direct-acting. Test them when order 2 arrives (BUILD_GUIDE §8.3, reservoir test): hold −60 kPa for 60 s with the pump off and watch the sensor.
  - Their orifice is small, so venting is slower. Lengthen `vent_s` if release lags.
- **The Z motor.** A single motor whose listing gives 1.5–2.0 A, ≤ 4 mH and an 8 mm lead saves about $70 over the VXB 3-pack. Measure 1–2 Ω per phase on arrival; reject anything around 10 Ω.
- **Rails.** The generic MGN12 rail is enough here. Pockets have ±1.85 mm of play and the slabs sit in rods, so you don't need HIWIN precision.
  - Clean the rail.
  - Count the balls in the carriage before installing it, because clones lose balls when the carriage slides off the rail.
  - Re-grease it.
- **Extrusions.** Amazon 20-series V-slot in stock lengths, cut at school, costs about the same ($130–150). Only do it if the school saw cuts square. Misumi's free cut to ±0.5 mm saves an afternoon of squaring.
- **Fasteners.** One large M5 T-nut and screw kit plus one M3 kit can replace the separate packs. Count first: about 130 M5 T-nuts (plus 6 for the Phase 0 jig) and 69 M3×10 are needed, and the kit must contain M5×10/12/16/20/25/30, M5×12 and M5×10 button heads and M3×8/10/12/16/20 (BOM.csv notes say where each goes). AFT charges a small-order fee under $75.
- **Small shop items.** The flag sheet ($11.99 + $13.49 delivery for a 12 × 12 in sheet) and the wood screws (Parts Express shipping) are cheaper as school offcuts or a local hardware-store pack.
- **Pi 5 2 GB.** It would probably run the software. [Guessing] It saves maybe $30–40, but the contract specifies 4 GB, so test memory use in `sim` before switching.

Where NOT to save

- **The cup and buffer.** Buy a genuine urethane pad and spare pads. The cup is the only part that touches your slabs. A cheap silicone cup can leave silicone residue rings on glossy plastic. [Likely] Run the 24 h marking test on a junk slab before any real inventory goes in (Phase 0B, then again on the machine, BUILD_GUIDE §13.2).
  - Buy exactly `ZPT25UUK20-B5-A10` (or the rotating `J20`). The flag, its guide and the sensor post are built on that buffer's catalog heights. [Certain, CAD] A generic AliExpress buffer or a Schmalz or Piab spring plunger has other heights, and the Piab bodies are not M10. [Likely]
- **The Z motor's specs.** 1.5–2.0 A, ≤ 4 mH, an 8 mm lead. A 0.4–0.6 A "integrated" motor stalls at the speed the firmware runs it. The carry-height sensor stops the run after a stall; it does not prevent one.
- **Valve type.** Never use pilot-operated, "servo-assisted" or diaphragm water valves. They need a pressure difference of 0.2–0.5 bar to work, so they won't seal or open reliably on vacuum. [Likely] The listing must say "direct acting" and give a vacuum rating, or a "0 bar" minimum.
- **The MOSFET modules.** Use modules specified for 3.3 V logic. The BOM's Adafruit #5648 is listed for 3–20 V logic and has its flyback diode on board (WIRING.md §4). IRF520 modules barely switch from a 3.3 V Pi pin. [Likely] If you substitute modules without an on-board diode (D4184 type), add a flyback diode across the pump and each valve coil (the conditional 1N5819 line).
- **The 24 V PSU and mains side.** Buy a Mean Well, not an unbranded "24 V 10 A" brick, and the closed mains box for the IEC inlet: a snap-in inlet on a 6 mm plywood panel leaves live tabs open. Have an instructor check the inlet, fuse, box and earth wiring before first power-up.
- **The laser.** Buy a ≤ 1 mW (Class 2) module. The KY-008 that comes with the receiver kit is usually sold as 5 mW (Class 3R), and the beam is on whenever the Pi is on, at face height when you bend over the stacks. [Likely]
- **The E-stop's second contact block.** A single block switching the 24 V capacitor inrush can weld closed without you noticing (WIRING.md §1.3). The AutomationDirect set has three.
- **The 4040 profile type.** It must be 20-series, slot 6, with two slots per face. Amazon's "4040 European standard" is often 40-series slot 8 with one slot per face, and the printed mounts won't fit it; so are Amazon's "4040 brackets". [Likely]
- **Dowel straightness.** Roll every dowel on a flat table and reject bowed ones. A bowed rod narrows the pocket at the top, where the cup lifts a full stack.

Required tools (not in the machine total)

Most of these are standard in a tech-school shop. The "buy" column is the price if the school doesn't have one.

Tool	Used for	Buy price if missing	At school?
Hand drill (cordless)	The 82 rod holes, drilled in place on the assembled machine (§5B), and the bed bolts	$60–100	Likely
Drill press	The rod-hole guide block, the drop arm's and the jig stub's cross holes	—	Likely
5/16 in brad-point bit, new and sharp, at least 4 1/2 in (115 mm) long	Rod holes, for the BOM's 5/16 in dowels (8 mm if you buy 8 mm dowels). The depth collar sits 57 mm above the cutting lips, and the chuck needs about 30 mm of shank above the collar [Likely]	$10	**Buy**
Depth-stop collar, 5/16 in	All rod holes the same 15 mm deep	$5	**Buy**
Drill guide (Milescraft DrillBlock type) or a drill press with long-table support	Holes square to the bed. The printed `drill_template` sets the position, not the angle.	$15–40	Likely
Metric hex keys, 1.5–10 mm, ball end	All socket screws. The probe board's front M3 screw needs the **2.5 mm ball end**, tilted about 20° (BUILD_GUIDE §7.6).	$15	Likely

Tool	Used for	Buy price if missing	At school?
Digital calipers, 150 mm	Measuring 5 of your slabs (`dims.py`), dowels, ring OD/ID, camera board holes, the switch levers; setting each switch 0.3 mm past its click (§6.8)	$20	Likely
Ferrule crimper (self-adjusting)	Screw-terminal wire ends	$25	Maybe
JST/Dupont crimper (IWISS SN-28B or SN-2549)	Sensor and GPIO connectors	$25	Maybe
Wire stripper, flush cutters	Wiring; cutting a long switch lever back to 20 mm (BUILD_GUIDE §6.8)	$20	Likely
Multimeter	Wiring checks, sensor voltages, beam receiver level, setting the carry-sensor trip	$30	Likely
Temperature-controlled soldering iron	Diodes, divider resistors, LED ring leads	$30–100	Likely
Heat-set insert tips (ruthex set fits 900M/T18 irons; TS100-style tips for a Pinecil)	M5 insert in `awl_adapter`, spares	$10–16	**Buy** unless the shop has them
Fine saw + miter box, or a bandsaw	Cutting 82 dowels to 236 mm	$15	Likely
Junior hacksaw or piercing saw, needle files (one about 1.5 mm thick), a vise	The probe flag and the carry flag (BUILD_GUIDE §7.4a, §7.6); clearing the 1.9 mm flag slot in `head_foot` (a standard 12.7 mm hacksaw blade does not fit it)	$15	Likely
7.2 mm (or 9/32 in) drill	Only if your left gate post was printed from an older STL with a 6.2 mm laser hole	$3	Likely
Jigsaw	41 × 41 mm notches in the bed for the uprights (CONTRACT)	$50	Likely
M5/M3 taps + tap handle; **1/8 NPT tap**	Extrusion end taps; reservoir caps	$8 (NPT)	Taps likely; NPT maybe
Dial indicator + magnetic base	Beam-twist check (DESIGN.md)	$25	Likely (machine shop)
Heat gun, clamps, square, level, 120/220 sandpaper	General; doming the rod tops	$25	Likely
Drills and small items			
Metric drill bits 1.8, 3.4, 5.2 and 5.5 mm	1.8: new board holes in the jig's bridge (§1.5); 3.4: M3 clearance holes and the carry flag (§3.4, §7.4a); 5.2: the probe flag (§7.6); 5.5: M5 clearance holes and the cross holes (§3.4, §1.5, §7.5)	$15	Likely
11/32 in drill	Tap drill for the 1/8 NPT reservoir ports (§8.2)	$6	Maybe
Sharp knife and steel rule	Cutting the pads (§5.5), the PU tube square (§8.2)	$10	Likely
Small spring scale, 0–500 g (or 0–5 N)	Belt tension (§6.6)	$8	Maybe
CA glue (superglue)	The awl pin (§3.4)	$5	Likely
Shims (card, thin wood) or a small lab jack	Raising the slab under the cup in Phase 0B (§1.10)	$0–25	Likely
microSD card reader	Flashing Raspberry Pi OS (§11.1)	$10	Likely
M2 flat washers, about 0.3 mm, a few	The camera-aim shim (§7.7): the M2 kit has screws and nuts only	$3	**Buy**

Tools you will probably have to buy: about $45–50. That is the brad-point bit, depth collar, heat-set tips, NPT tap, M2 washers and the spring scale. **If you had to buy every tool new, the total would be about $2,640** ($2,182 + about $460), so use the school's where you can.

How the quantities were derived

- **Extrusions.** These are `model.cut_list()` values (`cad/out/cut_list.csv`), plus an optional 2020 × 600 mm rear brace and, only if you build the optional light roof, two 2020 × ~500 mm roof posts:
  - 2040 × 1416 (2)
  - 4040 × 401 (2)
  - 4040 × 1416 (1)
  - 2040 × 340, both ends tapped M5 (1)
  - 2020 × 285 (1)
  - Phase 0 jig: 2020 × 312 (2) and the 2020 × 66 arm stub (1)
  - plus the Z lead screw, cut to 340 mm below the motor face, and two 1 mm aluminium flags: the probe flag blank, 24.4 × 55 mm, and the carry flag, 19.9 × 36 mm
- **Rods.** `len(layout.rods())` = 82. Each is 218 mm above the pad + 3 mm pad + 15 mm embedded = 236 mm, so 12 in (305 mm) stock leaves room to trim. Buy 100 so you can reject bowed ones.
- **Pads.** There are 13 pockets, and each pad is 83 × 138 mm.
- **M3 rail screws.** MGN12 rails have holes at 25 mm pitch: 1300 mm needs 52 and 330 mm needs 13. The carriages take 4 × M3 each on a 20 × 20 mm pattern. [Certain] The carry sensor adds 4 × M3×10 and 2 × M3×8.
- **M5 T-nuts, about 130.** That covers:
  - brackets and joining plates (about 80–90 nuts);
  - 20 for the printed mounts on the extrusions: x_motor_mount 4, x_idler_mount 2, x_bracket 4 (the spine), z_plate 2 (the drop arm), x_bumper_L 1, z_motor_mount tab 1, carry_sensor_mount 2, x_switch_mount 2, z_switch_mount 2. The z_motor_mount and z_end_cap screw into the spine's tapped end bores instead;
  - 16 bed bolts;
  - 6 feet (both base rails at each end and at midspan; the OpenBuilds feet bring their own).
- **Wood screws.** station_mount 4, gate posts 4 and the electronics panel.
- **Filament.** `print_list.json` totals 801 cm³ of solid parts in 27 prints from 25 STL files, including the Phase 0 jig and its occluder mount, the drill template, two fill gauges, the two carry-sensor parts and the two switch mounts. That is about 1020 g of PETG if printed solid, or about 720 g at the listed walls and infill. Add a Pi case, electronics trays, cable clips and reprints, and it comes to about 1.0–1.3 kg, so the list has 2 kg. [Likely]

Design gaps found while pricing (for the CAD and wiring owners)

These are not BOM problems, but they decide what gets bought.

Resolved since pricing (the CAD decisions in CONTRACT.md):

- The MGN12 rails sit on slots: the X rail centre is at Z = 410 on the 4040's upper front slot, and the Z rail is in the 2040 spine's right slot, on the cup axis. [Certain, `cad/out/checks.txt`]
- The bed has 41 × 41 mm notches at the two upright positions. [Certain, `model.py`]
- The light roof is optional and left out of the CAD (conditional lines in the CSV).
- The 2020 drop arm is 285 mm with two 5.5 mm cross holes, 19 and 84 mm from its lower end.
- The Z machinery sits above the highest point a held slab can reach, and the lead screw is cut to 340 mm. Printed end bumpers and a `z_end_cap` are hard stops on both axes. [Certain]
- The mains side has a closed box, the laser is Class 2, the E-stop has two contact blocks, and the MOSFET modules and limit switches match WIRING.md.
- The buffer code is `ZPT25UUK20-B5-A10` [Certain, SMC ZP catalog Table 2; Proax lists it], the probe flag, its guide and the barb elbow are in the CAD, and the jig posts and arm stub are in `cut_list.csv`.

- **The head camera now sees the face-down card window** [Certain, CAD frustum check]: the lens sits 5 mm toward the label from the cup axis and the foot was trimmed. Nothing to buy; the `camera_bracket` and `phase0_top` STLs changed, so print the current ones.
- **Both LED rings are in the CAD** [Certain, `cad/out/checks.txt`; Likely for the ring sizes]. The head ring (modelled 40 × 30 × 5 mm) sits in a 42.8 mm bore with three centring ribs, against a 33 mm lip, its LED face 1 mm under the lens front, and the barb elbow clears it by 3.75 mm at the full 20 mm stroke. The seat takes rings of 39.5–42.5 mm OD (shave the ribs for one over 40). The station seat is a 64 mm bore on a shelf that takes a 60 or a 63 mm ring. `station_mount`, `camera_bracket`, `phase0_base` and `phase0_top` changed: print the current ones.
- **The fastener margins are closed in the CAD**: the probe board takes M3×16 with a washer under the head and under the nut, and every Phase 0 jig wall an M5×10 passes through is 4 mm (one washer per screw).
- **The Z stall guard is designed** (carry-height sensor, above).
- **The homing switches have printed mounts** (`x_switch_mount`, `z_switch_mount`) [Certain, `cad/out/checks.txt`]: a finger on `x_bracket` and a chamfer on `z_plate` slide across each lever and hold it 0.6 mm past its click point to the hard stop. Nothing new to buy but the screws above (was open gap 7).

Still open:

1. **[Certain, CAD] There are only 18 mm between each rail end and the upright's inner face.** Use back-face brackets and flat joining plates on the outer ends (BUILD_GUIDE §4.3).
2. **[Likely] The laser receiver module is a 5 V board.** Divide its output before GPIO 5. It is also non-modulated, so room light can hold it HIGH. Test it with the room lights on.
3. **[Likely] Angel-eye LED rings are 12 V parts, sold by OD only.** Measure the ring OD, ID and thickness before printing the ring seats. Head: OD 39.5–42.5 mm (42.8 mm bore with ribs), ID about 24 mm up to OD − 3 (33 mm lip), 5 mm thick or less (each mm thicker takes 1 mm off the 3.75 mm barb clearance). Station: OD 59–63.5 mm (64 mm bore; some "60 mm" rings are 63 mm). No 40 mm COB listing is live: a 40 mm 12-SMD ring that fits is the fallback (check it for glare in Phase 0A).
4. **[Guessing] Some mounting patterns in the CAD are not verified.**
   - The head camera board's hole pattern is assumed to be 32 mm square (the ELP board is 38 × 38 mm), and its board-to-lens-front length 20 mm (`HEAD_LENS_PROJ`).
   - The optical endstop boards: the TCST2103 itself is from Vishay's drawing (19 mm flange holes, symmetric about the slot) [Certain], but the listing does not name the sensor, and the board's header position and component heights are unknown. This now matters twice: the probe and the carry-height sensor.
   - Measure these parts before printing `camera_bracket`, `phase0_top`, `head_foot` and `carry_sensor_mount`.
5. **[Certain, sourcing; resolved in CAD] The only Class 2 laser found is 7 mm**, so the left gate post's barrel hole is now 7.2 mm (the right post keeps 6.2 mm for the receiver).
6. **[Certain, forum report] BTT has shipped Rodents with an old FluidNC build.** Reflash the current FluidNC release before loading `firmware/fluidnc/config.yaml`. Also check the Rodent's limit and motor connector types (XH or other) before buying crimp housings.
7. **[Guessing] Which lever the homing switch 10-pack has.** The switch mounts are now in the CAD (resolved above), drawn for the Omron SS-5GL size with a short hinge lever. A lever that ends more than 23 mm from the hinge end must be cut back to 20 mm (BUILD_GUIDE §6.8); a roller lever does not fit. The listing was not opened: check the photos, or measure on arrival.

Main sources

- **Sourcing run, 22 Oct 2026:** every product link in BOM.csv; the per-line notes say which pages were opened ([Certain]) and which prices come from search text ([Likely]) or the old estimate.
- **Raspberry Pi 2026 prices:** Liliputing/Heise/TechSpot reports, The Register (Feb 2026)

- **Rodent:** BIQU product page
- **LRS-150-24:** TRC Electronics
- **Clippard EV-2-24:** Clippard
- **SMC ZPT/ZP codes and prices:** ZPT25UUK20-B5-A10 at Proax, ZP25UU at automationdistribution, SMC USA ZPT (J,K) family page with configurator and CAD
- **SMC ZP catalog (stroke table, buffer dimensions):** RS copy of the ZP catalog, pages 13-11-9 (Table 2) and 13-11-12
- **SMC M-5ALU-4 barb elbow:** automationdistribution
- **TCST2103 drawing:** Vishay doc 81147, drawing 6.550-5040.01-4
- **VXB Z motor 3-pack:** VXB
- **XGZP6847A100KPGN:** Digi-Key
- **E-stop set:** AutomationDirect GCX1131
- **Schmalz HT1:** Schmalz PFG 25 HT1
- **HIWIN MGN12 price tier:** DamenCNC MGNR12
- **Laser receiver behaviour:** Acxico KY-008 listing
- **D4184 module notes:** ProtoSupplies
- **FluidNC on Rodent:** V1E forum thread

SlabRail shopping list

The parts in BOM.csv, grouped into the three orders you place, with a direct link per line where the sourcing run found one (22 Oct 2026). Prices are per unit or per pack as listed; **a line cost excludes shipping, tax and duty**. Totals and the order plan are explained in BOM.md; the build steps in BUILD_GUIDE.md.

Confidence: **Certain** = the price was seen on the product page; **Likely** = a price from search text or a page that could not be re-opened; **Guessing** = an estimate (no page seen). **[ask school first]** = usually in a school shop: check before buying. A line marked NEEDS A BETTER LINK has no working product page yet.

Totals

Order	Parts	With the 15% allowance	Of which [ask school first]
1a: Phase 0A camera test	$408.90	**$470.23**	$68.67
1b: Phase 0B cup bench test	$135.16	**$155.43**	$12.00
2: the machine	$1,353.28	**$1,556.27**	$197.21
All three	**$1,897.34**	**$2,181.94**	$277.88

- **The 15% allowance ($284.60) is for shipping, duties and a few broken parts, and it is needed.** [Likely] Shipping already seen and NOT in the line costs: LiMo DDP $39.00 (both rails and the block), Digi-Key Marketplace $28.00 (vacuum sensors), MakerStock $13.49 (flag sheet), the endstop boards $5.70 = $86.19. Not shown anywhere yet: ELP express from China plus duty, Proax (Canada) shipping and brokerage, AFT's small-order fee (its order is under the $75 minimum), Woodpeckers (free only over $75), Parts Express and Misumi shipping. The US no longer has a de minimis duty exemption.
- With a borrowed school Pi, order 1a is $280.66 with the allowance; the Pi set ($164.85, $189.58 with the allowance) then moves to order 2.
- If the school supplies everything marked [ask school first], the whole list is about $1,862 with the allowance.
- **Buy order 2 only after both Phase 0 gates pass** (BUILD_GUIDE §1.9, §1.10). The qty-0 lines at the end are not in any total: buy one only when the test it names fails.
- **Before you order, check the lines marked NEEDS A BETTER LINK**, and get the Misumi quote first (its prices are estimates: Misumi blocks scripts, and cuts under 300 mm carry a flat minimum).

Order 1a: Phase 0A camera test

Rows tagged PHASE 0A in BOM.csv. **Also take from school stock (or a hardware store) what the jig needs from order-2 packs:**

- the screws: 6 × M5×10 with 6 M5 washers (one each) and 6 slot-6 M5 T-nuts, 1 × M5×30 + nylock, 2 × M3×16 + 4 washers + 2 nuts. No slot-6 T-nuts? A plain M5 hex nut slid into the slot from the open end of the post or stub works for the jig [Likely]. Or pull the KB-3D M5 T-nuts and the AFT M5×10 forward into this order (about $19);
- for the tray ledges: a strip of microfiber cloth, and spray adhesive or double-sided tape. Or pull the order-2 Pad facing and Spray adhesive lines forward ($35.96).

Consider ordering the two SMC cups and the two M-5ALU-4 elbows (order 1b, with the PTFE tape for the elbow's thread) now. The cup, buffer and elbow are the jig's real occluder, the elbow also clamps the probe flag, and Proax shows the cups back-ordered. That puts about $77 at risk if 0A fails.

Item	Qty	Link	Price each	Line cost	Notes
Phase 0 jig posts: HFS5-2020-312 (square ends, 312.0 +/- 0.5 mm)	2	us.misumi-ec.com	$6.00	$12.00	Guessing; check: could not open. Cut square to 312.0 +/- 0.5 mm. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
Phase 0 jig arm stub: HFS5-2020-66 (square ends)	1	us.misumi-ec.com	$4.00	$4.00	Guessing; check: could not open. Misumi charges a flat minimum under 300 mm, so $4 is probably low. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
M2 screw/nut kit: M2x6..M2x12 + nuts	1	Amazon B08MT9WWCT	$8.00	$8.00	Guessing; check: not checked. Price not seen (Amazon not opened); BOM estimate.
M2 standoff kit: M2 brass standoffs, assorted 3-20 mm, + M2 screws	1	Amazon B09X5WQTV1	$9.00	$9.00	Guessing; check: not checked. Price not seen; BOM estimate.
Raspberry Pi 5 4GB: Raspberry Pi 5, 4 GB	1	canakit.com	$110.00	$110.00	Certain; check: confirmed. Pick the 'Board Only' row. Adafruit #5812 is $130.
Pi power supply: Raspberry Pi 27 W USB-C PSU (US plug)	1	canakit.com	$12.95	$12.95	Certain; check: not checked. Not the CanaKit-brand 5 A unit on the same page.
microSD card: Raspberry Pi 64 GB A2 microSD, pre-loaded	1	canakit.com	$29.95	$29.95	Certain; check: not checked. cheaper official blank cards are out of stock.
Pi cooler: Raspberry Pi Active Cooler	1	canakit.com	$11.95	$11.95	Certain; check: not checked. retail is $11-15.
Station camera: Raspberry Pi Camera Module 3 (standard, not wide/NoIR)	1	canakit.com	$26.95	$26.95	Certain; check: not checked. Standard, not NoIR/Wide.
Pi 5 camera cable: 22-pin 0.5 mm to 15-pin 1 mm FPC, 500 mm (Raspberry Pi SC1130)	1	adafruit.com	$2.50	$2.50	Certain; check: not checked.
Head camera: ELP-USB8MP02G-L36: USB 8MP IMX179, M12 3.6 mm lens, UVC, manual focus; lens F2.8 or faster	1	webcamerausb.com	$55.00	$55.00	Certain; check: confirmed. Express shipping and US duty extra. Board 38 x 38 mm: measure the hole pattern before printing phase0_top (and later camera_bracket). Lens F2.8 or faster.
Head camera USB cable: USB 2.0 A-male to A-female extension, 3 m (x2, one spare; Adafruit #993)	2	adafruit.com	$6.95	$13.90	Certain; check: not checked. Standard jacket, not high-flex. 0A needs one; the second is the spare.
LED ring, head: COB 'angel eye' ring, 40 mm OD, white, 12 V (pair; 1 spare)	1	search: eBay: '40mm 12V COB angel eye halo ring white'	$12.00	$12.00	Guessing. NEEDS A BETTER LINK: the eBay listing found has ended; Amazon's 40 mm rings are 12-SMD, not COB. Find one before ordering 1a. **Fallback:** a 40 mm 12-SMD ring is OK if it fits the seat (OD 39.5–42.5 mm, ID about 24 mm up to OD − 3, 5 mm thick or less [Certain, CAD seat checks]; the seat's three ribs fit a 40.0 mm ring, so shave each by half of OD − 40 for a bigger one); check the head images for glare spots in Phase 0A [Guessing that discrete LEDs light the label as evenly].
LED ring, station: COB 'angel eye' ring, 60 mm OD, white, 12 V (pair; 1 spare)	1	Amazon B00W6ZXQOO	$12.00	$12.00	Likely; check: not checked. Price not seen (Amazon not opened). The seat is a 64 mm bore on a shelf: it takes a 60 or a 63 mm ring (OD 59–63.5 mm). Some '60 mm' COB rings are 63 mm OD. Measure OD and ID before printing phase0_base (and later station_mount).
12 V adapter (Phase 0A): 12 V 1 A wall adapter, 5.5 x 2.1 mm (Pololu #1471) + barrel-jack-to-screw-terminal pigtail (Adafruit #368)	1	pololu.com, adafruit.com	$14.95	$14.95	Certain; check: confirmed. Adafruit's own 12 V 1 A adapter is out of stock.

Item	Qty	Link	Price each	Line cost	Notes
Contact probe: Slotted optical endstop board with 3-pin cable, 2-pack (x2 = 4 boards: probe, carry-height sensor, 2 spares)	2	Amazon B07KNTR9GN	$2.54	$5.08	Certain; 2 packs = 4 boards (probe, carry-height sensor, 2 spares). +$5.70 delivery. The listing does not name TCST2103: measure the 3.1 mm slot and 19 mm holes before printing head_foot.
Probe flag stock: 1 mm soft aluminium sheet (1050, 3003 or 5052-H32), at least 50 x 60 mm (probe flag + carry flag)	1	Amazon B0DGZH5798	$11.99	$11.99	Certain [ask school first]. Needs >= 50 x 60 mm (probe flag + carry flag); +$13.49 delivery, so a school offcut is much cheaper.
Hookup wire: 22 AWG stranded hookup wire, 6 colours x 25 ft (PVC; fine for static wiring)	1	adafruit.com	$19.95	$19.95	Certain; check: confirmed. PVC (UL1007), not silicone. Fine for static wiring. [ask school first]
Heat shrink: Heat-shrink assortment, 280 pcs (Adafruit #4559)	1	adafruit.com	$9.95	$9.95	Certain; check: not checked. [ask school first]
PETG filament: PETG 1.75 mm, 2 x 1 kg: Elegoo PETG Pro black (structural) + Sunlu matte PETG black (camera parts)	1	us.elegoo.com, store.sunlu.com	$26.78	$26.78	Certain; check: confirmed. Structural prints. [ask school first]

Order 1a: $408.90 ($470.23 with the allowance).

Order 1b: Phase 0B cup bench test

Rows tagged PHASE 0B. Buy after 0A passes, except the cups and elbows (above). The push fittings kit is now an order-2 line: Phase 0B uses only the tube and the barb elbow.

Item	Qty	Link	Price each	Line cost	Notes
Tubing: 4 mm OD x 2.5 mm ID PU tube, 10 m, with a fittings kit (TAILONZ)	1	Amazon B07RK4MCKP	$13.99	$13.99	Certain; check: not checked. Kit's 1/8 and 1/4 fittings are BSPT, not NPT: keep them out of the reservoir cap.
Cup + spring buffer (primary): SMC ZPT25UUK20-B5-A10: 25 mm flat urethane pad, NON-ROTATING buffer (K), 20 mm stroke, M5 female vacuum port, M10x1 body	1	proax.ca, smcusa.com	$35.12	$35.12	Certain; check: confirmed (Proax: K non-rotating, 20 mm stroke, B5, A10). CAD 49.34 each, back-ordered, no lead time; shipping, duty and brokerage from Canada extra. Or ask a US SMC distributor for a written quote.
Spare cup pads: SMC ZP25UU (25 mm flat urethane pad, replacement)	4	automationdistribution.com	$4.10	$16.40	Certain; check: confirmed. Ask the distributor to confirm it fits the ZPT25 adapter.
Cup + spring buffer (spare): Second SMC ZPT25UUK20-B5-A10 (same code as above)	1	proax.ca	$35.12	$35.12	Certain. Same line as the primary: order qty 2 together.
Buffer-port fitting: SMC M-5ALU-4 barb elbow: M5 thread, for 4 x 2.5 mm soft urethane tube (x2, one spare)	2	automationdistribution.com	$2.09	$4.18	Certain; check: confirmed. Order with the cups: it is part of the Phase 0A occluder and clamps the flag.
PTFE thread tape: PTFE thread-seal tape, 1/2 in x 260 in (5-pack)	1	homedepot.com	$2.36	$2.36	Certain. The elbow's thread sealant in Phase 0B (BUILD_GUIDE §1.10); later the reservoir's fittings. It was part of the reservoir set.
Cup/slab cleaning: 99% isopropyl alcohol, 16 oz + Kimwipes (286)	1	Amazon B07NFSFBXQ, Amazon B075L9ZTPB	$12.00	$12.00	Likely [ask school first]. IPA $5.94; Kimwipes price not seen (~$6 [Guessing]).

Item	Qty	Link	Price each	Line cost	Notes
Hand vacuum pump (Phase 0B): Hand vacuum pump with gauge (CARSC brake-bleeder kit)	1	Amazon B07D2B13DF	$15.99	$15.99	Certain; check: not checked. Generic: whether it reaches -80 kPa is untested.

Order 1b: $135.16 ($155.43 with the allowance).

Order 2: the machine

Everything else with a quantity. Buy only after both Phase 0 gates pass. The carry-height sensor (the Z stall guard, BUILD_GUIDE §7.4a) needs no separate purchase except the M5×10 button heads: its board is the second one in the order-1a endstop packs, its flag is cut from the flag sheet, and its other screws and lead come from packs below. The two printed switch mounts (BUILD_GUIDE §6.8, §7.4) need nothing new either: 4 × M5×10 button heads from the same line, 4 M5 T-nuts, and 4 × M2×12 with nuts from the order-1a M2 kit. **Buy homing switches with the short hinge lever** (see that line).

Item	Qty	Link	Price each	Line cost	Notes
Base rail: HFS5-2040-1416 (20-series, slot 6, 2040, cut to 1416 mm)	2	us.misumi-ec.com	$24.00	$48.00	Guessing; check: could not open. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
X beam: HFS5-4040-1416 (20-series 4040, 8 slots = 2 per face, slot 6)	1	us.misumi-ec.com	$42.00	$42.00	Guessing; check: could not open. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
Upright: HFS5-4040-401	2	us.misumi-ec.com	$15.00	$30.00	Guessing; check: could not open. Option -TPW (both ends tapped) may tap only the centre bore; check the drawing. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
Z spine: HFS5-2040-340, BOTH ENDS TAPPED M5 (the two centre bores)	1	us.misumi-ec.com	$9.00	$9.00	Guessing; check: could not open. Tapping fee not included (or tap at school). Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
Head drop arm: HFS5-2020-285	1	us.misumi-ec.com	$6.00	$6.00	Guessing; check: could not open. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
Rear diagonal brace (if beam twist > 1 mm): HFS5-2020-600	1	us.misumi-ec.com	$7.00	$7.00	Guessing; check: could not open. Only used if beam twist > 1 mm; BOM keeps it as cheap insurance. Price is the BOM estimate (Misumi blocks scripts): get the Misumi quote.
X linear rail + carriage: MGN12H, 1500 mm rail + 1 block, cut to 1300 mm (write 'cut to 1300 mm' in the order remark)	1	limobearing.com	$37.96	$37.96	Certain; check: confirmed. One $39 DDP shipping charge covers all the LiMo lines (duty included). Preload not stated. Clean, count balls, grease.
Z linear rail + carriage: MGN12H, 350 mm rail + 1 block, cut to 330 mm (order remark)	1	limobearing.com	$12.43	$12.43	Certain; check: confirmed. Do not take the uncut 350 mm.
Spare MGN12H block: 1x MGN12H carriage block, same seller and series as the rails	1	limobearing.com	$5.71	$5.71	Certain; check: confirmed. One spare (a second is another $5.71).
X motor: NEMA17 48 mm, 59 Ncm, 2.0 A, 1.8 deg (StepperOnline 17HS19-2004S1)	1	omc-stepperonline.com	$9.62	$9.62	Likely; check: could not open. Price from search text; shipping extra. Not the -2004S2.

Item	Qty	Link	Price each	Line cost	Notes
Z motor with lead screw: NEMA17, 1.5-2.0 A/phase, <= 4 mH, >= 40 N.cm holding, integrated 400 mm Tr8x8 lead screw (8 mm LEAD) + brass T8 nut; CUT the screw to 340 mm below the motor face. Listed: VXB 3-pack, 1.7 A, 1.7 Ohm, 3.1 mH, 0.44 N.m	1	vxb.com, newegg.com	$99.99	$99.99	Certain; check: confirmed. Need 1 (2 spares). 1.7 A, 1.7 ohm, 3.1 mH (checker), 0.44 Nm: meets the spec. Brass nut not stated. Cut the screw to 340 mm.
GT2 belt kit: GT2 6 mm belt 5 m + 20T pulley 5 mm bore + 20T TOOTHED idler 5 mm bore (bearing)	1	Amazon B07JKT5BZQ	$14.99	$14.99	Likely; check: could not open. Confirm the idlers are TOOTHED in the photos; if smooth, add a toothed 20T 5 mm idler.
Rail grease: Lithium or PTFE grease (e.g. Super Lube 21030)	1	Amazon B0CTSB4XYX	$10.99	$10.99	Likely; check: not checked. Price from a search snapshot. [ask school first]
4040 corner brackets: 20-series (slot 6) inside L brackets, 2 holes per leg, 10-pack with T-nuts and screws: two per 4040 joint, one in each slot	1	Amazon B09ZXPX1KS	$18.00	$18.00	Guessing; check: could not open. Price not seen; BOM estimate. Check the ~40 mm leg length in the photos. Never buy '4040' Amazon brackets: 40-series, 8 mm slot, M8.
Flat joining plates: 2020 L plates, 60 x 60 x 4 mm, M5 holes at 20 mm, 10-pack	1	fabreeko.com	$19.99	$19.99	Certain; check: confirmed. Green/yellow/blue only.
2020 corner brackets: 20x20 L brackets slot 6 (20 pcs)	1	Amazon B08P9K7JY7	$14.99	$14.99	Likely; check: not checked. Price from search text.
M5 T-nuts: Slot-6 M5 HammerHead T-nuts, 20-series (150)	150	kb-3d.com	$0.09	$13.50	Certain; check: confirmed. Select Size = M5.
M5x10 socket head cap screws: ISO 4762 M5x10, 12.9 black oxide (100)	1	aftfasteners.com	$5.83	$5.83	Likely; check: WRONG, fixed. Checker price $5.83 (sourcer had $5.00). AFT: $75 minimum or small-order fee.
M5x12 SHCS: ISO 4762 M5x12, 12.9 (100)	1	aftfasteners.com	$6.00	$6.00	Likely; check: WRONG, fixed. Checker price $6.00 (was $5.13).
M5x30 SHCS: ISO 4762 M5x30, 12.9, partial thread (100)	1	aftfasteners.com	$8.29	$8.29	Likely; check: WRONG, fixed. Checker price $8.29 (was $7.90).
M5x25 SHCS: ISO 4762 M5x25, 12.9 (100)	1	aftfasteners.com	$8.12	$8.12	Likely; check: WRONG, fixed. Checker price $8.12 (was $6.96).
M5x16 + M5x20 SHCS: ISO 4762 M5x16 (100) + M5x20 (100), 12.9	1	aftfasteners.com, aftfasteners.com	$14.39	$14.39	Likely; check: confirmed.
M5x12 button head: M5x12 button head, ISO 7380, 12.9 (100)	1	Amazon B005DZZPW0	$10.00	$10.00	Guessing; check: not checked. Price not seen: sourcer estimates $8-12 for the 100-pack ($10 used). Only M5x12 fits the z_plate recess.
M5x10 button head (sensor and switch mounts): M5x10 button head, ISO 7380 (small pack)	1	search: 'M5x10 button head socket screw ISO 7380'	$6.00	$6.00	Guessing. 6 needed (buy a pack of 10 or more): 2 for the carry_sensor_mount, 2 for each switch mount.
M5 washers + nylock nuts: M5 DIN125 flat washers (100) + M5 DIN985 nylock (50)	1	kb-3d.com, kb-3d.com	$13.00	$13.00	Certain; check: not checked. [ask school first]
M3 T-nuts: Slot-6 M3 roll-in T-nuts, 20-series (100)	100	kb-3d.com	$0.17	$17.00	Certain; check: confirmed.
M3x10 SHCS (rails): M3x10 SHCS, LDO black nickel (2 x 50)	2	kb-3d.com	$1.99	$3.98	Certain; check: not checked.
M3 screw/nut assortment: M3x6, M3x8 (50 each), M3x12, M3x16, M3x20 (20 each) SHCS + M3 DIN934 nuts (50) + M3 DIN125 washers (200)	1	kb-3d.com, kb-3d.com, kb-3d.com	$19.15	$19.15	Certain; check: not checked. 7 KB-3D packs, $19.15 together. [ask school first]
Carry sensor small parts: 4 x M3x10, 2 x M3x8, 2 M3 washers + 2 nuts, 2 slot-6 M5 T-nuts	1	-	$0.00	$0.00	Certain.
Wood screws: #8 x 3/4 in pan-head wood screws (100)	1	parts-express.com	$3.29	$3.29	Certain; check: not checked. Shipping extra: buy any local 100-pack instead unless you order from Parts Express anyway. [ask school first]

Item	Qty	Link	Price each	Line cost	Notes
Heat-set inserts: CNC Kitchen heat-set insert combo, 200 (100 M3, 50 M4, 50 M5)	1	kb-3d.com	$26.99	$26.99	Certain; check: confirmed. No M2/M2.5. The M5 insert is 9.5 mm long vs the 9.0 mm pocket (sits ~0.5 mm proud). Only 1 M5 is needed: the single-size M5 pack is $12.40.
Rubber feet: Rubber feet set, 4-pack with drop-in tee nuts and M5x12 screws (x2)	2	us.openbuilds.com	$7.99	$15.98	Likely; check: could not open. Price from search text; store unreachable.
Threadlocker: Loctite 243 medium blue, 6 mL	1	kb-3d.com	$8.69	$8.69	Certain; check: confirmed. [ask school first]
M3 set screws: M3x6 cup-point set screws (20)	20	kb-3d.com	$0.09	$1.80	Certain; check: not checked.
Standoff kit: M2.5 + M3 nylon/brass standoffs, screws, nuts	1	Amazon B07CK7L2W6	$12.00	$12.00	Guessing; check: not checked. Price not seen; BOM estimate.
Awl pin: 3 x 30 mm steel or stainless dowel pin (hardened not needed), or a 3 mm drill-bit shank	1	Amazon B071G31FY9	$5.00	$5.00	Guessing; check: not checked. Price not seen. [ask school first]
Bed plywood: 3/4 in (17.9 mm actual) PureBond birch plywood, 2 x 8 ft project panel	1	homedepot.com	$60.17	$60.17	Certain; check: confirmed. The Woodcraft Baltic birch blank is no longer listed and no 12x60 in Baltic birch is sold online. 17.9 mm birch ply, voids possible; HD offers a free cut. Ask school for a >= 1.42 m Baltic birch offcut first.
Guide rods: 5/16 in (7.94 mm) x 12 in hardwood dowels, 100	100	woodpeckerscrafts.com	$0.20	$20.00	Certain; check: confirmed. $0.20 each at 100+. Free shipping only over $75. Drill 5/16 in, not 8 mm.
Pad foam: High-density EVA foam, 2 mm, black, 14 x 39 in sheet	1	thefoamory.com	$20.99	$20.99	Certain; check: not checked. One sheet cuts 28 pads (13 needed). Flatten for a day.
Pad facing: Tight-weave microfiber cloths (8-pack)	1	homedepot.com	$21.48	$21.48	Certain; check: not checked. Cheaper: EXPO 12x12 cloth x3 = $17.40 (weave not stated).
Spray adhesive: 3M Super 77 multipurpose spray adhesive, 14.1 oz (77-NAT)	1	homedepot.com	$14.48	$14.48	Certain; check: confirmed. URL replaced: the stored 77-DSC page redirects here (checker). [ask school first]
Wood glue: Titebond II, 16 oz	1	homedepot.com	$8.49	$8.49	Certain; check: not checked. [ask school first]
Electronics panel: 1/4 in PureBond birch plywood, 2 x 2 ft (5.2 mm actual)	1	homedepot.com	$12.77	$12.77	Certain; check: not checked. 5.2 mm actual. [ask school first]
Motion controller: BIGTREETECH Rodent V1.x (ESP32, 4x TMC2160, FluidNC)	1	biqu.equipment	$56.00	$56.00	Certain; check: confirmed. Back in stock. Reflash current FluidNC on arrival.
Controller USB cable: USB-A to USB-C data cable, 1 m (Adafruit #4474)	1	adafruit.com	$4.95	$4.95	Certain; check: not checked.
Vacuum pump: 24 V DC diaphragm vacuum pump, -65..-80 kPa, 5-12 L/min	1	Amazon B0891SR87D	$21.00	$21.00	Certain; check: could not open. Select the DC24V variant. Backup: VN-T1 DC24V $22.59.
Vacuum + vent valves: Clippard EV-2-24: 2-way NC, DIRECT-ACTING poppet, 24 VDC, rated to 28 inHg vacuum, 10-32 ports	2	clippard.com	$38.95	$77.90	Certain; check: confirmed. Rated 28 inHg vacuum on Clippard's data sheet. AHE lists $25.90 (sale).
Valve fittings: NITRA MS532-10N, #10-32 male to 4 mm push-to-connect, 5-pack	1	automationdirect.com	$6.50	$6.50	Certain; check: confirmed.
Check valve: SMC AKH04-00 check valve, 4 mm push-in	2	automationdistribution.com	$9.30	$18.60	Certain; check: confirmed. Arrow toward the pump.
Reservoir: 2 in x 2 ft Sch 40 PVC pipe (solid core, not foam core) + 2 Sch 40 socket caps + PVC cement	1	homedepot.com, homedepot.com, homedepot.com	$17.57	$17.57	Certain; check: confirmed. Cut 6 in. Not foam-core pipe. The PTFE tape is the order-1b line.
Push fittings kit: 4 mm x 1/8 NPT male straight push fittings, 10 (TAILONZ PC4-N1)	1	Amazon B091THC7QS	$11.99	$11.99	Certain; check: not checked. The reservoir cap's two 1/8 NPT ports; tees, unions and M5 male come with the tube kit.

Item	Qty	Link	Price each	Line cost	Notes
Sensor hose: Silicone tube 2 mm ID x 4 mm OD, 1 m	1	Amazon B08BRJS5GD	$5.89	$5.89	Certain; check: not checked.
Vacuum sensor: CFSensor XGZP6847A100KPGN (-100..0 kPa, 5 V, 0.5-4.5 V out)	5	digikey.com	$7.20	$36.00	Certain; check: confirmed. 5 for the $7.20 price break (2 needed). +$28 flat marketplace shipping, ~14 days.
ADC: ADS1115 16-bit I2C module (2-pack)	1	Amazon B0DYJ2VBKJ	$9.88	$9.88	Certain; check: not checked.
Carry-height sensor (Z stall guard): Second optical endstop board from the contact-probe packs; 10 k + 20 k divider to GPIO 6	1	-	$0.00	$0.00	Certain.
Carry flag: 1 mm aluminium, 19.9 x 36 mm, flat, two 3.4 mm holes	1	-	$0.00	$0.00	Certain.
Homing switches: Bare SPDT lever microswitches with C / NO / NC solder tabs (no endstop PCB), 10-pack	1	Amazon B07YQQ3R7W	$5.99	$5.99	Certain; check: not checked. Bare SPDT; wire C and NC. **Short hinge lever:** an Omron SS-5GL-size body (19.8 × 10.2 × 6.4 mm, holes 9.5 mm apart) with a flat lever ending at the body's end. The printed mounts take a lever up to 23 mm from the hinge end; cut a longer one back to 20 mm (BUILD_GUIDE §6.8). Not a roller lever. [Guessing which lever this pack has: check the photos]
Overfill beam: Non-modulated laser receiver module + KY-008 laser module (2 sets): use the kit's RECEIVER only	1	Amazon B082PFX8LK	$7.49	$7.49	Certain; check: not checked. Use the receivers only (the kit lasers are 5 mW).
Overfill laser (Class 2): 650 nm dot laser module, CLASS 2 (< 1 mW), 2.6-6 V, D7 x 21 mm (Quarton VLM-650-03 LPT)	1	Amazon B00ZR80S9S	$15.99	$15.99	Certain. 7 mm body: the left gate post prints with a 7.2 mm hole for it (BUILD_GUIDE sec. 5.9).
24 V PSU: Mean Well LRS-150-24 (24 V, 6.5 A)	1	trcelectronics.com	$20.39	$20.39	Certain; check: confirmed. Tariff included in the price.
Mains inlet: IEC C14 inlet with fuse holder + rocker switch (2-pack)	1	Amazon B008J80BJK	$9.00	$9.00	Guessing; check: not checked. Price not seen; BOM estimate. Check snap-in wall thickness vs the Hammond box.
Mains box: Hammond 1591XXTBK ABS box (123 x 83 x 61 mm) + PG7 cable gland + fully insulated 6.3 mm female disconnects	1	digikey.com, Amazon B08K8FB4KV, Amazon B07XJ6MJT9	$27.77	$27.77	Likely; check: confirmed. Hammond makes no 120x80x55; this is the nearest smooth-wall box.
Fuses: 5x20 mm slow-blow T3.15A (10)	1	Amazon B01EYJ4U1C	$6.00	$6.00	Likely; check: not checked. Price not seen; BOM estimate. [ask school first]
Power cord: IEC C13 power cord, 6 ft	1	trcelectronics.com	$5.17	$5.17	Certain; check: confirmed. Ships with the LRS-150-24. [ask school first]
Mains and earth wire: 18 AWG (0.75 mm2) 300 V+ wire, brown + blue + green/yellow, ~1 m each, plus M5 ring terminals and serrated washers	1	Amazon B0C49MP61P	$25.00	$25.00	Guessing; check: not checked. Prices not seen; $25 is the sourcer's estimate. Brown and blue variants not confirmed; serrated washers not sourced. US black/white/green MTW is easier to buy: WIRING.md §1.1 gives the colour mapping; the instructor decides. [ask school first]
E-stop: AutomationDirect GCX1131 22 mm latching E-stop (1 NC built in) + ECX1030-2 (2 x 1 NC blocks) + SA104SL 1-hole box	1	automationdirect.com, automationdirect.com, automationdirect.com	$36.75	$36.75	Certain; check: confirmed. Set = GCX1131 + ECX1030-2 + SA104SL, $36.75. Contacts rated DC-13 24 V 1.5 A. Gives 3 NC blocks; the BOM asks for 2 in series.
MOSFET switches: Adafruit MOSFET Driver #5648 (AO3406, 1N4007 flyback diode on board, signal 3-20 V logic)	5	adafruit.com	$3.95	$19.75	Certain; check: confirmed.
12 V supply for LED rings: 24 V to 12 V DC-DC buck, 3 A	1	Amazon B00CGV5NT4	$8.00	$8.00	Likely; check: not checked. Price not seen; BOM estimate. Verified alternative: Pololu D36V28F12, $24.95, in stock.

Item	Qty	Link	Price each	Line cost	Notes
Resistor kit: 1/4 W metal-film assortment incl. 220R, 2.2k, 10k, 20k	1	Amazon B07N1YYLZZ	$9.00	$9.00	Guessing; check: not checked. Price not seen. 220R confirmed; 2.2k and 20k not confirmed (two 10k in series work for 20k). [ask school first]
Capacitors: 100 nF ceramic capacitors (10)	1	adafruit.com	$1.95	$1.95	Certain; check: not checked. 100 nF only, not an assortment. [ask school first]
Lever connectors: WAGO 221-415, 5-port lever connectors (25)	1	automationdirect.com	$16.50	$16.50	Certain; check: confirmed.
Branch fuses: 3 inline ATC blade fuse holders (18 AWG) + ATC fuses 1 A, 3 A and 5 A (spares of each)	1	search: 'inline ATC fuse holder 18AWG'	$12.00	$12.00	Guessing. NEEDS A BETTER LINK (not sourced). 1 A ATC fuses are often missing from assortments.
Power wire: 18 AWG (or 16 AWG) stranded wire, red + black, about 25 ft each	1	homedepot.com, homedepot.com	$14.74	$14.74	Certain; check: not checked. 16 AWG PVC (60 V) instead of 18 AWG silicone: bus runs only; needs 1.5 mm2 ferrules. [ask school first]
Z motor cable (moving): 4-core 22 AWG shielded high-flex (drag-chain) cable, 5 m	1	search: 'TRVVP 4 core 22AWG shielded drag chain cable 5 m'	$14.00	$14.00	Guessing; price not seen. NEEDS A BETTER LINK. Shielded 4-core 22 AWG drag-chain cable, ~2.5 m used. Alternative (unshielded, sold by the foot): igus CF9-05-04 at KB-3D, $2.47/ft: order 9 ft, not qty 1.
Head signal cable (moving): 8-core 24 AWG shielded high-flex cable, 5 m	1	search: 'TRVVP 8 core 24AWG shielded drag chain cable 5 m'	$18.00	$18.00	Guessing; price not seen. NEEDS A BETTER LINK. Shielded 8-core 24 AWG; core 8 = carry sensor OUT. Alternative: two runs of the 4-core igus CF9-05-04 above (2 x 9 ft = $44, unshielded), then renumber the cores in WIRING.md sec. 11.
Carry sensor lead: 3-core, about 0.2 m (the endstop board's own 3-pin cable) + heat shrink	1	-	$0.00	$0.00	Certain.
Ferrules: Insulated ferrule kit, 800 pcs (0.5-10 mm2)	1	adafruit.com	$17.95	$17.95	Certain; check: confirmed. No 0.25/0.34 mm2: crimp 24-26 AWG ends into Dupont/XH instead. [ask school first]
JST-XH connectors: JST-XH 2.54 mm kit, 230 pcs (2-5 pin housings, pins, headers)	1	west3d.com	$5.99	$5.99	Certain; check: confirmed. Check the Rodent's connector types first.
Dupont connectors: Dupont 2.54 mm kit, 620 pcs	1	west3d.com	$6.99	$6.99	Certain; check: not checked.
I/O board: 5 x 7 cm proto-boards (3-pack, Adafruit #4786) + 6 x 5 mm 2-pin screw terminals (SparkFun PRT-08432)	1	adafruit.com, sparkfun.com	$9.55	$9.55	Certain; check: not checked. [ask school first]
Drag chain: Openable cable chain 15 x 30 mm inner, 1 m, with end mounts	1	search: Amazon: 'cable drag chain 15x30 1m openable'	$14.00	$14.00	Guessing. NEEDS A BETTER LINK (not sourced): openable links, 15 x 30 mm inner, 1 m, end brackets, R38 or larger.
Sleeving: Braided PET expandable sleeve, 10 mm (2 m)	2	west3d.com	$1.49	$2.98	Certain. Slide it on before fitting connectors, or use 10 mm spiral wrap.
Cable ties + mounts: 8 in cable ties + adhesive mounts (50 sets)	1	homedepot.com	$11.99	$11.99	Certain; check: not checked. [ask school first]
Carry sensor printed parts: carry_sensor_mount (6.2 cm3) + carry_flag_arm (17.2 cm3), PETG	1	-	$0.00	$0.00	Certain. Printed from cad/out/printed/.

Order 2: $1,353.28 ($1,556.27 with the allowance).

Only if a test fails (qty 0, not in any total)

Item	Buy only if	Link	About	Notes
Alternative to Misumi (not added to total)	Only if you cut and square stock V-slot at school	openbuildspartstore.com	$130–150	Guessing.
Genuine HIWIN X rail (upgrade)	Generic X rail fails the sec. 6.1 roll test or binds in sec. 13.1	damencnc.com	$420.00	DamenCNC EUR 245/m + cut + block, ex VAT: about $420 [Guessing on the exchange rate].
Z motor upgrade	Sec. 7.8 lift test still fails after the free fixes	omc-stepperonline.com	$55.00	No good match: the nearest (17E19S1684MB4-300RS) has a 300 mm screw, under the 311-356 mm window; ask for a custom screw.
Z counterweight	Sec. 7.8 hold test creeps	shop.sdp-si.com	$20.36	Carries about 0.66 kg [Likely], more than the head's CAD rollup of about 0.46 kg [Guessing] (DESIGN.md §4): weigh the head and pick a spring at or a little under its weight. Its mount is not in the CAD.
Light roof	Phase 0 or first runs show room-light glare	dollartree.com	$5.00	Likely.
Roof tape	Same as the light roof	homedepot.com	$12.00	Likely.
Roof posts	Same as the light roof (2 posts)	us.misumi-ec.com	2 × $7.00	Guessing.
Drag-chain USB cable (upgrade)	Head-camera dropouts the spare cable does not cure	l-com.com	$50.00	L-com CAUAAHFX-3M is A-A: add an A-female coupler. Rated 10,000 flex cycles, not drag-chain duty.
Vacuum pump upgrade (branded)	Generic pump fails the sec. 8.3 seal or reservoir test	search: 'miniature diaphragm vacuum pump 24V -80 kPa datasheet flow curve'	$200.00	Guessing, not sourced. Do NOT buy the VXB 'tiny 10 L/min' pump: it is probably the main pump itself.
Second cup material (marking test)	Urethane cup fails the 24 h marking test (buy 2)	schmalz.com	2 × $18.00 [Guessing]	Schmalz PFG 25 HT1-60 N007: price needs a Schmalz login, and an M10x1-to-G1/8 adapter nobody has sourced. Until one is found, a cup that marks means: stop and ask.
Branded E-stop (upgrade)	Monthly E-stop check finds the 24 V bus live	digikey.com	$175.00	Schneider XB5AS8444 (2 NC) + XALK01H7 box, about $155-195, 5-week lead time. Do not buy ZBE102 blocks for the AutomationDirect box.
Flyback diodes	Only if you swap the #5648 modules for diode-less ones	Amazon B07DHMKTXV	$6.00	Likely.
Stepper extension	X motor lead does not reach the Rodent (or make it)	search: 'stepper motor extension cable 1m'	$8.00	Likely.
Junk graded slabs	Acceptance tests (BUILD_GUIDE sec. 13): take them from your own stock	-	$0 from stock, or $150–400	Guessing.

Open problems

- **No product link yet** (estimates kept in the totals): the 40 mm COB head ring (order 1a, needed for the jig; a 12-SMD ring that fits the seat is the fallback), the branch fuse holders with 1/3/5 A fuses, the drag chain, the M5×10 button heads and the two shielded moving cables (the igus cable in their notes is unshielded, 4-core and sold by the foot).
- **Spec differences to act on:** the Class 2 laser is 7 mm (the left gate post now prints with a 7.2 mm hole); the bed is 17.9 mm birch ply, possibly with voids (drill a test hole); the dowels are 5/16 in (use a 5/16 in bit); the endstop boards do not name the TCST2103 and the ELP board's hole pattern is not given (measure both before printing the Phase 0 parts); the station seat takes a 60 or a 63 mm ring; the GT2 kit's idlers may be

smooth; Misumi's -TPW tapping may cover only one bore; the CNC Kitchen M5 insert sits 0.5 mm proud of the awl_adapter pocket; the AutomationDirect E-stop has three NC blocks (use at least two in series) rated 1.5 A DC-13; the hand vacuum pump is not proven to reach −80 kPa (Phase 0B needs −60 kPa); the Schmalz HT1 fallback cup needs an adapter nobody has sourced; the homing switch pack's lever length is not known (measure it, and cut any lever over 23 mm back to 20 mm).

- **Prices not seen on a live page:** the Misumi lines (get the quote), and the lines whose Notes say "price not seen" (their BOM estimates are used).

Research summary

Confidence tags: **[Certain]** = primary source, **[Likely]** = strong inference or several secondary sources, **[Guessing]** = filling a gap. Research date: October 2026. Many sites blocked direct page fetches, so most facts come from search-engine extracts of the cited pages.

1. Does this machine already exist?

No. [Likely] No product at any price takes 200–300 graded slabs, feeds them itself, and returns cert numbers.

- **Raw-card sorters are built for raw cards only.** TCGplayer Roca Sorter/Sifter, PhyzBatch, SortSwift, CardMill and the open-source mault all exclude slabs. Roca's FAQ says no toploaders, and PhyzBatch says unsleeved cards only. [Certain]
- **Sheet-fed scanners don't take slabs.** Ricoh/Fujitsu advises against feeding slabs, and Card Dealer Pro lists the fi-8170 as "Slabs: No". [Certain]
- **Every slab-aware tool needs a person to present slabs one at a time or lay them out for a photo.** This covers the PSA app, Courtyard, CollX, Kronozio, Card Ladder, Card Dealer Pro, Misprint and Slabox. [Certain]
- **The closest mechanical precedent is graded-slab vending machines.** VMFS advertises "elevator delivery". That proves one-slab-at-a-time handling can be built, but it isn't a scanner. [Likely]

The no-build option, about $350–$650

1. Buy a hands-free 2D presentation scanner: Zebra DS9308 (about $320–440) or Honeywell Genesis XP 7680g. [Certain] Get a 2D imager, not a laser scanner. PSA fronts carry an ITF or Code128 barcode, and PSA and CGC backs a QR (§4). [Certain, real labels]
2. Configure it to send Enter after each scan.
3. Scan into a spreadsheet, or into Card Dealer Pro's "Search Cert Numbers" field.
4. Realistic speed is 20–30 slabs/min, so 250 slabs is about 8–13 minutes. [Likely] Handling is the bottleneck, not the scanner.

What the machine buys you

- Those 10–15 minutes per 250 slabs, plus a front and back photo of every slab.
- The photos matter. Fake labels reuse real cert numbers, and swapped-card slabs still pass a cert check. [Likely]

2. PSA slab physical facts (design inputs)

Item	Value	Confidence
Outer size	80.3 W × 135.5 H × 6.25–6.9 T mm. PSA published 80.3 × 135.1 × 6.86 for its "thinner holder for thicker cards". The 2024 holder has an "identical exterior".	Certain/Likely
Design max	82.6 × 136.5 × 7.0 mm (PSA). CGC 7.5 T, BGS/SGC 8.5 T, PSA thick 10.25 T.	Likely
Mass	42–45 g (old holder), 50–60 g (2024+ holder). Design max 65 g.	Likely
Surface	Flat, glossy, **no stacking ridges** (slabs slide on each other). Scuffs easily. Edges smooth and sonically welded. Corners chip on impact.	Certain/Likely
Label	About 70 × 20 mm at the top of the front, above the card window.	Likely
Front code	1D barcode with only the cert number: **ITF** (always 8 digits) on 2016-and-older and 2017+ "lighthouse" labels, **Code128** on current labels.	Certain, real labels (§4)
Back code	2016 labels: ITF only. 2017+ lighthouse: **QR** `http://psacard.com/cert/<n>` + ITF. Current labels: **QR** `https://www.psacard.com/cert/<n>/` + Code128. Before 2016: none.	Certain, real labels (§4)

Item	Value	Confidence
Cert format	8 digits until about Nov 2024, 9 digits since. 2026 certs are about 100M–160M.	Likely

Consequences for the design:

- **300 slabs = about 2 m of plastic and 15–18 kg.** No single stack can hold them.
- **No slab may be slid out from under a loaded stack.** Face-on-face sliding under load is the main scuff mechanism.
- **The ITF is always the cert zero-padded to 8 digits; 9-digit certs never use ITF** (they use Code128). The software strips leading zeros before comparing reads and treats an ITF of any other length as a misread. [Certain, real labels]
- **ITF can "short read"** (a glare-cut `289412` was seen), so a lone 1D read is never trusted without a second agreeing source.

3. Cert → card details (optional)

- **Endpoint:** `GET https://api.psacard.com/publicapi/cert/GetByCertNumber/{cert}` with an `Authorization: bearer <token>` header. [Certain, from the Swagger spec]
- **Quota:** documented as **100 calls/day** on free accounts. [Certain]
- **Approval gate:** since about July 2026, several developers report valid tokens refused with `403 "Access to this API is limited to approved customers"`. The fix is to email collectors-apis@collectors.com. [Certain that it happened, Likely general]
- **Bulk alternatives:** Card Hedge API (1,000+ cert lookups/day, 100 per call, from about $49/mo). [Likely] Card Dealer Pro's paste-a-list-of-certs import. [Certain]
- **The machine never needs the network.** The cert list comes straight from the barcodes. `tools/enrich.py` is a separate, resumable, quota-metered step.

4. Real label formats (verified on public scans, Oct 2026)

What was looked at. [Certain] 171 images of graded Pokémon slabs, listed in a local index (not in the repository; no image was copied into it). 128 of them are "good": a real Pokémon slab at 800 px or more. They form 64 slabs, each with a front and a back image:

- PSA 31: 10 current labels, 11 2017+ "lighthouse" labels, 3 from 2016, 7 from before 2016;
- CGC 31: 19 2023+ standard, 5 Pristine gold, 5 2020–23 "OG" blue, 2 Certified Authentic;
- 1 BGS and 1 SGC;
- 36 English and 28 Japanese cards.

The good images are 116 studio scans and 12 seller images, from **fanaticscollect.com** (116) and **myslabs.com** (12). The other 43 images (phone photos, PSA and CGC guide pages, press images, renders and mock-ups) were used only to collect more payload strings, from **cgccards.com**, **cgcgrading.com** (guides and news), **allvintagecards.com**, **forums.collectors.com** and **legitapp.com**. Every code was decoded with the machine's own decoder (zxing-cpp) and parser.

What each label carries [Certain for this set; the payload strings are in `software/tests/test_real_payloads.py`]:

Label	Back label	Front label
PSA current (9-digit certs, some 8)	QR `https://www.psacard.com/cert/<n>/` + Code128 `<n>`	Code128 `<n>`

Label	Back label	Front label
PSA 2017+ lighthouse (8-digit)	QR `http://psacard.com/cert/<n>` (some the https form) + ITF	ITF
PSA 2016	ITF only	ITF
PSA before 2016	no code	ITF
CGC 2023+ (10 digits; one 13-digit cert, 140 + 10)	QR `https://www.cgccards.com/certlookup/<n>/<grade>/` (any case, slash optional)	short-bar Code128 `<n>`
CGC 2020–23 OG	QR on `cgccomics.com/certlookup/<n>/<grade>` + Code128 of 26–28 digits whose last 10 are the cert	no code
CGC Certified Authentic	no code	QR `…/certlookup/<n>/0_0/` + Code128 `<n>`
BGS, SGC	no code	no code (the printed cert only)

- **Every one of the 54 QR payloads with a cert agreed with the printed cert.** [Certain]
- **The ITF is the cert zero-padded to 8 digits.** All 48 ITF payloads were 8 digits, including three 7-digit certs on guide photos. 9-digit certs never used ITF. [Certain for the set; Likely in general: a reholdered 7-digit cert could sit on a QR + ITF label] On the oldest labels the printed number is zero-padded too: OCR read `02074012` and `02458772`. [Certain, 8 extra old-label images]
- **ITF appeared only on PSA labels.** CGC uses Code128, and the set's 1 BGS and 1 SGC slab showed no code at all; a sweep of 216 BGS, SGC and CGC frames found no ITF of any length. [Certain for the set] But BGS and SGC are reported to print a code on some holders: Beckett's FAQ, quoted second-hand, mentions "a bar coded serial number", and SGC says the cert can be read from a QR code on newer holders. Their symbology is unknown. [Likely] The machine's 8-digit ITF rule (an ITF-8 confirmed by OCR names PSA) assumes no other grader prints an 8-digit ITF. [Guessing]
- **CGC 2023+ backs never carried a Code128** (0 of 26) and **graded CGC fronts never a QR** (0 of 26). OG fronts carried no code (0 of 10). [Certain for the set, Likely in general]
- **Grade `0_0` was on both Authentic fronts**, and their backs decoded nothing. One official CGC render of a 2023+ back also shows a `0_0` QR (a placeholder cert), so "0_0 = front" is [Likely], not certain; calling a slab face-up by mistake only stops the run.
- **A CGC QR without a cert** (`…/certlookup/`, `cgccomics.com/verify`) appeared only on press images and mock-ups of back labels [Likely]: a hint, not proof.

What a code seen from above proves about the slab's side (the machine's face-down rule uses only the [Certain] rows): a PSA QR, a graded CGC QR and the CGC 26–28-digit barcode are printed only on back labels (face-down); a CGC `0_0` QR and a CGC 10- or 13-digit Code128 only on front labels (face-up); a PSA ITF or Code128 is on both, so it proves nothing. Applied to these images: 50 of 64 backs proven face-down, 22–24 of 64 fronts proven face-up, 0 false proofs either way; the 14 backs left unproven are exactly the slabs with no back-label code. [Certain]

Failure modes seen on real labels [Certain]:

- **A phantom Code128:** zxing-cpp returned `09435764` as a thin band inside the real `27435764` at full resolution only (absent at 10 other scales from 0.5× to 2×).
- **A glare-truncated ITF:** `289412` read from `28941258`.
- **A silent wrong cert from the old parser:** its CGC rule cut the 13-digit cert to 12 digits and called it `ok`; nothing else on the slab contradicted it.
- **OCR misreads:** a first digit 5 read as 9, other digit runs on the label or the card, a BGS `0019865140` read as `0079865140`. OCR is never the only evidence.
- **Phone photos through the plastic lost the QR** on both lighthouse backs tried (ITF only): the closest thing here to the machine's camera, and the case where the face-down proof fails.

- **Tesseract timeouts came from OpenMP threads:** on a loaded 4-CPU computer every OCR call with the default threads timed out at 30 s; with `OMP_THREAD_LIMIT=1` a station frame took about 3 s and the label band alone about 1.4 s.

Where the card sits [Likely ±0.5 mm, scale from the card itself, 63 × 88 mm; `cad/layout.py HOLDERS`]: measured on the good studio-scan backs, the card's top edge lies 97.2–100.0 mm (median 98.7) from the holder's far (non-label) end on 22 PSA slabs and 93.8–96.2 mm (median 95.2) on 27 CGC slabs, so a CGC card top sits about 3.5 mm nearer the far end. The face-down card window is placed from these numbers (the far end is what sets the worst case: a slab at the cup end of its pocket play has its far end on the wall). The PSA median length came out at the nominal 135.5 mm, which checks the method. The CGC holders measured about 133.9 × 79.4 mm, but most CGC scans are cropped a few pixels inside the slab edge, so that probably reads short: CGC publishes about 3.15 × 5.30 in (80.0 × 134.6 mm) [Likely]. The card-top distance is not affected. A second measurement of every good back (card bottom to far edge, plus 88 mm) found one PSA back the set above leaves out, `fc_18805593`, at 96.29 mm, and put the CGC minimum at 94.18 (0.4 mm on the safe side). The PSA one would cut the PSA window margin from about 3.7 to about 2.8 mm: harmless, because the CGC worst case sets the window. [Likely] An earlier pass, on 27 PSA backs, gave the card top 34.8–37.8 mm from the label-end edge (`CARD_TOP_FROM_EDGE`) and guessed CGC 2.5–3 mm further in from 15 backs; the per-grader numbers above replace that guess. The simulator now uses them too (`sim.HOLDERS`, the same as the CAD's): a PSA card top 34.8–38.3 mm from the label-end edge (both PSA measurements), a CGC one 37.7–40.1 mm on its 133.9 mm holder with 2.55 mm of play. [Certain, `tests/test_carry_sensor.py` checks it against the CAD's export]

Read rates at the machine's resolution [Certain for flat scans]: PSA codes read on every slab; the CGC 2023+ front Code128 was the weakest symbol (16 of 19 standard labels; very short bars and a tight quiet zone); a synthetic glare spot on the label cost 2 of 19 standard and 1 of 5 gold CGC back QRs and some PSA front barcodes.

Limits. [Certain] These are studio scans and seller photos, not the machine's cameras: flat, evenly lit, with no slab-plastic reflections, curvature, defocus or lens-edge effects, so real read rates will be lower [Likely]. Some groups are small (2 Authentic, 3 PSA 2016, 5 OG, one 13-digit CGC cert), so "never seen" is [Certain] for this set and [Likely] in general. The machine's parsing and face-down rules were written from these same slabs, so "0 wrong certs and 0 false proofs" here shows the rules fit them, not that they hold on new labels: Phase 0 on your own slabs, and `python -m tools.real_eval --phase0`, is the test that counts.

SlabRail software

Runs on the Raspberry Pi 5 in the machine. It moves the head through the FluidNC controller, reads every slab's label, and writes `batches/<id>/certs.csv`. Everything also runs on any PC against a simulator (`--sim`), so you can learn the commands before the machine exists.

[Certain] None of this has driven real hardware yet. It is tested against the simulator, which raises an error on any unsafe move. Expect to fix small things during the first supervised runs, with a mechatronics instructor next to you.

Install

On the Pi (Raspberry Pi OS Bookworm 64-bit), from this `software/` folder:

```
bash setup_pi.sh          # apt packages, I2C on, venv, config/machine.toml
sudo reboot
source .venv/bin/activate
```

The script installs numpy, OpenCV, picamera2 and gpiozero from **apt**. The venv uses `--system-site-packages` so it can see them. Do not `pip install numpy` or `opencv-python` on the Pi. A pip numpy hides the apt one, and picamera2 then breaks. `bash setup_pi.sh --service` also starts the web page at every boot.

On a PC (for the simulator and tests):

```
python3 -m venv .venv && .venv/bin/pip install -r requirements-dev.txt
.venv/bin/python -m pytest -q tests        # 300 tests (1 skipped without SLABRAIL_REAL_IMAGES), about 16-21 minutes
```

First-time order

Do these in order. Each step depends on the one before it.

Step	Command	What you do
1	`python -m tools.gen_config`	Writes `config/machine.toml` from `cad/layout.py` (setup_pi.sh already did this).
2	`python -m slabscan selftest -i`	Checks every part. `-i` adds the hand checks: push the cup up, press a slab on the cup, block the beam.
3	`python -m slabscan home`	X goes to its left switch, Z to its top switch. Keep a hand on the e-stop the first time.
4	`python -m slabscan mark`	**Before the rods go in.** Awl adapter on. For each pocket, the awl drops into the drill template's two dimples. You clamp the template and drill 8 rod holes. Pockets 0 and 12 come last, because one of their dimples is past the X travel: two 5/16 in dowel offcuts through the template into the neighbour's holes set them. `--only 0,12` continues after a break.
5	Glue the rods, lay the pads	
6	`python -m slabscan calibrate z`	Cup on, pocket 0 empty. Probes the bare pad 4 times and writes `pad_trigger_c`. It also checks that the contact sensor triggers when you push the cup up by hand.
7	`python -m slabscan calibrate x`	Awl on. First mark the centres of pockets 0 and 12 with a tape cross on each pad (BUILD_GUIDE 12.2). Centre the awl over pocket 0, then over pocket 12: first by eye at the safe height, then low over the tape cross. Writes `offset_x` and prints the scale error across the row.
8	`python -m slabscan focus head`	One slab face-down in pocket 0, pocket 12 empty. It first checks the camera's aim over pocket 12 (`camera aim: OK` or `FAIL`, picture in `focus/focus_head_empty.jpg`). Then the cup rests on the slab: turn the M12 lens until the score peaks, then lock the lens nut.
9	`python -m slabscan focus station --save`	One slab in pocket 0. The machine picks it up and holds it over the station camera. It sweeps `LensPosition`, saves the sharpest value that decodes, then puts the slab back.

Step	Command	What you do
10	`python -m slabscan run --name test`	First real batch, on junk slabs, with you watching.

Calibration values go to `config/calibration.toml`. That file overrides any key in `machine.toml`, and `gen_config` never overwrites it. Put your own settings there too, for example `serial_port`, `cam_head`, `custom_regex`, `front_photo` or `lid_switch = true`. Two of them change what can be `ok`: `custom_regex` types every ITF by your pattern, so the 8-digit ITF rule never fires and every pre-2016 PSA slab is `needs_review`; and `psa_cert_digits` (lengths without leading zeros) must include 7 if you have pre-2016 slabs, or every 7-digit cert is `needs_review`.

Exposure. The LED rings switch on just before each photo. After switching on, the software waits `light_settle_s` (0.3 s, **[Guessing]**) so auto exposure can settle. If the first photo of a slab comes out too bright or too dark, give the station camera a fixed exposure once focus is set, for example `cam_station = { type = "picamera", index = 0, lens_position = 7.1, exposure_us = 8000, gain = 1.0 }`. Tune the numbers with `focus station`.

Commands

All commands are `python -m slabscan <command>`. Add `--sim` to use the simulator, as in `python -m slabscan --sim jog`.

Command	Does
`selftest [-i] [--motion]`	Checks the controller, the probe pin (a cut wire reads TRIGGERED because the wiring is fail-safe), the vacuum sensor, pump, valves (and, with `-i`, seal, leak and vent), the LED rings, both cameras (saves an image of each), the overfill beam, the lid switch, the carry-height sensor (GPIO 6; with `--motion` it homes, steps the head down past `carry_sensor_c0` over pocket 0 and reports where the sensor trips), tesseract, disk space and, in learned mode, the camera aim the last run measured over an empty pocket (WARN until one has). Exit code 1 if anything fails.
`home`	Homes both axes and prints the position.
`jog [--no-rods]`	Typed moves, `x 300`, `x +2`, `z -1`, `p 3`, `station`, `up`, `probe`, plus `keys` for arrow keys. `help` lists everything.
`calibrate z` / `calibrate x`	See above.
`focus head|station [--save] [--no-motion]`	Live sharpness score (Laplacian variance of the sharpest tiles) plus what decodes. With `--no-motion` you hold the slab in view, for example in the Phase 0 jig (then the camera aim is not checked).
`mark [--only 0,12] [--dry-run]`	Drill-template walk (above). `--dry-run` prints the order and positions.
`run [--name NAME] [--front-photo always|fallback] [--all-face-down] [--yes]`	One batch. It refuses to start before `calibrate z` and `calibrate x` have been run. `--yes` skips the start question. `--all-face-down` turns the face-down check off for this batch (below).
`resume BATCH_DIR [--accept-heights] [--face-down-ok]`	Continues a stopped batch (below). `--face-down-ok`: you checked the slab the face-down check stopped on. It is refused unless the last stop was that face-down stop, and it confirms that one slab only.
`sim [--slabs N] [--mix MIX] [--eras E] [--face-down-check] [--jp-rate R] [--sim-head cad|cad_r2|open|none] [--sim-head-tilt DEG]`	Full simulated batch. Prints the summary and a time estimate. `--mix` sets the label mix per grader and era, e.g. `psa:current=0.6,psa:2017+=0.2,cgc:2023+=0.2` (kinds: `psa:current|2017+|2016|old`, `cgc:2023+|og|authentic`, `bgs:`, `sgc:`; default: mostly PSA, some CGC, a few BGS); `--eras` is a PSA-only override. `--face-down-check` keeps the check on and counts the slabs it would stop at (a simulated operator confirms each). `--jp-rate` gives that share of cards the Japanese back. `--sim-head`: `cad` (the default) is today's head (lens 5 mm toward the label, card window clear), `cad_r2` the round-2 head whose foot hides the card window, `open` a hypothetical narrow head. `--sim-head-tilt DEG` tilts the simulated head camera toward the cup (a mis-aimed camera; it works with every command that takes `--sim`).
`web [--port 8080]`	Phone page (below).

Command	Does
`phase0 [--sim N] [--out DIR] [--face-up N] [--save] [--force]`	The Phase 0 camera test (same as `python -m tools.phase0`): reads 30-50 of your slabs in the Phase 0 jig before you buy any motion parts. The read-rate report is written as soon as you type `q`. The face-up pass is off unless you ask for it (`--face-up` defaults to 0); a GO for the learned face-down check needs 15 face-up slabs. `--save` writes its threshold on a GO. Each run needs its own `--out` folder: one that already holds a `phase0.csv` is refused, and nothing in it changes (`--force` overwrites it).
`refs [--drop CERT | --drop-operator]`	Lists the learned face-down check's card-back references, with the batch, position and cert each came from. `--drop CERT` deletes one slab's reference, for example after a wrong confirmation; `--drop-operator` deletes every reference you confirmed and keeps the back-QR ones.

Jog safety rules (enforced in software):

- Nothing moves before homing.
- Every downward move is a probe move that stops on contact.
- Below the rod tops, the head moves sideways only within ±5 mm of a pocket centre. The head is 70 mm wide and a pocket is 84 mm.
- A move is never started while the probe already reads triggered.
- No move or probe target goes below machine Z −7.5. The firmware's soft limit is −8, and the CAD keeps the Z carriage on its rail down to −12. A probe that would reach further is cut short there.
- After an E-stop the next command homes first (Z straight up, then X) and is not run itself.

`--no-rods` lifts the sideways rule. Use it only while the bed has no rods.

Running a batch

Loading rules:

- Slabs go **face-down**, **label end toward you**, in pockets 1–11.
- No stack goes above the MAX line on the fill gauge.
- Pocket 0 and pocket 12 stay empty.

Face-down check (`face_down_check` in the config). Before the cup touches a slab, the frame from above must prove the slab is face-down. Otherwise the run stops with nothing touched, and you confirm it with `resume --face-down-ok`.

- `"stop"`: only a back-label code proves it (a PSA back QR, a graded CGC back QR, the CGC 2020-23 back barcode). Older PSA labels, CGC Authentic, BGS and SGC stop every time.
- `"learned"` (the default): a back-label code, or a card window that matches the card backs of earlier code- or operator-confirmed slabs (`slabscan/facedown.py`). The references live in `config/card_backs.npz` (not in git; `refs` lists them, `refs --drop CERT` deletes one, and deleting the file starts over). It only becomes active when all of these hold:
  - Phase 0 wrote `face_down_threshold` and `face_down_threshold_sig` (`phase0 --face-up 15 --save` on a GO). A threshold measured for another card window, or typed in without its sig, is not used. No threshold below 0.55 is accepted.
  - 5 references exist, and 5 slabs proven face-down on this machine showed the card window clear: at most 10% of it may look the same on every slab as over an empty pocket.
  - Until then it acts like `"stop"`.
  - Today's CAD head (lens 5 mm toward the label, trimmed foot) leaves the card window clear [Certain, in the CAD and the simulator]. A head that hides the window (the round-2 head) is detected and refused, and so is one that hides only part of it [Certain, sim: a foot edge 3 mm or more inside it; an edge 1 mm inside is not seen, and it lifts face-up scores from 0.37 to 0.52].

    - A slab is accepted only when **two** references (two different slabs) of one back design match it, not just the best one [Certain, 64 real slabs: one odd back, a card sitting low in its holder, matched a blue Water-type front at 0.56-0.62; no second back came above 0.44]. Blue Water-type fronts are the closest real match to a card back: put one in Phase 0's face-up pass.
    - The camera's aim, checked in every run on the view over an empty pocket: only the pad may show in the middle of the card window and 1 mm beyond its cup-side edge. A tilt toward the cup brings the cup pad and the foot in there. Then the learned check accepts nothing and learns nothing that run, and the start line, every stop, the summary and `selftest` say to re-aim the camera (BUILD_GUIDE §7.7). `focus head` looks over the empty pocket 12 first and says the same. A verdict counts only for the run that took it. A resume looks again over an empty pocket at its survey. If no pocket is empty, the learned check accepts nothing and learns nothing until the run empties its first source pocket and looks over that one. So a camera knocked while the run was stopped is seen. [Certain, sim, round-3 card window: a camera knocked 1° while stopped is caught, with and without an empty pocket at the resume.] [Certain, sim, round-3 card window: a square camera passes, 0.5° toward the cup is caught; the camera-locked check alone let 0.6–1.0° through: only 0–2% of the window was camera-locked. With today's smaller window a head part enters at about 1.3°, and the check fires from about 1.3° (a verifier measured 1.30° and 1.33°).]
    - What the aim check does not see [Certain, sim and geometry]:
        - A head part as dark as the pad shows no edge, and the 10% camera-locked rule does not catch it either while it covers a strip of the window (a dark foot over 9–26% of the window read 0.3–0.9% camera-locked).
        - It checks only a tilt toward the cup. A tilt away from the cup, or a roll along the row, is not checked.
        - Its thresholds come from the simulator's flat pad. Lint, dust or glue on a real pad can make it fire. That is safe (the learned check stays off), but it needs a real empty-pocket picture from Phase 0 to tune.
    - A back design known only from your confirmations (no back-code slab) is used once three of your confirmations agree. A code-proven or confirmed slab is learned from only after it was placed: not if the read station saw a back-label code from below (it was face-up after all), and a code proof only with an 'ok' read of that same cert (a neighbour's back QR in view gives another cert).
- `"off"` (or `run --all-face-down`): no check. A face-up slab gets the cup on its front.

The machine:

- homes,
- probes every pocket,
- then moves stack 1 onto pocket 0, stack 2 onto pocket 1, and so on, reading each slab on the way.

Afterwards each stack sits **one pocket to the left, in reverse order**. Pocket 12 holds the slabs to check by hand: unreadable ones, numbers read only by OCR (no grader), conflicts, or graders other than PSA and CGC. Only a `needs_review` row of PSA or CGC may stay in the output stack. A grader guessed from a barcode's length counts, so another grader's slab whose barcode has a PSA or CGC format can end up there as `needs_review` too. The same holds for an `ok` from an 8-digit ITF: another grader's slab with an 8-digit ITF whose printed number OCR reads would come out `ok PSA` [Guessing that none exists: none was seen in the real set, which has 1 BGS and 1 SGC slab; BGS and SGC are reported to print some code on some holders, type unknown]. Put any non-PSA slabs you own through Phase 0 (`python -m tools.real_eval --phase0`) before trusting `ok PSA` on slabs with no QR.

What makes an `ok` (`slabscan/certs.py`):

- A QR carries its own error correction. A barcode (ITF, Code128) needs a second printed code (the other label side, or the QR) or OCR of the printed digits to agree.
- Two barcodes alone (no QR) can agree on the same misread, because both sides carry the same pattern. So the front OCR also checks every barcode-only `ok`. If it reads other digits, the slab is `needs_review`; if it reads

nothing, the slab stays `ok`.

- Two frames of one printed code are one source. A thin false read inside a real barcode (a phantom) is dropped only when a different code or OCR confirms the other number, and then from every frame of that barcode. Only the thin read can be dropped. If the confirmed number is the thin one, the slab is a `conflict`.
- The grader must come from a code that names it (a cert URL), from two printed codes in its format, or from an 8-digit ITF whose digits OCR confirms. That last rule names PSA: in the real set only PSA labels carry ITF (CGC uses Code128; its 1 BGS and 1 SGC slab showed no code). It is what makes a pre-2016 PSA slab (front ITF only) `ok`, with the note "barcode confirmed by OCR; grader PSA from its 8-digit ITF". An ITF alone, or OCR alone, is never `ok`; an ITF of any other length is a misread. A code in no grader's format (a QR or other code with a bare number) never confirms a grader, not even beside an ITF. Any other bare number gets its grader from its length only, and OCR confirms its digits, not its grader: a slab whose only code is one front Code128 is `needs_review` even when OCR agrees.
- A PSA Code128 is 8 or 9 digits, and a 10-digit CGC Code128 never starts with 0 (every real one). A Code128 outside those formats (a 7-digit number, a BGS cert like 0019865140) is not read as a cert; it is listed as unparsed.

Carry-height sensor (Z stall guard). Before every sideways move the head must read "at carry height" on GPIO 6 (and "below it" at every grip, which catches a dark sensor). A Z stall that left the head more than about 2.5 mm low stops the run ("Z stall suspected …"): a held slab is lowered straight onto its own pocket, or held on the pump for you. `selftest --motion` measures where it trips. `jog`, `calibrate` and `mark` do not read it.

Stopping

- Ctrl-C once, or Stop on the web page, finishes the current slab and parks with nothing in the cup.
- Ctrl-C twice is a hard stop: the motors halt where they are and nothing moves after that. A slab in the cup stays held by the pump until you hold it and type `vent`. The next run homes first.
- Use the e-stop for anything dangerous.

E-stop. It cuts the 24 V supply: the controller, the motors, the pump and both valves. The controller's USB port disappears. The software then:

- stops at once and moves nothing more (it can't: the motors have no power),
- shuts the valves from the Pi side, so nothing switches on when the power comes back,
- tells you what to do. If a slab was in the cup, the closed valves keep it there for a while. Hold it, pull it off gently and put it face-down on top of the pocket the message names.

Twist the E-stop to release it, then resume (command line or web page). The software reconnects to the controller and homes before anything else moves. It does not trust the controller's position after a reboot, because FluidNC's unlock command (`$X`) marks the axes as homed without homing them (**[Certain]**, FluidNC source `ProcessSettings.cpp`, `disable_alarm_lock`).

When the machine stops on its own, it prints what to do. Examples: a cup that won't seal, a stack taller than expected, lost vacuum, or a full exception pocket. If a slab was still in the cup, the machine first puts it back on top of the stack it came from, so it gets read again on resume. If the vacuum was lost, it does not vent and does not move (venting could drop the slab). It tells you which stack to put the slab back on.

Resume: `python -m slabscan resume batches/<id>`.

- The state is rebuilt from `journal.jsonl` and `certs.csv`.
- Every pocket is probed again and compared with what the journal predicts.
- You may empty the exception pocket before resuming. Counting there restarts from what is left. An empty pocket also lets the resume check the camera's aim at once. With no empty pocket, the learned face-down check accepts nothing until the run empties its first source pocket: every slab without a back-label code stops for you until then (the stop says "the head camera's aim is not checked this run yet").

- Any other change of more than 4.5 mm (under one slab) stops the resume and lists the pockets. A missing or extra slab must be put back first.
- If it looks like one slab moved without being recorded (a power cut mid-move), the message says which slab to move back.
- `--accept-heights` takes the new heights instead. The slab-count check at the end then flags the difference.

Output (`batches/<id>/`)

File	Contents
`certs.csv`	One row per slab: cert, grader, status (`ok`, `needs_review`, `no_read`, `conflict`, `other_grader`), which pocket it came from and where it is now (`out_pocket`, `out_index` counted from the bottom), image paths, and `face_down`: how it was proven face-down (`qr` a PSA or CGC back QR, `code` the CGC 2020-23 back barcode, `operator`, `learned(0.93)` or `off`). Written row by row, so a crash loses nothing.
`certs.txt`	Only the `ok` certs, one per line, in machine order. Paste it into a spreadsheet or Card Dealer Pro.
`report.html`	Every slab with its label photo and a Face-down column (learned rows in bold: check their photos). Opens on a phone.
`summary.json`	Counts, duplicates, the per-stack slab count check, `itf_mode` (always `zero_pad`: the ITF is the cert padded to 8 digits) and `itf_evidence` (slabs where QR and ITF agreed), mean cycle time, and `face_down` (mode, counts, every learned row, the references, and in learned mode `camera_aim`: `ok`, the failed check's message, or "not checked this run ..." after a resume that never looked over an empty pocket). A failed aim check also prints a `CAMERA AIM:` summary line.
`journal.jsonl`	Machine log used by resume, including the face-down events (`face_down_stop` on the stop line, `face_down_confirmed`, `face_down_reference`, `face_down_learned`, `face_down_baseline` with its camera-aim verdict `aim`, `face_down_ok_unused`).
`images/`	Back photo of every slab, plus a front photo when the station camera was used. In learned mode also `face_down_window.jpg` (the run's first slab with the card window drawn) and `face_down_empty.jpg` (the view over an empty pocket).
`stop.json`	Why the last run stopped (deleted when a run finishes).

Web page

`python -m slabscan web` serves `http://<pi-address>:8080/`. On it you can:

- start, stop and resume runs,
- see live counts, the current pocket, the last label photo and the last 10 slabs,
- see the stop message,
- download `certs.csv` and `certs.txt` for every batch.

`python -m slabscan --sim web` runs the page against the simulator.

> **LAN only.** The page has **no login**. Anyone on your Wi-Fi can start or stop the machine. Use it on your own home or shop network only, and never port-forward it or expose it to the internet. The page sends an `X-SlabRail` header that other websites can't add. That blocks drive-by requests from your own browser, but it is not a password.

Card details (optional)

`python -m tools.enrich batches/<id>` adds year, set, card, grade and population from the PSA Public API. The result goes to `certs_enriched.csv`. You never need it: `certs.csv` is complete on its own.

1. Get a token at psacard.com/publicapi.
2. `export PSA_API_TOKEN=...`

How it behaves:

- Free accounts get about 100 calls a day. The tool counts calls and stops at `--daily-limit`. Run it again the next day and it continues.
- Answers are cached in `batches/psa_cache.sqlite`.
- New tokens are often refused with "403 limited to approved customers". The tool stops cleanly and tells you to email collectors-apis@collectors.com.
- CGC certs are skipped, because the PSA API cannot look them up.

Simulator and timing

`python -m slabscan sim --slabs 330` runs a full load in about 5 minutes on a PC. It prints:

- the survey time,
- the seconds per slab,
- the total for a full load. The 23 Oct 2026 run (seed 7, default mix, `--face-down-check`, after the 8-digit ITF rule and the card position per grader) gave 75 s of homing and survey, then 14.2 s per slab: about 79 min for 330 slabs; `ok` 324, `needs_review` 6: 5 BGS slabs read only by OCR (in pocket 12) and one pre-2016 PSA slab whose OCR read other digits (in the output stack). The other 20 pre-2016 PSA slabs are `ok` from their front ITF and OCR. No `ok` has a wrong cert or grader [Certain, sim: checked against the simulated slabs]. Face-down: 287 by back QR, 41 by the learned card-back match (lowest score 0.76), 2 confirmed (the first slabs, before 5 references existed); with back-label codes only, 43 would have stopped. Without `--face-down-check` the check is off and the summary says how many slabs would have stopped.
- The simulated card sits where it does in real holders, per grader (`sim.HOLDERS`, from the CAD's `layout.HOLDERS`): a PSA card top 34.8–38.3 mm from the label-end edge, a CGC one 37.7–40.1 mm from the edge of its shorter holder, so about 3.4 mm nearer the cup. Other graders use PSA's [Guessing].

Motion uses a trapezoid acceleration model. Camera frame times and the Pi's decode speed are **[Guessing]** values, listed at the top of `slabscan/cli.py`. Phase 0 measures the real ones.

Code map

Module	Role
`runner.py`	The batch: survey, transfer one slab, stop/secure, resume (State rebuilt from the journal).
`certs.py`, `decode.py`	Payload parsing, multi-read consensus, zxing and OCR decoding.
`motion.py`	G-code sender for FluidNC (`G38.3` probing, status parsing).
`hw.py`, `camera.py`	Pi GPIO (vacuum, valves, LEDs, beam, lid, carry-height sensor), ADS1115, USB and Pi cameras.
`factory.py`	Builds real or simulated backends from the config.
`ops.py`	jog, calibrate, focus, mark, selftest.
`cli.py`, `web.py`	Command line and web page.
`facedown.py`	Learned face-down check: card-window descriptor, card-back references, the accept rule.
`simworld.py`, `sim.py`	Simulator with a crash monitor, synthetic labels, card backs and fronts, and the head parts in the head camera's view.
`config.py`, `results.py`, `psa_api.py`	Config TOML, batch output files, PSA API client.

Module	Role
`tools/gen_config.py`, `tools/enrich.py`, `tools/phase0.py`, `tools/decode_sweep.py`, `tools/bench_tests.py`, `tools/real_eval.py`	Config generator, PSA enrichment, Phase 0 camera test, decode sweep, the acceptance bench tests, and the real-label check: the run's reading order, consensus and face-down rule on photos of real slabs (`index.json` or `--phase0 DIR`; `--machine` rescales them to the machine's cameras); exit 1 on a wrong cert in an `ok` verdict or a false face-down proof.

SlabRail CAD

Parametric CadQuery model of the whole machine. `layout.py` holds the master dimensions. The model, the exports, the drawings, the geometry checks and the machine software's config all read from it. Everything in `out/` is generated: change the Python, then regenerate. Do not edit the exports by hand.

Nothing has been built yet. The model is checked against itself (177 geometry checks, below), not against real parts.

Units

- Millimetres everywhere: the Python, the STEP, STL and DXF files, `print_list.json` and `cut_list.csv`. Volumes in `print_list.json` are cm³.
- The STEP files are AP214 (`AUTOMOTIVE_DESIGN`), written by Open CASCADE. The DXF declares millimetres (`$INSUNITS` 4).
- The GLB files are the only exception. glTF expects metres and +Y up, so their root node scales by 0.001 and turns the model Z-up to Y-up. The mesh data inside is still CAD millimetres on the CAD axes (see `export_glb.py`).

Coordinate system and origin

Taken from `layout.py` and confirmed by measuring the model:

Axis	Direction	Zero
X	Left to right along the pocket row (the gantry beam's direction).	The left interior edge of cell 0, which is the inside surface of the leftmost guide rods (`rod_00` spans X −8 to 0).
Y	Front (−) to back (+). The operator stands at −Y; slab labels point to −Y.	The pocket row's centre line, which is also the cup's fixed Y. There is no Y axis.
Z	Up.	**The top surface of the pocket floor pads.**

Reference heights and positions, measured on the model:

- Pad top Z 0, bed (plywood) top Z −3, bed underside −21, base-rail underside −41, feet bottom −49.
- The 14 cells are 92 mm apart (84 mm pocket + 8 mm wall line). Cell 7 is the read station (X 686, no pocket), so pocket 0 is at X 42, pocket 6 at 594, pocket 7 at 778, and pocket 12, the exception pocket, at 1238.
- The X beam's front face (the rail mounting face) is at Y 107, and its centre is at Z 400.
- Machine coordinates use the same frame. **Machine X** is the X of the suction-cup axis. **c** (machine Z) is the height of the cup face above Z 0. In the `home` assembly the cup face is at Z 248, which is c = 248.
- Overall envelope in every pose: X −78 to 1358, Y −117.5 to 172, Z −49 to 671. The frame and the Z motor set it. The Z motor rides on the X carriage but does not move in Z.

Poses

`build.POSES` defines four assembly poses. Each one is a call to `model.machine(cup_x, c, held_slab, fill)`. "Full" means 30 slabs (`STACK_MAX`).

Pose	Cup X	c	Slab on the cup	Stacks	What it shows
`home`	134 (pocket 1)	248 (`C_SAFE_EMPTY`, travel height)	no	start of a run: pockets 1–11 full, 0 and 12 empty	The machine as loaded. This is not the homing position: the X home switch trips at cup X 19 and the Z top switch at c = 258.

Pose	Cup X	c	Slab on the cup	Stacks	What it shows
`pick_bottom`	1146 (pocket 11)	6.6 (one slab, `SLAB_T`)	no	pockets 1–10 full, pocket 11 holds 1, 0 and 12 empty	The lowest pick: the cup on the last slab of a pocket between full neighbours.
`carry`	88 (between pocket 0 at 42 and pocket 1 at 134)	248 (`C_SAFE_LOADED`)	yes	pocket 0 holds 12, pocket 1 holds 17, pockets 2–11 full	Loaded travel over the rods.
`station`	686 (station, cell 7)	171.6 (`STATION_READ_C`)	yes	pockets 0–2 full, pocket 3 holds 12, pocket 4 empty, pockets 5–11 full	Reading the slab's front label over the upward camera. The slab's front is 140 mm above the station lens.

With nothing pressing on it, the cup hangs 2 mm (`CUP_HANG`) below c. A held slab's underside is therefore at c − 2 − 6.6. `checks.py` uses many more poses than these four: every pick, place, travel height, hard stop and the Phase 0 jig.

Part names

Every part in an assembly is a `model.Part` with a `name`, `color`, `kind` (below) and `group`. The group is `static`, `carriage` (moves in X) or `head` (moves in X and Z). In the assembly STEP each part is one named, coloured product. In the GLB each part is one named mesh, with kind, group and note in the node's glTF `extras`.

- Subsystem prefixes: `x_` is the X axis (beam, rail, carriage, motor, idler, belt, switch, bracket) and `z_` is the Z axis (spine, rail, carriage, motor, lead screw, plate, end cap, switch).
- `_L` / `_R` mean the left (−X) and right (+X) side.
- Numbered parts: `pad_N`, `stack_N` and `labels_N` are pocket N (0–12). `rod_00`… `rod_81` are the 82 guide rods. `base_rail_0`/`_1` are the two base rails.
- Fastener heads (63 of them) are named by group: `fx_…` static (18), `fc_…` on the carriage (31), `fh_…` on the head (10), plus the probe board's `probe_m3…` and `probe_nut…` (4). Only heads that could foul another part are modelled. In the GLB they are merged into one mesh, `fasteners`, whose extras list the 63 names.
- `held_slab` and `held_label` are the slab on the cup, in the `carry` and `station` poses.
- Counts: 239 parts in `home` and `pick_bottom`, 243 in `carry`, 241 in `station`.

Printed parts have the same name in the assembly and in `out/printed/`, with three exceptions:

- `station` in the assembly is `station_mount` in the files.
- `belt_clamp_L` and `belt_clamp_R` are two prints of `belt_clamp`.
- Some files are not in the machine assemblies: the build and loading tools (`awl_adapter`, `drill_template`, `fill_gauge`) and the Phase 0 test jig (`phase0_base`, `phase0_occluder_mount`, `phase0_top`, `phase0_tray`).

What each part is (bought, cut, printed, wood, sheet)

The `kind` field of every part:

kind	Count (home)	What it is
`purchased`	37	Bought parts: aluminium extrusions, MGN12 rails and carriages, NEMA 17 motors, T8 lead screw, belt and pulleys, cameras, LED rings, switches, slotted optical sensors, vacuum cup, spring buffer, feet. **Modelled from catalog or listing dimensions as simplified envelopes: accurate where they meet another part, not in their detail.** The extrusions and the lead screw are also cut to length (`out/cut_list.csv`). `gate_beam` is the overfill laser beam, drawn as a thin red transparent rod.
`made`	2	Cut from 1 mm aluminium sheet: `probe_flag` (flat blank in `out/probe_flag_blank.dxf`/`.svg`, bent 90° and drilled) and `carry_flag` (flat). Blank sizes are in `out/cut_list.csv`.

kind	Count (home)	What it is
`printed`	19	3D printed. Material and settings are in each part's note and in `out/printed/print_list.json`. There are 25 printed files (27 prints) including tools and the Phase 0 jig.
`wood`	83	`bed_plywood` (18 mm plywood) and the 82 guide rods: hardwood dowels, 5/16 in (7.94 mm) in the cut list, modelled at 8.0 mm (`ROD_D`).
`foam`	13	Pocket floor pads, 3 mm EVA + microfiber (`PAD_T`).
`slab`	22	Graded slabs and their labels at the nominal PSA size (80.3 × 135.5 × 6.6, `dims.py`), there to check clearances. They are not parts to make.
`fastener`	63	Screw and nut heads that could foul something. Not every fastener in the machine is modelled.

Not in the CAD: frame corner brackets and joining plates, cables, the vacuum tube, the electronics panel, and the optional light roof and rear brace (`model.machine(roof=True)` adds a roof envelope; no export uses it).

Files

Source (`cad/*.py`)

File	What it does
`layout.py`	Master dimensions: pockets, rods, heights, travel, hard stops, gantry stack-up, camera window, carry sensor. `python3 layout.py` prints the pockets and runs the static layout checks.
`dims.py`	Slab dimensions (nominal and max).
`model.py`	Builds every part from `layout.py` and lists the printed parts (`printed_catalog()`), the cut list and the probe-flag blank.
`build.py`	Exports `out/` (below). `build.assembly(pose)` is the part list for a pose.
`export_glb.py`	Writes the GLB viewing files and checks them with trimesh.
`checks.py`	The 177 geometry checks.
`drawings.py`	The dimensioned drawing set (`out/drawings/`).
`render_docs.py`, `render_parts.py`	The PNG images in `docs/img/`.

Outputs (`cad/out/`)

File	What it is
`assembly/slabrail_<pose>.step`	The full assembly in each pose (`home`, `pick_bottom`, `carry`, `station`). One named, coloured product per part, in machine coordinates. About 4.9–5.0 MB each.
`assembly/slabrail_home.glb`, `slabrail_carry.glb`	The same assemblies as meshes for viewing (0.25 mm / 0.3 rad tessellation, about 1.9 MB each). Use them to look at the design. Measure on the STEP files.
`printed/<part>.step`, `printed/<part>.stl`	The 25 printed parts, one solid each, in print orientation, standing on Z 0 with the bounding-box corner at the origin. These files are not in machine coordinates. STL tessellation: 0.05 mm / 0.1 rad.
`printed/print_list.json`	Per printed part: `qty`, `size_mm` (bounding box), `volume_cm3`, and a print note (material, orientation, walls, infill).
`cut_list.csv`	Stock cut to length: profile, use, length_mm, qty. Covers extrusions, the lead screw, the two aluminium flag blanks, the Phase 0 jig posts and stub, and the 82 dowels.
`probe_flag_blank.dxf`	Flat outline of the probe flag (mm).
`probe_flag_blank.svg`	The probe flag drawing with dimensions, bend line, hole and steps. Print at 100 % and check the 10 mm bar.
`checks.txt`	Output of the last `checks.py` run: one line per check (PASS/FAIL, value, limit, the closest pair).
`drawings/SlabRail_drawings.pdf`	The drawing set. `drawings/png/` holds previews and `drawings/dims.json` lists every dimension shown and its source.

File	What it is
`heads_cad.json`	Head shadow boxes and camera offsets for the software simulator. Not a review file.
`renders/`	OpenSCAD renders and their temporary STLs (not in git).
`export.log`, `render_docs.log`	Logs of earlier runs (not in git). Not needed.

Opening the files

- **STEP** (assemblies and printed parts): Fusion 360 (open or upload the file), Onshape (Import), FreeCAD (File > Open). Parts arrive with their names and colours. The assemblies contain 241–245 solids each.
- **GLB**: any glTF viewer. Drop the file on https://gltf-viewer.donmccurdy.com, open it in Windows 3D Viewer, or import it into Blender (File > Import > glTF 2.0). Slabs are semi-transparent.
- **STL**: any slicer. Millimetres, already in print orientation.
- **DXF / SVG**: any CAD or vector program. The SVG is 1:1 at 100 % print scale.

Regenerating

Run everything from the `cad/` folder. You need Python 3 with CadQuery. `export_glb.py`'s self-check also needs trimesh, `drawings.py` needs matplotlib, and the renders need OpenSCAD and `xvfb-run`. Tested here with CadQuery 2.8.0 and trimesh 5.1.1.

Command	What it does
`python3 build.py`	Re-exports the printed STEP/STL and `print_list.json`, `cut_list.csv`, the probe-flag DXF/SVG, `heads_cad.json` and the four assembly STEPs, **then renders the PNGs in `out/renders/`**. Without OpenSCAD it stops with "openscad not installed" after the exports are written. `--renders` renders only.
`python3 export_glb.py`	Writes `slabrail_home.glb` and `slabrail_carry.glb` and checks them (mesh count, bounds against the STEP). Name poses to export others: `python3 export_glb.py station`.
`python3 checks.py > out/checks.txt`	Runs the 177 geometry checks (about 12–17 min; 16 min 43 s on 27 Oct on a shared 4-core machine) and prints one line per check. Exit code 1 if any check fails.
`python3 drawings.py`	Writes the drawing set to `out/drawings/`. Re-measures every dimension and exits 1 if one disagrees with the model.

Every STEP header carries a timestamp, so `build.py` changes every STEP file in git even when the geometry has not changed. Re-export only after changing the model.

Checks

`checks.py` runs **177 checks; all 177 pass** [Certain, re-run on 27 Oct 2026 after the mesh fix below; the numbers are in `out/checks.txt`, unchanged from the 23 Oct run]. It measures exact minimum distances and overlap volumes (OpenCascade) at every pick, place, travel, station, hard-stop and Phase 0 pose: all-pairs interference, the held slab against everything above it, rail margins, camera sight lines, the face-down card window, LED rings, the probe, the carry sensor and home switches, and printed-part walls. `docs/DESIGN.md` ("Verification") summarises the results.

Not checked (`docs/DESIGN.md`): belt slack, cables and the vacuum tube, the real shapes of bought parts, loads and deflection, the side load the strikers put on the switch levers, and the corner brackets and joining plates.

Export check (27 Oct 2026)

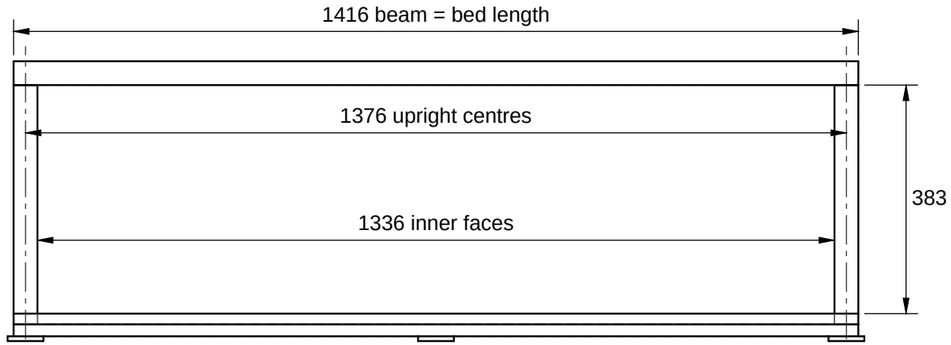
All exports were re-made on 27 Oct 2026 after the mesh fix below, then checked against the current model:

- **Assembly STEP files are current.** For each pose, the STEP re-imported with CadQuery has the same number of solids as `build.assembly(pose)` (home 241, pick_bottom 241, carry 245, station 243). The bounding box and volume of every solid also match. Compared with the previous export, the only changed solid is `station` (`station_mount`): +14.3 mm³.
- **GLB files**: 177 meshes (home) and 181 (carry), 55,814 and 55,862 triangles. Each is one mesh per non-fastener part plus the merged `fasteners` mesh. Bounds reloaded with trimesh equal the STEP bounding box (0.0 mm difference).
- **Printed parts**: all 25 STLs are closed (watertight), one body each, with no non-manifold edges. Each STL's bounding box equals `print_list.json` `size_mm` within 0.05 mm (`size_mm` is rounded to 0.1 mm; the largest difference, 0.05 mm, is on `x_bracket` and `z_motor_mount`), and each STL's volume rounds to the listed `volume_cm3`. All 25 STEPs are one valid solid. `print_list.json` did not change.

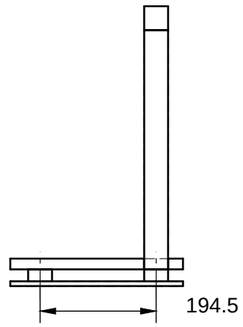
Mesh fix. The earlier `station_mount`, `phase0_base` and `phase0_tray` STLs were not clean meshes. The STEPs were valid; the cause was in the model, and `model.py` now avoids it without moving any functional feature:

- `camera_mast()` (`station_mount`, `phase0_base`): the four 5 mm M2 bosses (centres at X ±10.5) touched the inside of the mast wall (X ±13) along a line, which left four non-manifold edges. Each boss now has a 3 mm wide web that runs 0.5 mm into the wall. The web lies at |X| ≥ 12.5, outside the 25 mm camera board, so the board sits on the same boss tops. +14.3 mm³ per part.
- `phase0_tray`: the 45° lead-in taper started on the recess wall, which left four 0.2 mm sliver faces at the recess corners (the STL split into 5 bodies). The taper now starts 0.5 mm lower and 0.5 mm inset, inside the recess, so the same 45° faces run from Z 3 to the top. The lead-in's rounded corners are 0.5 mm larger in radius (+0.6 mm³ in all).

The camera, hole positions, mast outline, recess size and depth, the lead-in angle and every size and volume in `print_list.json` are unchanged.



FRAME, FRONT (1:10)



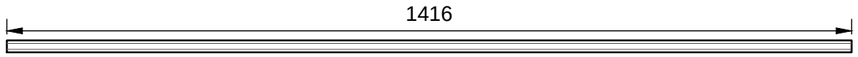
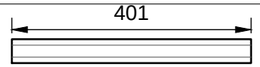
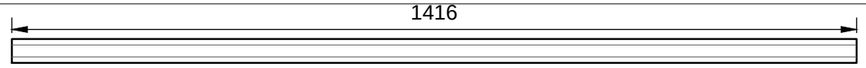
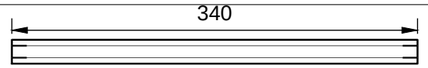
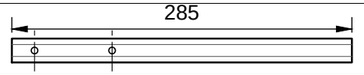
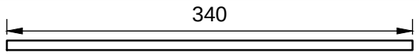
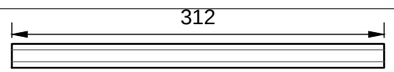
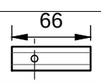
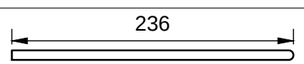
RIGHT (1:10)

Frame set-up (BUILD_GUIDE 4.5; tolerances quoted)

Base rails: centre distance (RIGHT view) +/-0.5 at both ends; diagonals equal within 1. Upright inner faces (FRONT) +/-0.5, equal at the bottom and the top [Certain, CAD]. Bed top to beam bottom +/-0.5, equal at both ends [Certain, CAD]. Uprights stand on the back base rail through notches in the bed (D-03); the beam sits on the uprights, flush with their outer faces. Joints (L brackets, back joining plates) are NOT in the CAD: BUILD_GUIDE 4.3 [Guessing on the exact layout].

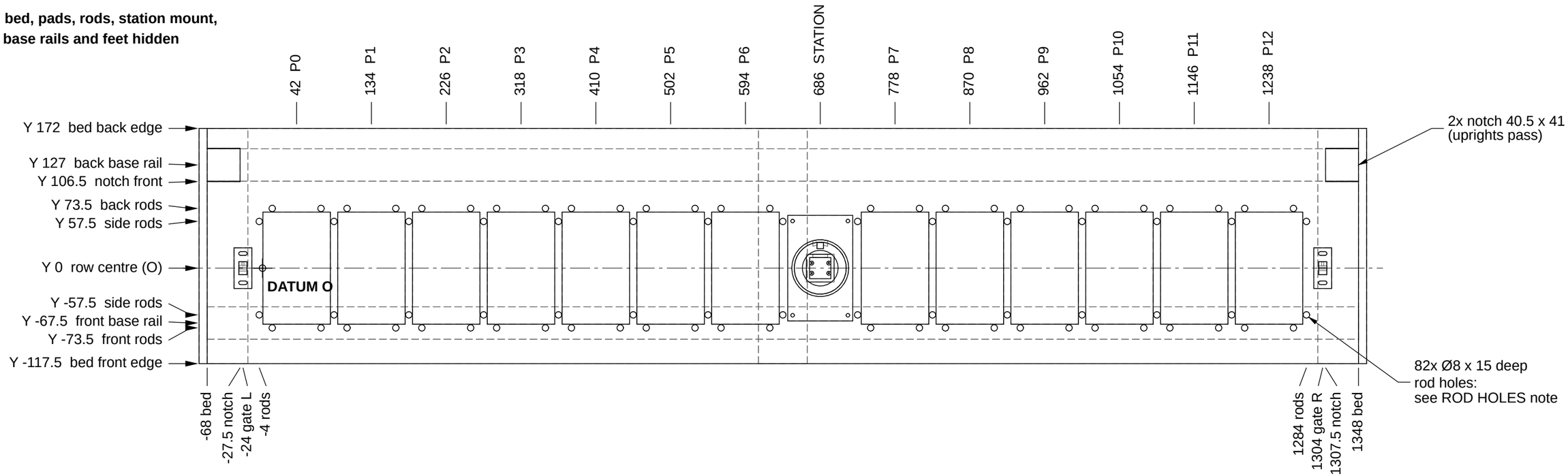
CUT LIST (cad/out/cut_list.csv; extrusion cuts +/-0.5 with square ends, BUILD_GUIDE 4.1)

Extrusions: 20-series, slot 6 (Misumi HFS5, BOM.csv)

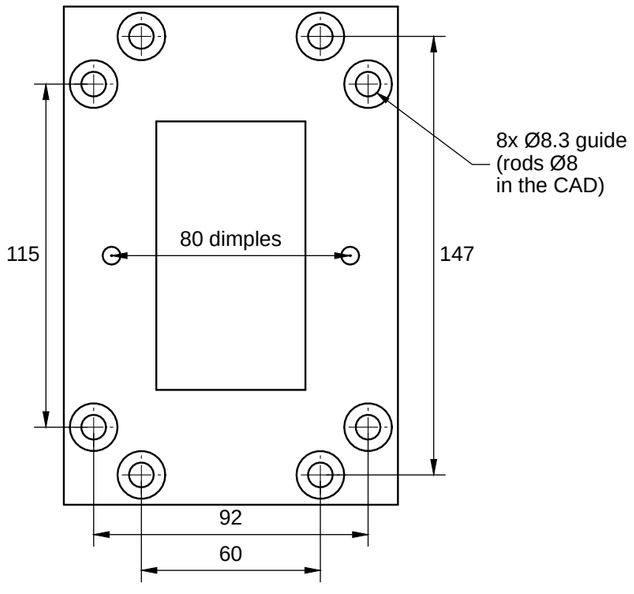
Item	Bar (length from the CAD; scale per row)	Profile / stock	Use, ends, holes	Qty	Tolerance
C-01		2040	base rail, lying flat under the bed; square ends	2	+/-0.5
C-02		4040	upright; square ends	2	+/-0.5
C-03		4040	X beam, on top of the uprights; square ends	1	+/-0.5
C-04		2040	Z spine; BOTH ends: 2x M5 tapped, the two centre bores (20 apart), for 2x M5x12 button heads (bottom) and 2x M5x16 (top), BUILD_GUIDE 7.1, 7.3	1	+/-0.5
C-05		2020	head drop arm; 2x Ø5.5 THRU across the arm (machine X), 19 and 84 from the end in head_foot's socket: match-drill (BUILD_GUIDE 7.5)	1	+/-0.5
C-06		T8x8	T8x8 Z screw, length below the motor face: cut the motor's 400 mm screw; 311-356 works (cut_list.csv)	1	311-356
C-07		1 mm Al sheet	probe flag blank: D-06	1	D-06
C-08		2020	Phase 0 jig post; cut square	2	+/-0.5
C-09		2020	Phase 0 jig arm stub; cross hole 19 from one end, match-drilled through head_foot	1	+/-0.5
C-10		1 mm Al sheet	carry-sensor flag: D-06	1	D-06
C-11		5/16 in dowel	guide rod; dome the top R2-3 (ROD_DOME_R; BUILD_GUIDE 5.4), chamfer the bottom lightly	82	PROPOSED +/-0.5

SlabRail - graded-slab batch scanner	**SHEET D-02 of D-17**
Frame and extrusion cut list	SCALE 1:5, 1:10 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

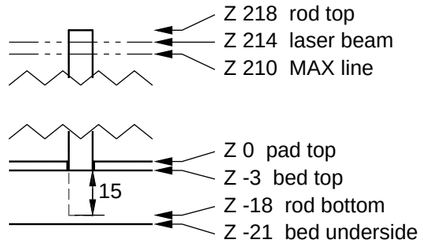
PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails and feet hidden



DETAIL C (1:2): one cell's rod holes = drill_template
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)



DETAIL D (1:2): rod in the bed,
front, broken



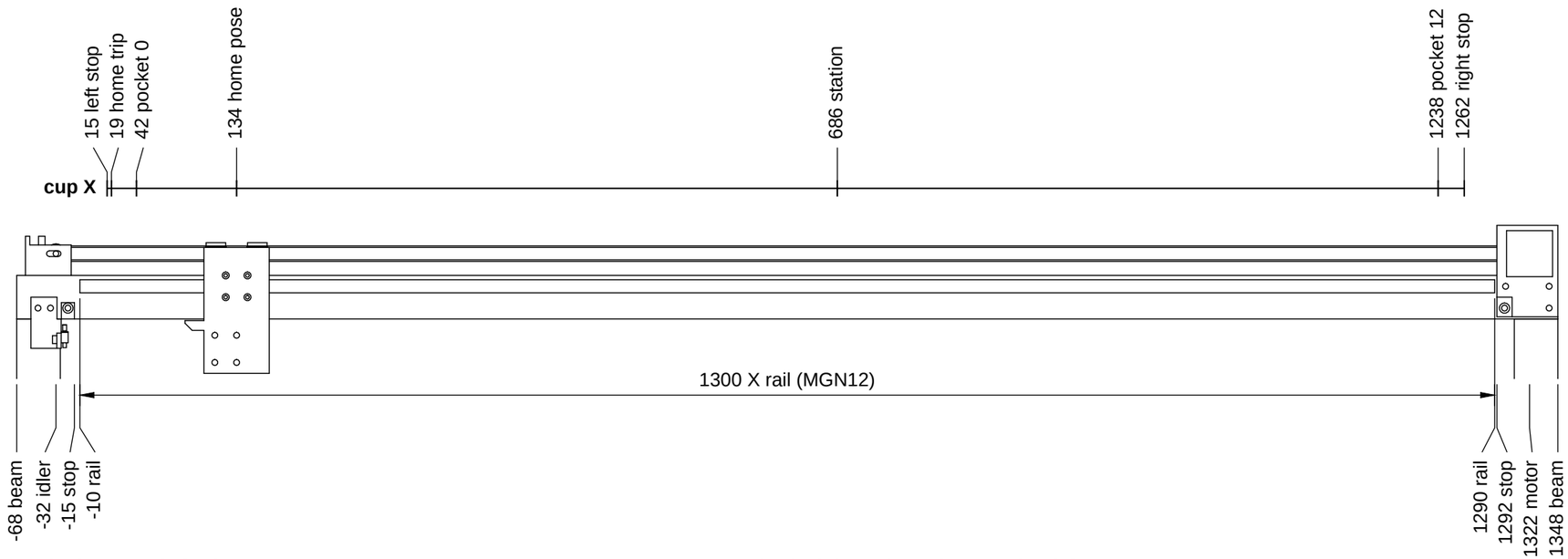
Bed and pads

Item	Size (from the CAD)	Note
Bed, plywood (W)	1416 x 289.5 x 18	cut on site; test-drill for voids first (BUILD_GUIDE 5.1)
Pads (EVA + microfiber), 13	83 x 138 x 3	spray adhesive, after the rods (5.9)
Rods (dowels), 82	D-02 item C-11	glued; plumb within 1 over 200 [Guessing target] (5.7)

Datum O = machine origin, X 0 and Y 0 (the pocket-row centre line), as the software uses them; X ordinates from O. The rod holes are NOT laid out from the bed edges: the machine drills them through drill_template, so beam skew and belt stretch are built in (BUILD_GUIDE 5.6). Station mount: cell 7, four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

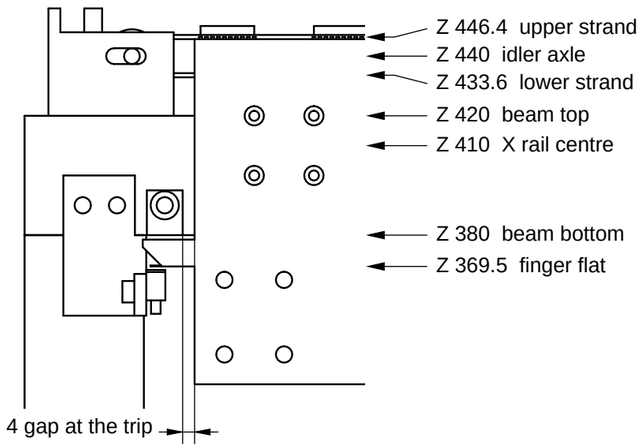
ROD HOLES: 82x Ø8 (CAD ROD_D; drill with the brad-point bit that matches the dowels: 5/16 in for the BOM's 7.94 dowels), 15 deep (ROD_EMBED), square to the bed (guide block, BUILD_GUIDE 5.6). Side rods on the cell wall lines, first X -4, then every 92, at Y ±57.5; front/back rods in pairs 60 apart about each pocket centre, at Y ±73.5; none front/back in the station cell 7.

SlabRail - graded-slab batch scanner	**SHEET D-03 of D-17**
Bed, pockets and guide rods	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**



FRONT (1:5)

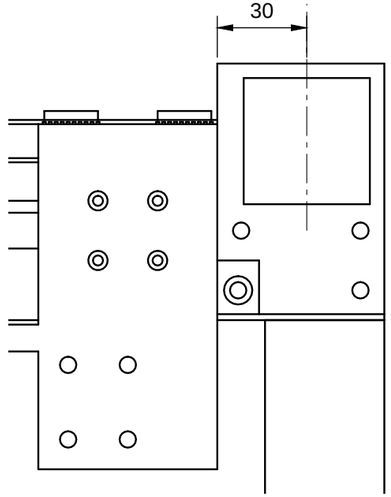
DETAIL A (1:2): left end, carriage at the home-switch trip
(cup X = X_HOME_SWITCH; front view)



Click at cup X 19 (+/-2, BUILD_GUIDE 6.8)

FRONT: X axis with the carriage at home (cup over pocket 1); Z axis and head not shown. cup X = machine X of the suction-cup axis. Belt: GT2 loop over the beam top, both ends clamped on the x_bracket tab (BUILD_GUIDE 6.5).

DETAIL B (1:2): right end, carriage on the right stop
(cup X = X_HARD_STOP[1]; front view)



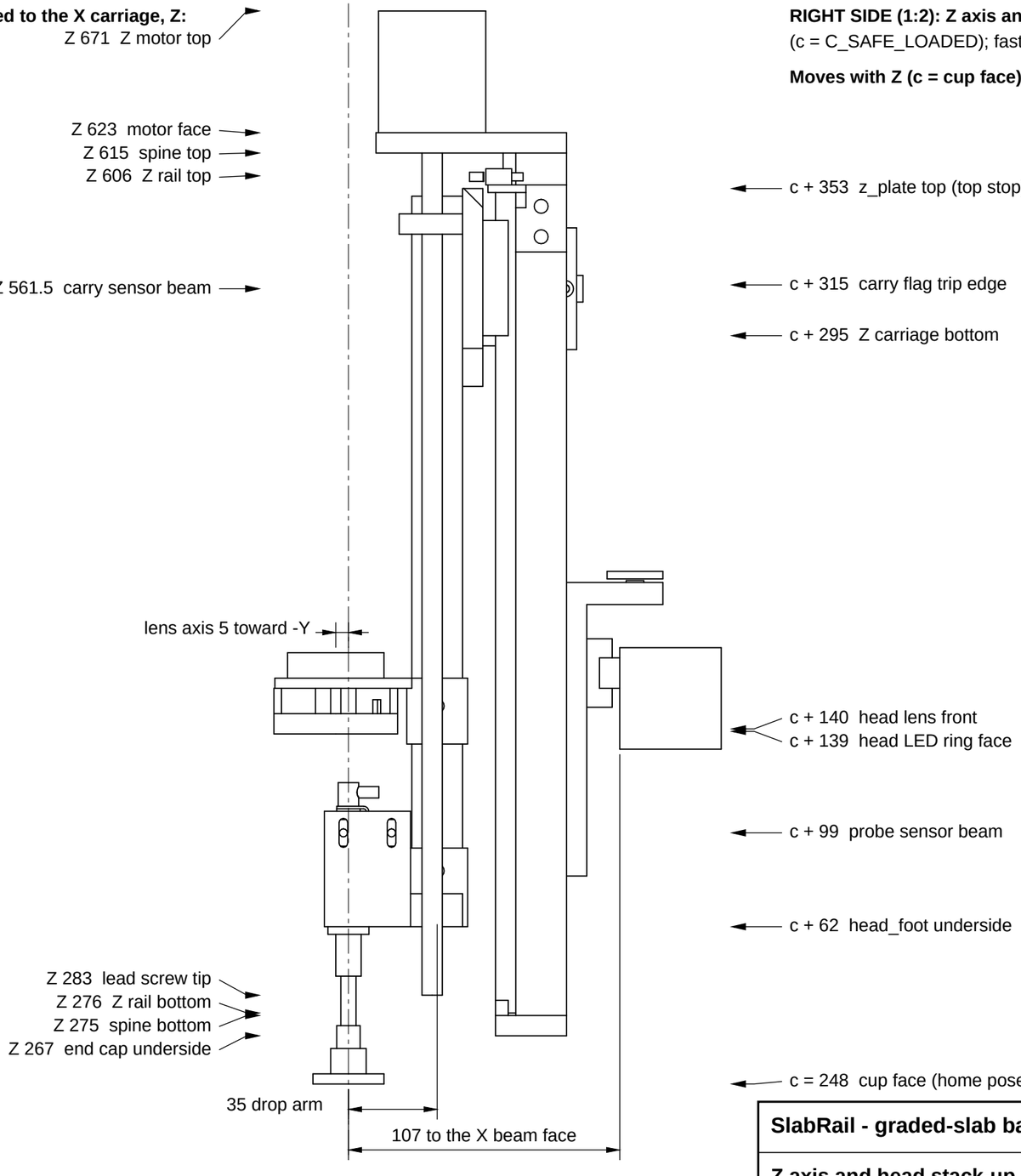
x_bracket lands on the mount's left end

Y stack-up, cup axis to the X beam's front face

Constant (layout.py)	mm	Part
ARM_TO_CUP	35	drop-arm centre to cup axis
EXT20 / 2	10 (of 20)	drop arm (2020): half = 10
Z_PLATE_T	8	z_plate
MGN12_H	13	Z carriage (MGN12H)
EXT20	20	Z spine (2040, 20 deep)
X_BRACKET_T	8	x_bracket
MGN12_H	13	X carriage (MGN12H)
= Y_CUP_TO_BEAM	107	sum: cup axis to beam front face

CAD check: beam front face at Y 107

SlabRail - graded-slab batch scanner	**SHEET D-04 of D-17**
X axis	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

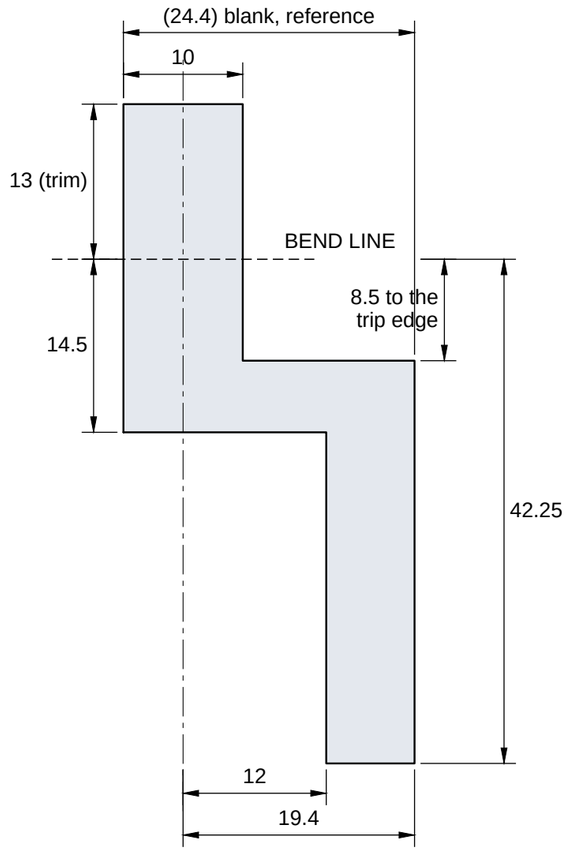


Z-height table (values read from cad/layout.py)

Constant	Value	Meaning
SLAB_T	6.6	c at pick contact = top of the stack: one slab (pose pick_bottom) = SLAB_T; n slabs = n x SLAB_T, nominal
CUP_HANG	2	the free cup hangs this far below c; the probe trips after this much buffer travel
PROBE_TRIP_TOL	0.5	trip tolerance on CUP_HANG (BUILD_GUIDE 7.6)
PROBE_WINDOW	12	max probing travel past the predicted contact
BUFFER_STROKE	20	spring buffer stroke (SMC ZPT25UUK20-B5-A10)
STATION_LENS_Z	23	station lens front above the pad top
STATION_READ_C	171.6	c at the station read = STATION_LENS_Z + CAM_TO_SLAB + SLAB_T + CUP_HANG
CAM_TO_SLAB	140	both cameras: lens front to the slab face
HEAD_CAM_DY	-5	head lens axis, Y from the cup axis (negative = toward the label end)
POST_H	218	guide-rod tops above the pad top
C_SAFE_LOADED	248	carry height, loaded and empty (C_SAFE_EMPTY is the same)
CARRY_SENSOR_C0	246.5	carry sensor: beam blocked from this c up to the top stop
CARRY_BAND	1	carry sensor band: within it either reading is allowed
C_MAX	250	top of the software's Z travel
Z_SWITCH_C	258	Z top (home) switch trip
C_HARD_TOP	262	top hard stop: z_plate meets z_motor_mount
C_MIN	-12	lowest c the design allows
C_HARD_BOTTOM	-14	bottom hard stop: Z carriage skirts on the z_end_cap lips
Z_CARRIAGE_ABOVE_CUP	295	Z carriage bottom edge above the cup face
Z_SPINE_BOTTOM	275	spine bottom end above the pad top (set +/-1, BUILD_GUIDE 7.1)
Z_SCREW_DX	24	lead screw axis, X from the cup axis
Z_SPINE_OFFSET_X	-10	Z spine centre, X from the cup axis

SlabRail - graded-slab batch scanner	**SHEET D-05 of D-17**
Z axis and head stack-up	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

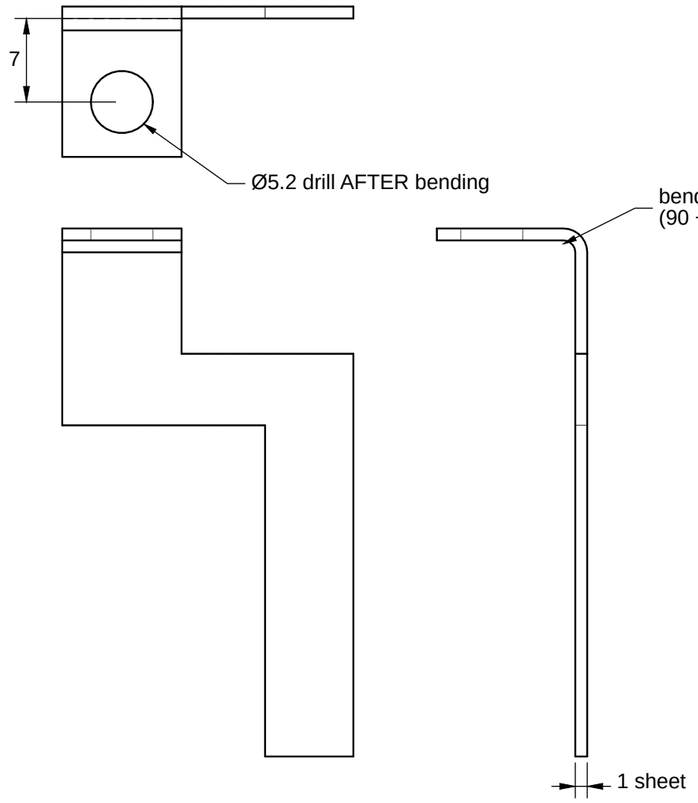
PROBE FLAG BLANK (2:1), flat, before bending
u from the neck centre line (= hole line), v from the bend line



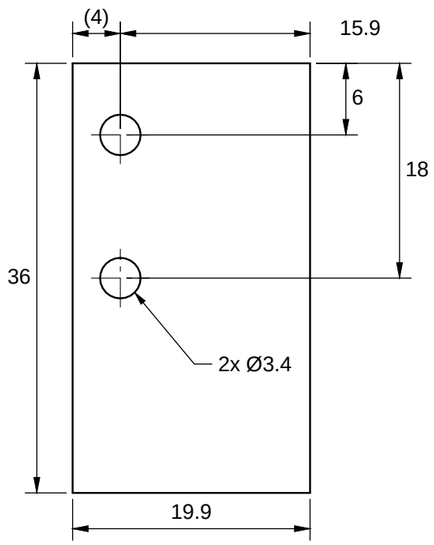
Probe flag (BUILD_GUIDE 7.6; build.py probe_flag_blank.svg notes)

Aluminium sheet, 1 thick (cut_list C-07). 1. Cut and file to the outline; the blade's top edge is the trip edge: straight, square. 2. This view shows the face toward the buffer. Neck in the vise, jaws at the BEND line; bend the tab 90° TOWARD YOU. 3. THEN drill on the centre line, 7 from the plate's inner face. 4. Trim the tab to 2 past the hole. 5. Deburr: it rides in head_foot's guide slot and the sensor slot. Outline: cad/out/probe_flag_blank.dxf (1:1 print: .svg).

PROBE FLAG AS BENT (2:1), for reference
(head frame: X along the row, Z above the cup face)



CARRY FLAG (2:1), flat
(seen from the front; top edge = trip edge)



Carry flag (BUILD_GUIDE 7.4a)

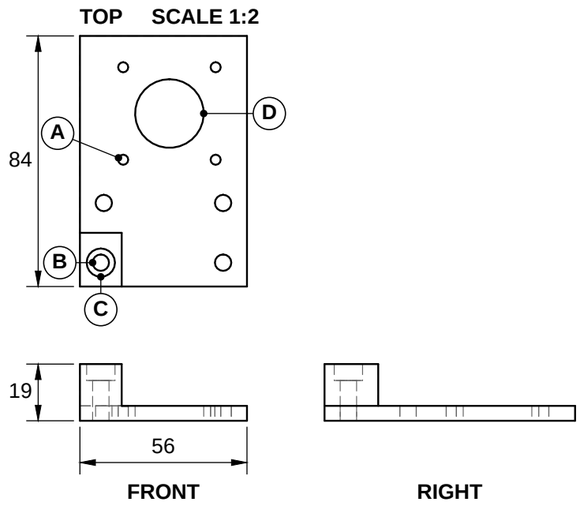
Aluminium sheet, 1 thick, flat (cut_list C-10). File the top edge straight. Fix to carry_flag_arm's leg end with 2 x M3x8, inner edge toward the spine, top edge flush with the arm's top.

Tolerance: outlines PROPOSED +/-0.2 (no doc states one); bend 90 +/-0.5° (BUILD_GUIDE 7.6). The trip points are set by adjustment on assembly (probe 7.6, carry 7.4a), not by these outlines.

SlabRail - graded-slab batch scanner	**SHEET D-06 of D-17**
Flat parts (probe flag, carry flag)	SCALE 2:1
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_motor_mount

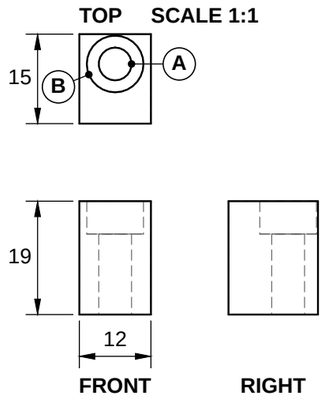
Printed part (P), qty 1, cad/out/printed/x_motor_mount.step. Print: PETG, beam face down (as exported), 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, centres spread over 41 x 20 (X x Y), not a grid: M5 clearance
C	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk
D	Ø23 THRU, axis Z: NEMA 17 pilot clearance
E	Left end = right X hard stop face, 30 from the motor axis; motor axis 20 above the beam top

x_bumper_L

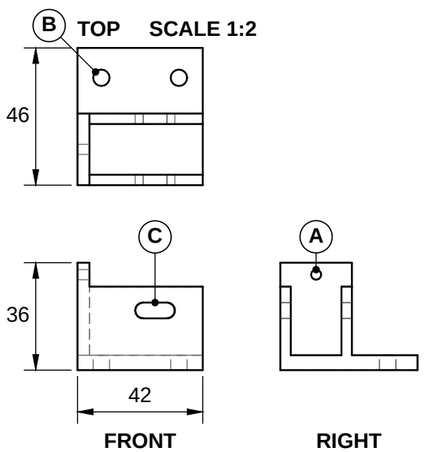
Printed part (P), qty 1, cad/out/printed/x_bumper_L.step. Print: PETG, beam face down (as exported), 4 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Z: M5 clearance
B	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk

x_idler_mount

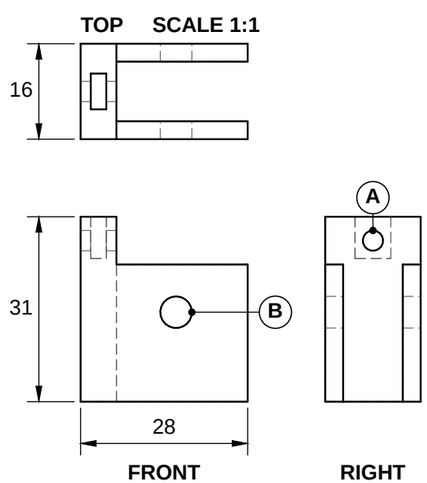
Printed part (P), qty 1, cad/out/printed/x_idler_mount.step. Print: PETG, foot down, 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU, axis X: M3 clearance
B	2x Ø5.5 THRU, axis Z, 26 apart in X: M5 clearance
C	2x slot 5.3 wide x 13.3 long, axis Y: M5 idler axle (tension travel)

x_idler_slider

Printed part (P), qty 1, cad/out/printed/x_idler_slider.step. Print: PETG, as exported, 5 walls, 50% infill



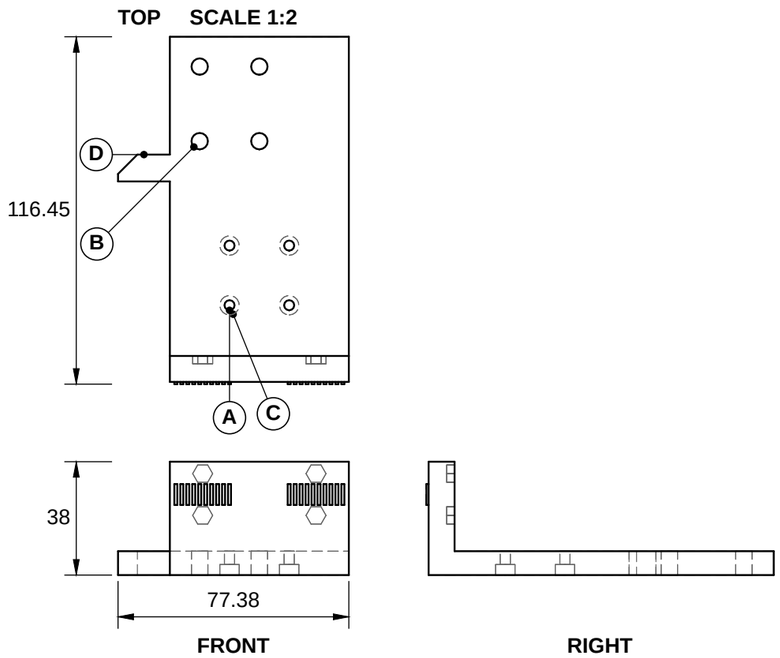
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU (2 walls on one axis), axis X: M3 clearance
B	Ø5.3 THRU (2 walls on one axis), axis Y: M5 axle
C	M3 nut slot 2.6 wide, from the top, in the bridge

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-08 of D-17**
Printed parts: X drive	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_bracket

Printed part (P), qty 1, cad/out/printed/x_bracket.step. Print: PETG, print flat on its spine face (as exported), 6 walls, 50% infill; the finger off its left edge works the X home switch



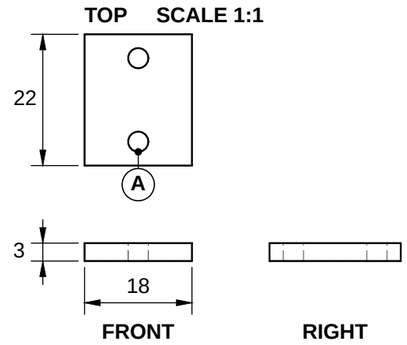
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, on 20 x 25 (X x Y): M5 clearance
C	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
D	X-switch finger 17.38 past the left edge; flat underside at machine Z 369.5; 45° lead-in 6.5 deep
E	GT2 rack on the tab floor, both sides: teeth at 2 pitch (belt teeth down, belt_clamp cap on top)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

belt_clamp

Printed part (P), qty 2, cad/out/printed/belt_clamp.step. Print: PETG, flat, 100% infill

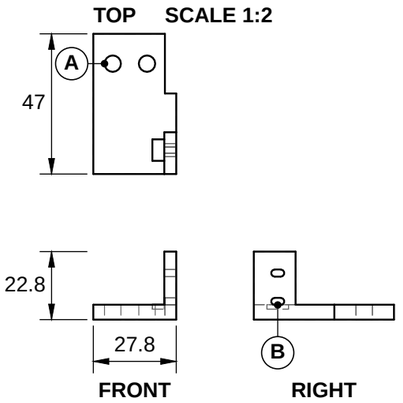
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø3.4 THRU, axis Z, 14 apart in Y: M3 clearance



x_switch_mount

Printed part (P), qty 1, cad/out/printed/x_switch_mount.step. Print: PETG, front plate down (as exported), 4 walls, 40% infill; X home switch on 2 x M2x12, 2 x M5x10 button heads

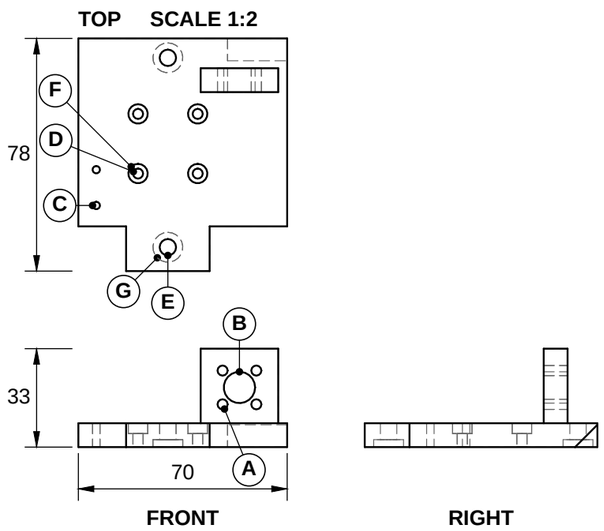
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 11.5 apart in X: M5 clearance
B	2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2



SlabRail - graded-slab batch scanner	**SHEET D-09 of D-17**
Printed parts: X carriage	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

z_plate

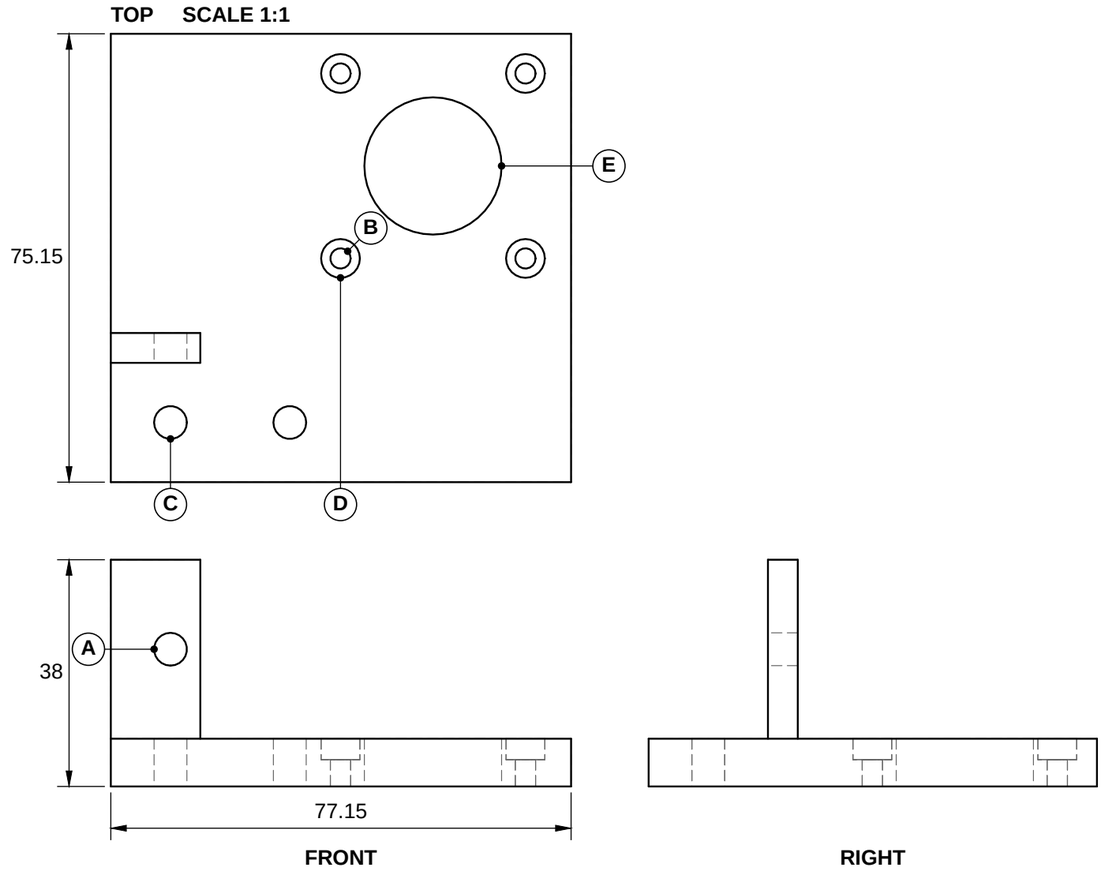
Printed part (P), qty 1, cad/out/printed/z_plate.step. Print: PETG, carriage face down, nut shelf up, 6 walls, 50% infill; the switch chamfer at the top right starts at the bed (45 deg)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Y, on 11.31 x 11.31 (X x Z): M3 clearance
B	Ø10.5 THRU, axis Y: T8 nut body clearance
C	2x Ø2.5 THRU, axis Z, 12 apart in Y: carry_flag_arm, M3x10 self-tapping
D	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
E	2x Ø5.5 THRU, axis Z, 63.5 apart in Y: M5 clearance
F	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
G	2x Ø10 x 2.5 deep, axis Z, 63.5 apart in Y: M5 button-head recess
H	Z-switch chamfer 45°, 7.5 x 7.5, 20 long at the right end of the top back edge
I	Top edge (Z top hard stop) 58 and T8 flange shelf top 48 above the Z carriage's bottom edge

z_motor_mount

Printed part (P), qty 1, cad/out/printed/z_motor_mount.step. Print: PETG, motor face down, tab up, 5 walls, 40% infill



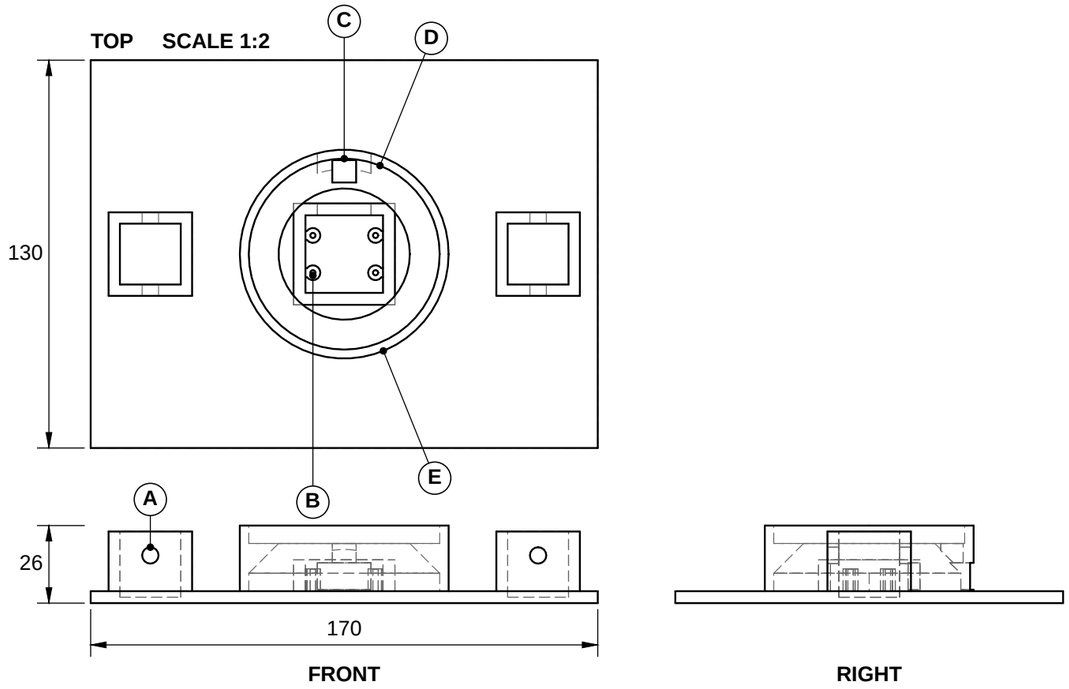
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Y: M5 clearance
B	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
C	2x Ø5.5 THRU, axis Z, 20 apart in X: M5 clearance
D	4x Ø6.5 x 3.5 deep, axis Z, on 31 x 31 (X x Y): M3 head counterbore
E	Ø23 THRU, axis Z: NEMA 17 pilot clearance
F	Plate underside sits on the spine top and is the Z top hard stop; tab hole 15 below it (left slot)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-10 of D-17**
Printed parts: Z carriage plate, Z motor mount	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

phase0_base

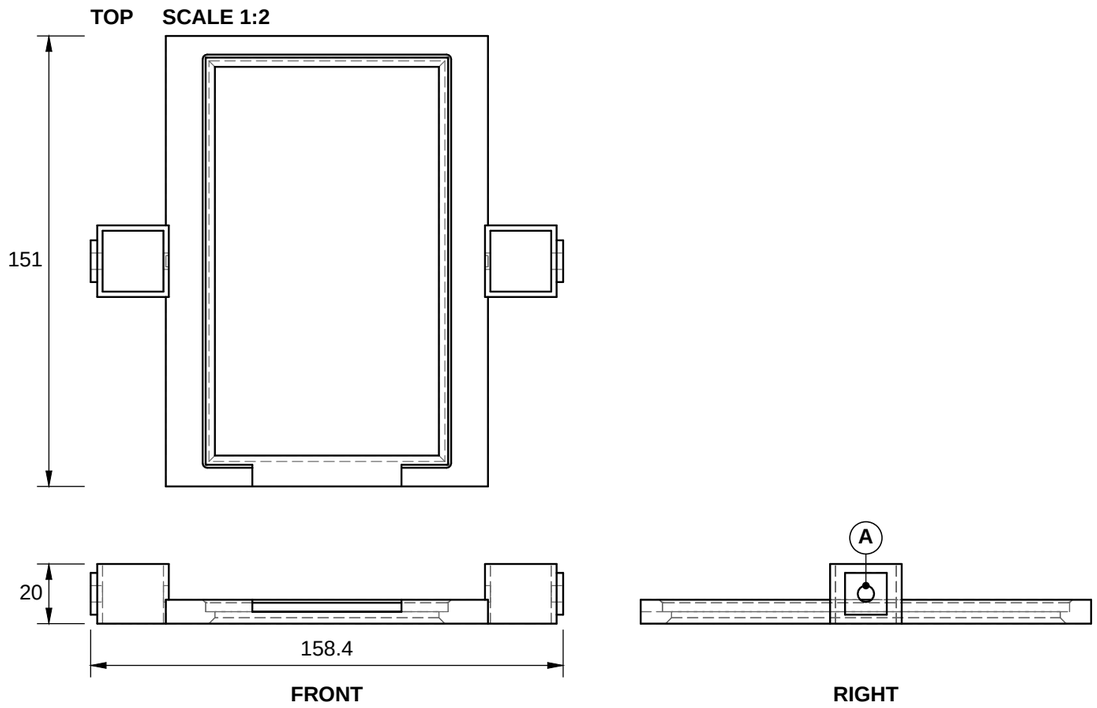
Printed part (P), qty 1, cad/out/printed/phase0_base.step. Print: PLA/PETG black (Phase 0 test jig)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU (2 walls on one axis), axis Y, 130 apart in X: M5 clearance
B	4x Ø1.8 x 7.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping
C	Ø64 x 6 deep, axis Z: hollow under the ring shelf (45° cone roof)
D	Ø64 x 6 deep, axis Z: station LED ring seat bore
E	Ø70 x 22 long (outside), axis Z: ring seat tube
F	Post sockets 20.4 square, floor 2 above the underside, centres 130 apart
G	Ring shelf 20 and socket cross holes 16 above the underside

phase0_tray

Printed part (P), qty 1, cad/out/printed/phase0_tray.step. Print: PLA/PETG (Phase 0 test jig), ledges down



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (4 walls on one axis), axis X: M5 clearance
B	Slab recess 81.3 x 136.5, 4 deep; opening 75.1 x 130.3; the ledges carry the slab's front face
C	Post clamps 20.4 square, centres 130 apart

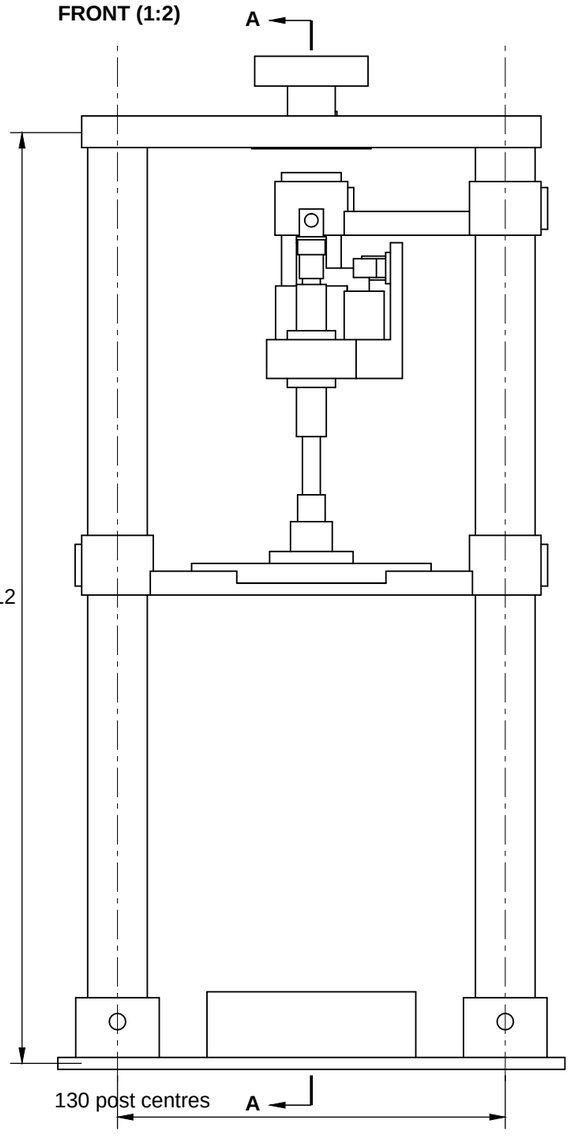
Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-15 of D-17**
Printed parts: Phase 0 jig (base, tray)	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

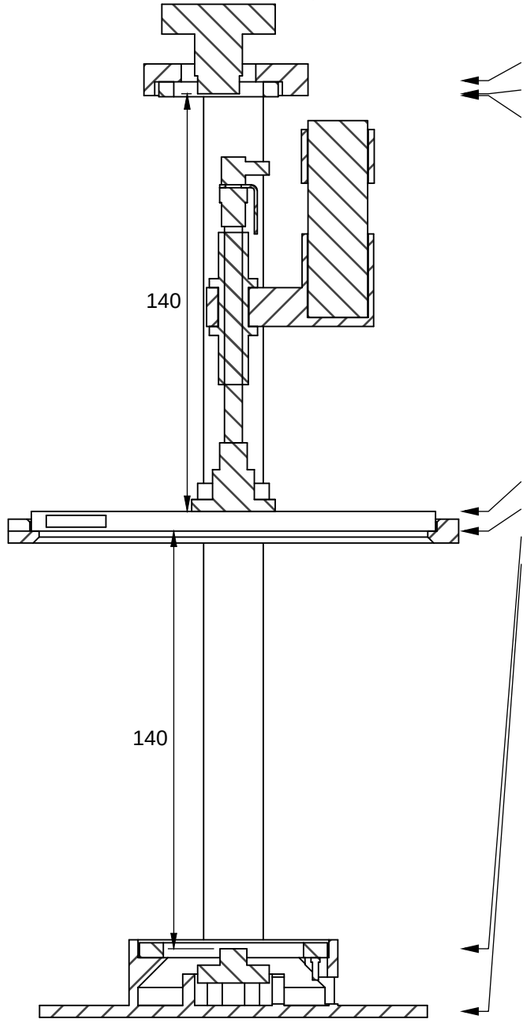
PHASE 0A TEST JIG: one slab between the two cameras, as on the machine (BUILD_GUIDE 1.5); fasteners not shown

Heights above the base underside

FRONT (1:2)



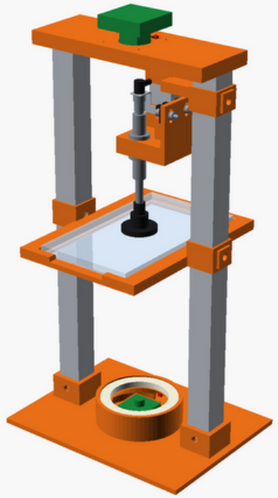
SECTION A-A (1:2)
on both lens axes, seen from +X; cut faces hatched



Z 314 post tops
Z 309.6 head lens front
Z 309 bridge underside

Z 169.6 slab back
Z 163 slab front face (on the ledges)
Z 23 station lens front
Z 2 post bottoms (socket floor)

ISO (render; not to scale)



Part	Type	Qty	Sheet
phase0_base (camera mast, ring seat, post sockets)	P	1	D-15
phase0_tray (the slab on edge ledges)	P	1	D-15
phase0_top (bridge: head camera, head ring)	P	1	D-16
phase0_occluder_mount	P	1	D-16
head_foot, the machine's own (occluder)	P	1	D-12
2020 post, 312, cut square +/-0.5 (C-08)	C	2	D-02
2020 arm stub, 66 (C-09)	C	1	D-02
probe flag (C-07)	S	1	D-06
cameras, LED rings, SMC buffer + cup + elbow, endstop board	B	-	BOM.csv

The posts set the bridge height: cut them square. Both lens-to-slab distances (RIGHT view) equal CAM_TO_SLAB = 140 on the machine (set +/-2, BUILD_GUIDE 1.5). Phase 0 is the go / no-go test before any motion parts are bought.

SlabRail - graded-slab batch scanner	**SHEET D-17 of D-17**
Phase 0 test jig	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**